



GE VERNOVA

ADMS

16.25.1

DNAF Standard Applications Analyst Guide

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Change History

Date	Change Description
May 31, 2023	ADMS 3.16: • Updated section Scaling Loads. • Updated chapter Protection Validation. • Significantly updated Fault Location and subsections, including reordering sections, updating algorithm naming (Radial and Meshed), and various descriptions and configurations information. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
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1. Introduction

GE's Advanced Distribution Management System (ADMS) allows distribution operators and engineers to monitor, control, analyze, and plan the operation of the power distribution system. A key component of the DMS is the Distribution Network Analysis Functions (DNAF), which form an integrated package of network analysis applications. The core network analysis engine is the Distribution Power Flow (DPF). This application generates a large amount of output data that cannot be easily observed by the operator. For this reason, several distribution analysis functions are utilized to analyze the results of the power flow and present summarized information to the operator.

DNAF applications are executed in real-time and study modes. The DPF is also run in the Distribution Operations Training Simulator (DOTS).

The real-time mode utilizes the current power system conditions (SCADA and Advanced Metering Infrastructure measurements) in evaluating the system. These measurements consist of analog and status measurements, including switch statuses. The status measurements can also be a multiphase point which has the individual phase statuses contained in one point. The study mode solves operational studies, using data retrieved either from a saved case, from a real-time snapshot of the system, or from a nominal snapshot of the system.

Voltages and angles from the Energy Management System (EMS) State Estimator solution can also be used in the real-time DPF to initialize the distribution source voltage locations. These locations may be on the source side of the station transformer or at locations in the subtransmission network.

The network analysis functions include the following:

- **Distribution Power Flow (DPF):** This function finds the complex voltages at all nodes and the power flows through all branch devices in the distribution system.

The following functions are executed together with the DPF.

- **Bus Load Allocation (BLA):** This function provides estimates for the loads (kW and kVAR) at the distribution feeder nodes.
- **Limit Monitor (LIM):** This function checks for branch flow overloads, flow imbalances (for three-phase switching devices), voltage imbalances (for three-phase loads), and voltages that are outside of acceptable operating limits.
- **Power Quality Index Analysis (PQI):** This function determines the extent of feeder and individual load voltage violations (including voltage imbalance).
- **Loss Analysis (LA):** This function calculates a device's technical loss and aggregates the losses by feeder and station.
- **Protection Validation (PRV):** This function calculates phase-to-ground, double-phase-to-ground, and three-phase fault currents for each application protection zone, where an application protection zone is bounded by feeder circuit breakers and field reclosers.
- **Short Circuit Calculation (SCC):** This function calculates single phase-to-ground, double

phase-to-ground, phase-to-phase, and three-phase currents for each selected location (power transformer, line, and switching devices).

- **Fault Location (FL):** This function uses fault Amp measurements and fault indicator status measurements to determine potential distribution feeder fault locations.
- **Fault Isolation and Service Restoration (FISR):** This function generates a switching sequence to isolate faulted sections and to restore service to the non-faulted sections.
- **Automated Feeder Reconfiguration (AFR):** This function provides plans to unload overloaded segments. AFR also generates Return to Normal (RTN) switching plans.
- **Planned Outage Switching (PLOS):** This function provides plans to isolate a component, minimizing the number of interrupted customers. This function is available only in study mode.
- **Fault Isolation and Service Restoration Contingency Analysis (FISRCA):** This function executes a FISR analysis for every switchable section in the station. This function is available only in study mode.
- **Load and Volt/VAR Management (LVM):** This function optimizes and coordinates capacitors, voltage regulating transformers (regulators and LTCs), and generator and DER VAR output to maintain system operation within voltage limits. LVM is used as a least-intrusive demand management tool.

This document covers the standard DNAF applications: DPF, BLA, LIM, PQI, LA, PRV, SCC, and FL. For details on the advanced DNAF applications: FISR, AFR, PLOS, FISRCA, and LVM, refer to the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

1.1. Terminology

For a list of terms and acronyms, and their definitions, refer to the *ADMS Glossary of Terms*.

1.2. Network and Device Modeling

The Distribution Network Analysis Functions use the same connectivity model that supports other ADMS functions, including the Network Topology Processor (NTP). The network model consists of topology objects including lines, cables, switching devices (disconnect switches, fuses, breakers, reclosers, sectionalizers, permanent jumpers, and network protectors), distribution transformers, regulators, auto-transformers, capacitors, and other devices.

Network connectivity is represented by device node connections (branch objects such as lines have two nodes, and shunt devices such as shunt capacitors have one node). If two topology devices are connected, each device specifies one common connection node.

Most of the power flow-related device data (including impedance data, limit data, and other information) is given in an associated device model. The device models include conductor models, switch models, series reactor models, and transformer models. These models are loaded into ADMS, and each individual device in the substation references a model (multiple devices may

reference the same model). For more information about device modeling, refer to the *ADMS Modeling Overview and Converter User Guide*.

2. DNAF Configuration Editor

The DNAF Configuration Editor (DCE) allows a user to easily modify the DNAF configuration file, `DNAF_PARAMS.xml`. The `DNAF_PARAMS.xml` file provides key application data, including convergence tolerances, specific application parameters, and other data that is necessary for the applications. The file is read into the ADMS server upon startup and after an online update.

2.1. Using the DNAF Configuration Editor

The editor tool has a separate tab for each DNAF application. All DNAF Configuration Editor tabs that configure standard applications are described in [Displays Overview](#).

2.1.1. Starting the DNAF Configuration Editor

The DNAF Configuration Editor can be started by double-clicking the executable file (`DNAFConfiguration.Editor.exe`) from Windows Explorer, via a desktop shortcut, or from the Start menu (Start> All Programs> Eterra> Distribution> Tools> Configuration Tools> DNAF Configuration Editor). After the application starts, the window shown in figure below appears.

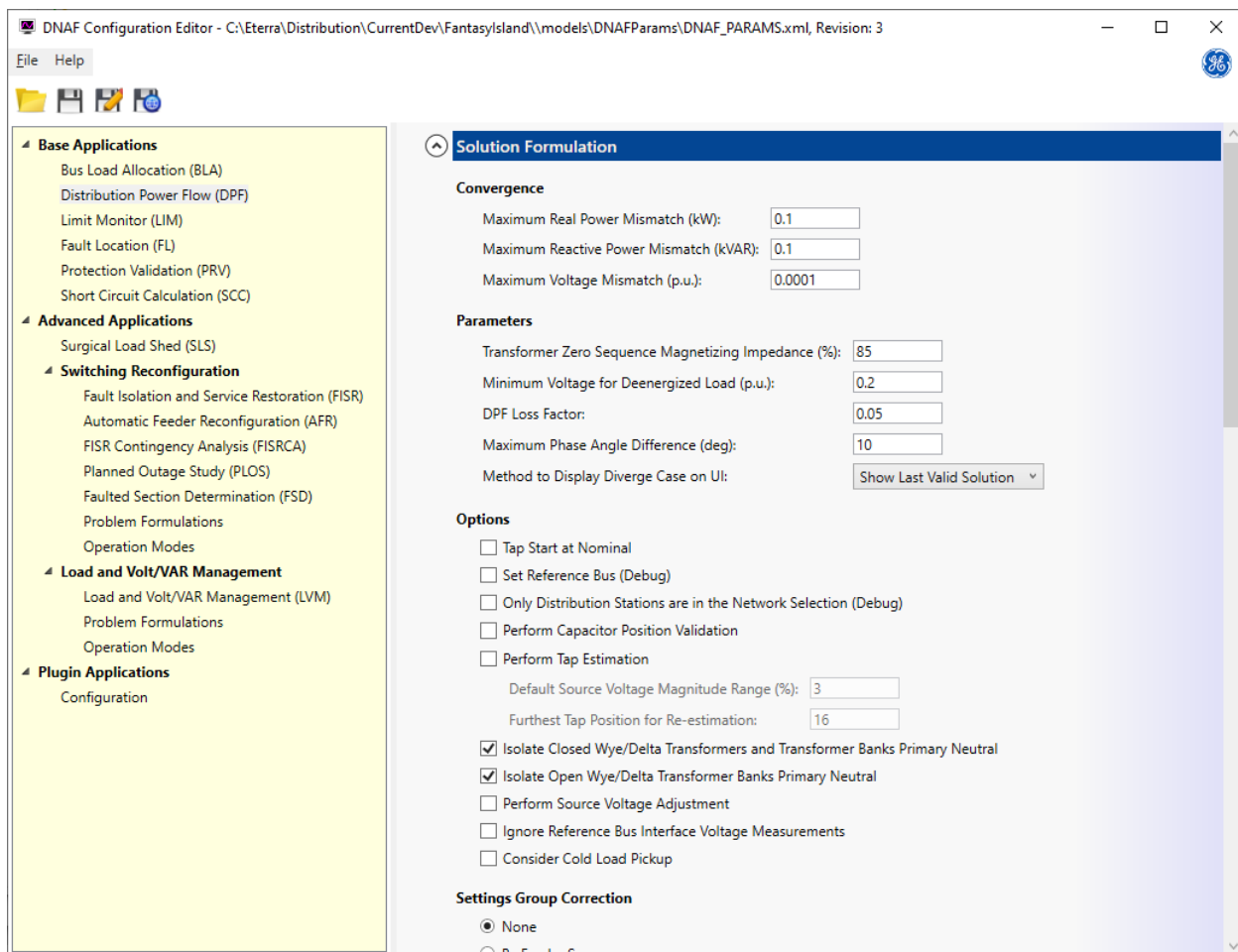
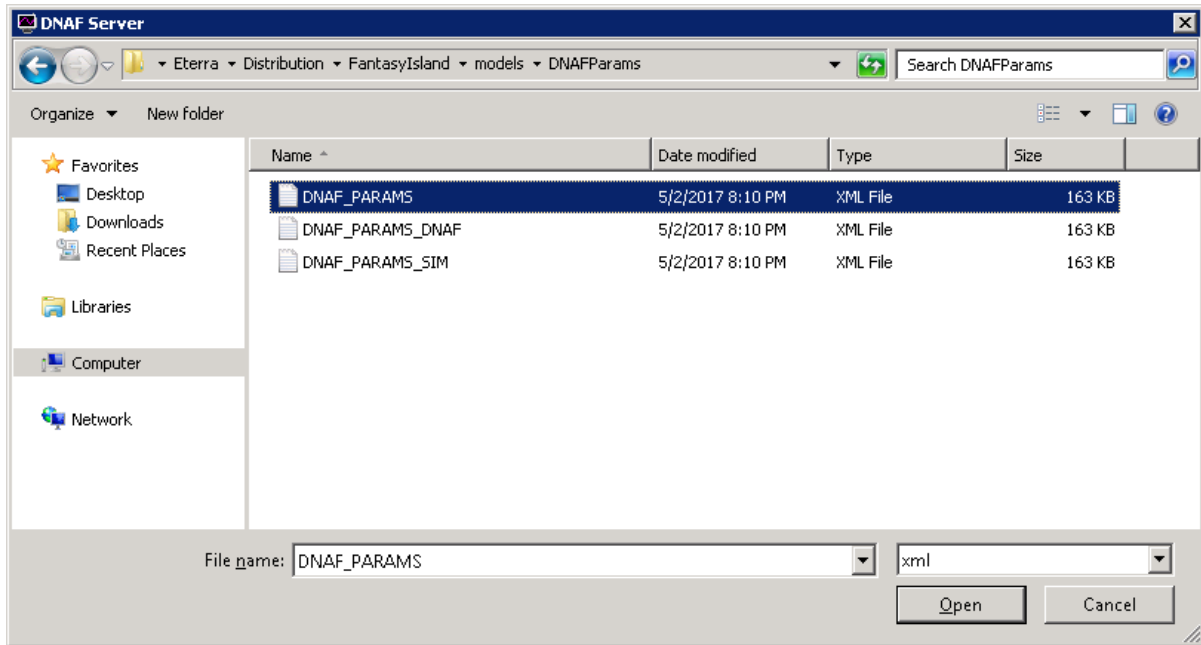


Figure 1. DNAF Configuration Editor

The next step is to load the appropriate DNAF_PARAMS.xml file. This file can reside on the local disk or on a mapped network drive.

To load a DNAF_PARAMS.xml file:

1. From the File menu, click Open (or, click the Open File toolbar icon).
2. Navigate to the appropriate directory, and then double-click the DNAF_PARAMS.xml file, or select the file and click the Open button.



Note

Only files with "DNAS_PARAMS" in their names can be loaded into the editor.

Alternatively, if you defined paths to configuration files for different operation modes (see [Configuration Options](#)), on the File menu, choose one of these options:

- Open Configuration for > DNAS Server
- Open Configuration for > Simulator Server
- Open Configuration for > Study Server

The DNAS_PARAMS.xml file is opened under the selected file location.

2.1.2. Changing Parameters

To edit parameters, select the desired application page from the navigation panel on the left and make the necessary edits.

2.1.3. Saving Your Changes

To save all edits, use one of these methods:

- To save changes locally, click the Save/Save As toolbar button, or select Save/SaveAs from the File menu.

This saves the latest version of the DNAS_PARAMS.xml file.

- To save the edits and immediately apply the changes to the running server, click the Save

Online toolbar button, or select Online Save from the File menu.

The online save function creates a copy of the previous DNAF_PARAMS.xml file with the name DNAF_PARAMS.xml.old, and then saves the new file as DNAF_PARAMS.xml.

2.1.4. Importing Older DNAF_PARAMS.csv Files

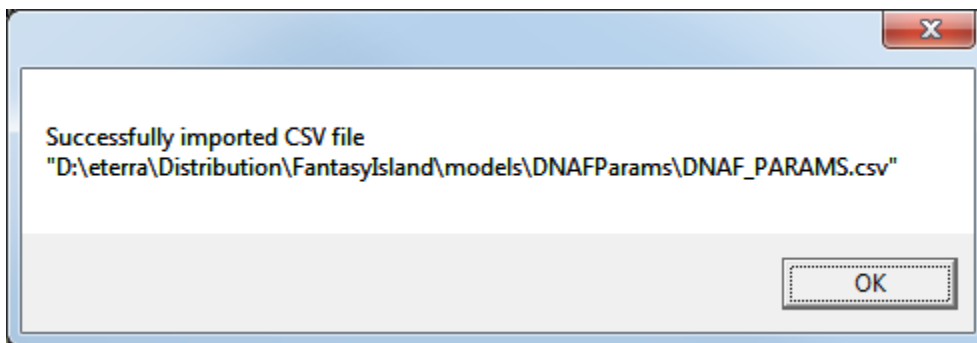
To open an older parameter file that uses comma-separated value format (for example, DNAF_PARAMS.csv):

1. On the File menu, click Import CSV File.

A standard Open dialog box opens.

Navigate to the appropriate directory and double-click the DNAF_PARAMS.csv file, or select the file and click Open.

2. After you successfully perform an import, the following message appears:



2.1.5. Configuration Options

On the File menu, click Options to open the Options window (see figure below).

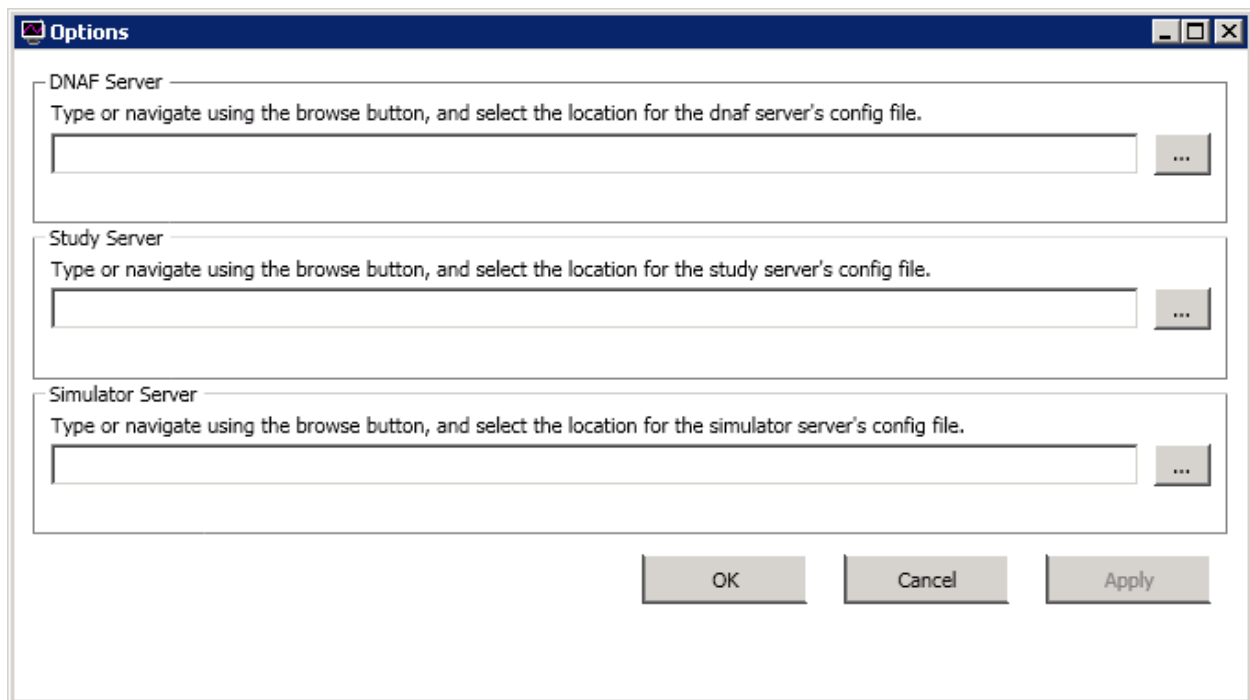


Figure 2. Configuration File Paths

On the Options window, you can configure the paths to the DNAF_PARAMS.xml files for the DNAF server, study server, and simulator server setups.

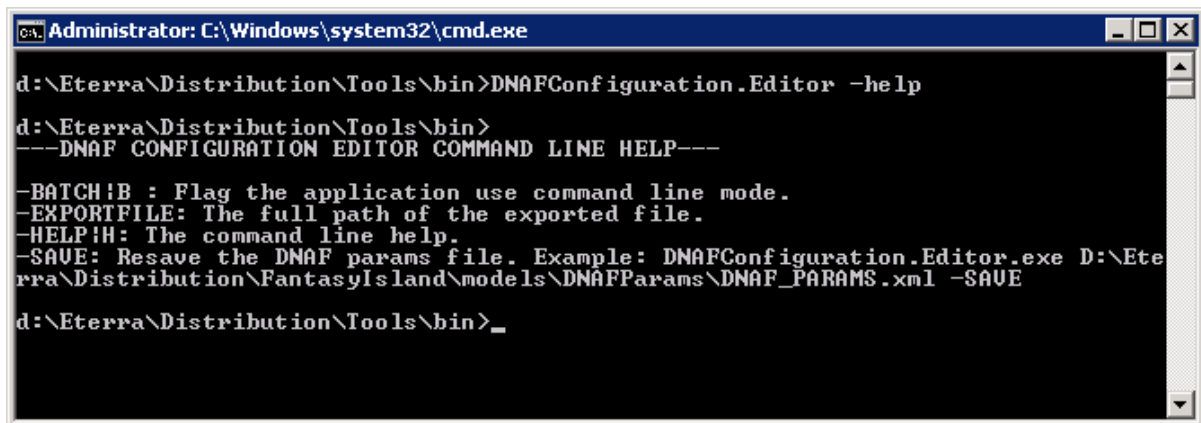
2.1.6. Command-Line Parameters

The DNAF Configuration Editor command-line parameters are as follows:

- **-HELP (or -H):** Shows help information.

For Example:

```
>DNAFConfiguration.Editor -HELP
```



- **-BATCH (or -B):** Specifies that the application runs in command-line mode (with no user interface).
- **-SAVE:** Resaves the DNAF parameters file.

For example:

```
>DNAFConfiguration.Editor D:\Eterra\Distribution\FantasyIsland\models\DNAFParams\DNAFParams.xml
-SAVE
```

- **-EXPORTFILE:** Exports all the parameters' tooltips to an XML file, which can be opened in Microsoft Excel. Figure below shows an example XML file opened in Microsoft Excel.

For example:

```
>DNAFConfiguration.Editor -B -EXPORTFILE >D:\DCEParametersTooltips.xml
```

ApplicationName	ParamDescription	ParamTooltips
Fault Isolation and Service Restoration (FISR)	Default Problem Formulation	Default problem formulation index.
Fault Isolation and Service Restoration (FISR)	Associated Limit Strategy	Associated limit strategy index.
Fault Isolation and Service Restoration (FISR)	P Mismatch (kW)	Convergence - real power mismatch (in kW).
Fault Isolation and Service Restoration (FISR)	Q Mismatch (kVAR)	Convergence - reactive power mismatch (in kVAR).
Fault Isolation and Service Restoration (FISR)	Voltage Mismatch (p.u.)	Convergence -voltage mismatch (in pu).
Fault Isolation and Service Restoration (FISR)	Number of Selected Top Plans	Number of top ranked plans to be fully evaluated during
Fault Isolation and Service Restoration (FISR)	Maximum Plan Duration (min)	Maximum plan duration in minutes. Used to compute a
Fault Isolation and Service Restoration (FISR)	Manual Switch Operation Time (min)	The time in minutes it takes to operate a manual switch.
Fault Isolation and Service Restoration (FISR)	Automatic Switch Operation Time (min)	The time in minutes it takes to operate a automatic swit

Figure 3. Exported Parameter Tooltips

2.2. Displays Overview

This section provides an overview of the DNAF Configuration Editor's displays where you can configure the standard DNAF applications.

2.2.1. BLA Configuration

The Bus Load Allocation (BLA) function provides estimates for the loads (kW and kVAR) at the distribution feeder nodes. [Bus Load Allocation](#) provides detailed information regarding BLA.

This display contains the configuration parameters for BLA. For the detailed list of the configuration parameters, refer to [BLA Configuration](#).

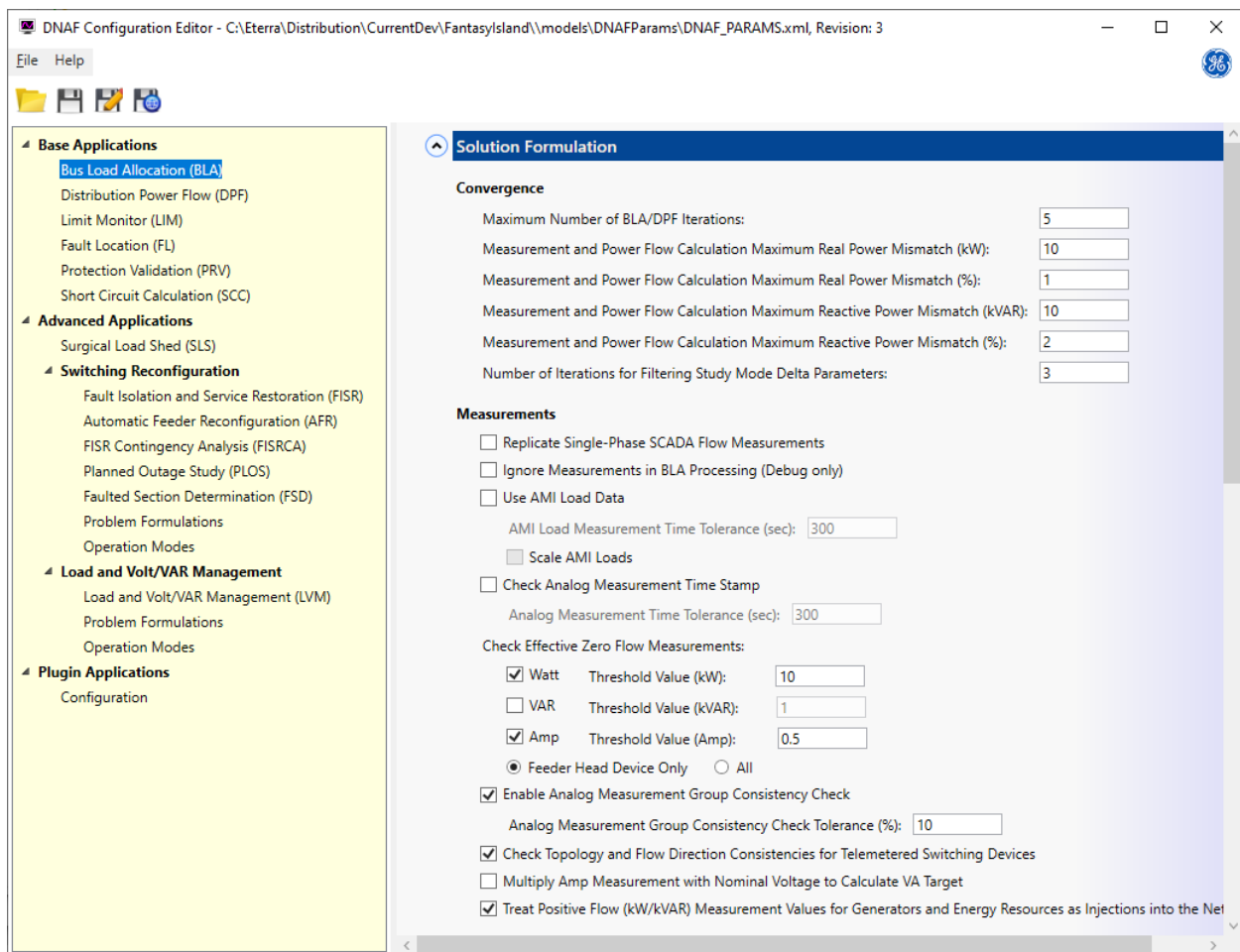


Figure 4. BLA Configuration

2.2.2. DPF Configuration

The Distribution Power Flow (DPF) function finds the complex voltages at all nodes and the power flows through all branch devices in the distribution system. [Distribution Power Flow](#) provides detailed information regarding DPF.

This display contains the configuration parameters for DPF. For the detailed list of the configuration parameters, refer to [DPF Check Before Operate](#).

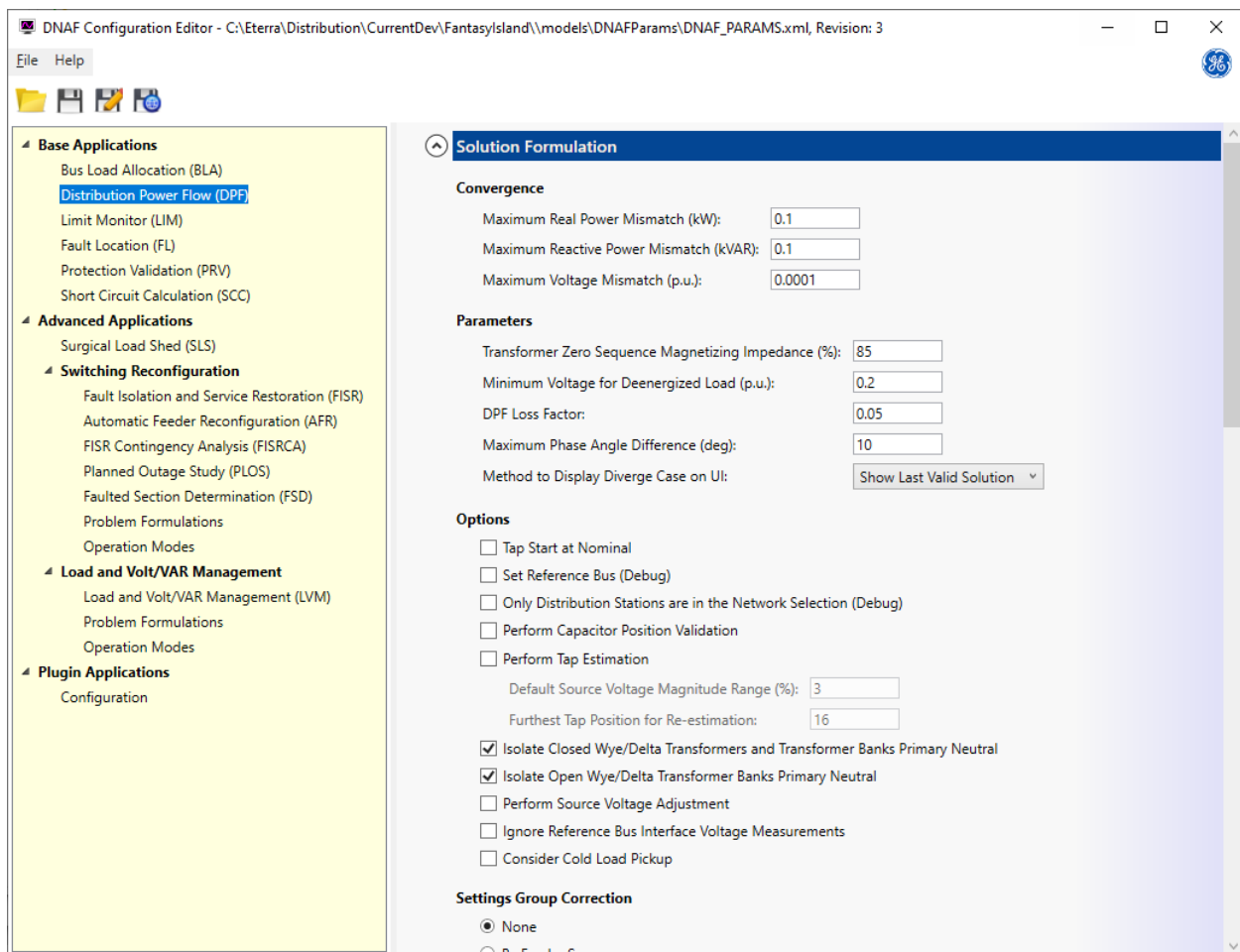


Figure 5. DPF Configuration

2.2.3. LIM Configuration

The Limit Monitor (LIM) function checks for branch flows and voltages that are outside of acceptable operating limits. Switching device flow imbalance and voltage imbalance for three-phase loads are also calculated in this function. [Limit Monitor](#) provides detailed information about LIM.

This display contains the configuration parameters for LIM. For the detailed list of the configuration parameters, refer to [LIM Configuration](#).

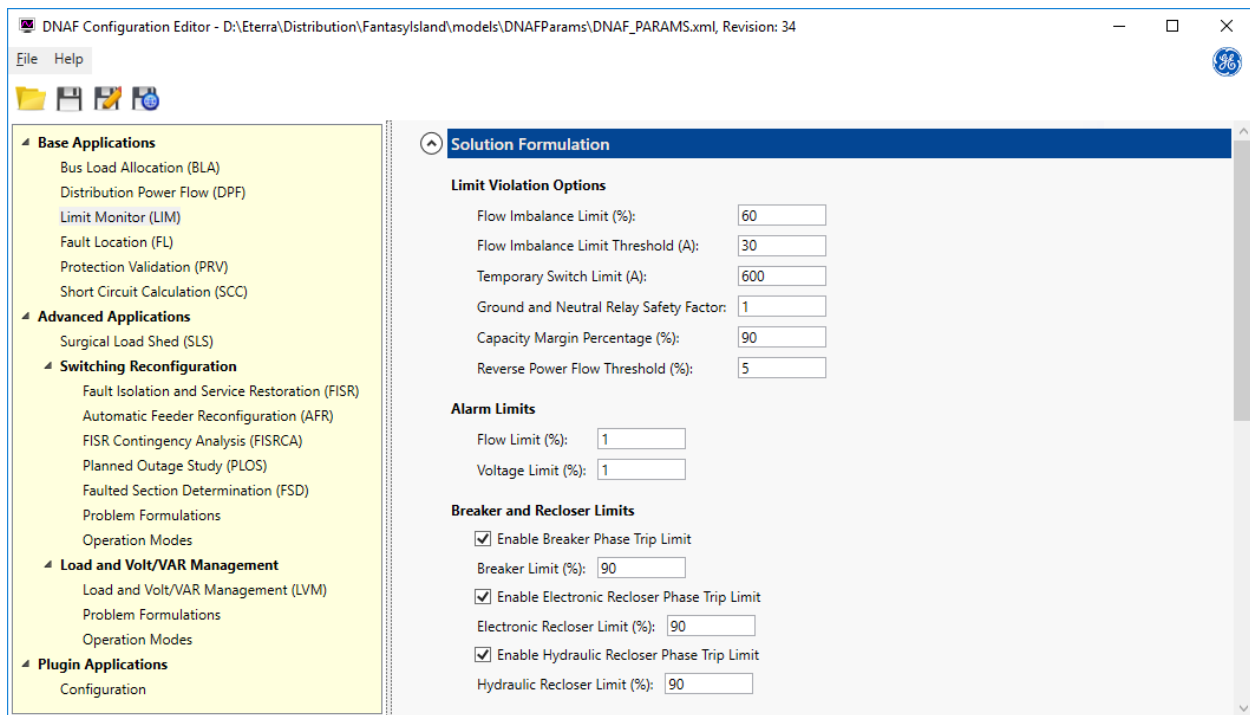


Figure 6. Limit Monitor Configuration

2.2.4. FL Configuration

The Fault Location (FL) function uses fault Amp measurements and faulted phase status measurements to determine potential fault locations along distribution feeders. [Fault Location](#) provides detailed information about FL.

This display contains the configuration parameters for FL. For the detailed list of the configuration parameters, refer to [FL Configuration](#).

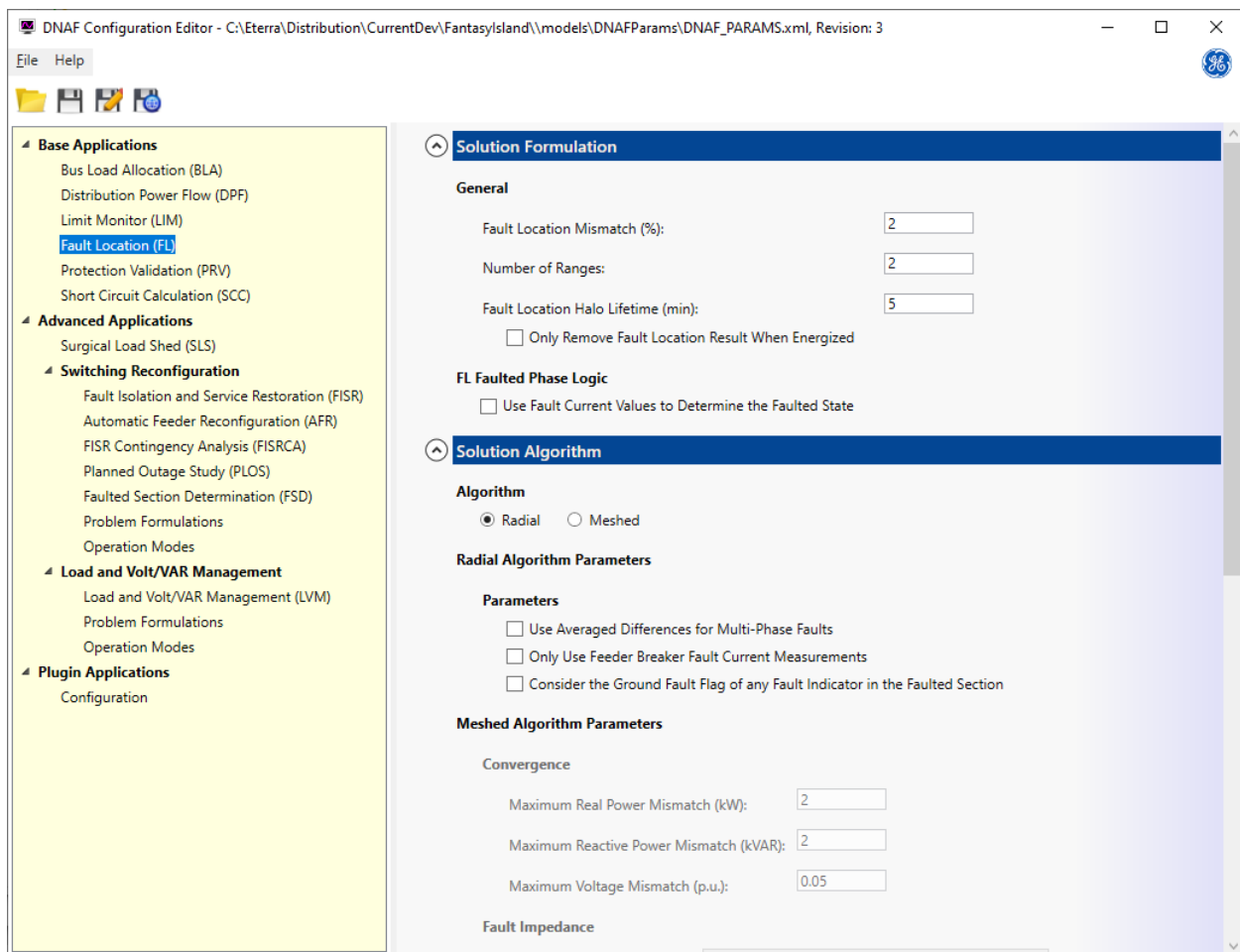


Figure 7. Fault Location Configuration

2.2.5. PRV Configuration

The Protection Validation (PRV) function calculates phase-to-ground, double-phase-to-ground, and three-phase fault currents for each application protection zone, where an application protection zone is defined by feeder circuit breakers and field reclosers. [Protection Validation](#) provides detailed information about PRV.

This display contains the configuration parameters for PRV. For the detailed list of the configuration parameters, refer to [PRV Configuration](#).

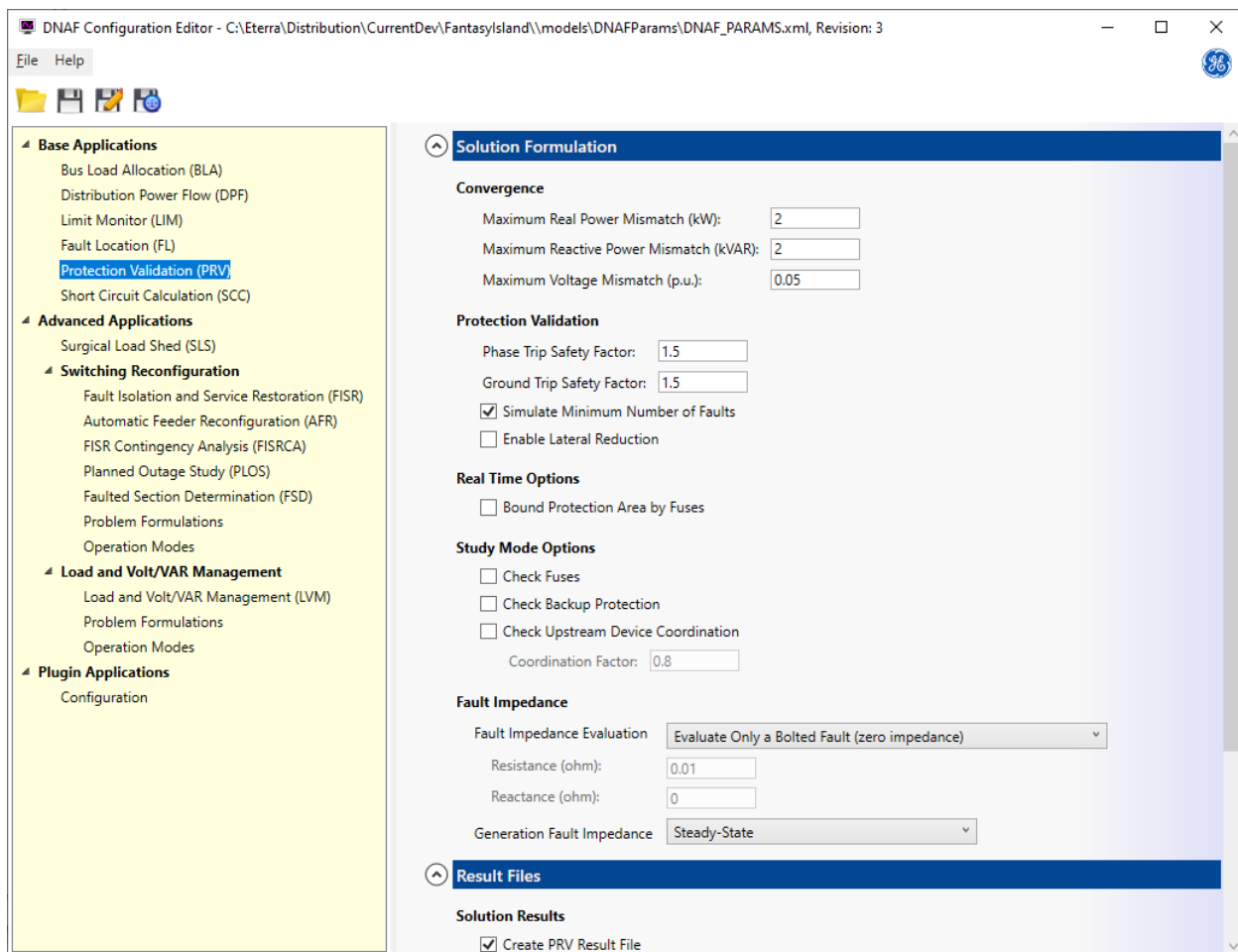


Figure 8. Protection Validation Configuration

2.2.6. SCC Configuration

The Short Circuit Calculation (SCC) function calculates single phase-to-ground, double phase-to-ground, phase-to-phase, and three-phase currents for a user-selected location (power transformer, line, and switching devices). [Short Circuit Calculation](#) provides detailed information about SCC.

This display contains the configuration parameters for SCC. For the detailed list of the configuration parameters, refer to [SCC Configuration](#).

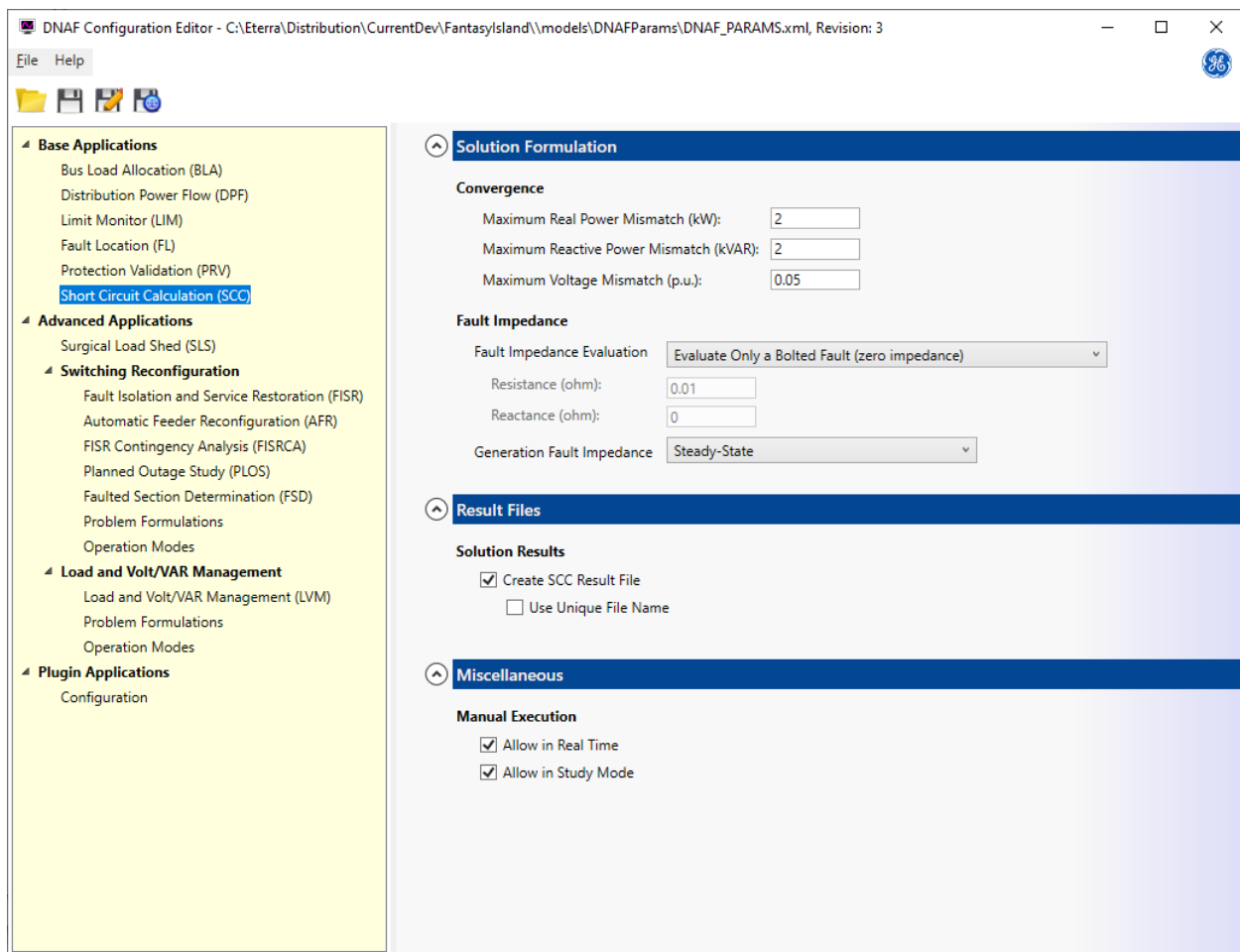


Figure 9. Short Circuit Calculation Configuration

3. Functionality and Concepts

3.1. Network Topology Processor

The Network Topology Processor (NTP) is primarily responsible for determining distribution system electrical connectivity. The distribution network is generally modeled from the data management point of view. Specifically, the data model provides hierarchical views such as system-substation-feeder-equipment. These logical views allow the information to be organized in a way familiar to most utility distribution operators.

However, this asset ownership-based logical model is not suitable for many distribution analysis applications. These applications usually select relevant parts of the network model for study. When selecting components from the distribution network model, the applications are more concerned about network electrical connectivity than about asset ownership. To support these distribution analysis applications, the concept of "network selection" (or "study area") is introduced.

NTP runs each time a network topology change is detected, so that it may evaluate switch positions and temporary network modifications (such as cuts or jumpers), to determine which parts of the network are energized. It begins from known energization points (generators, sub-transmission buses, and so on) and steps through all connected electrical equipment – stopping only when an open switching device is encountered. During this process, dynamic, or energizing, feeder ownership is determined based on the dynamic switch state.

3.1.1. Model Conversion

When creating a new station model file, the Converter application runs a subset of NTP functions to determine static network connectivity. When the model is initially created, electrical components typically do not have an associated feeder since that information depends on the branch connectivity and nominal state of the switches. However, feeder information can optionally be provided as a string in the static model for validation purposes only (the data is not used or stored in the station model files).

During the conversion process, the NTP application is executed to determine which components are connected to a given feeder. This execution sets the nominal feeder ownership for each component.

3.1.2. Network Selection

A network selection is a subset of the distribution network data model selected for a particular study. It can be thought of as the study area of interest within the network model.

Components in the network selections are either electrically connected or disconnected through an open point, such as a currently open switching device. The network selection defines the scope of study by grouping network components that are relevant to the ongoing study.

When creating the network selection, network connectivity, not asset ownership, is usually the selection criterion. For example, a network selection may contain electrically connected components that belong to different substations. The concept of network selection successfully provides a logical grouping view specifically suitable for distribution analysis applications.

It is important to note that different distribution analysis applications may have distinct network selection criteria. They may also have different study interests and scopes, even though they are triggered by the same event. For example, when the real-time power flow is triggered by a status change, it only needs to be concerned with the part of the network that is electrically connected to that switch. This may involve one or two feeders or an entire station, depending on the current topology. However, the Fault Isolation and Service Restoration (FISR) application extends its study area by including the adjacent feeders that are not currently connected— but that might provide electrical paths for restoring the de-energized portion of the feeder.

A universal definition of the term "network selection" is not sufficient to meet the various demands of distribution analysis applications. Network selection is very application-dependent; therefore, several network selection methods have been developed to address this issue. Note that, to more clearly illustrate the network selection methods, the provided examples are substation/feeder-oriented and do not include subtransmission modeling.

The network selection determination methods are defined as follows:

Method 1

The network selection determination in method 1 includes an entire substation. Every topology object belonging to this substation is included in the network selection. Some of the components that belong to other substations but that are electrically connected to the substation of interest are also included in the network selection. A network selection at this level may contain more than one connectivity island.

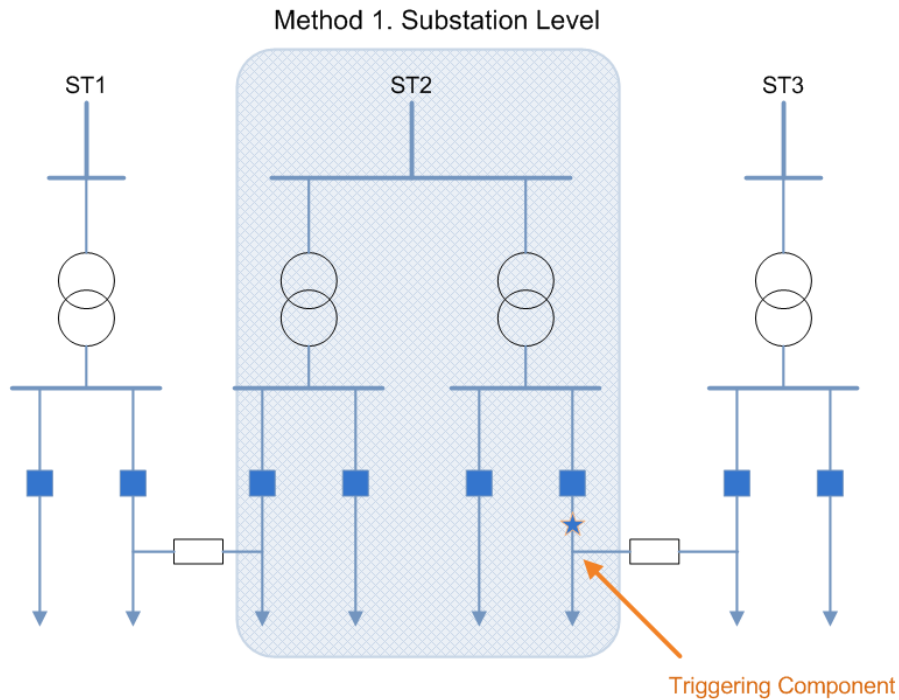


Figure 10. Network Selection Method 1

Method 2

The network selection determination in method 2 traces from the selected feeder to all neighboring feeders, which are a specified depth from the selected feeder (N-tie levels). For example, for a depth of two, network selection searches for all adjacent feeders that are within two feeder tiepoint connections of the selected feeder. All associated feeders are included in the network selection. Some of the components that belong to other substations but that electrically connect this substation are also included in the network selection. As shown in figure below, when a fault occurs on a feeder belonging to station ST2, all feeders of station ST2 and the first feeder of station ST3 are included in the network selection because they are within two normally open tie switches from the selected feeder in ST2. The feeders of ST1 and the second feeder of ST3 are not included because they do not meet the two-tieswitch criterion.

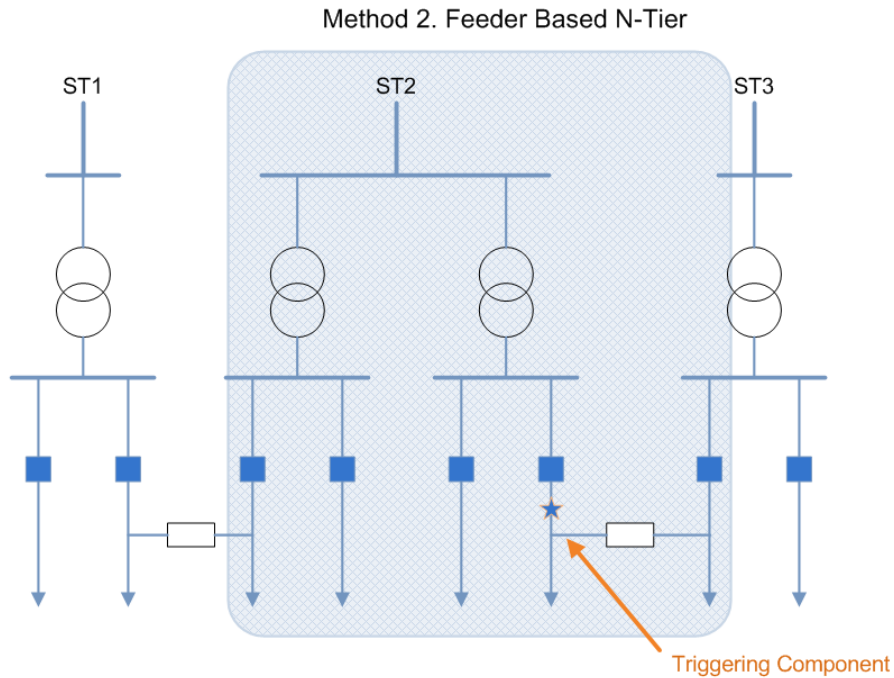


Figure 11. Network Selection at Method2 with a Depth of Two

Method 3

The network selection determination in this method traces from the selected feeder to all neighboring stations, which are a specified depth from the selected feeder (N-tie levels). For example, for a depth of two, this method searches for all adjacent feeders that are within two feeder tiepoint connections of the selected feeder. All components in associated stations are included in the network selection. Figure below shows the network selection with depth of two in method 3.

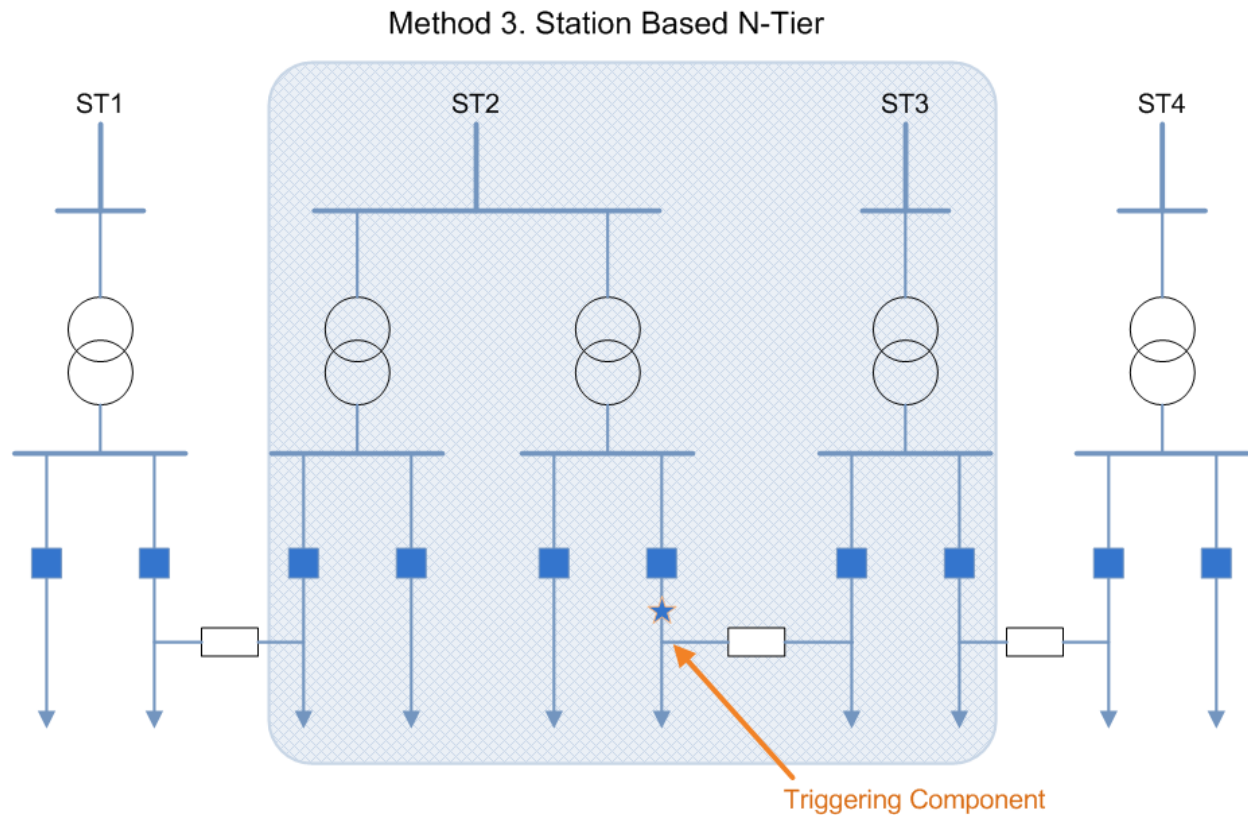


Figure 12. Network Selection at Method3 with a Depth of Two Rules

The network selection method is pre-configured for each application to ensure a quick and effective solution. This simplifies the use of the network selection mechanism.

The following table summarizes these rules.

Table 1. Network Selection Assignment Rules

Network Selection Level	Applied Distribution Analysis Functions
Method 1	Distribution Power Flow (DPF)
Substation Level	Protection Validation (PRV)
	Load and Volt/VAR Management (LVM)
	Short Circuit Calculation (SCC)
Method 2	Fault Isolation and Service Restoration (FISR)
Feeder Based N-Tier	Planned Outage Study (PLOS)
	Fault Isolation and Service Restoration – Contingency Analysis (FISR-CA)
Method 3	Automatic Feeder Reconfiguration (AFR)
Station Based N-Tier	

3.1.3. Boundary Processing

To maximize performance, the Network Topology Processor processes multiple stations concurrently. Often, electrical networks span multiple stations, or may have components in one station that are dependent on the network status of another station (such as with a sub-fed station). In these situations, the static boundaries of the station are respected – the NTP processor does not "cross" to the attached station during its processing. Instead, the NTP processor updates the station boundary information for the attached station based on the electrical connectivity of the station being processed. It then signals that the attached station needs to be re-solved to account for the updated external branch information.

This station boundary information is not just limited to energization/de-energization states. The division of the network between two (or more) stations prevents what is occurring on one side of the station boundary from being seen on the other side. Because of this, transferring information across the station boundary also includes such information as the status of connected fault indicators, whether a fault occurred downstream, or whether the line is fed from a looped section.

3.2. Subtransmission Model

The transmission/subtransmission system model is the networked portion of the power system that is adjacent to the distribution circuits. The loading and voltage level of the system significantly depends on the operating conditions of the connected distribution system. The transmission/subtransmission model includes the connectivity and characteristics of the substation transformers and transmission/subtransmission lines.

The reasons for including the subtransmission model in the DMS model include:

- Monitoring flow imbalance on subtransmission lines in the real-time Distribution Power Flow.
- Checking that LVM and all reconfiguration functions respect subtransmission voltage and flow limits. ADMS can only monitor limits for devices that are explicitly included in the DMS model.
- Representing transmission losses in the DNAF optimization applications (LVM, AFR).

To model the interface between the defined subtransmission network and the distribution network, the concept of a distribution equivalent reference bus interface is introduced. A distribution equivalent reference bus object corresponds to a node or bus in the EMS balanced, single-phase network model used for transmission and subtransmission. In the DMS, the reference bus is modeled as a generator with equivalent impedance. The impedance values defined for the reference bus interface objects are used only if all these values are non-zero for every interface in the substation. Otherwise, the equivalent impedance values defined for the substation object itself are used. The EMS calculated State Estimator voltages and angles can optionally be applied at these reference buses for use by the DNAF.

The network applications will only look at a study interface if it is required for energization of a sub-fed station. Otherwise, only modeled reference bus interfaces are used in the network applications.

The subtransmission and distribution models are considered in solving all network analysis functions, including the Distribution Power Flow, Load and Volt/VAR Management, and all reconfiguration functions. Since distribution systems are normally loosely coupled, it is possible to use some equivalent technique during the process.

To ensure efficient solution processing, lumped load equivalents are used to represent feeders and substations that are not impacted. This means that the detailed analysis can be focused on a smaller, related part of the network, while still considering the effects of the larger network. The method for implementing the subtransmission model is discussed below for each application.

Distribution Power Flow: The entire subtransmission network connected to the distribution substation of interest is solved for each execution of DPF. The other distribution stations that are connected to the subtransmission network are translated into lumped load equivalents, as illustrated in figure below. If the distribution station of interest is connected to another station through a closed tie switch and if there is a loop due to this abnormal condition, then the full model for both distribution substations is used. The Limit Monitor (LIM), Loss Analysis (LA), and Power Quality Index (PQI) functions are evaluated for all fully modeled stations (these functions are not solved for the equivalent stations), and the LIM is executed for all subtransmission lines and buses in the network selection.

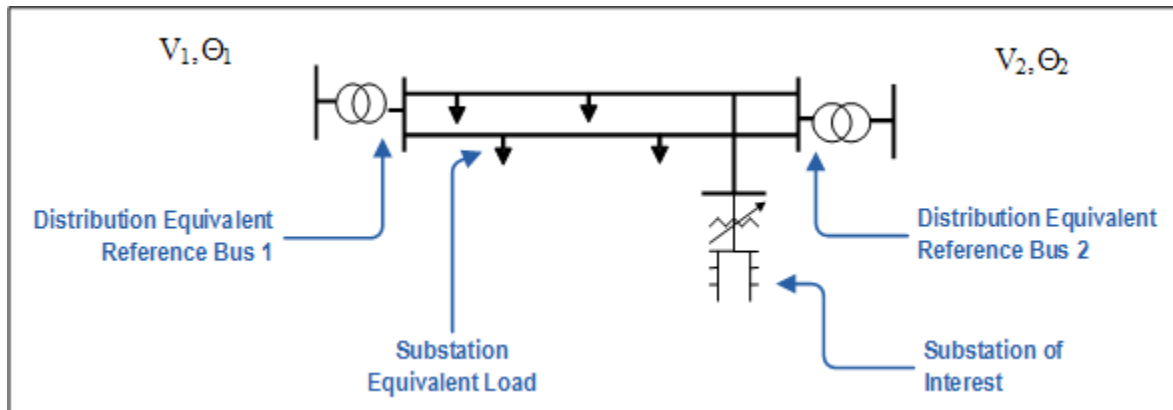


Figure 13. Equivalent Model of Small Subnetworks for DNAF Studies

Protection Validation (PRV): Same as DPF.

Short Circuit Calculation (SCC): Same as DPF.

Load and Volt/VAR Management (LVM): Same as DPF.

Fault Isolation and Service Restoration (FISR): Similar to DPF. The entire subtransmission network is modeled with lumped load equivalents for the stations that are not involved in the restoration.

Automatic Feeder Reconfiguration (AFR): The real-time AFR function is treated exactly like DPF. The study mode AFR does not reduce neighboring stations to equivalents, but it uses the fully detailed model for the neighboring stations.

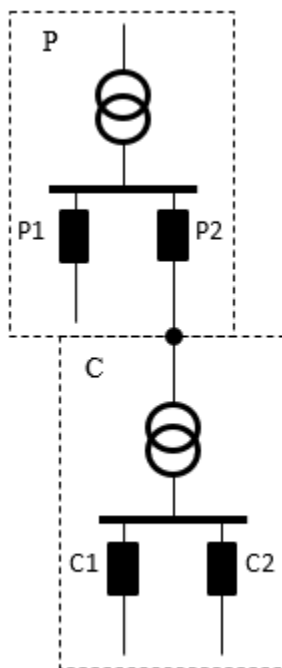
Planned Outage Switching (PLOS): Same as FISR.

Fault Isolation and Service Restoration– Contingency Analysis (FISRCA): Same as FISR.

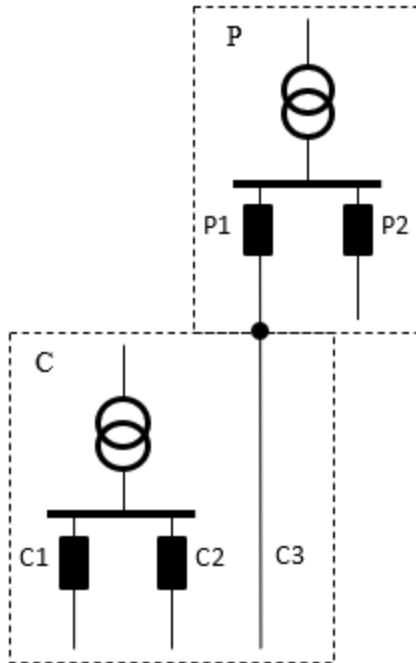
3.3. Sub-fed Station Configurations

For some distribution network configurations, the whole station or part of the station is energized from another station that is typically operated at a higher voltage level. This is referred to as a sub-fed configuration. Six supported sub-fed station configurations are illustrated below. The black dots in these figures represent the station boundary switches.

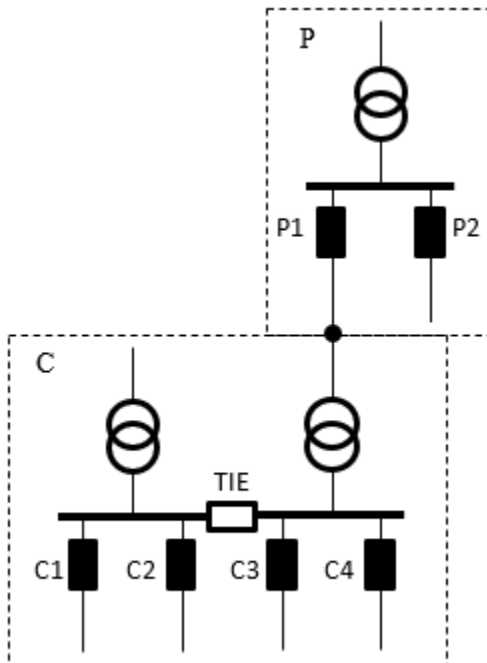
1. One feeder of the parent station energizes the whole child station. In the figure below, feeder P2 in station P feeds the entire station C.



2. One feeder of the parent station energizes part of a feeder in the child station. In the figure below, feeder P1 in station P energizes feeder C3 in station C.

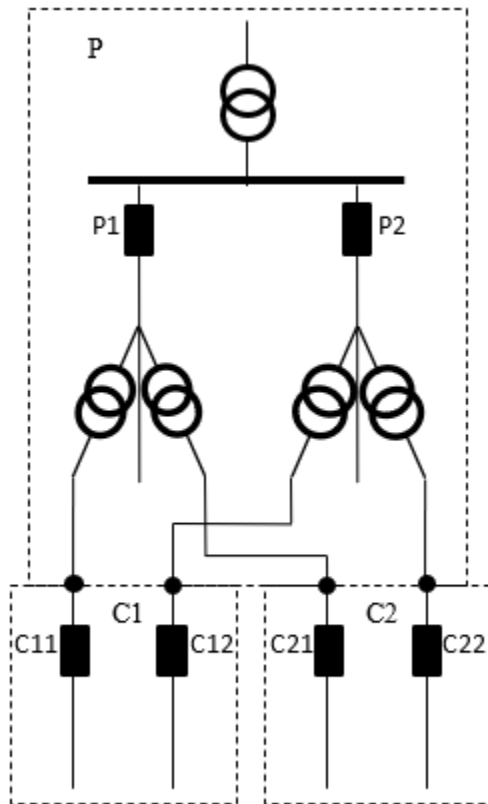


In some cases, there is an open tie breaker between the bus of the sub-fed feeders and the bus of the non-sub-fed feeders. In the figure below, feeder P1 of station P feeds feeders C3 and C4 in station C, feeders C1 and C2 of station C are non-sub-fed feeders, and there is an open tie breaker TIE between the load side buses.

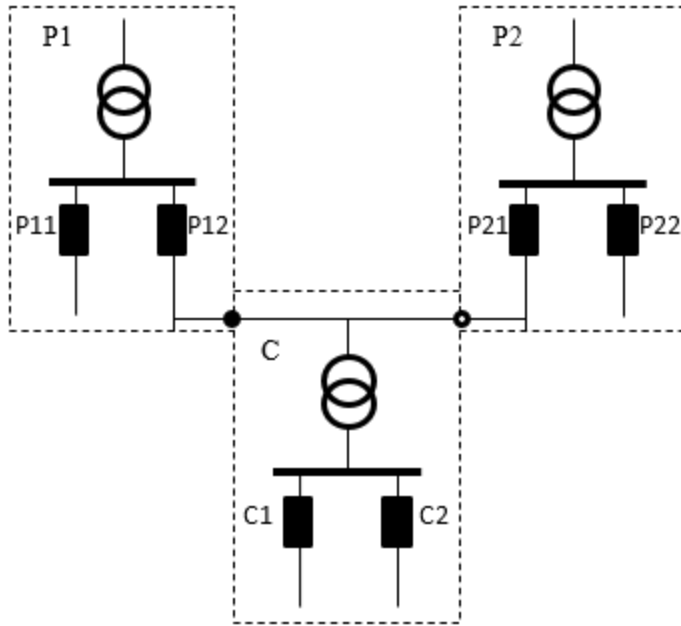


- One feeder of the parent station energizes two feeders from two different child stations. In the figure below, feeder P1 of station P energizes feeder C11 in station C1 and feeder C21 in station C2, and feeder P2 of station P energizes C12 in station C1 and feeder C22 in station C2. The

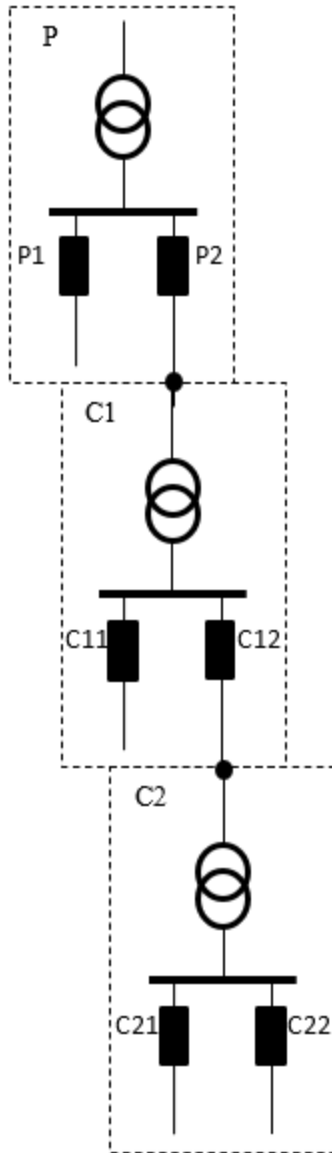
power transformers of the child stations are at the end of the energizing feeders in the parent station.



4. The child station can potentially be fed from more than two parent stations. In the figure below, station C is currently energized from feeder P12 in station P1. Station C can also be energized from feeder P21 in station P2 or both feeders P12 and P21 if both of the boundary switches are closed.



5. Multiple layers of sub-fed stations. In the figure below, station C1 is energized from feeder P2 in station P, and feeder C12 in station C1 energizes station C2.



6. Passthrough sub-fed stations. The sub-fed child station is energized as expected from a parent station, and only the parent station feeder enters into the station internals and passes through on the high voltage side to energize another child station. In the figure below, feeder P2 in station P first energizes a passthrough sub-fed station C1, followed by a normal sub-fed station C2.

There are also cases where there are two passthrough sub-fed stations like C1 before reaching a normal sub-fed station like C2.

3.4. DNAF Alarms

In real-time operation, the DNAF can issue alarms to the Alarm subsystem. These alarms include Failure to Complete alarms (the application was not able to solve), Return to Normal alarms (an

application previously unable to solve has recently solved), and specific application results alarms.

All DNAF-generated alarms are distinguished from SCADA-generated alarms by assigning them a unique category. The presentation of DNAF alarms (for example, color) is also configurable.

The DNAF System Configuration display is responsible for enabling or disabling application-specific alarms. Each type of alarm can be enabled or disabled on a station-by-station basis. This allows run-time control of alarm output.

DNAF alarms are generated by the network applications and sent to the PROXYSRV application (to show the alarms in the Habitat system activity log). DNAF alarms follow the user's area permission (Area of Responsibility or AOR).

Note

The Habitat ALARM server parameters must be properly configured in the server configuration file to show ADMS alarm events. For more details, refer to the *ADMS Software Installation and Maintenance Guide*.

The following links provide a list of DNAF-specific alarms:

- [DPF Alarms](#)
- [PRV Alarms](#)
- [FL Alarms](#)

For FISR, AFR, and LVM Alarms, refer to the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

For more information about the structure and format of DNAF alarms, see [DNAF Alarm Format](#).

3.5. DNAF Results

Each time a DNAF application runs, it returns a result status and can generate a result file containing execution details.

3.5.1. Result Status

The following subsections list and describe different result statuses for DNAF applications.

3.5.1.1. Basic Status

- Basic status for all applications:
 - **Unsolved:** The application has not been solved yet.
 - **Solving:** The application is being solved now.

- **Solved:** The application is solved.
- Basic status specific to advanced applications (LVM, FISR, AFR, PLOS, and FISRCA):
 - **Solved (No Plan):** An advanced application that generates plans (for example, reconfiguration or LVM) has completed successfully but did not generate any plans. See the result file for more information.
- Basic status specific to reconfiguration applications (FISR, AFR, PLOS, and FISRCA):
 - **Solved (No Plan) – No Tie Switches:** A reconfiguration application has completed successfully but did not generate any plans due to no available (eligible) tie switches. See the result file for more information.
 - **Solved (No Plan) – Violations on All Plans Found:** A reconfiguration application has completed successfully, but the generated plans have violations. The problem formulation does not allow plans with violations. See the result file for more information.
 - **Solved (No Plan) – Non-Converged or Scheduled Loading BLA Islands:** A reconfiguration application has completed successfully, but the generated plans have non-converged or scheduled loading BLA islands. See the result file for more information.
- Basic status specific to the Protection Validation (PRV) application:
 - **Solved - No Violations:** The application is solved with no violations.
 - **Solved - Interruption Violation:** The application is solved with interruption violations.
 - **Solved - Sensitivity Violation:** The application is solved with phase or ground fault sensitivity violations.
 - **Solved - Pick-up Violation:** The application is solved with phase or ground pick-up violations.
 - **Solved - Violations:** The application is solved with interruption, pick-up, and sensitivity violations.

3.5.1.2. Suspect Status

- Suspect status for all applications:
 - **Suspect – Data:** The application solved successfully but contained suspect data, such as defaulted source voltages or suspect capacitors, loads, generators, or energy resources. Detailed tabular data and result files provide more details about the suspicious data items.
 - **Suspect – BLA Model Change:** The application solved, but it was necessary to modify one or more data parameters for BLA to converge.
 - **Suspect – BLA Not Converged:** The core analysis solved, but BLA did not converge. This indicates potential problems with system loading or measurements.
- Suspect status specific to FISRCA:
 - **Suspect – Partial Solution:** This applies to FISRCA only and indicates that FISR results were

only found for part of the solution space.

3.5.1.3. Execution Rejected Status

- Rejected status for all applications:
 - **Rejected – Permission:** The application could not run because of a lack of permissions. To fix this issue, check that application functional permission exists and/or AOR permissions are correct.
 - **Rejected – No Manual Trigger:** The application could not run because manual application triggering has been disabled in the server configuration.
 - **Rejected – Application is disabled:** The application could not run because the DNAF application has been disabled. To fix this issue, check the DNAF System Configuration display.
 - **Rejected – Station is not DNAF Capable:** The application could not run because the station data model is not DNAF capable. To fix this issue, check the Converter log file and correct the modeling data.
 - **Rejected – Adjacent Station is not DNAF Capable:** The application could not run because a neighboring station data model is not DNAF capable. To fix this issue, check the modeling data.
- Rejected status specific to the LVM application:
 - **Rejected – LVM Voltage Safety:** The LVM application could not solve because of voltage safety limits. If the maximum difference for any node voltage between the DPF-calculated value and the valid measurement value is larger than the threshold indicated in the LVM page of the DNAF Configuration Editor, the LVM request will be rejected.
 - **Rejected – Prerequisite Not Met:** The LVM application could not solve because prerequisite conditions are not met. Check the settings in the "Auto Disabling Checks" section on the LVM page of the DNAF Configuration Editor.
 - **Rejected – Application Configuration Validation Error:** The LVM application could not solve because there are no eligible devices. Check the settings in the "Eligible Controls" section on the LVM page of the DNAF Configuration Editor.
- Rejected status specific to the FISR application:
 - **Rejected – Faulted Feeder is Disabled:** FISR could not run because Fault Location functionality has been disabled on the fault feeder.
 - **Rejected – Faulted Section is Energized:** The application could not run because the faulted section has become energized.
 - **Rejected – Unsolved Fault Not Found:** The application could not run because no fault location could be found. Verify that a fault exists in the station of interest.
 - **Rejected – No Outage Device:** The FISR application cannot run because the selected device is not associated with the substation where FISR runs.

- Rejected status specific to the AFR application:
 - **Rejected – No Overloaded Components:** AFR could not run because no overloaded components were found.
- Rejected status specific to reconfiguration applications:
 - **Rejected – No Valid DPF Solution:** Reconfiguration application could not run because there is no valid power flow solution. Check DPF status.
 - **Rejected – Too Many Feeders for Reconfiguration:** Reconfiguration application could not run because there are too many feeders in the network selection. Check the Maximum Number of Feeder parameter and the adjacency level for the corresponding reconfiguration application in the DNAF Configuration Editor.
 - **Rejected – Fault Not on Feeder Main:** Reconfiguration application could not run because the fault is on a lateral.
- Rejected status specific to the Fault Location application:
 - **No Result – Missing Phase Target:** The Fault Location application cannot run. The fault current measurement data is available, but the phase fault indicator status data is missing.
 - **No Result – Missing Current:** The Fault Location application cannot run. The fault indicator status data is available, but the fault current measurement data is missing.
 - **No Result – Missing Current and Phase Target:** The Fault Location application cannot run. Both the fault indicator status data and the fault current measurement data are missing. This status typically occurs when the Fault Location application is invoked manually.
 - **No Result – Missing Fault Branch or Fault Type:** The Fault Location application cannot run. The faulted branch is missing or the fault type is not determined.
- Rejected status specific to the PLOS application:
 - **Rejected – No Outage Device:** The PLOS application cannot run because the selected device is not associated with the substation where PLOS runs.
 - **Rejected – Outage Not on Feeder Main:** The PLOS application could not run because the outage is on a lateral.
- Rejected status specific to the SCC application:
 - **Rejected – Selected Device Not Branch Device:** The SCC application cannot run because the selected device is a shunt device.

3.5.1.4. Execution Failed

- Failed Status for all applications
 - **Failed – DPF Not Converged:** The application failed because the power flow solution could not converge. This can happen due to excessive loading, unrealistic voltages, model data errors, bad settings on locally regulating devices, or other reasons.

- **Failed – Server Communication:** The application could not solve due to bad communications between the main and DNAF servers. This could be caused by a misconfiguration of server IP addresses and ports or if the server is down.
- **Failed – Model Error:** The application could not solve due to a failure in building the DPF model.
- **Failed – Internal:** An internal data error was discovered during execution and the solution was aborted.
- **Failed – Unknown:** An unknown error occurred.
- Failed Status for FL application
 - **Failed – Upstream Feeder Breaker Not Found:** The FL application cannot run because the upstream feeder breaker is missing in the model file.

3.5.2. Result Files

To assist in testing and debugging, all DNAF applications can generate result files that contain key application results. Depending on the specific application, result files can be uniquely named so that the next application execution does not overwrite the previous results. Some applications also include the option to generate more detailed/comprehensive output.

Result file options are configurable through the DNAF Configurator application. Each application tab contains a Result Files section with options. For more information about the available result files, refer to [DNAF Result Files](#).

3.6. Manual Entry Files

To assist in testing and debugging, two manual entry files can be created and used by the DNAF applications: `Manual_Entry_Init.dat` and `Manual_Entry.dat`. These files must be placed in the server directory which is defined by the server parameter `DNAF_INPUT_PATH` in the DNAF server configuration file.

- **Manual_Entry_Init.dat:** Manual entries made through the user interface (UI) have a higher priority than manual entries in this file. This file is commonly used to set up demo scenarios. This file is read only once, when the server is started.
- **Manual_Entry.dat:** The manual entries in this file have a higher priority than manual entries through the UI, and they overwrite the manual entries specified through the UI. This file is read every time a DNAF function is executed.

For more information, refer to [Manual Entry Files](#).

3.7. Protection Settings

Protective devices such as relays and fuses are designed to protect the distribution system from damage in the case of faults. Overcurrent relays are typically used to trip breakers and reclosers so that faulted sections of the system can be isolated. A typical relay installation provides separate settings to detect ground and phase faults. Modern digital relays can be configured so that different groups of such settings can be created and stored in the device, such that different settings groups can be enabled depending on system conditions.

The ADMS system provides a simplified representation of overcurrent protection functionality which can be used to provide data for the Protection Validation (PRV) application and to read and control SCADA measurements associated with the protective devices.

The ADMS system stores the following parameters for overcurrent protection data:

- Phase trip current for a settings group
- Ground trip current for a settings group
- Provision for multiple settings groups (where only one is active at a time)
- Provision for settings group templates which apply to multiple devices
- Ground trip capability and enabling

Note that the simplified protection data in the ADMS system does not represent the following real-world protection data:

- Fast and slow reclose behavior (only lockout trip currents are modeled)
- Detailed time-current curve data
- Impedance relay data or related functionality (for example, load encroachment)

3.7.1. Protection Setting Objects

Various objects are modeled in the ADMS system to represent the protection data.

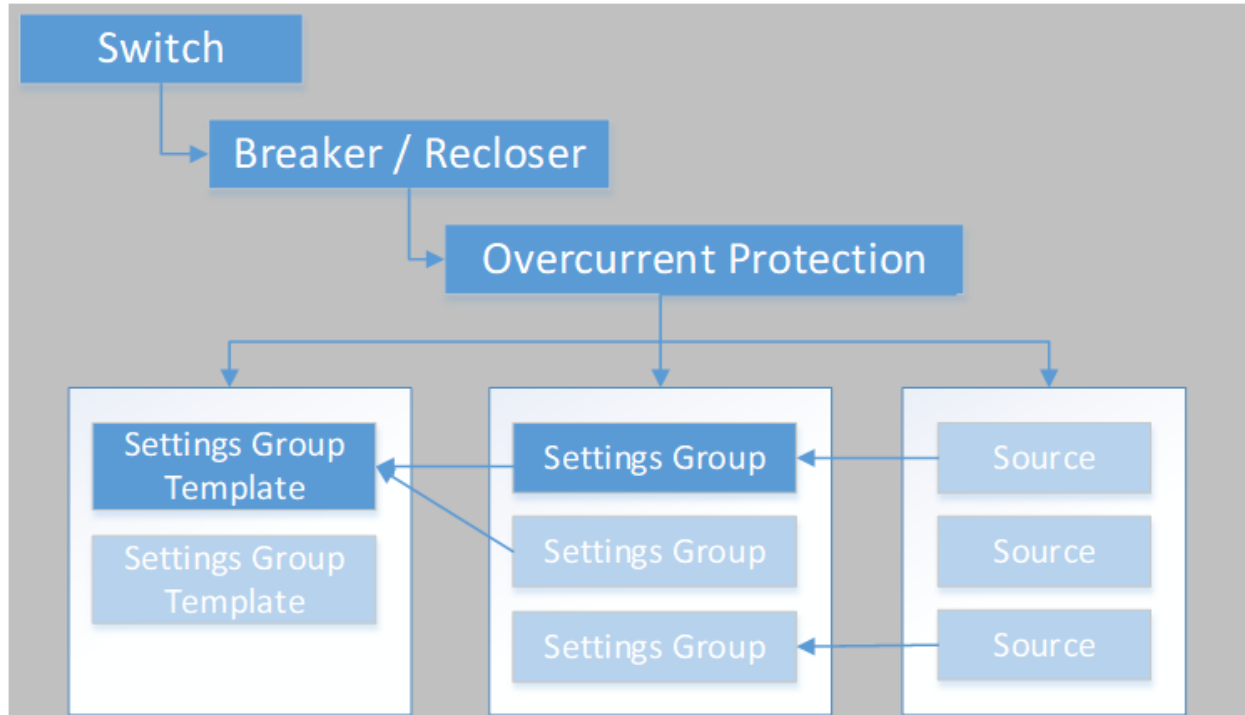


Figure 14. Protection Settings

Switch - A basic switch object represents any device that can be opened to break electrical flow.

Breaker / Recloser - For breakers, reclosers, and sectionalizers, an extra object is modeled to represent the special parameters associated with these objects.

Overcurrent Protection - An optional overcurrent protection object can be associated with a breaker/recloser/sectionalizer type of switch. Each overcurrent protection object must contain at least one protection settings group.

Settings Group Template - A collection of settings groups can be defined in a settings group template. In addition to listing supported settings groups for a switching device (or a group of switching devices), settings group templates also identify allowed transitions between supported settings groups (see figure below).

If a settings group template is defined for a switch, this data will override any device-specific data that may be defined.

Protection Settings Template Information - TEST1 (Supported Transitions Between Settings Groups)												
Settings Group Name	Value	Phase Trip (Amps)	Ground Trip (Amps)	Mode	SG1	SG2	SG3	SG4	SG5	SG6	SG7	SG8
SG1	1	200	100	Recloser	Yes	Yes	Yes	Yes	No	No	No	No
SG2	2	200	150	Recloser	Yes	Yes	Yes	Yes	No	No	No	No
SG3	3	480	280	Recloser	Yes	Yes	Yes	Yes	No	No	No	No
SG4	4	500	300	Recloser	Yes	Yes	Yes	Yes	No	No	No	No
SG5	5	600	600	Switch	Yes	Yes	Yes	Yes	Yes	No	No	No
SG6	6	400	400	Custom	No	No	No	No	No	Yes	No	No
SG7	7	280	280	Sectionalizer	Yes	Yes	Yes	Yes	No	No	Yes	No
SG8	8	200	200	Sectionalizer	Yes	Yes	Yes	Yes	No	No	No	Yes

Figure 15. Settings Group Template Example

If adaptive relaying is configured to use the source supply approach (see below), then the overcurrent protection object also contains a list of feeders associated with each settings group.

3.7.2. Sources of Protection Setting Data

Protection setting data can be loaded into the system model from a number of possible sources:

- **Static Model Data:** The XML data files that define the contents of each station can be used to define two sets of phase and ground trip data for each recloser (PhTrip, Grdtrp, AltPhTrip, and AltGrdTrp) or breaker (PhaseTrip, GroundTrip, AltPhaseTrip, and AltGroundTrip). This is the default source of data, which is used if the other optional sources of data do not exist or are not populated for a device. Data defined in these fields is assigned to two settings groups with the default names of S1 and S2, respectively.
- **RelayAndImpedanceData.csv:** This optional file contains a mixture of source impedance data and relay trip setting currents. If a relay device is defined in this file, it contains data for primary and alternate trip currents. The data overrides the trip currents defined in the static model data as described above.
- **ProtectionSettings.csv:** This optional file contains detailed protection data including the following:
 - Name and description of settings group template.
 - List of up to 8 settings groups in the settings group template, including name, mode, phase trip, ground trip and allowed transitions for each settings group in the settings group template.
 - Name, phase trip, and ground trip currents for up to 8 settings groups.
 - A SCADA index for each settings group which associates it with a specific measurement.
 - Optional source data which maps a station and feeder supply to a specific settings group. See the Source Supply option in the adaptive relaying strategies described below for more details.

If this file contains an entry for a specific protection device, it overrides any data specified in either the static model file or the RelayAndImpedanceData.csv file.

This file also contains the definition of any Settings Group Templates, including the constituent settings group data and the allowable transitions between settings groups. If a settings group template is defined for a specific switch, this data will override any device-specific data specified in the first part of the file.

- **Real-Time Measurements:** There are a number of different measurement types associated with protection, settings groups, and templates. Any settings specified in SCADA will override the associated static model data. See [Protection Setting Measurements](#) for a description of the different SCADA measurement types and options.

3.7.3. Protection Setting Measurements

Each breaker or recloser can have a variety of SCADA measurement points associated with protection data. These measurements define data such as the current settings group, the nominal settings group, the trip setting for a settings group, and the current settings group template. In some cases, there are multiple ways to configure SCADA depending on the customer needs and device capabilities.

3.7.3.1. Active Settings Group

There are three different methods that DMS can use to read/write the active settings group for a device. There is also an externally defined point which allows a field device to specify the active settings group.

Multiple Status Measurements

The first option is to use eight status points:

```
StatusSettingGroupApplied1 – StatusSettingGroupApplied8
```

One of these points is set to True for the active settings group and the others are set to False representing the inactive settings groups. A corresponding set of control points are defined which allow DMS to change the active group by setting the appropriate point:

```
CtrlEnableSettingGroup1 – CtrlEnableSettingGroup8
```

Single Analog Measurement

The second option is to use a single analog point:

```
MeasProtectionSettingGroup
```

The active setting group is read and rounded to an integer value between 1 and (up to) 8 representing the different settings groups defined.

The corresponding control point for DMS control is:

```
SetPntProtectionSettingsGroup
```

The active group can be changed by writing a value 1-8 corresponding to the (up to) 8 possible settings groups.

Single Status Measurement

The last option is suitable for cases where there are only two possible settings groups defined, a primary and an alternate settings group. The active group is determined by reading the point:

```
StatusSettingGroupPrimAlt
```

Where:

- True = Alternate settings group is active.
- False = Primary settings group is active.

The active group can be changed in DMS by the following pair of control points:

```
CtrlEnableAltSetting  
CtrlDisableAltSetting
```

The active group can be changed by sending enable or disable alternate setting group control point. `CtrlDisableAltSetting` control is used to activate Primary setting group; `CtrlEnableAltSetting` control is used to activate Alternate setting group.

External Setting

The `ActiveSettingGroup` measurement is an analog that can be used by the customer or external device to override the current settings group defined in DMS. The value will be defined in the same way as `MeasProtectionSettingGroup` described above.

3.7.3.2. Nominal Settings Group

The nominal / default setting group is defined as part of the model for either the device or the settings group template. However, the `NominalSettingGroup` point in SCADA can be used to define the nominal setting in real-time.

The choice of nominal settings group is considered when the system topology returns to normal after switching or reconfiguration application topology changes.

3.7.3.3. Phase Trip Currents

The phase trip current for settings groups is defined as part of the model, but can be overridden in SCADA to change values in DMS real-time. There are two different ways of setting the phase trip current.

Setting Group Specific Phase Trip

A set of eight analog measurements, `PhaseTripSettingGroup1` – `PhaseTripSettingGroup8`, allow the phase trip current of each of the (up to) 8 settings groups to be set in real-time.

Generic Phase Trip

The alternate method of specifying a real-time phase trip current is to use a single measurement analog `MeasActivePickup`, which represents the phase trip value of the currently active settings group.

3.7.3.4. Active Settings Group Template

The ActiveSettingsGroupTemplate measurement analog is used to determine the currently active template for a device. The analog (integer) value should match the value specified in the template definition in the ProtectionSettings.csv file in the modeled data.

3.7.4. Adaptive Relaying Strategies

The term adaptive relaying refers to the ability of protective devices to change settings to adapt to new system conditions. For example, switching on a system may affect the fault levels seen by a downstream recloser. This may require trip currents on the recloser to be changed.

In the ADMS system, adaptive relaying is implemented as selection of the best settings group for a protective device. Two options are available for evaluating which settings group should be selected; these are described below. The preferred approach can be selected in the reconfiguration settings in the DNAF Configuration Editor.

3.7.4.1. Source Supply

One adaptive relaying technique is to determine which settings group should be active based on which feeder is supplying the protection device. The Source (station and feeder) that might feed a device is mapped to a particular settings group during the model building using data in the ProtectionSettings.csv file.

Figure below shows an example of how feeders might be mapped to settings groups for a sample recloser.

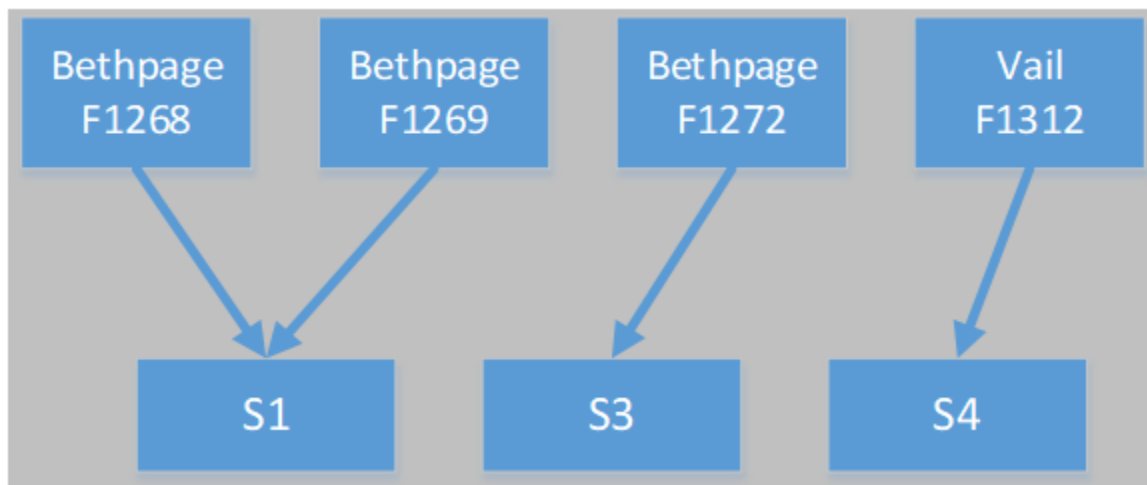


Figure 16. Mapping Feeders to Protection Settings

As a switching is performed on the system (or as switching plans are generated) and the dynamic source changes, the appropriate settings group can be determined from this data.

3.7.4.2. Minimum Phase Trip

The other option for adaptive relaying is to choose the active protection settings group with the highest allowable phase trip current. Using this technique, the active protection settings group is selected by finding the settings group with the highest phase trip current that is below the minimum phase fault current on the protected zone (the settings group's phase trip current is multiplied by the appropriate safety factor). The active protection settings group's phase trip current should also be above the estimated load current (multiplied by the appropriate safety factor).

3.7.5. Using the Protection Settings in DNAF Applications

3.7.5.1. Protection Validation

The protection validation application is a consumer of protection settings data. It uses the trip settings from the active protection settings group to run a series of checks and determine if protection is valid.

3.7.5.2. Reconfiguration Applications

The AFR and FISR applications can create steps to change the active settings group for a protection device as part of the plan moves suggested by the analysis.

Each application can be configured in the DNAF Configuration Editor to generate the settings group steps and to determine which type of adaptive relaying to use.

3.7.5.3. Return to Nominal

When the system returns to a normal state after manual switching or reconfiguration, the protection system can be configured to restore the active setting group for each device to its proper setting.

After each powerflow solution on a station, the DMS will check the protection settings on that station as configured in the DNAF Configuration Editor DPF tab.

None

No checks are performed. The active protection setting group is not changed.

By Feeder Source

This option is used in conjunction with the Source Supply adaptive relaying approach. The feeder supplying the protection device is checked to ensure that the appropriate setting group is active. If it is found to be incorrect, DMS will change the setting group and raise an alarm to inform the user about the change.

By Nominal Setting

This option is used in conjunction with the Minimum Phase Trip adaptive relaying approach. Each feeder is evaluated to determine if there is any off-nominal switching. If all switching is nominal then each protection device on the feeder is checked to ensure that the nominal setting group is active. The preferred nominal setting group is determined by checking either the NominalSettingGroup measurement if it exists and is valid, or by defaulting to the value of the setting group template or individual setting group model.

3.7.5.4. Arc Suppression Coils

Regulating Arc Suppression Coils (ASC) or Petersen Coils modify the neutral grounding reactance of affected power transformers. These reactances are updated every time DPF runs based on topology so the effects are automatically included in Short Circuit, Protection Validation, and reconfiguration applications.

For more details, see [Arc Suppression Coil Grounding Impedance](#) and [Arc Suppression Coils](#).

4. Bus Load Allocation

The ADMS load estimation procedure, also called Bus Load Allocation (BLA), allocates loads (kW and kVAR) at distribution load nodes. The BLA function allocates loads to buses without telemetered metering, automatically accounting for other measurements, to ensure consistency between allocated and metered loads.

Load estimation techniques can be divided into two general types: state estimation and allocation procedures. State estimation techniques are commonly used in EMS to obtain the best estimate for the bus voltages and angles.

For distribution systems, state estimation procedures have been proposed in the literature, but several problems arise in their practical application. These problems include a lack of telemetered buses (a vast number of pseudo-measurements are required). In addition, many measurements are in Amperes only, and there are difficulties in parameter tuning. Due to these problems with state estimation procedures, a simpler load allocation procedure is implemented. This procedure scales the loads so that they match the available real-time flow measurements.

The BLA procedure uses the real-time measurements from SCADA and AMI, along with pre-defined customer information, to scale the distribution system loads to the real-time measurements. The expected value at a given date/time is obtained by multiplying the nominal load by a scaling factor that reflects the energy usage at a specific time. The allocation procedure requires minimum iterations with the power flow to maximize performance.

The BLA function allocates loads (kW and kVAR) at distribution nodes to support other analysis functions. Loads are allocated using the load model data, and they take into account the actual or predicted feeder loads and the relevant day type and time of day. The BLA function ensures that the load model and current network topology are consistent with metered flows, current breaker statuses, and metered loads along the distribution feeders.

The input data from the BLA procedure is taken from the Network Topology Processor (NTP) function and the distribution system model. The NTP determines the substation and feeder connectivity, and the distribution system model provides all load data, including the typical load shapes.

The BLA function has the following capabilities:

- Can allocate loads for both radial and meshed distribution systems
- Incorporates system losses during the allocation procedure
- Multiple load models associated with one distribution node are supported
- Three-phase and single-phase load models associated with a three-phase node are supported

The BLA algorithm is capable of estimating loads to be restored after an outage (either forced or planned), as a function of the allocated load and cold load pick-up models that account for outage duration and type of load.

The BLA algorithm performs the following two major tasks:

- Forms measurement islands and performs measurement validation
- Allocates loads

4.1. Forming Measurement Islands and Measurement Validation

An "analog measurement group" is a collection of analog measurements at one location in the network. One exception is that voltage measurements can be in a different location than the flow measurement location but there must be zero impedance between the voltage and the flow measurement locations for the voltage measurement to be included in the analog measurement group. These measurements can be a combination of single-phase and three-phase measurements. For example, one typical analog measurement group at the secondary side of a power transformer includes three single-phase current (Amp) measurements, one three-phase real power (Watt) measurement, one three-phase reactive power (VAR) measurement and one three-phase apparent power (VA) measurement.

A "measurement island" is a collection of components bounded by one or more analog measurement groups for the purpose of allocating load. Technically speaking, a measurement island is bounded by reference bus interface objects, upstream measurement islands, downstream measurement islands, or feeder and lateral ends. All non-telemetered and conforming loads inside of a measurement island are scaled up or down based on the island's effective net Watt/VAR or VA measurement values. Telemetered, manually entered, non-conforming, and AMI meter loads do not participate in the load allocation process, therefore they are excluded from measurement islands.

The following subsections explain how a measurement island is formed.

4.1.1. Analog Measurement Groups

For any analog measurement group to be used by BLA to form a measurement island, there must be enough valid single-phase or three-phase measurements in the group to be used to calculate the effective single-phase Watt/VAR or VA measurement values for all phases. These measurements may include Watt, VAR, VA, Amp, Voltage (V) and Power Factor (PF) measurements.

Single-phase measurement means three individual phase measurements: one for phase A, one for phase B, and another for Phase C. Single-phase measurement can be a flow measurement or a voltage measurement. Three-phase measurement is supported only for flow (one measurement that is the total for all phases).

The measurement group used by BLA should include at least one of the following valid measurement combinations:

- G1: Single-phase Watt, VAR
- G2: Single-phase Watt, VA, PF

- G3: Single-phase VAR, VA, PF
- G4: Single-phase VA, PF
- G5: Single-phase Watt, Amp, V, PF
- G6: Single-phase VAR, Amp, V, PF
- G7: Single-phase Amp, V, PF
- G8: Single-phase Watt, VA, and three-phase PF
- G9: Single-phase VAR, VA, and three-phase PF
- G10: Single-phase VA, and three-phase PF
- G11: Single-phase Watt, Amp, V, and three-phase PF
- G12: Single-phase VAR, Amp, V, and three-phase PF
- G13: Single-phase Amp, V, and three-phase PF
- G14: Single-phase Amp, and three-phase Watt, VAR
- G15: Single-phase Watt, Amp, and three-phase VA, PF
- G16: Single-phase VAR, Amp, and three-phase VA, PF
- G17: Single-phase Amp, and three-phase VA, PF
- G18: Three-phase Watt, VAR
- G19: Single-phase VA
- G20: Single-phase Amp, V
- G21: Single-phase Amp, and three-phase VA
- G22: Single-phase Amp
- G23: Three-phase VA
- G24: Three-phase Amp

In the above combinations, the single-phase measurements are required and used for all phases. There are three special cases:

- When there are three valid single-phase Amp and PF measurements as well as one valid single-phase V measurement available at one location, and there are no valid single-phase Watt and VAR measurements, the effective V measurement values of other two phases are estimated so that the single-phase effective Watt and VAR measurement values can be estimated and used by BLA.
 - If there is no previous DPF solution, the V measurement values are replicated to the other two phases.
 - If there is a previous DPF solution, the difference between the measured voltage and the DPF calculated voltage on the measured phase is calculated, and this difference is applied to the DPF calculated voltages at the other two phases to estimate the effective V measurement values used by BLA.

For example, there is a valid single-phase V measurement on Phase A of 7.22 kV, and the

previous DPF calculated voltages are 7.20 kV on Phase A, 7.24 kV on Phase B, and 7.21 kV on Phase C. So, the difference between the measured voltage and DPF calculated voltage on Phase A is 0.02 kV, making the estimated effective V measurement values on phases B and C 7.26 kV and 7.23 kV, respectively.

- The single-phase measurement values at one phase can be replicated to other non-telemetered phases at the same location.

The DNAF Configuration Editor's BLA tab has an option to specify whether to replicate the single-phase measurement values to other non-telemetered phases when there are single-phase measurements only defined at one phase in the model at a given location. If this option is checked, the single-phase flow measurement values are replicated.

☒ Replicate Single-Phase SCADA Flow Measurements

- The flow (Watt, VAR, or Amp) measurements used by BLA are affected by Check Effective Zero Flow Measurements options in the DNAF Configuration Editor's BLA tab, as shown below.

The Feeder Head Device Only/All radio buttons specify whether BLA checks for effective zero value flow measurements at specified locations:

- If Feeder Head Device Only is selected, only flow measurements at feeder head devices are checked.
- If All is selected, all flow measurements are checked.

The three check boxes specify whether BLA checks the effective zero value of valid Watt, VAR, or Amp measurements at the location specified by the Feeder Head Device Only/All radio buttons. If any option is checked and the absolute value of a corresponding measurement value is smaller than the value specified for the associated Threshold Value, BLA does not use this measurement.

Check Effective Zero Flow Measurements:

☒ Watt	Threshold Value (kW):	10.0
☐ VAR	Threshold Value (kVAR):	1.0
☒ Amp	Threshold Value (Amp):	0.5
☒ Feeder Head Device Only ☐ All		

Measurement groups can also be formed from manual feeder head flows entered on the Feeder Input for Station grid tabular display. Users can enter single-phase Amps and a power factor that is used for all phases to be used by BLA:

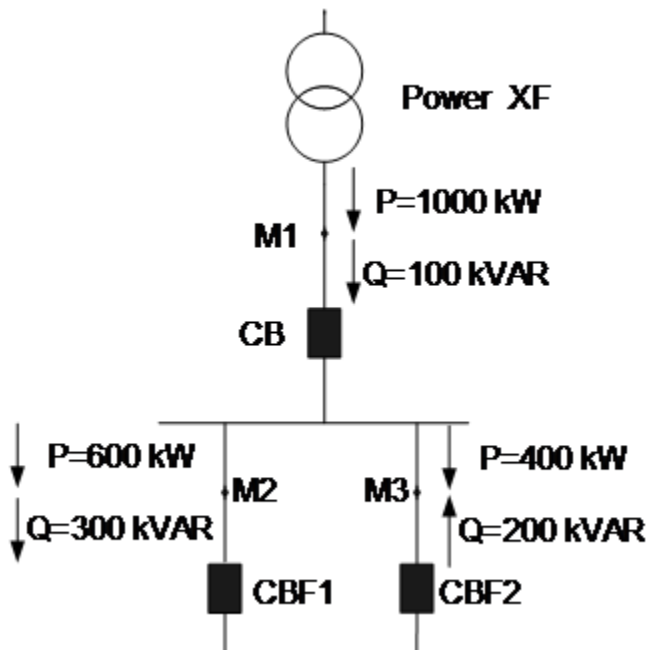
- G25: Manual feeder head single-phase Amp, PF.

If this manual feeder head measurement group is formed, all other measurement groups in the same feeder are not used by BLA.

4.1.2. Effective Device for a Measurement

When modeling an analog measurement, the user can specify the Effective Device ID for that measurement. With the effective device and the measurement node, BLA now knows the exact terminal where the measurement is attached, and which effective device is being measured.

For a real or reactive power measurement whose effective device is a branch, flow is positive into the branch and negative out of the branch. For example: the system diagram below has one power transformer, Power XF, which feeds two feeders. Power XF and circuit breaker CB are directly connected. CBF1 and CBF2 are the circuit breakers of the feeders.



There are three measurement groups in this system:

- Measurement M1 is at the connection node of Power XF and CB.
 - If its effective device is Power XF, the measurement values are negative, because from that node both the real and reactive power are out of Power XF. The measurement values are -1000 kW and -100 kVAR.
 - If its effective device is CB, the measurement values are positive, because from that node both the real and reactive power are into CB. The measurement values are 1000 kW and 100 kVAR.
- Measurement M2 is at the upstream node of CBF1. If its effective device is CBF1, the measurement values are positive, because from that node both the real and reactive power are into CBF1. The measurement values are 600 kW and 300 kVAR.
- Measurement M3 is at the upstream node of CBF2. If its effective device is CBF2:
 - From that node, the real power is into CBF2. Therefore, the measurement value is positive; it is 400 kW.

- However, the reactive power is out of CBF2 for that node. Therefore, the measurement value is negative; it is -200 kVAR.

4.1.3. Analog Measurement Group Consistency Checks

Before the measurement values are used by BLA, they may also receive an additional consistency check, which looks at the consistency between real power, reactive power, apparent power, voltage, current, and angle. The consistency checks that are executed depend entirely on the measurements that are present in a measurement group.

Two consistency check parameters are specified in the DNAF Configuration Editor:

- **Analog Measurement Group Consistency Check:** This option specifies whether the analog measurement group consistency check is performed. If the option is not selected, analog measurement group consistency check is not performed and BLA uses all analog measurement groups with valid flow measurements.
- **Analog Measurement Group Consistency Check Tolerance:** Specifies the maximum percentage difference allowed for the analog measurement group consistency check.

The consistency between real power, reactive power, apparent power, voltage, current, and angle measurements presented in a measurement group is checked against this parameter. If any percentage difference is larger than this value, the measurement group is not consistent.

☒ Enable Analog Measurement Group Consistency Check

Analog Measurement Group Consistency Check Tolerance (%):

The first step of the consistency check is to find the erroneous measurement values. These measurement values include negative VA measurement value, negative voltage measurement value, negative Amp measurement value, and the PF measurement whose absolute value is larger than 1.0. These measurement values are not used by BLA. The "Used in BLA" column in the "ANALOG MEASUREMENT GROUPS" section in the DPF result file shows whether a message has been used in BLA.

For the measurement values that passed the first step, the following checks are performed if applicable:

- C1: If valid single-phase Watt, VAR, and PF measurements exist, the sign consistency among them is checked by equation 4.1.

$$Watt_{1ph-meas} \cdot VAR_{1ph-meas} \cdot PF_{1ph-meas} \geq 0.0 \quad (\text{Equation 4.1})$$

In the equation above, $PF_{1ph-meas}$ is negative if it is leading.

- C2: If valid single-phase Watt, VAR, and PF measurements exist, the difference between the absolute value of the Watt measurement and the effective Watt measurement value calculated from Watt, VAR, and PF measurements should be within the tolerance. It is checked by equation

4.2.

$$\left| \sqrt{Watt_{1ph-meas}^2 + VAR_{1ph-meas}^2} \cdot PF_{1ph-meas} - |Watt_{1ph-meas}| \right| \leq Tol \cdot |Watt_{1ph-meas}| \quad (\text{Equation 4.2})$$

Where:

Tol is the maximum percent difference allowed (specified in Analog Measurement Group Consistency Check Tolerance text box).

- C3: If valid single-phase Watt, VAR, and PF measurements exist, the difference between the absolute value of the VAR measurement and the effective VAR measurement value calculated from Watt, VAR, and PF measurements should be within the tolerance. It is checked by equation 4.3.

$$\left| \sqrt{Watt_{1ph-meas}^2 + VAR_{1ph-meas}^2} \cdot \sqrt{1 - PF_{1ph-meas}^2} - |VAR_{1ph-meas}| \right| \leq Tol \cdot |VAR_{1ph-meas}| \quad (\text{Equation 4.3})$$

- C4: If valid single-phase Watt, VAR, and VA measurements exist, the difference between the VA measurement and the effective VA measurement value calculated from Watt and VAR measurements should be within the tolerance. It is checked by equation 4.4.

$$|VA_{1ph-meas} - \sqrt{Watt_{1ph-meas}^2 + VAR_{1ph-meas}^2}| \leq Tol \cdot VA_{1ph-meas} \quad (\text{Equation 4.4})$$

- C5: If valid single-phase VA, Amp, and V measurements exist, the difference between the VA measurement and the effective VA measurement value calculated from Amp and V measurements should be within the tolerance. It is checked by equation 4.5.

$$|VA_{1ph-meas} - V_{1ph-meas} \cdot I_{1ph-meas}| \leq Tol \cdot VA_{1ph-meas} \quad (\text{Equation 4.5})$$

- C6: If valid single-phase Watt, VAR, Amp, and V measurements exist, the two effective VA measurement values can be calculated: one from Watt and VAR measurements and the other from Amp and V measurements. The difference between them should be within the tolerance. It is checked by equation 4.6.

$$\left| \sqrt{Watt_{1ph-meas}^2 + VAR_{1ph-meas}^2} - V_{1ph-meas} \cdot I_{1ph-meas} \right| \leq Tol \cdot V_{1ph-meas} \cdot I_{1ph-meas} \quad (\text{Equation 4.6})$$

- C7: If valid single-phase Watt, VA, and PF measurements exist, the difference between the absolute value of the Watt measurement and the effective Watt measurement value calculated from VA and PF measurements should be within the tolerance. It is checked by equation 4.7.

$$\left| |VA_{1ph-meas} \cdot PF_{1ph-meas}| - |Watt_{1ph-meas}| \right| \leq Tol \cdot |Watt_{1ph-meas}| \quad (\text{Equation 4.7})$$

- C8: If valid single-phase VAR, VA, and PF measurements exist, the difference between the

absolute value of the VAR measurement and the effective VAR measurement value calculated from VA and PF measurements should be within the tolerance. It is checked by equation 4.8.

$$\left| \left| VA_{1ph-meas} \cdot \sqrt{1 - PF_{1ph-meas}^2} \right| - |VAR_{1ph-meas}| \right| \leq Tol \cdot |VAR_{1ph-meas}| \quad (\text{Equation 4.8})$$

- C9: If valid single-phase Watt, Amp, V, and PF measurements exist, the difference between the absolute value of the Watt measurement and the effective Watt measurement value calculated from Amp, V, and PF measurements should be within the tolerance. It is checked by equation 4.9.

$$\left| \left| V_{1ph-meas} \cdot I_{1ph-meas} \cdot PF_{1ph-meas} \right| - |Watt_{1ph-meas}| \right| \leq Tol \cdot |Watt_{1ph-meas}| \quad (\text{Equation 4.9})$$

- C10: If valid single-phase VAR, Amp, V, and PF measurements exist, the difference between the absolute value of the VAR measurement and the effective VAR measurement value calculated from Amp, V, and PF measurements should be within the tolerance. It is checked by equation 4.10.

$$\left| \left| V_{1ph-meas} \cdot I_{1ph-meas} \cdot \sqrt{1 - PF_{1ph-meas}^2} \right| - |VAR_{1ph-meas}| \right| \leq Tol \cdot |VAR_{1ph-meas}| \quad (\text{Equation 4.10})$$

- C11: If valid single-phase Watt and VA measurements exist, the VA measurement should be larger than the absolute value of the Watt measurement within the tolerance. It is checked by equation 4.11.

$$VA_{1ph-meas} \geq (1 - Tol) \cdot |Watt_{1ph-meas}| \quad (\text{Equation 4.11})$$

- C12: If valid single-phase VAR and VA measurements exist, the VA measurement should be larger than the absolute value of the VAR measurement within the tolerance. It is checked by equation 4.12.

$$VA_{1ph-meas} \geq (1 - Tol) \cdot |VAR_{1ph-meas}| \quad (\text{Equation 4.12})$$

- C13: If valid single-phase Watt, Amp, and V measurements exist, the effective VA measurement value calculated from Amp and V measurements should be larger than the absolute value of the Watt measurement within the tolerance. It is checked by equation 4.13.

$$V_{1ph-meas} \cdot I_{1ph-meas} \geq (1 - Tol) \cdot |Watt_{1ph-meas}| \quad (\text{Equation 4.13})$$

- C14: If valid single-phase VAR, Amp, and V measurements exist, the effective VA measurement value calculated from Amp and V measurements should be larger than the absolute value of

the VAR measurement within the tolerance. It is checked by equation 4.14.

$$V_{1ph-meas} \cdot I_{1ph-meas} \geq (1 - Tol) \cdot |VAR_{1ph-meas}| \quad (\text{Equation 4.14})$$

- C15: If valid single-phase Watt and PF measurements exist and the PF measurement value is 0, the value of the Watt measurement should be 0.
- C16: If valid single-phase VAR and PF measurements exist and the PF measurement value is 1, the value of the VAR measurement should be 0.

There are three scenarios after the applicable checks.

- All applicable checks are passed.

In this scenario, BLA uses all the measurement values in this measurement group.

- Some of the applicable checks fail and one or more obvious wrong measurement values are found.

In this scenario, BLA does not use this wrong measurement value, but uses other correct measurement values in this measurement group. For example, assume there are five measurements in a measurement group: Watt measurement of 400 kW, VAR measurement of 300 kVAR, PF measurement of 0.5, Amp measurement of 100 A, and V measurement of 5 kV. The tolerance is 5%. The following applicable checks are conducted:

C1: $400 \cdot 300 \cdot 0.5 \geq 0.0$, passed;

C2: $|\sqrt{400^2 + 300^2} \cdot 0.5| - 400| = 150 \leq 0.05 \cdot 400 = 20$, failed;

C3: $|\sqrt{400^2 + 300^2} \cdot \sqrt{1 - 0.5^2}| - 300| = 133 \leq 0.05 \cdot 300 = 15$, failed;

C6: $|\sqrt{400^2 + 300^2} - 5 \cdot 100| = 0 \leq 0.05 \cdot 5 \cdot 100 = 25$, passed;

C9: $||5 \cdot 100 \cdot 0.5| - 400| = 150 \leq 0.05 \cdot 400 = 20$, failed;

C10: $||5 \cdot 100 \cdot \sqrt{1 - 0.5^2}| - 300| = 133 \leq 0.05 \cdot 300 = 15$, failed;

C13: $5 \cdot 100 = 500 \geq (1 - 0.05) \cdot 400 = 380$, passed;

C14: $5 \cdot 100 = 500 \geq (1 - 0.05) \cdot 300 = 285$, passed;

The PF measurement value is found to be not correct, so BLA does not use it. BLA uses other measurement values in this measurement group.

- Some or all applicable checks fail and there is no obvious single wrong measurement value.

In this scenario, this measurement group is not used by BLA. For example, assume there are

three measurements in a measurement group: Watt measurement of 400 kW, VAR measurement of 300 kVAR, and VA measurement of 600 kVA. The tolerance is 5%. Then the checks C11 and C12 are passed, and C4 fails. There is no obvious wrong measurement value, so BLA does not use this measurement group.

4.1.4. Analog Measurement Group Effective Measurement

If a measurement group is used by BLA, the effective single-phase flow measurements are calculated to be used for scaling the loads. In an analog measurement group, there may be a set of redundant measurements that BLA can use after the consistency check. In this case, there are several methods for calculating the effective single-phase flow measurements. It is preferred to calculate the effective single-phase Watt and VAR measurement values. If effective single-phase Watt or effective single-phase VAR measurement values cannot be determined, the effective single-phase VA measurement values are determined.

The BLA uses the priority order from G1 to G25 for processing measurements. The following methods from M1 to M25 are the calculations used by BLA to determine the effective Watt, VAR, or VA measurement values:

- M1: Calculate single-phase effective Watt and VAR from G1 (single-phase Watt, VAR).

$$Watt_{\text{effe}} = Watt_{1\text{ph-meas}} \quad (\text{Equation 4.15})$$

$$VAR_{\text{effe}} = VAR_{1\text{ph-meas}} \quad (\text{Equation 4.16})$$

- M2: Calculate single-phase effective Watt and VAR from G2 (single-phase Watt, VA, PF).

$$Watt_{\text{effe}} = Watt_{1\text{ph-meas}} \quad (\text{Equation 4.17})$$

$$VAR_{\text{effe}} = \text{sign} \cdot VA_{1\text{ph-meas}} \cdot \sin(\cos^{-1} PF_{1\text{ph-meas}}) \quad (\text{Equation 4.18})$$

Where:

$$\text{sign} = \begin{cases} 1 & \text{if } Watt_{1\text{ph-meas}} \cdot PF_{1\text{ph-meas}} \geq 0 \\ -1 & \text{if } Watt_{1\text{ph-meas}} \cdot PF_{1\text{ph-meas}} < 0 \end{cases}$$

- M3: Calculate single-phase effective Watt and VAR from G3 (single-phase VAR, VA, PF)

$$Watt_{\text{effe}} = \text{sign} \cdot VA_{1\text{ph-meas}} \cdot |PF_{1\text{ph-meas}}| \quad (\text{Equation 4.19})$$

Where:

$$sign = \begin{cases} 1 & \text{if } VAR_{1ph-meas} \cdot PF_{1ph-meas} \geq 0 \\ -1 & \text{if } VAR_{1ph-meas} \cdot PF_{1ph-meas} < 0 \end{cases}$$

$$VAR_{effe} = VAR_{1ph-meas} \quad (\text{Equation 4.20})$$

- M4: Calculate single-phase effective Watt and VAR from G4 (single-phase VA, PF).

$$Watt_{effe} = VA_{1ph-meas} \cdot |PF_{1ph-meas}| \quad (\text{Equation 4.21})$$

$$VAR_{effe} = VA_{1ph-meas} \cdot \sin(\cos^{-1}PF_{1ph-meas}) \quad (\text{Equation 4.22})$$

- M5: Calculate single-phase effective Watt and VAR from G5 (single-phase Watt, Amp, V, PF).

$$Watt_{effe} = Watt_{1ph-meas} \quad (\text{Equation 4.23})$$

$$VAR_{effe} = sign \cdot V_{1ph-meas} \cdot Amp_{1ph-meas} \cdot \sin(\cos^{-1}PF_{1ph-meas}) \quad (\text{Equation 4.24})$$

Where:

$$sign = \begin{cases} 1 & \text{if } Watt_{1ph-meas} \cdot PF_{1ph-meas} \geq 0 \\ -1 & \text{if } Watt_{1ph-meas} \cdot PF_{1ph-meas} < 0 \end{cases}$$

- M6: Calculate single-phase effective Watt and VAR from G6 (single-phase VAR, Amp, V, PF).

$$Watt_{effe} = sign \cdot V_{1ph-meas} \cdot Amp_{1ph-meas} \cdot |PF_{1ph-meas}| \quad (\text{Equation 4.25})$$

Where:

$$sign = \begin{cases} 1 & \text{if } VAR_{1ph-meas} \cdot PF_{1ph-meas} \geq 0 \\ -1 & \text{if } VAR_{1ph-meas} \cdot PF_{1ph-meas} < 0 \end{cases}$$

$$VAR_{effe} = VAR_{1ph-meas} \quad (\text{Equation 4.26})$$

- M7: Calculate single-phase effective Watt and VAR from G7 (single-phase Amp, V, PF).

$$Watt_{effe} = V_{1ph-meas} \cdot Amp_{1ph-meas} \cdot |PF_{1ph-meas}| \quad (\text{Equation 4.27})$$

$$VAR_{effe} = V_{1ph-meas} \cdot Amp_{1ph-meas} \cdot \sin(\cos^{-1}PF_{1ph-meas}) \quad (\text{Equation 4.28})$$

- M8: Calculate single-phase effective Watt and VAR from G8 (single-phase Watt, VA, and three-phase PF). The calculated effective VAR value is approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{effe} = Watt_{1ph-meas} \quad (\text{Equation 4.29})$$

$$VAR_{effe} = sign \cdot VA_{1ph-meas} \cdot \sin(\cos^{-1}PF_{3ph-meas}) \quad (\text{Equation 4.30})$$

Where:

$$sign = \begin{cases} 1 & \text{if } Watt_{1ph-meas} \cdot PF_{3ph-meas} \geq 0 \\ -1 & \text{if } Watt_{1ph-meas} \cdot PF_{3ph-meas} < 0 \end{cases}$$

- M9: Calculate single-phase effective Watt and VAR from G9 (single-phase VAR, VA, and three-phase PF). The calculated effective Watt value is approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{effe} = sign \cdot VA_{1ph-meas} \cdot |PF_{3ph-meas}| \quad (\text{Equation 4.31})$$

Where:

$$sign = \begin{cases} 1 & \text{if } VAR_{1ph-meas} \cdot PF_{3ph-meas} \geq 0 \\ -1 & \text{if } VAR_{1ph-meas} \cdot PF_{3ph-meas} < 0 \end{cases}$$

$$VAR_{effe} = VAR_{1ph-meas} \quad (\text{Equation 4.32})$$

- M10: Calculate single-phase effective Watt and VAR from G10 (single-phase VA, and three-phase PF). The calculated effective Watt and VAR values are approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{effe} = VA_{1ph-meas} \cdot |PF_{3ph-meas}| \quad (\text{Equation 4.33})$$

$$VAR_{effe} = VA_{1ph-meas} \cdot \sin(\cos^{-1} PF_{3ph-meas}) \quad (\text{Equation 4.34})$$

- M11: Calculate single-phase effective Watt and VAR from G11 (single-phase Watt, Amp, V, and three-phase PF). The calculated effective VAR value is approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{effe} = Watt_{1ph-meas} \quad (\text{Equation 4.35})$$

$$VAR_{effe} = sign \cdot V_{1ph-meas} \cdot Amp_{1ph-meas} \cdot \sin(\cos^{-1} PF_{3ph-meas}) \quad (\text{Equation 4.36})$$

Where:

$$sign = \begin{cases} 1 & \text{if } Watt_{1ph-meas} \cdot PF_{3ph-meas} \geq 0 \\ -1 & \text{if } Watt_{1ph-meas} \cdot PF_{3ph-meas} < 0 \end{cases}$$

- M12: Calculate single-phase effective Watt and VAR from G12 (single-phase VAR, Amp, V, and three-phase PF). The calculated effective Watt value is approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{\text{effe}} = sign \cdot V_{1\text{ph-meas}} \cdot Amp_{1\text{ph-meas}} \cdot |PF_{3\text{ph-meas}}| \quad (\text{Equation 4.37})$$

Where:

$$sign = \begin{cases} 1 & \text{if } VAR_{1\text{ph-meas}} \cdot PF_{3\text{ph-meas}} \geq 0 \\ -1 & \text{if } VAR_{1\text{ph-meas}} \cdot PF_{3\text{ph-meas}} < 0 \end{cases}$$

$$VAR_{\text{effe}} = VAR_{1\text{ph-meas}} \quad (\text{Equation 4.38})$$

- M13: Calculate single-phase effective Watt and VAR from G13 (single-phase Amp, V, and three-phase PF). The calculated effective Watt and VAR values are approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{\text{effe}} = V_{1\text{ph-meas}} \cdot Amp_{1\text{ph-meas}} \cdot |PF_{3\text{ph-meas}}| \quad (\text{Equation 4.39})$$

$$VAR_{\text{effe}} = V_{1\text{ph-meas}} \cdot Amp_{1\text{ph-meas}} \cdot \sin(\cos^{-1} PF_{3\text{ph-meas}}) \quad (\text{Equation 4.40})$$

- M14: Calculate single-phase effective Watt and VAR from G14 (single-phase Amp, and three-phase Watt, VAR). The calculated effective Watt and VAR values are approximate because it is assumed that the power factor is the same for all three phases. The kVA allocation is used; the kVA target is calculated based on the calculated effective Watt and VAR values.

Note

Single-phase capacitor/DER injections increase the power factor deviation for phases, which makes the calculated effective Watt and VAR values less accurate.

$$Watt_{\text{effe}} = Watt_{3\text{ph-meas}} \cdot \frac{Amp_{1\text{ph-meas}}}{\sum_{i=1}^3 Amp_{1\text{ph-meas},i}} \quad (\text{Equation 4.41})$$

$$VAR_{\text{effe}} = VAR_{3\text{ph-meas}} \cdot \frac{Amp_{1\text{ph-meas}}}{\sum_{i=1}^3 Amp_{1\text{ph-meas},i}} \quad (\text{Equation 4.42})$$

- M15: Calculate single-phase effective Watt and VAR from G15 (single-phase Watt, Amp, and three-phase VA, PF). The calculated effective VAR value is approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{\text{effe}} = Watt_{1\text{ph-meas}} \quad (\text{Equation 4.43})$$

$$VAR_{\text{effe}} = sign \cdot VA_{3\text{ph-meas}} \cdot \sin(\cos^{-1} PF_{3\text{ph-meas}}) \cdot \frac{Amp_{1\text{ph-meas}}}{\sum_{i=1}^3 Amp_{1\text{ph-meas},i}} \quad (\text{Equation 4.44})$$

Where:

$$sign = \begin{cases} 1 & \text{if } Watt_{1ph-meas} \cdot PF_{3ph-meas} \geq 0 \\ -1 & \text{if } Watt_{1ph-meas} \cdot PF_{3ph-meas} < 0 \end{cases}$$

- M16: Calculate single-phase effective Watt and VAR from G16 (single-phase VAR, Amp, and three-phase VA, PF). The calculated effective Watt value is approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{effe} = sign \cdot VA_{3ph-meas} \cdot |PF_{3ph-meas}| \cdot \frac{Amp_{1ph-meas}}{\sum_{i=1}^3 Amp_{1ph-meas,i}} \quad (\text{Equation 4.45})$$

Where:

$$sign = \begin{cases} 1 & \text{if } VAR_{1ph-meas} \cdot PF_{3ph-meas} \geq 0 \\ -1 & \text{if } VAR_{1ph-meas} \cdot PF_{3ph-meas} < 0 \end{cases}$$

$$VAR_{effe} = VAR_{1ph-meas} \quad (\text{Equation 4.46})$$

- M17: Calculate single-phase effective Watt and VAR from G17 (single-phase Amp, and three-phase VA, PF). The calculated effective Watt and VAR values are approximate because it is assumed that the power factor is the same for all three phases.

$$Watt_{effe} = VA_{3ph-meas} \cdot |PF_{3ph-meas}| \cdot \frac{Amp_{1ph-meas}}{\sum_{i=1}^3 Amp_{1ph-meas,i}} \quad (\text{Equation 4.47})$$

$$VAR_{effe} = VA_{3ph-meas} \cdot \sin(\cos^{-1} PF_{3ph-meas}) \cdot \frac{Amp_{1ph-meas}}{\sum_{i=1}^3 Amp_{1ph-meas,i}} \quad (\text{Equation 4.48})$$

- M18: Calculate single-phase effective Watt and VAR from G18 (three-phase Watt, VAR). The calculated effective Watt and VAR values are approximate because it is assumed that three phases are balanced.

$$Watt_{effe} = \frac{Watt_{3ph-meas}}{3} \quad (\text{Equation 4.49})$$

$$VAR_{effe} = \frac{VAR_{3ph-meas}}{3} \quad (\text{Equation 4.50})$$

- M19: Calculate single-phase effective VA from G19 (single-phase VA).

$$VA_{effe} = VA_{1ph-meas} \quad (\text{Equation 4.51})$$

- M20: Calculate single-phase effective VA from G20 (single-phase Amp, V).

$$VA_{effe} = V_{1ph-meas} \cdot Amp_{1ph-meas} \quad (\text{Equation 4.52})$$

- M21: Calculate single-phase effective VA from G21 (single-phase Amp, and three-phase VA). The calculated effective VA value is approximate because it is assumed that the power factor is the same for all three phases.

$$VA_{\text{effe}} = VA_{3\text{ph-meas}} \cdot \frac{Amp_{1\text{ph-meas}}}{\sum_{i=1}^3 Amp_{1\text{ph-meas},i}} \quad (\text{Equation 4.53})$$

- M22: Calculate single-phase effective VA from G22 (single-phase Amp). The "Multiply Amp Measurement with Nominal Voltage to Calculate VA Target" check box in DCE specifies the voltage value used to calculate the effective VA value. If this option is selected, the nominal voltage is used. Otherwise, the DPF calculated voltage is used.

$$VA_{\text{effe}} = V \cdot Amp_{1\text{ph-meas}} \quad (\text{Equation 4.54})$$

- M23: Calculate single-phase effective VA from G23 (three-phase VA). The calculated effective VA value is approximate because it is assumed that three phases are balanced.

$$VA_{\text{effe}} = \frac{VA_{3\text{ph-meas}}}{3} \quad (\text{Equation 4.55})$$

- M24: Calculate single-phase effective VA from G24 (three-phase Amp). The "Multiply Amp Measurement with Nominal Voltage to Calculate VA Target" check box in DCE specifies the voltage value used to calculate the effective VA value. If this option is selected, the nominal voltage is used. Otherwise, the DPF calculated voltage is used.

$$VA_{\text{effe}} = \frac{V \cdot Amp_{3\text{ph-meas}}}{3} \quad (\text{Equation 4.56})$$

- M25: Calculate single-phase effective Watt and VAR from G25 (manual feeder head single-phase Amp, PF). The manually entered power factor is used for all phases and the $V_{1\text{ph-calc}}$ is the single-phase voltage calculated in the previous BLA iteration, or nominal voltage if it is the first BLA iteration.

$$Watt_{\text{effe}} = V_{1\text{ph-calc}} \cdot Amp_{1\text{ph-man}} \cdot |PF_{\text{man}}| \quad (\text{Equation 4.57})$$

$$VAR_{\text{effe}} = V_{1\text{ph-calc}} \cdot Amp_{1\text{ph-man}} \cdot \sin(\cos^{-1} PF_{\text{man}}) \quad (\text{Equation 4.58})$$

4.1.5. No Load Measurement Island

Some measurement islands do not have any load in them. The most common such measurement islands are the station internal measurement islands that are upstream of feeders. BLA retains these station internal measurement islands.

Occasionally, there are some no-load measurement islands in feeders. BLA can merge these measurement islands with the neighboring measurement islands, based on the setting of the

parameter shown below. The parameter is found on the BLA tab in DCE.

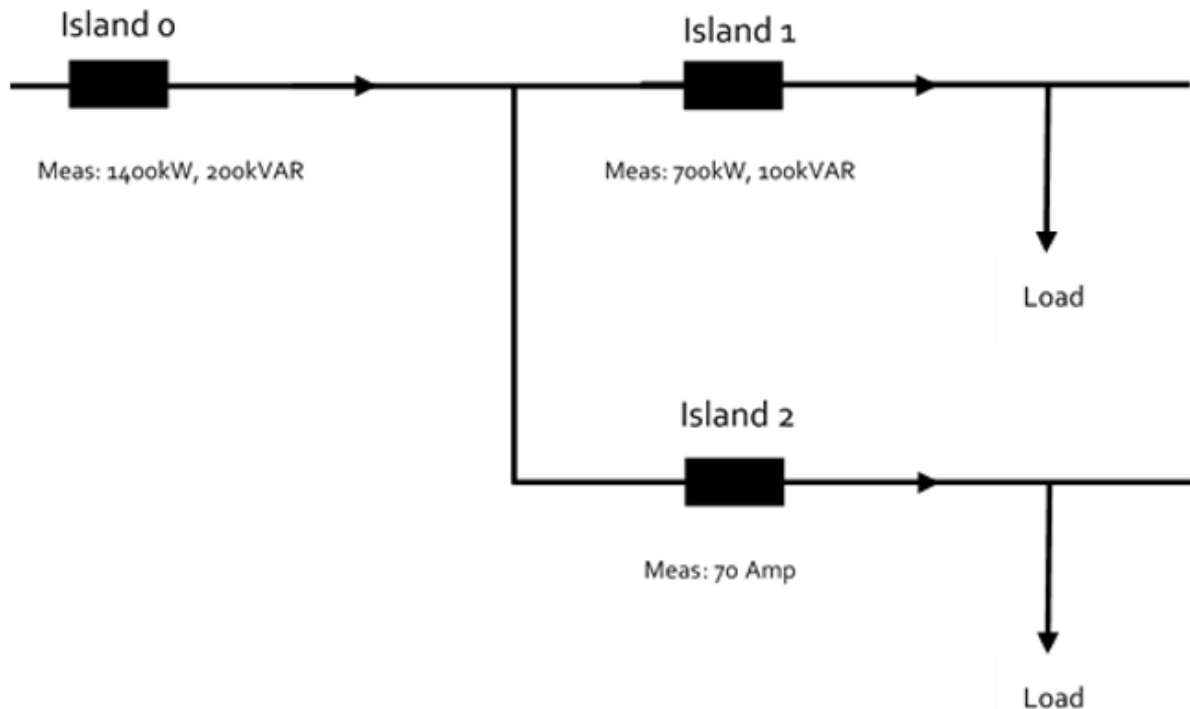
☒ Merge No Load Measurement Islands

If this option is not selected, BLA retains these no-load measurement islands.

If this option is selected, BLA merges these no-load measurement islands with the neighboring measurement islands based on the following two rules:

- If the measurement island is bounded by neighboring measurement islands with measurement groups of different measurement sets (such as Watt/VAR vs. Amp only):
 - The measurement island will be merged into the neighboring island.
 - The measurement group with the least complete measurement set between them and this measurement group will not be used by BLA.

For example, in the simple system diagram below, measurement island0 has two neighboring measurement islands: Island1 and Island2. There is a valid single-phase Watt and VAR measurement group (700kW, 100kVAR) between Island0 and Island1, and there is a valid single-phase Amp measurement group (70Amp) between Island0 and Island2. Island0 will be merged into Island2 and the single-phase Amp measurement group (70Amp) between Island0 and Island2 will not be used by BLA.



- If the measurement island is bounded by neighboring measurement islands with measurement groups of the same measurement set, this measurement island will be merged to one of the downstream neighboring islands. The measurement group between them is not used by BLA.

4.1.6. Less Accurate Downstream Measurement Island

In some BLA/DPF scenarios, an upstream measurement island has valid single-phase Watt and VAR measurements, and the downstream islands only have valid Amp or VA measurements. In the DNAF Configuration Editor BLA tab, the "Method for Handling Less Accurate Island" field deals with downstream islands that have less complete measurement groups. The user can decide how to process the less complete measurement groups by selecting either the Merge Downstream Island or Retain Measurement Islands option.

4.1.6.1. Merge Downstream Island

If the Merge Downstream Island option is selected, BLA merges the downstream island into the upstream island and forms one larger island. Thus, the measurement group between these two islands is not used by BLA.

If a downstream island is more accurate than an upstream island (for example, it has a more complete measurement set), no merging is performed. The upstream island uses kVA allocation and the downstream island uses the kW/kVAR method.

4.1.6.2. Retain Measurement Islands

If the Retain Measurement Islands option is selected, BLA retains the existing upstream and downstream measurement islands. BLA must estimate a power factor for the island with the downstream measurement group to allow the upstream island net Watt and VAR flows to be calculated. This power factor is calculated from the upstream measurement group and any real and reactive injection from capacitors, generators, and Distributed Energy Resources (DERs) in these two measurement islands. The assumption for this is that the loads' power factor is the same in these two islands, which is approximate.

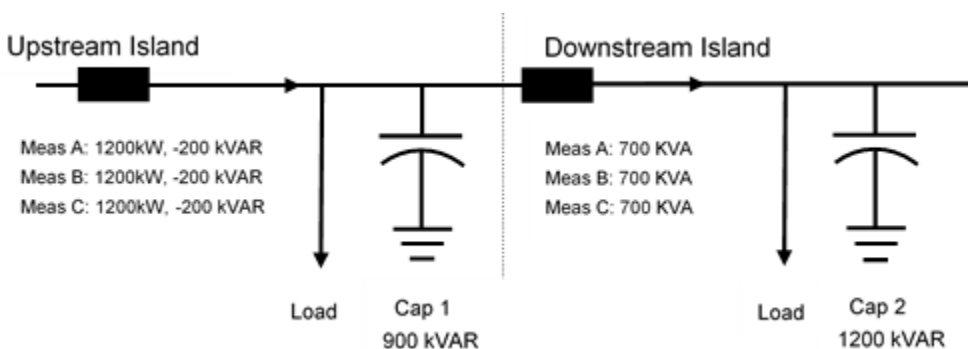


Figure 17. Sample Two-Island System

Three additional steps are performed before BLA allocates the loads. These steps are illustrated using the two-island system shown in figure above.

The upstream measurement island has single-phase Watt and VAR measurements, and the downstream island has single-phase VA measurements. There is one 300 kVAR per phase (900 kVAR

three-phase) capacitor Cap 1, in the upstream island, and there is one 400 kVAR per phase (1200 kVAR three-phase) capacitor, Cap 2, in the downstream island. Both capacitors are on in this example.

Step 1: Calculate the power factor for both islands.

BLA uses the upstream island's W measurements and adjusts it by the real power output of connected generators and DERs in both islands from the last BLA iteration to calculate the total real power of all loads in both islands. In this example, the total real power for all loads is 1200 kW per-phase.

BLA uses the upstream island's VAR measurements and adjusts it by the reactive power output of connected capacitors, generators and DERs in both islands from the last BLA iteration to calculate the total reactive power of all loads in both islands. In this simple example, it is assumed that the voltages across all nodes in the system are 1.0 per unit, so the per phase reactive power of all loads is 300+400-200= 500 kVAR.

Then, BLA calculates the effective power factor of all loads in both islands. In this example, the power factor is $\frac{1200}{\sqrt{1200^2+500^2}} = 0.923$ lagging.

Step 2: Apply power factor to the downstream measurement island.

BLA uses the effective power factor calculated in step 1 denoted by pf to calculate the effective Watt and VAR flow in the downstream island.

The downstream island's total per phase apparent power should be equal to the single-phase VA measurement denoted by S_{meas} at the head of the downstream island. If the measurement at the head of the downstream island is not single-phase VA but single-phase Amp, the Amp measurement is multiplied by the nominal voltage at the measurement node to get the equivalent single-phase VA measurement value.

If P_e denotes the per phase effective Watt flow into the downstream island, S_{load} denotes the total per phase load apparent power in the downstream island, and P_{inj} denotes the per phase real power output from the connected generators and DERs in the downstream island, then:

$$P_e = S_{load} \cdot pf - P_{inj} \quad (\text{Equation 4.59})$$

If Q_e denotes the per phase effective VAR flow into the downstream island, and Q_{inj} denotes the per phase reactive power output from the connected capacitors, generators and DERs in the downstream island, then:

$$Q_e = S_{load} \cdot \sqrt{1 - pf^2} - Q_{inj} \quad (\text{Equation 4.60})$$

Since $S_{meas}^2 = P_e^2 + Q_e^2$ by solving this equation, the total per phase load can be calculated by the following equation:

$$S_{load} = \sqrt[2]{1 - pf^2} \cdot Q_{inj} + pf \cdot P_{inj} + \sqrt[2]{S_{meas}^2 - P_{inj}^2 \cdot (1 - pf^2) - Q_{inj}^2 \cdot pf^2 + 2 \cdot P_{inj} \cdot Q_{inj} \cdot pf \cdot \sqrt[2]{1 - pf^2}} \quad (\text{Equation 4.61})$$

With equation 4.61, S_{load} can be determined; with equations 4.59 and 4.60, P_e and Q_e can be determined, respectively.

In the example, the per phase apparent power load is calculated as follows:

$$S_{load} = \sqrt[2]{1 - 0.923^2} \cdot 400 + \sqrt[2]{700^2 - 400^2 \cdot 0.923^2} = 748.6 \text{ kVA}$$

The per phase real power flow in the downstream island is:

$$P_e = 748.6 \cdot 0.923 = 691.0 \text{ kW}$$

The per phase reactive power flow in the downstream island is:

$$Q_e = 748.6 \cdot \sqrt[2]{1 - 0.923^2} - 400 = -111.9 \text{ kVAR}$$

Step 3: Calculate effective upstream measurement island flow.

BLA calculates the effective net per phase real power flow in the upstream island by subtracting the effective per phase real power flow in the downstream island from the single-phase Watt measurements of the upstream island. In the example, the net per phase real power flow in the upstream island is $1200 - 691.0 = 509.0 \text{ kW}$.

BLA subtracts the effective per phase reactive power in the downstream island from the single-phase VAR measurements of the upstream island to get the net per phase reactive flow in the upstream island. In the example system, the net per phase reactive power in the upstream island is $-200 - (-111.9) = -88.1 \text{ kVAR}$.

4.1.7. Measurement Island Consistency Checks

The initial value for each load is taken as the product of the nominal load (either P and PF, or P and Q) and the scaling factor taken from the appropriate load schedule for the appropriate time. A "load schedule" or profile represents the normalized real and reactive power values over a 24-hour period. The load schedules are categorized based on day type (weekday, weekend, holiday, others) and season.

After the measurement islands have been formed, using the equivalent measurements determined

from SCADA and measurement consistency validation, the measurement islands are subjected to one or two additional checks: the nominal island load check and the net island load check. The checks are done for apparent power and also for real or reactive power if the valid effective real or reactive power measurements can be formed. The results of these consistency checks do not impact BLA or any DNAF applications. It is only an indication of how accurately the load model approximates the present loading given by the SCADA measurements.

4.1.7.1. Nominal Island Load Check

The nominal island load check sums up all the scheduled nominal loads in this island and all its downstream islands. It also sums up the variation of all these loads based on their load category accuracy (or variability). Then it calculates the high and low range of the island load from the sum of the total load and its variation. The nominal Island load check also calculates the high and low range of the equivalent island measurements at the island head measurement group from the effective measurement values and their associated accuracy. If there is an overlap between the load range and the measurement range, then the island has passed the nominal load check.

The simplified single-phase two-island system shown in figure below illustrates this check:

- The upstream island has measurement 1200 kVA and 5% accuracy.
- The downstream island has measurement 700 kVA and 5% accuracy.
- There is one load in the upstream island, with load value 250kVA and 40% accuracy.
- There is one load in the downstream island, with load value 600kVA and 20% accuracy.

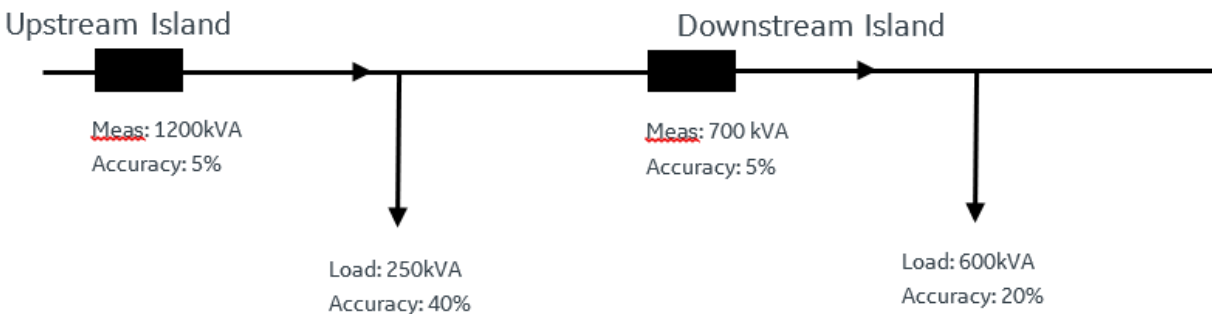


Figure 18. Sample Single-Phase Two-Island System

For the upstream island:

- The high and low range of the equivalent island measurements are $1200 \times (1+5\%) = 1260$ kVA and $1200 \times (1-5\%) = 1140$ kVA, respectively.
- The high and low range of the load are $250 \times (1+40\%) + 600 \times (1+20\%) = 1070$ kVA and $250 \times (1-40\%) + 600 \times (1-20\%) = 630$ kVA, respectively.

Since there is no overlap between the measurement range (1140-1260) and the load range (630-1070), the upstream island has failed the nominal load check.

For the downstream island:

- The high and low range of the equivalent island measurements are $700 \times (1+5\%) = 735$ kVA and $700 \times (1-5\%) = 665$ kVA, respectively.
- The high and low range of the load are $600 \times (1+20\%) = 720$ kVA and $600 \times (1-20\%) = 480$ kVA, respectively.

Since there is an overlap between the measurement range (665-735) and the load range (480-720), the downstream island has passed the nominal load check.

All islands that have loads inside them and do not contain a reference bus interface object have this check performed.

4.1.7.2. Net Island Load Check

The net island load check is similar to the nominal island load check, except that only the loads inside the island are compared with the equivalent net island flow measurements. The net measurement is processed so that it represents the net load inside the island.

The check is illustrated using the simplified single-phase two-island system shown in figure above. For the upstream island:

- The high and low range of the equivalent island measurements are $(1200-700)(1+5\%) = 525$ kVA **and** $(1200-700)(1-5\%) = 475$ kVA, respectively.
- The high and low range of the load are $250 \times (1+40\%) = 350$ kVA and $250 \times (1-40\%) = 150$ kVA, respectively.

Since there is no overlap between the measurement range (475-525) and the load range (150-350), the upstream island has failed the net load check.

For the downstream island:

- The high and low range of the equivalent island measurements are $700 \times (1+5\%) = 735$ kVA and $700 \times (1-5\%) = 665$ kVA, respectively.
- The high and low range of the load are $600 \times (1+20\%) = 720$ kVA and $600 \times (1-20\%) = 480$ kVA, respectively.

Since there is an overlap between the measurement range (665-735) and the load range (480-720), the downstream island has passed the net load check. For this example, the nominal and net island checks are identical for the downstream island.

4.1.8. BLA Consistency Checks on Switching Devices

In BLA, consistency checks are performed for telemetered switching devices. The switching device is checked for inconsistencies between the device status and the network topology. Also, the device analog measurements are checked for consistency, including current, voltage, real power, reactive power, and apparent power.

For the flow measurements, either the forward measurement or the reverse measurement is considered when checking for inconsistencies. To determine which one to use, the following rules are applied.

- The forward measurement and the reverse measurement should be modeled on opposite terminals of the switching device. Under nominal conditions, the flow measurement is modeled on the upstream node, and the reverse flow measurement is modeled on the downstream node. It is expected that the effective device is modeled for these measurement. (For related information, see [Effective Device for a Measurement](#))
- When the forward measurement is modeled on the "from" node of the switching device under nominal conditions, it is expected that the "from" node is at the upstream side and that the "from" node DPF flows should be positive under nominal conditions. In this situation the forward flow measurement is considered.
- When the forward measurement is modeled on the "to" node of the switching device, it is expected that the "to" node is at the upstream side and that the "to" node DPF flows should be negative under nominal conditions. In this situation the forward flow measurement is considered.
- When the reverse measurement is modeled on the "from" node of the switching device, it is expected that the "from" node is at the downstream side and that the "from" node DPF flows should be negative under nominal conditions. In this situation the reverse flow measurement is considered.
- When the reverse measurement is modeled on the "to" node of the switching device, it is expected that the "to" node is at the downstream side and that the "to" node DPF flows should be positive under nominal conditions. In this situation the reverse flow measurement is considered.
- For abnormal conditions, when a forward measurement is modeled on the "from" side and the DPF flow results are negative, the reverse measurement is respected. Similarly, the forward measurement is respected when a reverse measurement is modelled on the "to" side and the DPF flow results are negative.

The switching devices are verified for the following inconsistencies:

- If the switching device is open, the measured flow (current, real power, reactive power, and apparent power) through the open switching device must be zero.
- If the measurement value is greater than the effective value set in the DNAF configuration, no flow should be observed on the deenergized phases.
- If the measurement value is greater than the effective value set in the DNAF configuration, no neighboring switching device should be open and deenergizing the downstream network (for example: an open cut, an open switch).
- If there are customers fed by energized switching devices, the measured flow through the device must be greater than zero.
- For real power flow and reactive power flow, having the effective device modeled, a sign mismatch should not be observed between the measurement value and the respective DPF calculated flow value of the switching device. (For related information, see [Effective Device for a](#)

Measurement)

4.2. Load Allocation

If a valid measurement island has been formed and it has loads, the loads are scaled to match the island net load calculated from the effective kW and kVAR, or kVA measurement values. If a measurement island does not have any loads, BLA convergence is not checked for that island.

Figure below shows the load allocation flowchart.

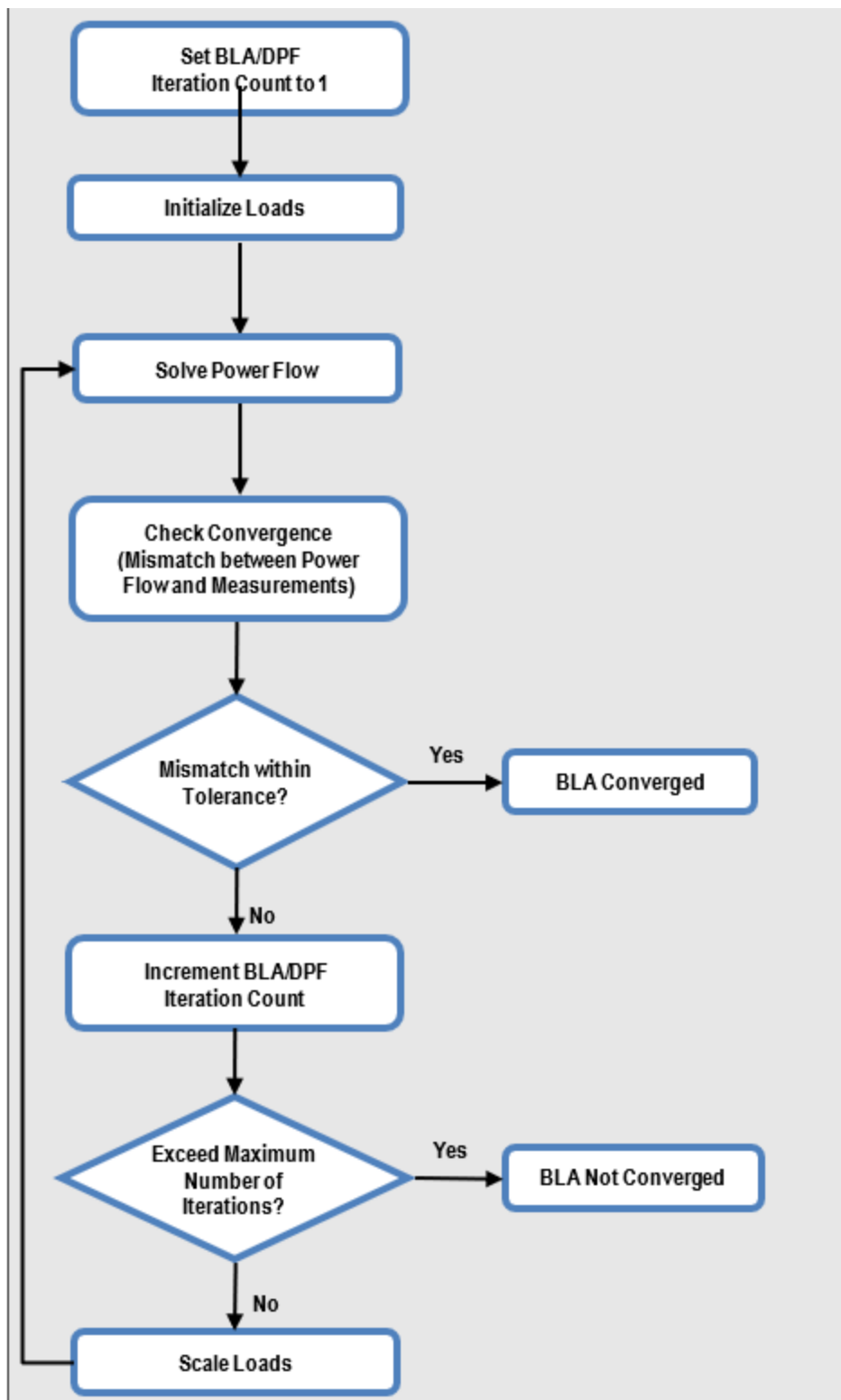


Figure 19. BLA Scaling Loads Flowchart

Before scaling the load, the load allocation type must be determined since the load scaling and mismatch check is based on kW and kVAR values or kVA values.

4.2.1. Load Allocation Method

The power flow calculated flow values and the measurements for a BLA measurement island are compared and the non-telemetered and conforming loads are then scaled up or down, based on either kW and kVAR values, or kVA values. There are three load allocation methods: kW and kVAR, kVA, and Amp and PF. The method to use is decided by the bounding analog measurement groups and their associated effective measurements (the measurement island has effective single-phase Watt and VAR measurements or effective single-phase VA measurements).

Analog Measurement Groups describes all the measurement groups that can be used to form a measurement island. Among them, G1 through G7 and G25 can determine accurate effective single-phase Watt and VAR, and VA measurement values; G8 through G18 can determine approximate effective single-phase Watt and VAR, and VA measurement values; G19 through G24 can only determine effective single-phase VA measurement values.

Among all the measurement groups bounding a measurement island, the following logic is applied to decide which method to use:

- kW and kVAR:

If the measurement group closest to the reference bus can get accurate effective single-phase Watt and VAR measurement values for all three phases (G1 through G7), and all other measurement groups do not include any manual feeder head single-phase Amps and PF measurement groups (G25), kW and kVAR load allocation method is used.

- kVA

If the measurement group closest to the reference bus cannot get accurate effective single-phase Watt and VAR measurement values for all three phases, because any phase has only approximate Watt and VAR measurement values (G8 through G18) or effective single-phase VA measurement values (G19 through G24), the kVA load allocation method is used.

- Amp and PF

The Amp and PF load allocation method is used in these cases:

- The measurement group closest to the reference bus is the manual feeder head single-phase Amps and PF measurement group (G25).
- The measurement group closest to the reference bus can get accurate, effective single-phase Watt and VAR measurement values for all three phases (G1 through G7), and all other measurement groups include at least one manual feeder head single-phase Amps and PF measurement group (G25).

4.2.2. Load Schedule

The instantaneous real (P) and reactive (Q) load values for every load point in a distribution substation are not known. Therefore, load schedules are used to estimate the load value for

different times of the day, different day types, and different seasons. Although the load allocation algorithm adjusts the load values to match the flow measurements, a reasonable initial estimate of the peak or average load values can provide a good starting point for the load allocation algorithm and increase the likelihood of BLA convergence. For this purpose, static and dynamic load schedules are used.

Static load schedules are defined in the `globals.csv` file as a Schedule object with a `SCHEDTYPE_SCHED` field value of 1. Different load schedules are defined for different day types and different seasons. If a station has no loads, such as a station that contains a hot standby feeder (feeder breaker is open), the DPF solves successfully and load allocation is not performed.

The ratio of the real power scheduled values at any time of the day to the peak or average real power value defined in the load object is defined in the load schedule.

Figure below shows an example of a residential real power load schedule for one day.

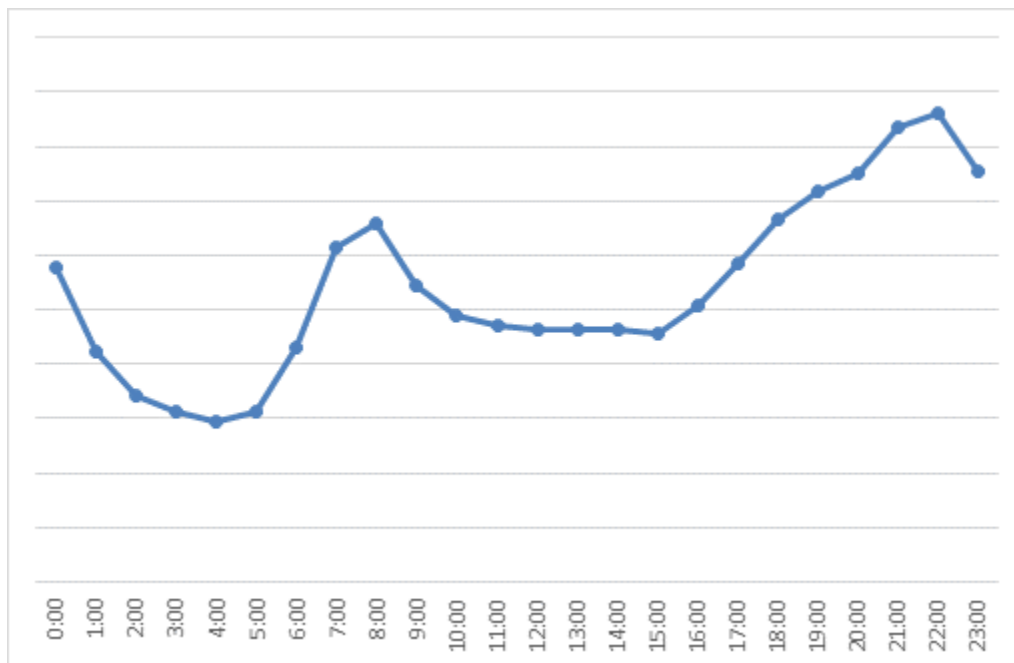


Figure 20. Real Power Load Schedule

If the reactive power load schedule is defined, the ratio of the reactive power scheduled values at any time of the day to the peak or average reactive power value defined in the load object is defined in the load schedule. If the reactive power load schedule is not defined, the power factor schedule must be defined. In this case, the reactive power load scheduled values can be calculated from the real power scheduled values and the scheduled power factor at that time. Most of the time, power factor schedules are used.

In addition, the system supports dynamic schedules for loads, generators, and DERs (real and reactive power). Dynamic schedules can be provided by an external system for load points and can be used in power flow solutions. If no manual entry values or SCADA analog measurements are available, the dynamic real power and reactive power load values override the static scheduled values. In case that only the real power is provided by dynamic schedule, and static real and

reactive power schedules are available for the device (either load, generator, or DER), the following procedure is applied to compute a dynamic reactive power value:

- Compute a scheduled power factor value: the static real and reactive schedules are used to compute a scheduled power factor at the specified study time.
- The computed power factor along with the dynamic real power are then used to compute a dynamic reactive power value.

The system also supports dynamic schedules at the feeder level. If no feeder head manual entry values or SCADA analog measurements are available, the dynamic schedule point values are used in the BLA application to form the feeder head measurement island.

4.2.3. Load Profile Multiplier Schedule

For each load category object, an optional load profile multiplier schedule can be defined. This schedule may have a single schedule point, in which case the multiplier is the same for all hours of the day; or multiple points may be defined, resulting in different multipliers based on the time of day.

The value that is determined from the schedule and the latest DPF execution time is multiplied by the load profile value to generate the effective load profile value. For the initial load allocation, the nominal real power is multiplied by the real power profile multiplier value and the real power profile value. If the load category's family of load profiles define a reactive power value, the nominal reactive power value is multiplied by the reactive power profile multiplier and the reactive power profile value. If a power factor value is defined in the schedule, the reactive power profile multiplier is not applied.

The load profile multiplier schedule applies only to the native load. The DER component of the load defined in the embedded generation schedule for load is not multiplied by the load profile multiplier.

4.2.4. Cold Load Pickup (CLPU) Schedule

The increased CLPU demand is due to different factors, including transformer magnetizing inrush current, capacitor inrush current, starting surges from induction motors, loss of load diversity of process and thermostatically controlled loads, etc. Most of these factors last a short time (seconds or shorter). They only affect protection and are not considered in power flow. Loss of load diversity persists for prolonged periods of time and increases the thermal loading of the equipment, so it is considered in power flow.

The linear decay model shown in figure below is used to model the CLPU demand.

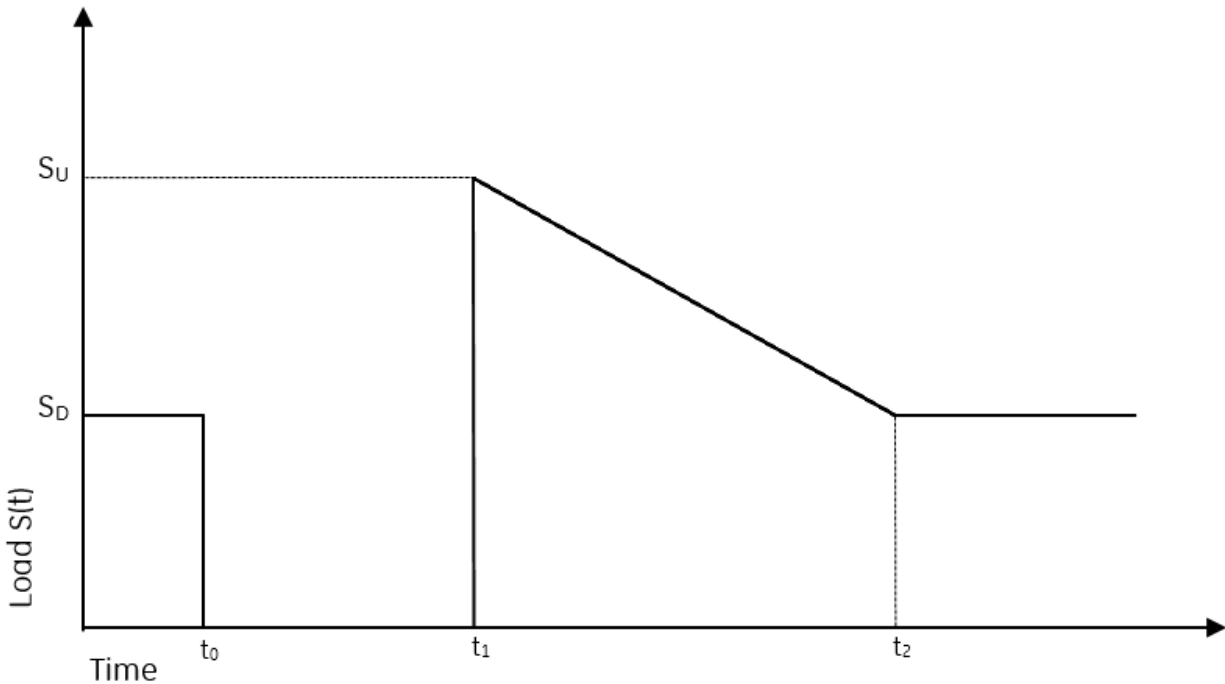


Figure 21. Linear Decay CLPU Model

In figure above, S_D is the diversified load before the outage happens at time t_0 . S_U is the un-diversified load after the load is restored at t_1 , and it decreases linearly and is close to the diversified load S_D at t_2 .

To support this CLPU model, the cold load pickup schedule in the load schedule must be defined in the global.csv file. This schedule defines the cold load peak real power loading ratio (P_U/P_D) and decay for different outage durations in hours. For every schedule point, the X value is the outage duration in hours.

The first Y schedule point value is the cold load peak loading ratio (P_U/P_D) for the outage duration.

- If the load's power factor schedule is used, the reactive power peak loading can be calculated from the real power peak load and the power factor.
- If the load's reactive power schedule is used, the reactive power peak loading can be calculated by multiplying the diversified reactive power load with this cold load peak loading ratio.

The second Y schedule point value is the duration in hours from the time load is restored to the time after which the CLPU ends and the load value decreases to the level of diversified load S_D , shown in interval t_1 to t_2 in figure above.

In this schedule, if the outage duration is between the two adjacent schedule points, linear interpolation is used to calculate the peak loading ratio and CLPU duration for that outage duration. Then the CLPU multiplier can be calculated. This CLPU multiplier is applied to the load initialization.

The following example illustrates how the CLPU multiplier is calculated. Table below shows a CLPU schedule.

Table 2. Cold Load Pickup Schedule

Outage Time (hours)	Peak Loading Ratio	CLPU Duration (hours)
0	1.0	0
0.5	1.2	1.0
1.0	1.4	2.0
1.5	1.6	3.0
2.0	1.8	4.0
2.5	2.0	5.0
3.0	2.2	6.0

Assume a load is outaged for 2.2 hours before it is restored, so the peak loading ratio is $1.8 + (2.0 - 1.8) * (2.2 - 2.0) / (2.5 - 2.0) = 1.88$, and the CLPU duration is $4.0 + (5.0 - 4.0) * (2.2 - 2.0) / (2.5 - 2.0) = 4.4$ hours. If a power flow is run 1 hour after restoration, the CLPU multiplier at that time is $1.0 + (1.88 - 1.0) * (4.4 - 1.0) / 4.4 = 1.68$.

DPF/BLA can be configured to consider CLPU effect. This is done using the Consider Cold Load Pickup option in the DNAF Configuration Editor, as shown below. If this option is selected, the CLPU multiplier will be calculated and applied to load initialization.

☐ Consider Cold Load Pickup

4.2.5. Load Initialization

The first step of scaling loads is to run the initial power flow. For the initial power flow, the initial real and reactive power of all loads is determined in the order below:

1. For loads with manually entered values, these values are used.
2. For telemetered loads, measured values are used.
3. For non-conforming loads, the nominally scheduled values are used.
4. For primary meter loads, the nominally scheduled values are used if the Adjust Primary Meter Load During Load Allocation option in DCE is not enabled.

i Note

Loads in items 1 to 4 above cannot be adjusted by BLA during load allocation. Therefore, they are not included in any measurement island.

5. For loads with smart meters, meter measurement values are used if the Use AMI Load Data option in DCE is enabled.
 - If the Scale AMI Loads option in DCE is enabled, meter measurement values are adjusted by BLA during load allocation to match a SCADA measurement value on the upstream device.
 -

If the Scale AMI Loads option in DCE is disabled, meter measurement values are not adjusted by BLA during load allocation and are not included in any measurement island.

6. For all loads in the measurement island, the nominally scheduled values are used if this is the first DPF execution after the server starts or the Start with Solved Loads from Previous DPF Execution option in DCE is not enabled.
7. For all loads in the measurement island, the nominally scheduled values are used if at least one load in the measurement island was not allocated by BLA in the last DPF execution.
8. For all loads in the measurement island, the nominally scheduled values are used if at least one load in the measurement island was de-energized before and this is the first DPF execution after the load becomes energized.
9. For all loads in the measurement island, the nominally scheduled values are used if BLA attempted to allocate negative load to at least one load which cannot be negative (generating power) in this measurement island in the last DPF execution.
10. For all loads in the measurement island, the nominally scheduled values are used if BLA attempted to allocate load greater than the Maximum Load Percent during Load Allocation (%) specified in DCE to at least one load in the measurement island in the last DPF execution.
11. For all loads in the measurement island, the nominally scheduled values are used if at least one load in the measurement island was in a measurement island for which BLA was partially converged in the last DPF execution.
12. For all loads in the measurement island, the nominally scheduled values are used if BLA attempted to shift the load in any phase to a value greater than the Maximum Three-Phase Load Shift When Three-Phase Balanced Loads are Balanced (%) specified in DCE for at least one balanced three-phase load in the measurement island in the last DPF execution.
13. For all other cases, load values solved by the previous DPF execution are used.

4.2.6. Check Convergence

After the power flow has been executed, the difference between the net load calculated from the island measurements and the net load calculated from the power flow is computed. If the difference is within a given tolerance, the BLA and DPF have converged. If the tolerance is not reached, then the BLA and DPF iterate until the tolerance is met or the maximum number of BLA/DPF iterations has been reached. If the tolerance is not met within the maximum number of iterations, BLA is not converged.

The parameters in the DNAF Configuration Editor's BLA tab, shown in figure below, determine BLA convergence.

Base Applications Bus Load Allocation (BLA) Distribution Power Flow (DPF) Limit Monitor (LIM) Fault Location (FL) Protection Validation (PRV) Short Circuit Calculation (SCC) Advanced Applications Surgical Load Shed (SLS) Switching Reconfiguration	Solution Formulation Convergence Maximum Number of BLA/DPF Iterations: 5 Maximum Real Power Mismatch (kW): 10 Maximum Real Power Mismatch (%): 1 Maximum Reactive Power Mismatch (kVAR): 10 Maximum Reactive Power Mismatch (%): 2
---	--

Figure 22. BLA Convergence Parameters

If the kW and kVAR or Amp and PF load allocation type is used, both the kW and kVAR mismatches need to be within the tolerance either in actual value or percentage to achieve BLA convergence. If kVA load allocation type is used, the kVA mismatches need to be within the tolerance either in actual value or percentage to achieve BLA convergence. The kVA mismatch tolerance is calculated from the kW and kVAR mismatch tolerance specified in figure above:

$$kVA_{tol} = \sqrt{kW_{tol}^2 + kVAR_{tol}^2} \quad (\text{Equation 4.62})$$

In any of the following six conditions, the sum of the three single-phase mismatches is checked against the island mismatch tolerances described above.

- One of the bounding measurement groups only has three-phase measurements.
- All loads in the measurement island are three-phase loads downstream of a delta connected distribution transformer, such as an open wye-delta or three-phase closed wye-delta distribution transformer.
- There is at least one delta connected step transformer in the measurement island.
- The percentage of nominally scheduled real power of balanced three-phase loads to the total loads' nominally scheduled real power in the measurement island is above the threshold specified by Maximum Threshold for Treating Three-Phase Loads as Single-Phase Loads (%) parameter on the DNAF Configuration Editor's BLA tab.
- The percentage of loads connected to delta distribution transformers to the total loads' nominally scheduled real power in the measurement island is above the threshold specified by the Maximum Threshold for Treating Three-Phase Loads as Single-Phase Loads (%) parameter on the DNAF Configuration Editor's BLA tab.
- There is at least one three-phase generator in the measurement island. This condition does not apply to DERs.

If none of the above conditions occurs, single-phase mismatches are checked, and the measurement island is converged when all three mismatches for real power and all three mismatches for reactive power are within their respective tolerances.

4.2.7. Scaling Loads

The loads are scaled based on the difference in the sum of the connected loads and the island single-phase effective measurement values. This difference determines the amount of load that must be added to or removed from all loads in the measurement island. The loads are then adjusted based on their load category accuracy (or variability); loads with a smaller accuracy/variability value are not changed as much as loads with a higher accuracy/variability value.

The following equation is used for the load scaling algorithm:

$$\Delta = \text{Mismatch} \cdot \frac{\text{Load} \cdot \text{Acc}^2}{\sum (\text{Load} \cdot \text{Acc}^2)} \quad (\text{Equation 4.63})$$

Where:

Mismatch is the mismatch between the measured and calculated flow.

Acc is the load accuracy/variability.

The new load value for this BLA iteration is the allocated value from the last BLA iteration plus the calculated *Delta* value.

The loads are scaled up or down based on the cases described in the following sections.

4.2.7.1. Three-Phase Load Scaling

If one of the bounding measurement groups only has three-phase measurements, the three-phase scaling method is used. In this method, the three-phase load value that needs to be adjusted for all loads in the measurement island is calculated first, using equation 4.64:

$$\Delta_{3ph} = \text{Mismatch}_{3ph} \cdot \frac{\text{Load}_{3ph} \cdot \text{Acc}^2}{\sum (\text{Load}_{3ph} \cdot \text{Acc}^2)} \quad (\text{Equation 4.64})$$

Where:

Delta_{3ph} is the three-phase load value that needs to be adjusted for a load.

Mismatch_{3ph} is the three-phase total mismatch between the measured and calculated flow for the measurement island.

Load_{3ph} is the current three-phase load value for a load.

Acc is the load accuracy/variability.

Then for every load, the load value that needs to be adjusted is calculated for individual phases based on the ratio of load values of individual phases using equation 4.65:

$$\Delta_{1ph} = \Delta_{3ph} \cdot \frac{\text{Load}_{1ph}}{\sum_{ph=A}^c \text{Load}_{ph}} \quad (\text{Equation 4.65})$$

Where:

Δ_{1ph} is the load value that needs to be adjusted at individual phase for a load.

$Load_{1ph}$ is the current load value at one phase for a load.

4.2.7.2. Single-Phase Load Scaling

For all the other cases, the single-phase load scaling method is used. This method includes three steps:

Step 1: Scale loads downstream of a delta connected distribution transformer

If a distribution transformer is delta connected in the primary winding or a secondary winding, the total three-phase mismatch of the loads downstream of delta connected distribution transformers is calculated by equation 4.66:

$$Mismatch_{\Delta} = Mismatch_{total} \cdot \frac{Load_{\Delta}}{Load_{total}} \quad (\text{Equation 4.66})$$

Where:

$Mismatch_{\Delta}$ is the three-phase mismatch of all the loads downstream of delta connected distribution transformers in the measurement island.

$Mismatch_{total}$ is the total three-phase mismatch for the measurement island.

$Load_{\Delta}$ is the three-phase load value of all the loads downstream of delta connected distribution transformers in the measurement island.

$Load_{total}$ is the three-phase load value of all the loads in the measurement island.

After $Mismatch_{\Delta}$ is calculated, it is used as input $Mismatch_{3ph}$ in Equation 4.64. Then by using equations 4.64 and 4.65, the loads downstream of a delta connected distribution transformer are adjusted in the same way as scaling a three-phase load, described in [Three-Phase Load Scaling](#).

Step 2: Scale Three-Phase Loads

Three-phase loads sometimes are scaled as single-phase loads, as shown in figure below.

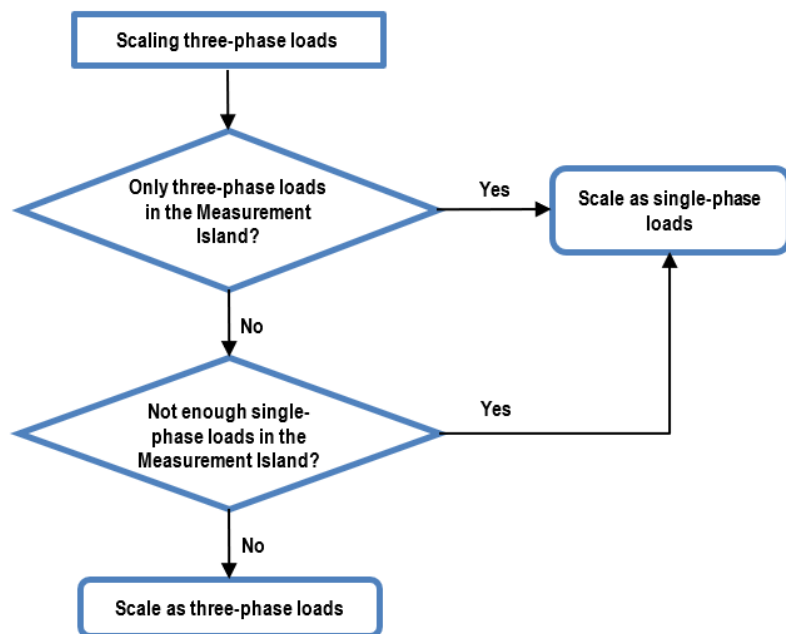


Figure 23. Scaling Three-Phase Loads

In figure above, there is enough single-phase load if this ratio:

$$\frac{\text{Total apparent power of all single – phase loads not downstream of delta connected winding distribution transformers}}{\text{Total apparent power of all loads in the measurement island}}$$

is below the Minimum Confidence Deviation Ratio for Determining if Three-Phase Load Balancing is Utilized (%) for any phase. This % value is set on the DNAF Configuration Editor’s BLA tab.

If scaling as single-phase loads, the individual phases for all three-phase loads in this measurement island are scaled independently (not forced to be balanced) when scaling the loads. The three-phase loads are not scaled in this step. Instead, they are scaled in step 3 as single-phase loads.

If three-phase loads are scaled in this step, all three phases are scaled up or down by the same Δ_{1ph} . The average mismatch of the three-phase loads at any phase is calculated using equation 4.67:

$$\text{Mismatch}_{1ph} = \frac{\text{Mismatch}_{total}}{3} \cdot \frac{\text{Load}_{3ph}}{\text{Load}_{total}} \quad (\text{Equation 4.67})$$

Where:

Mismatch_{1ph} is the single-phase mismatch of all the three-phase loads in the measurement island.

Mismatch_{total} is the total three-phase mismatch for the measurement island.

Load_{3ph} is the three-phase load value of all the three-phase loads in the measurement island.

$Load_{total}$ is the three-phase load value of all the loads in the measurement island.

This calculated $Mismatch_{1ph}$ value and equation 4.68 are used to calculate the individual phase load to be adjusted.

$$\Delta_{1ph} = Mismatch_{1ph} \cdot \frac{Load_{1ph} \cdot Acc^2}{\sum (Load_{1ph} \cdot Acc^2)} \quad (\text{Equation 4.68})$$

Where:

Δ_{1ph} is the single-phase load value that needs to be adjusted for a three-phase load.

$Mismatch_{1ph}$ is the single-phase mismatch of all the three-phase loads in the measurement island.

$Load_{1ph}$ is the single-phase load value for a three-phase load.

Acc is the load accuracy/variability.

Step 3: Scale Single-Phase Loads

In this step, all single-phase loads and three-phase loads not scaled in step 2 are scaled. $Mismatch_{1ph}$ is calculated for every phase by subtracting the mismatches in steps 1 and 2 from the total island mismatch (Watt and VAR, or VA).

For single-phase loads and three-phase loads not scaled in step 2, the $Mismatch_{1ph}$ and equation 4.63 are used to calculate the Δ_{1ph} value for every phase. If a three-phase load has the same nominal load value (kW) for each phase, the load shift percentage is calculated for all phases by equation 4.69:

$$\text{Shift \%} = \left| \frac{\Delta_{1ph}}{Load} \right| \cdot 100 \quad (\text{Equation 4.69})$$

Where:

$Load$ is the load allocated in the previous BLA iteration. If the load shift percentage in any phase is larger than the threshold specified by the Maximum Three-Phase Load Shift When Three-Phase Balanced Loads are Balanced (%) parameter set in the DNAF Configuration Editor's BLA tab, the load in this phase is only adjusted by the value of Δ_{1ph} multiplied by the Maximum Three-Phase Load Shift When Three-Phase Balanced Loads are Balanced (%) value.

4.2.8. Acceleration and Deceleration

To reduce the number of BLA/DPF iterations (improving the overall performance of the DPF), acceleration and deceleration parameters may be used when adjusting the loads during the BLA/DPF outer loop processing. These factors may slightly increase or decrease the mismatch between the measured and calculated flow during each BLA iteration. If the model has a low quality solution (voltages are far from nominal) or there is a steep slope in the load to voltage dependency curve (see [Load Representation](#)), the acceleration/deceleration factors can assist in preventing load

convergence oscillation which means the difference in the measured and calculated island flows never falls inside of the acceptable range specified by the BLA Maximum P and Q Mismatch parameters.

4.2.8.1. Acceleration

Three acceleration factor parameters specified in DCE can be used to slightly increase the mismatch between the measured and calculated flow value during each BLA iteration.

- Real and reactive power acceleration factor

This parameter specifies the real, reactive, and apparent power acceleration ratio used during load allocation. The loads are scaled based on the mismatch between the island measurements and the power flow calculations. This acceleration ratio is multiplied by the mismatch to determine the amount of load that must be added to or reduced from all loads in the measurement island in the second and later BLA iterations in either of the following two situations:

- All three phases of the loads are allocated together.
- Loads are allocated by individual phase, and BLA is not converged for any phase.

The real and reactive power acceleration ratio should be slightly larger than 1; a typical value is 1.05.

- Fast real power acceleration factor

This parameter specifies the real power fast acceleration ratio to improve slow real power convergence problems. This value can only be applied to the load in the island when each phase is scaled separately. This ratio can be applied in the second and later BLA iterations.

BLA compares the real or apparent power mismatch between the island measurements and the power flow calculations between the current and the previous BLA iterations. If the sign of the two mismatches has not changed, the mismatch of the current iteration is smaller than the mismatch of the previous iteration, and the percentage change of the two mismatches is smaller than the Percent Change to Apply Fast Acceleration for Real Power Mismatch (%) parameter specified in the DCE, this real power fast acceleration ratio is multiplied by the real or apparent power mismatch of the current iteration to determine the amount of load that must be added to or reduced from all loads in the measurement island.

The fast real power acceleration ratio should be no less than the real and reactive power acceleration ratio; a typical value is 1.25

- Reactive power consecutive iteration acceleration factor

This parameter specifies the reactive power consecutive iteration acceleration factor, which may be applied during the second and third BLA iteration when each phase of the load is scaled separately. During these two iterations, if the ratio of the mismatch of the current iteration and the previous iteration is larger than the Reactive Power Consecutive Iteration Mismatch Ratio specified in DCE for all three phases, this specified value is multiplied by the reactive power

mismatch of the current iteration to determine the amount of load that must be added to or reduced from all loads in the measurement island.

The reactive power consecutive iteration acceleration factor should be larger than 1; a typical value is 1.75.

4.2.8.2. Deceleration

Two deceleration factor parameters specified in DCE can be used to slightly decrease the mismatch between the measured and calculated flow value during each BLA iteration.

- Load deceleration factor

This parameter specifies the real or apparent power deceleration ratio used during load allocation. The loads are scaled based on the mismatch between the island measurements and the power flow calculations. This deceleration ratio is multiplied by the mismatch to determine the amount of load that must be added to or reduced from all loads in the measurement island in either of the following two situations:

- Checking mismatch in the first BLA iteration.
- Checking mismatch in the second and later BLA iterations, if BLA is converged for at least one phase when loads are allocated by individual phase.

The load deceleration ratio should be between 0 and 1; a typical value is 0.9.

- Reactive power deceleration factor

This parameter specifies the reactive power deceleration ratio used during load allocation. The loads are scaled based on the mismatch between the island measurements and the power flow calculations. This deceleration ratio is multiplied by the mismatch to determine the amount of load that must be added to or reduced from all loads in the measurement island in either of these situations:

- Checking mismatch in the first BLA iteration.
- Checking mismatch in the second and later BLA iterations, if BLA is converged for at least one phase when loads are allocated by individual phase.

The reactive power deceleration ratio should be between 0 and 1; a typical value is 0.8.

4.2.9. Study Mode Load Allocation

In study mode, the historical load data (determined from real-time power flow executions) and temperature adjustment schedules can be used to allocate the loads.

4.2.9.1. Historical Load Data

During the real-time power flow executions, historical correction factors are updated and stored for a configurable number of daily time periods, for each day type, and for each load. Initially, the system is configured to store one historical correction value per day type per load.

Note that, while the number of historical values stored for each load is configurable (number of time windows with start and end times per load category), the administrator must be aware of potential impacts to system performance and resource usage as the number of stored historical values increases. If this number increases, the performance impact increases proportionally to the overall size of the model.

4.2.9.2. Temperature Adjustment Schedules

The temperature adjustment schedule provides a scaling factor based on the current temperature. For each load category, there is one associated temperature adjustment schedule. Figure below shows a sample temperature adjustment schedule.

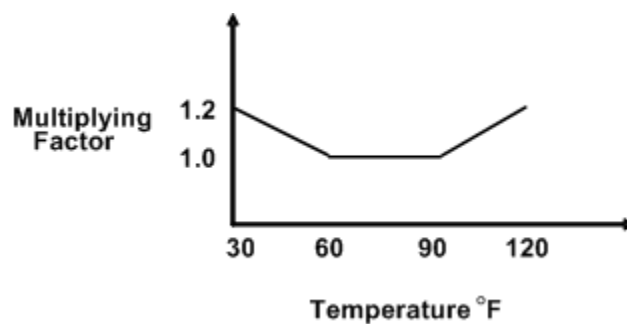


Figure 24. Sample Temperature Adjustment Schedule

For the schedule shown in figure above, the nominal load is scaled by approximately 1.0 when the temperature is approximately 75 degrees (no change to the existing load value). However, when the temperature is cooler than 60 degrees or warmer than 90 degrees, the load multiplying factor increases.

If there is a temperature adjustment schedule for the load, the temperature effects are taken out of the stored load scaling factor. For example, if the load is increased by 5% due to high temperatures, the adjusted scaling factor is reduced by 5%. This prevents including the temperature effects twice when performing a power flow in the study mode.

4.3. Incorporating Load Measurements from Advanced Metering Infrastructure

BLA can optionally utilize load measurements coming from the Advanced Metering Infrastructure (AMI) to form measurement islands. This is enabled by checking the "Use AMI Load Data" option from the DNAF Configuration Editor BLA tab. Associated with this option is the parameter "AMI

Load Measurement Time Tolerance" which is used to determine whether AMI load measurements are recent or not and whether they can be meaningfully used in BLA. In addition, the "Scale AMI Loads" option specifies whether AMI loads are scaled to match a SCADA measurement value on the upstream device.

To use the AMI measurements in the BLA/DPF solution, the following parameters must be set in the DNAF server configuration file:

- DNAF_AMI_DPF_PREOLVE_READ_ENABLE

Set to True (=1) for the DNAF server to read AMI data before every DPF execution.

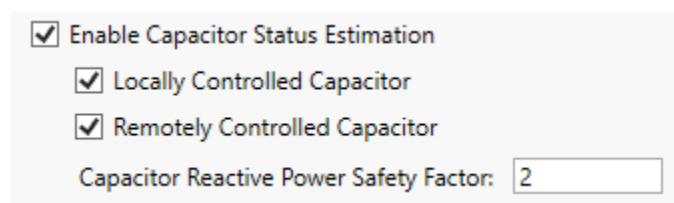
- DNAF_AMI_DPF_PREOLVE_READ_WAIT_TIME

Set to a reasonable value in seconds, depending on communication latency in the AMI infrastructure.

If the DNAF server is configured for DPF pre-solve AMI read and a DPF solution is requested for a station (due to a periodic trigger, an event trigger, or a user request), then before solving DPF, all load-profile enabled AMI meters are identified in the network selection, and a read request is sent out for those meters. The DPF execution then waits for DNAF_AMI_DPF_PREOLVE_READ_WAIT_TIME seconds. During this waiting period, further DPF requests for the station are declined. At the end of the waiting time, DPF execution is resumed for the station. When DPF execution is resumed, AMI load measurements that are within the AMI Load Measurement Time Tolerance (in seconds) are considered recent and are used by BLA for load allocation.

4.4. Capacitor Status Estimation

If after two BLA iterations BLA still cannot converge, BLA can optionally change the status of locally controlled and/or remotely controlled capacitors that do not have a valid status measurement. The DNAF Configuration Editor's BLA tab has options to specify whether this is allowed and which types of capacitors are allowed.



The screenshot shows a configuration window with the following options:

- ☒ Enable Capacitor Status Estimation
 - ☒ Locally Controlled Capacitor
 - ☒ Remotely Controlled Capacitor
- Capacitor Reactive Power Safety Factor:

If the Enable Capacitor State Estimation to be Performed in BLA and Enable Locally Controlled Capacitor options are both selected, BLA can change the status of locally controlled capacitors that do not have a valid status measurement.

Eligible locally controlled capacitors should be in the status not changed zone. For example, for a voltage-controlled capacitor in a BLA not converged measurement island, the three operating ranges are defined as follows:

- The capacitor is turned on when the voltage is less than 0.9 per unit.
- The capacitor's status is not changed when the voltage is greater than 0.9 per unit and smaller than 1.1 per unit.
- The device is turned off when the voltage is greater than 1.1 per unit.

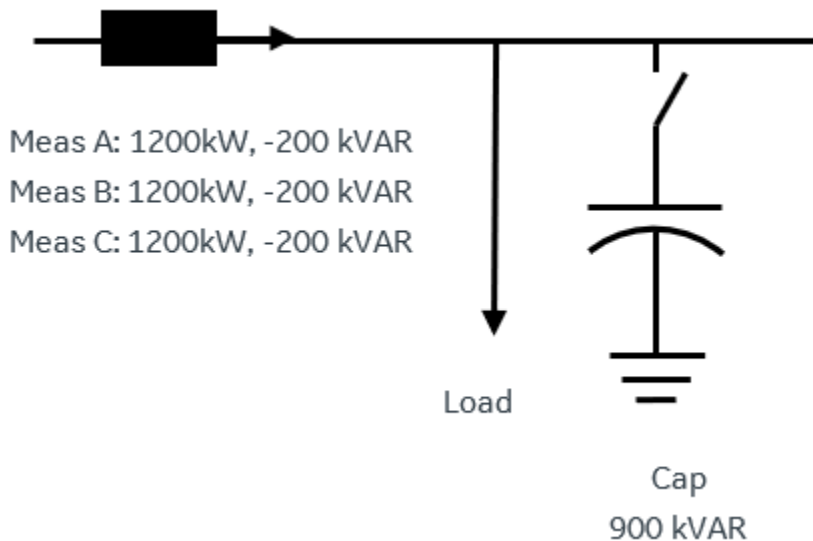
If the voltage is 0.88 per unit and the capacitor is turned on, BLA cannot change its status. If the voltage is 0.99 per unit, BLA can change its status whether it is currently turned on or off.

If the Enable Capacitor State Estimation to be Performed in BLA and Enable Remotely Controlled Capacitor options are both selected, BLA can change the status of remotely controlled capacitors that do not have a valid status measurement.

For any eligible capacitor, its status may be changed by BLA after the second iteration if any one of the following two conditions is met:

1. The capacitor size is less than the difference between the calculated and measured reactive power for the measurement island multiplied by a safety factor. This safety factor is specified by the Capacitor Toggle Q Safety Factor in DCE.

The following is an illustration of a simplified one-island system with one capacitor.



The island has a measurement of 1,200 kW into the island and 200 kVAR out of the island. There is only one load that cannot generate reactive load, and one 300 kVAR per phase capacitor that is in off status in the island. The capacitor is eligible for BLA to change status and the Capacitor Toggle Q Safety Factor is 2.0. In the first two iterations, BLA can only allocate nearly 0 kVAR for the load, so the difference between the calculated and measured reactive power for this measurement island is about 200kVAR. The capacitor size is 300 kVAR per phase, which is smaller than $200 * 2.0 = 400$ kVAR, so BLA may turn it on after two iterations.

2. The difference between the calculated and measured reactive power does not change between the two iterations. For the island in the illustration above, the difference between the calculated

and measured reactive power in the two iterations is 200kVAR, so in the third iteration, BLA may turn the capacitor on.

4.5. Substation BLA Solution Status

BLA has the following solution statuses for a substation:

- **BLA Scheduled Loading:** BLA is not performed in any measurement islands in the substation and the loads are assigned their scheduled values.
- **BLA Partial Scheduled Loading:** BLA is performed at not all of the measurement islands in the substation, and BLA is converged for all the performed measurement islands. The loads in the measurement islands, where BLA was not performed, are assigned their scheduled values.

Note that if BLA is not converged in any of the performed measurement islands, the status for this substation is BLA Partially Converged.

- **BLA Converged:** All measurement islands in the substation have converged. This means that the calculated and measured flow values match within the user-specified tolerances.
- **BLA Converged - Model Adjusted:** All measurement islands in the station have converged but the algorithm adjusted the model to achieve convergence.

The model adjustment encompasses several items including changing the state of locally controlled capacitors (BLA option in DCE) and artificial injection of VARs (BLA option in DCE).

- **BLA Partially Converged:** One or more islands in the substation do not achieve BLA convergence (the measured and calculated Watt/VAR flows are outside of their allowable tolerances).

The total number of measurement islands in the substation in which the loads need to be allocated and the number of measurement islands that BLA is converged are displayed in the DNAF Solution pop-up display and related grid tabular displays.

- **BLA Partially Converged - Model Adjusted:** One or more islands do not achieve BLA convergence (the measured and calculated Watt/VAR flows are outside of the allowable tolerances) even though artificial model adjustments were made.

The total number of measurement islands in the substation in which the loads need to be allocated and the number of measurement islands that BLA is converged is displayed in the DNAF Solution pop-up display and related grid tabular displays.

- **Neighbor Station BLA Partially Converged:** This status occurs if multiple stations are solved in a single power flow solution but the electrically tied station does not achieve BLA convergence in one or more measurement islands. The other stations that do achieve BLA convergence are considered BLA Converged.
- **Neighbor Station BLA Partial Converged - Model Adjusted:** This status occurs if multiple stations are solved in a single power flow solution but the electrically tied station does not achieve BLA convergence in one or more measurement islands even after model adjustments

have been made. The other stations that do achieve BLA convergence are considered BLA Converged.

4.6. BLA Configuration

This section describes the Bus Load Allocation configuration parameters.

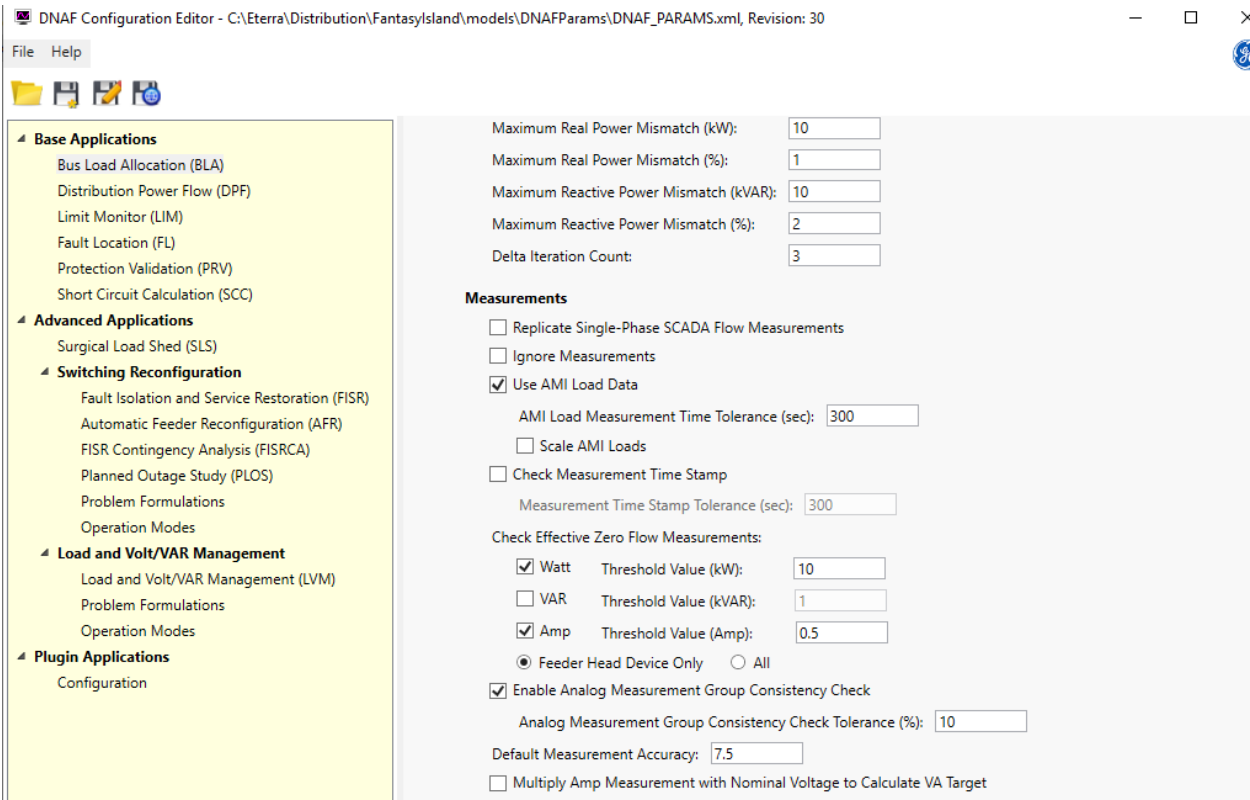


Figure 25. BLA Configuration

The BLA configuration parameters are provided in table below. All listed parameters are contained by the <Params xsi:type = "BLAParams"> element in the DNAF_PARAMS.xml configuration file.

Table 3. BLA Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Convergence			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Number of BLA/DPF Iterations	MAXITER_BLA <MaxIter>	Text box	This value specifies the maximum number of BLA/DPF iterations that can be performed to check the BLA convergence. This value excludes the initial BLA/DPF iteration, where BLA solves for the nominal voltage. The BLA is the outer control loop for the DPF application. A BLA iteration is performed after a DPF iteration until BLA converges or the maximum number of BLA/DPF iterations is reached. Multiple BLA/DPF executions are needed to account for losses and load to voltage dependencies. BLA checks the per phase mismatch between the calculated flow and measured flow through the measured branch devices at the end of every BLA/DPF iteration. If the real power mismatch is smaller than the tolerances specified in the Measurement and Power Flow Calculation Maximum Real Power Mismatch (kW) or the Measurement and Power Flow Calculation Maximum Real Power Mismatch (%), and the reactive power mismatch is smaller than tolerances specified in the Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (kVAR), or the Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (%), for all measurement islands that contain one or more loads and this maximum iteration count has not been exceeded, BLA is considered to be converged. Otherwise, if the count of iterations set by this parameter has been reached, and the mismatch for any measurement island that has loads is still larger than the applicable tolerances specified in the Measurement and Power Flow Calculation Maximum Real Power Mismatch (kW), Measurement and Power Flow Calculation Maximum Real Power Mismatch (%), Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (kVAR), or Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (%) parameters, BLA is not converged. Note The initial BLA/DPF iteration always occurs when BLA solves for the nominal voltage. This iteration is not counted by the Maximum Number of BLA/DPF Iterations parameter. Therefore, the actual number of BLA/DPF iterations can be 1 + Maximum Number of BLA/DPF Iterations. The maximum number of DPF iterations is defined by the IterMax parameter in alg.dat.

The allowable range is 1-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Measurement and Power Flow Calculation Maximum Real Power Mismatch (kW)	MAXABSPMISMA TCH_BLA <MaxPMismatch 2>	Text box	This value specifies the per phase maximum absolute kW mismatch between the calculated real power flow and the real power flow measurement for each measurement island that contains one or more loads. BLA checks this parameter when the kW and kVAR load allocation method is used. If the kVA load allocation method is used, BLA checks the maximum apparent power mismatch in kVA calculated from this parameter and the Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (kVAR) parameter. The real power is converged if either the absolute kW mismatch is less than the tolerance specified by this parameter or the percentage mismatch is less than the tolerance specified by the Measurement and Power Flow Calculation Maximum Real Power Mismatch (%) parameter. The allowable range is 0-100.
Measurement and Power Flow Calculation Maximum Real Power Mismatch (%)	MAXPMISMATCH _BLA <MaxPMismatch >	Text box	This value specifies the per phase maximum percentage mismatch between the calculated real power flow and the real power flow measurement for each measurement island that contains one or more loads. BLA checks this parameter when the kW and kVAR load allocation method is used. If the kVA load allocation method is used, BLA checks the maximum apparent power mismatch in kVA calculated from this parameter and the Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (%). The real power is converged if either the absolute kW mismatch is less than the tolerance specified by the Measurement and Power Flow Calculation Maximum Real Power Mismatch (kW) parameter or the percentage mismatch is less than the tolerance specified by this parameter. The allowable range is 0-20.
Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (kVAR)	MAXABSQMISM ATCH_BLA <MaxQMismatch 2>	Text box	This value specifies the per phase maximum absolute kVAR mismatch between the calculated reactive power flow and the reactive power measurement for each measurement island that contains one or more loads. BLA checks this parameter when the kW and kVAR load allocation method is used. If the kVA load allocation method is used, BLA checks the maximum apparent power mismatch in kVA calculated from this parameter and the Measurement and Power Flow Calculation Maximum Real Power Mismatch (kW) parameter. The reactive power is converged if either the absolute kVAR mismatch is less than the tolerance specified by this parameter or the percentage mismatch is less than the tolerance specified by the Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (%) parameter. The allowable range is 0-100.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (%)	MAXQMISMATCH_H_BLA <MaxQMismatch>	Text box	This value specifies the per phase maximum percentage mismatch between the calculated reactive power flow and the reactive power measurement for each measurement island that contains one or more loads. BLA checks this parameter when the kW and kVAR load allocation method is used. If the kVA load allocation method is used, BLA checks the maximum apparent power mismatch in kVA calculated from this parameter and the Measurement and Power Flow Calculation Maximum Real Power Mismatch (%) parameter. The reactive power is converged if either the absolute kVAR mismatch is less than the tolerance specified by the Measurement and Power Flow Calculation Maximum Reactive Power Mismatch (kVAR) parameter or the percentage mismatch is less than the tolerance specified by this parameter. The allowable range is 0-20.
Number of Iterations for Filtering Study Mode Delta Parameters	DELTACNT_BLA <DeltaCount>	Text box	This value specifies the number of iterations used for calculating a rolling average of the study mode load correction factors. If the Use Study Correction Factors option is selected in the DNAF Solution pop-up display, the load correction factors (if defined) are used in study mode to perform the initial load allocation. The study correction factor is the difference between the initially allocated load value (based on load schedules) and the final power flow value. It is calculated for every load for every regular day type for each user-defined interval by the real time DPF. A typical value is 3. The allowable range is 1-10.
Measurements			
Replicate Single-Phase SCADA Flow Measurements	REPLICATESINGL EPHASESCADAFL OWMEAS_BLA <ReplicateSingle PhaseSCADAFlo wMeasurements>	Check box	This specifies whether BLA replicates a flow measurement value to the other non-telemetered phases when there is only one single-phase flow measurement defined in the model at a given location. If this option is selected, the flow measurement value is replicated and used in the BLA flow mismatch calculations.
Ignore Measurements in BLA Processing (Debug only)	IGNOREMEAS_B LA <IgnoreMeas>	Check box	This specifies whether BLA uses analog measurements. If it is selected, BLA does not use any analog measurements. This is typically used only rarely, for debugging purposes.
Use AMI Load Data	<UseAmiLoadDat a>	Check box	This specifies whether BLA uses the load data from AMI. If this is selected and the difference between the evaluation time and the AMI real and reactive power measurement timestamp is less than the AMI Load Measurement Time Tolerance, AMI real and reactive power measurements are used to set the load real and reactive power values.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
AMI Load Measurement Time Tolerance	<AmiLoadMeasurementTimeTolerance>	Text box	This value specifies the maximum difference (in seconds) between the evaluation time and the AMI load measurement timestamp. If the time difference is less than this value, the AMI real power and reactive power measurements are used for load allocation. Otherwise, the AMI load measurement is not used in the BLA/DPF solution. The allowable range is 0-3600.
Check Analog Measurement Time Stamp	<ScaleAMILoads>	Check box	This specifies whether BLA allocates loads with AMI measurements. The allocation only takes effect when BLA uses the load data from AMI and when the difference between the evaluation time and the AMI real and reactive power measurement timestamp is smaller than the AMI Load Measurement Time Tolerance. If selected, BLA sets the initial real and reactive power values of the loads that have AMI data as the AMI real and reactive power measurements. Then, BLA scales these loads in the BLA load allocation process. If not selected, BLA does not scale the loads that have AMI data. BLA sets the load real and reactive power values to the AMI real and reactive power measurements.
Check Analog Measurement Time Stamp	<MeasurementCheckTimeStamp>	Check box	This specifies whether a recency check is performed on analog measurements, which are used in load allocation. A flag is defined on a measurement to specify if it requires the field timestamp check. If it is True while this check box is checked, the recency check is performed on that measurement. When an analog fails this recency check, it is excluded from the formation of measurement islands in BLA.
Analog Measurement Time Tolerance (sec)	<MeasurementCheckTimeStampTolerance>	Text box	This value specifies the maximum difference (in seconds) between the evaluation time and SCADA time measurement timestamp. If the time difference is greater than this number, the measurement is not used to form measurement islands in BLA. The allowable range is 0-3600.
Check Effective Zero Flow Measurements: Watt	<CheckEffectiveZeroWattMeasurements>	Check box	This specifies whether BLA checks the effective zero value real power measurement at the location specified by the Feeder Head Device Only/All setting. If this option is selected and the absolute value of a valid Watt measurement is less than the value specified in the associated Threshold Value (kW) text box, BLA does not use this Watt measurement for load allocation.
Threshold Value (kW)	<EffectiveZeroWattMeasurements>	Text box	This value specifies the threshold value when checking for effective zero value real power measurements for both forward and reverse power flow conditions. If the Check Effective Zero Flow Measurements — Watt option is selected and the absolute value of a valid Watt measurement is less than the value specified by this parameter, BLA does not use this Watt measurement for load allocation. The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Check Effective Zero Flow Measurements: VAR	<CheckEffectiveZeroVARMeas>	Check box	This specifies whether BLA checks the effective zero value reactive power measurement at the location specified by the Feeder Head Device Only/All setting. If this option is selected and the absolute value of a valid VAR measurement is less than the value specified by the associated Threshold Value (kVAR) text box, BLA does not use this VAR measurement for load allocation.
Threshold Value (kVAR)	<EffectiveZeroVARMeas>	Text box	This value specifies the threshold value when checking for effective zero value reactive power measurements for both forward and reverse power flow conditions. If the Check Effective Zero Flow Measurements — VAR option is selected and the absolute value of a valid VAR measurement is smaller than the value specified by this parameter, BLA does not use this VAR measurement for load allocation. The allowable range is 0-10.
Check Effective Zero Flow Measurements: Amp	<CheckEffectiveZeroAmpMeas>	Check box	This specifies whether BLA checks the effective zero flow (Amp) measurement at the location specified by the Feeder Head Device Only/All setting. If this option is selected and the absolute value of a valid Amp measurement is less than the value specified in the Threshold Value (Amp) text box, BLA does not use this Amp measurement for load allocation.
Threshold Value (Amp)	<EffectiveZeroAmpMeas>	Text box	This value specifies the threshold value when checking for effective zero value flow (Amp) measurements for both forward and reverse power flow conditions. If the Check Effective Zero Flow Measurements — Amp option is checked and the absolute value of a valid Amp measurement is smaller than the value specified by this parameter, BLA does not use this Amp measurement for load allocation. The allowable range is 0-10.
Check Effective Zero Flow Measurements: Feeder Head Device Only/All	<CheckEffectiveZeroFlowMeasDevType>	Option button	This setting specifies whether BLA checks for effective zero value flow (Watt, VAR, or Amp) measurements at specified locations: • If Feeder Head Device Only is selected, only flow measurements at feeder head devices are checked. • If All is selected, all flow measurements are checked.
Enable Analog Measurement Group Consistency Check	<EnableAMGConsistencyCheck>	Check box	This specifies whether to perform an analog measurement group consistency check. If the option is selected, the analog measurement group consistency check is performed. If the option is not checked, the analog measurement group consistency check is not performed and all the analog measurement groups with valid flow measurements are used by BLA to form a measurement island.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Analog Measurement Group Consistency Check Tolerance (%)	<AMGConsistencyCheckTolerance>	Text box	This value specifies the maximum percent difference allowed for analog measurement group consistency check. The consistency between real power, reactive power, apparent power, voltage, current, and angle measurements presented in a measurement group is checked against the parameter. If any percentage difference is greater than this parameter, the measurement group is not used by BLA to form a measurement island. The allowable range is 0-20.
Check Topology and Flow Direction Consistencies for Telemetered Switching Devices	<CheckTelemeteredSwitchConsistencies>	Check box	This check box specifies whether BLA performs topology and flow direction consistency checks for telemetered switches. If this option is selected, the topology and flow direction consistency checks are performed on analog measurements: voltage (V), current (Amp), real power (KW), reactive power (KVAR), and apparent power (KVA). If this option is not selected, the consistency checks are not performed. Telemetered switches are considered as BLA-consistent.
Multiply Amp Measurement with Nominal Voltage to Calculate VA Target	<MultiplyAmpMeasurementWithNominalVoltage>	Check box	When there is no valid real power, reactive power, apparent power, and voltage measurement in a measurement group, but there is a valid Amp measurement, an effective apparent power value can be calculated by multiplying the Amp measurement by a voltage. This check box indicates whether the nominal voltage or the DPF calculated voltage is used to calculate apparent power value. If this option is selected, the nominal voltage is used.
Treat Positive Flow (kW/kVAR) Measurement Values for Generators and Energy Resources as Injections into the Network	<PositiveDGDERFlowMeasurementValueAsInjection>	Check box	This specifies whether BLA treats positive kW or kVAR flow measurement values for generators/DERs as injections into the network.
Acceleration			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Load Deceleration Factor - Load Scaling Parameter	LDSCALEDEACCEL_BLA <LoadDeaccFactor or>	Text box	This value specifies the real or apparent power deceleration ratio used during load allocation. The loads are scaled based on the mismatch between the island measurements and the power flow calculations. This deceleration ratio is multiplied by the mismatch to determine the amount of load that must be added to or reduced from all loads in the measurement island in either of these situations: • Checking mismatch in the first BLA iteration. • Checking mismatch in the second and later BLA iterations, if BLA is converged for at least one phase when loads are allocated by individual phase. This applies to real and apparent power. The load deceleration ratio should be between 0 and 1. A typical value is 0.9. The allowable range is 0.1-1.
Reactive Power Deceleration Factor (First Iteration Only)	QDEACCEL_BLA <QDeAccelFactor >	Text box	This value specifies the reactive power deceleration ratio used during load allocation. The loads are scaled based on the mismatch between the island measurements and the power flow calculations. This deceleration ratio is multiplied by the mismatch to determine the amount of load that must be added to or reduced from all loads in the measurement island in either of these situations: • Checking mismatch in the first BLA iteration. • Checking mismatch in the second and later BLA iterations, if BLA is converged for at least one phase when loads are allocated by individual phase. The reactive power deceleration ratio should be between 0 and 1. A typical value is 0.8. The allowable range is 0-1.
Real and Reactive Power Acceleration Factor (after First Iteration)	PQACCEL_BLA <PQAccelFactor>	Text box	This value specifies the real, reactive, and apparent power acceleration ratio used during load allocation. The loads are scaled based on the mismatch between the island measurements and the power flow calculations. This acceleration ratio is multiplied by the mismatch to determine the amount of load that must be added to or reduced from all loads in the measurement island in the second and later BLA iterations in either of these situations: • All three phases of the loads are allocated together. • Loads are allocated by individual phase, and BLA is not converged for any phase. The real and reactive power acceleration ratio should be slightly larger than 1. A typical value is 1.05. The allowable range is 1-2.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Fast Acceleration Factor to Improve Slow Real Power Convergence Problems	PFASTACCEL_BLA <PFastAccel>	Text box	This value specifies the real power fast acceleration ratio to improve slow real power convergence problems. This value can only be applied to the load in the island when each phase is scaled separately. This ratio can be applied in the second and later BLA iterations. BLA compares the real or apparent power mismatch between the island measurements and the power flow calculations between the current and the previous BLA iterations. If the sign of the two mismatches has not changed, the mismatch of the current iteration is smaller than the mismatch of the previous iteration, and the percentage change of the two mismatches is smaller than the Percent Change to Apply Fast Acceleration for Real Power Mismatch (%) parameter, this real power fast acceleration ratio is multiplied by the real or apparent power mismatch of the current iteration to determine the amount of load that must be added to or reduced from all loads in the measurement island. The fast real power acceleration ratio should be no less than the real and reactive power acceleration ratio. A typical value is 1.25. The allowable range is 1-2.
Percent Change to Apply Fast Acceleration for Real Power Mismatch (%)	PFASTOLERANCE_BLA <PFastTolerance>	Text box	This value specifies the percent change of the real or apparent power mismatches between iterations for determining if the Fast Real Power Acceleration Ratio parameter is applied for an island real or apparent power mismatch when each phase of at least one load in that island is scaled separately. BLA compares the real or apparent power mismatch between the island measurements and the power flow calculations between the current iteration and the previous iteration. If the signs of the two mismatches have not changed, the mismatch of the current iteration is smaller than the mismatch of the previous iteration, and the percentage change of the two mismatches is smaller than the Percent Change to Apply Fast Acceleration for Real Power Mismatch, the Fast Real Power Acceleration Ratio is multiplied by the real or apparent power mismatch of the current iteration to determine the amount of load that must be added to or reduced from all loads in the measurement island. A typical value is 10.0%. The allowable range is 0.1-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Reactive Power Consecutive Iteration Mismatch Ratio	QDELTAMISRATIO_BLA <QDeltaMisRatio>	Text box	This value specifies the reactive power consecutive iteration mismatch ratio that is used in the second and third BLA iterations when each phase of the load is scaled separately. During these two iterations, if the ratio of the mismatch of the current iteration and the previous iteration is larger than this specified value for all three phases, the Reactive Power Consecutive Iteration Acceleration Factor parameter is multiplied by the reactive power mismatch of the current iteration to determine the amount of load that must be added to or reduced from all loads in the measurement island. The reactive power consecutive iteration mismatch ratio should be larger than 0. A typical value is 0.3. The allowable range is 0.01-1.
Reactive Power Consecutive Iteration Acceleration Factor	QDELTAACCELFACTOR_BLA <QDeltaAccFactor>	Text box	This value specifies the reactive power consecutive iteration acceleration factor that may be applied during the second and third BLA iteration when each phase of the load is scaled separately. During these two iterations, if the ratio of the mismatch of the current iteration and the previous iteration is larger than the Reactive Power Consecutive Iteration Mismatch Ratio for all three phases, this specified value is multiplied by the reactive power mismatch of the current iteration to determine the amount of load that must be added to or reduced from all loads in the measurement island. The reactive power consecutive iteration acceleration factor should be larger than 1. A typical value is 1.75. The allowable range is 1-5.
Load Allocation			
Start with Solved Loads from Previous DPF Execution	<StartWithPreviousDPFSolveLoads>	Check Box	This specifies whether BLA starts with solved load values from the previous DPF execution when BLA allocates loads using measurements: • If the option is selected, BLA starts with solved load values from the previous DPF execution. • If the option is cleared, BLA starts with nominally scheduled load values.
Adjust Primary Meter Load during Load Allocation	ADJUSTPRIMMETERLOAD_BLA <AdjustPrimaryMeterLoad>	Check box	This specifies whether a primary meter load is adjusted in the load allocation process: • If this option is selected, the primary meter load is adjusted like all other non-telemetered loads. • If this option is cleared, the load is set to its nominally scheduled value and is not adjusted in the load allocation process (it is effectively treated as a SCADA telemetered load).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Threshold for Treating Three-phase Loads as Single-phase Loads (%)	MAXTHREEPHASE_SINGLE_BLA <MaxThreePhaseLoad>	Text box	This value specifies the maximum threshold (percentage) of balanced three phase loads and loads connected to delta winding distribution transformers to the total load in the measurement island. This parameter is used to decide how the island flow mismatches are calculated to check for the convergence of a measurement island. If the percentage of balanced three-phase loads and loads connected to delta (secondary winding) distribution transformers to the total load is above this threshold, the sum of three single-phase mismatches is calculated and checked against the BLA maximum allowable mismatch to decide BLA convergence. Otherwise, three single phase mismatches are calculated and checked. A typical value is 75%. The allowable range is 0-100.
Minimum Confidence Deviation Ratio for Determining if Three-phase Load Balancing is Utilized (%)	MINONEPHASE_LOAD_BALANCE <OnePhaseLoadBalance>	Text box	This value specifies the minimum percentage threshold of single phase loads not connected to delta (secondary winding) distribution transformers to the total load in the measurement island. If the percentage of total apparent power of all single-phase loads not connected to delta (secondary winding) distribution transformers to the total apparent power of all loads in the measurement island is below this threshold for any phase, the individual phases for all three-phase loads in this measurement island are scaled independently (not forced to be balanced) when scaling the loads. A typical value is 5%. The allowable range is 0.1-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Three-phase Load Shift When Three-phase Balanced Loads are Balanced (%)	MAXTHPHLDSHI FT_BLA <MaxThrPhLdShi ft>	Text box	This value specifies the maximum allowable three phase load shift threshold (percentage). This threshold is only applied to three-phase loads that have the same nominal real and reactive power specified for each phase, the loads are not connected to delta (secondary winding) distribution transformers, and all three phases of the load are scaled independently during any BLA iteration. The load shift percentage is specified as the real, reactive, or apparent power to be shifted (either increased or reduced) at this BLA iteration to the real, reactive, or apparent power already allocated for this load in the previous BLA iteration. The load shift percentage is calculated for all three phases. If the load shift percentage in one phase is larger than this specified threshold, the load in this phase is shifted by this load shift threshold in percentage multiplied by the load shift calculated at this BLA iteration. For example, the load shift percentage is 45% and the kVA allocation method is used for one measurement island. In the previous BLA iteration, the apparent power allocated for a three-phase load in this island is 100 kVA. In this BLA iteration, it is suggested to increase the load by 50 kVA, which is larger than $45\% \times 100 = 45$ kVAR, so the load is increased by $45\% \times 50 = 22.5$ kVAR. A typical value is 45%. The allowable range is 0.1-100.
Three-phase Load Maximum Imbalance (%)	MAXIMBAL_BLA <MaxLoadImbal >	Text box	This value specifies the maximum threshold (percentage) of three-phase loads to total loads in the measurement island. • If the kW load allocation method is used, the load is calculated as the real power multiplied by the square of the accuracy. • If the kVA allocation method is used, the load is calculated as the apparent power multiplied by the square of the accuracy. If the average percentage of three-phase loads to total loads of all phases is larger than this value in this measurement island, the individual phases for all three-phase loads in this measurement island are scaled independently (not forced to be balanced). A typical value is 75%. The allowable range is 0.1-100.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Small Island Injection Flow Safety Factor to Avoid Negative Power Loads	LOWLDSAFETY_BLA <LowLoadSafety>	Text box	This value specifies the small island injection flow safety factor to avoid allocating negative real and reactive power to any regular load that does not inject energy into the network. Consider the following example: • The total load needs to be reduced for a measurement island that only contains regular loads in the BLA load allocation iteration; and, • The ratio of the island's mismatch multiplied by this safety factor to the island's net power from the power flow results is larger than 1.0. In this case, the actual mismatch used to scale the loads is divided by this ratio. A typical value is 1.1. The allowable range is 1-10.
Maximum Load Percent During Load Allocation (%)	MAXLOADPERCENT_BLA <MaxLoadPercent>	Text box	This value specifies the maximum loading percentage allowed during BLA load allocation process. If the adjusted load value during the BLA process is greater than the product of the nominal value and the maximum percentage, the load value is adjusted down to the product result. A typical value is 200.0. The allowable range is 0-1000.
Power Factor			
Maximum Load Power Factor - Load Scaling Parameter	MAXLOADPOWERFACTOR_BLA <MaxPowerFactor>	Text box	This value specifies the maximum load power factor allowed for any regular load that does not inject energy into the network. It is used primarily for balancing conventional reactive load when VAR flow is near 0.0. A typical value is 0.9995 lagging. The allowable range is 0-1.
Minimum Power Factor Percentage of Nominal Scheduled Power Factor (%)	MINPOWFACTORE_BLA <MinPowFactPercent>	Text box	This value specifies the minimum power factor allowed when allocating loads. It is specified as a percentage of the nominally scheduled power factor for each load. This parameter prevents the reactive load from getting scaled lower than the nominal power factor multiplied by this parameter and divided by 100. A typical value is 75%. The allowable range is 0-100.
Model Adjustment			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Enable Capacitor State Estimation to be Performed in BLA	REMOTECAPSWITCH_BLA <AllowTurnOnCap>	Check box	This specifies whether to allow BLA to change the status of locally controlled and/or remotely controlled capacitors that do not have a valid status measurement after the second BLA/DPF iteration, so that reactive load allocation can be successfully completed.
Enable Locally Controlled Capacitor Status Estimation to be Performed in BLA	<AllowTurnOnLocalCap>	Check box	This specifies whether to allow BLA to change the status of locally controlled capacitors that do not have a valid status measurement after the second BLA/DPF iteration, so that reactive load allocation can be successfully completed. Eligible locally controlled capacitors should be in the status not changed zone. For more information, refer to Distribution Power Flow.
Enable Remotely Controlled Capacitor Status Estimation to be Performed in BLA	<AllowTurnOnRemoteCap>	Check box	This specifies whether to allow BLA to change the status of remotely controlled capacitors that do not have a valid status measurement after the second BLA/DPF iteration, so that reactive load allocation can be successfully completed.
Capacitor Status Estimation Reactive Safety Factor	CAPSWITCHSAFETY_BLA <CapToggleQSafetyFactor>	Text box	If the Enable Capacitor State Estimation to be Performed in BLA option is checked, this value specifies the safety factor used in deciding whether to change the status of locally controlled and/or remotely controlled capacitors after the second iteration. If the capacitor size is less than this safety factor multiplied by the difference between the calculated and measured reactive power for the measurement island, the capacitor status may be changed by BLA. A typical value is 2.0. The allowable range is 0-10.

Measurement Islands

Method for Handling Less Accurate Island	LESSACCDOWNISLAND_BLA <LessAccDownMethod>	Drop-down list	This selection specifies the method for dealing with downstream islands that have less complete measurement sets. For example, if a measurement island at the head of the feeder has individual phase kW and kVAR measurements but a downstream island has only individual phase kVA or Amp measurements, the analyst can either select Merge Downstream Island to merge the downstream island into the upstream island (form one larger measurement island), or select Retain Measurement Islands to retain the existing measurement islands. If the islands are not merged, BLA must estimate a power factor angle for the island with the less complete measurement set to allow the net upstream island flows to be calculated.
Merge No Load Measurement Islands	<MergeNoLoadIsland>	Check box	This check box specifies whether BLA merges no-load measurement islands in feeders with the neighboring measurement islands. If the option is checked, no-load measurement islands in feeders are merged with the neighboring measurement islands.

Additional Load Adjustment

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Enable Temperature Adjustment	ENABLETMPSCH ED_BLA <EnableTempAdj Sch>	Check box	This check box specifies whether temperature adjustment schedules are used to update the real and reactive power load multipliers for associated load categories before BLA is started.
Use Load Forecast Data	USELFDATA_BLA <LoadForecast>	Check box	This check box specifies whether load forecast data is used. The application code related to this field is under development.
DPF Minimum Periodic Check			
Maximum Loading Difference for a Periodic DPF Execution (%)	<TriggerLoadPer cent>	Text box	This value specifies the maximum percent change in the total nominally scheduled load of every load category in each feeder. The parameter is used to determine if a BLA/DPF periodic execution is required when the power flow is scheduled to run at the DPF minimum solution time period. The DPF minimum solution time period is defined by DNAF_DPF_MIN_PERIOD in the appropriate server configuration file. If the load percent change is below this specified threshold, DPF is executed. DPF is always be executed at the maximum solution time period, which is defined by DNAF_DPF_MAX_PERIOD in the appropriate server configuration file. A typical value is 2.0%. The allowable range is 0-10.
Old and Deprecated Parameters			
Function Name	NAME_BLA <Name>	Text box	This value specifies the application name in the DNAF Solution pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCENUM_BLA <FuncEnum>	Text box	This value specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_BLA	Text box	This value represents the application Index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Enable Real-Time Track Delta	ENABLERTRACK DELTA_BLA <EnableRTTrack Delta>	Check box	This field is now deprecated.

5. Distribution Power Flow

The Distribution Power Flow (DPF) application is an integral part of the DMS. The DPF finds a phase domain solution to the unbalanced power system network model, which includes the individual bus phase voltages, network power flows, and network losses. DPF is capable of modeling and solving any unbalanced distribution system configurations, including networked (meshed) systems and radial systems.

The DPF algorithm accounts for the following items:

- Non-linear load-to-voltage dependencies
- Automatic control actions of locally controlled capacitors and voltage regulators
- Distribution transformer voltage drops
- Voltage-dependent distribution transformer coil and core losses
- Line and underground cable charging
- Power transformers, including fixed tap or LTC devices

5.1. DPF Methodology

The DPF provides a phase domain solution to unbalanced power system network models. Solution input and output are in terms of phase quantities. There are no assumptions of a radial network, and radial networks are solved just as an interconnected network. The solution handles multiple PV buses and any physically realizable combination of network connections. The DPF provides a solution based on convergence tolerances for real power, reactive power, and voltage.

5.1.1. Nodal Admittance Matrix Based Power Flow

The power flow solution is obtained using a nodal admittance matrix analysis technique, where the unknowns are the phase bus voltages. Individual phase voltages are treated as independent variables. Symmetrical component analysis is not used during the solution, but some sequence quantities are computed and output in the final states of an analysis.

The basic methodology iterates to solve for voltages using the following equation:

$$Y \cdot V = I \quad \text{(Equation 5.1)}$$

Where:

Y is a sparse phase-oriented admittance matrix

V is the vector of complex phase voltages

I is a vector of complex phase current injections

The current injections are not always given as part of the input data; therefore, they may need to be computed from other input data. For example, a constant power load can be converted into a current injection by assuming a voltage value. The transformation from constant power to constant current is adjusted each iteration until convergence is achieved.

The iteration-based power flow method solves meshed networks and handles the typical non-linearity of constant power loads. Additional special logic is added to appropriately model PV (constant real power and constant voltage magnitude) generator buses. Both balanced and unbalanced generator buses or reference buses are supported. For more details about reference bus interface objects and related sub-transmission models, see [Functionality and Concepts](#).

The solution methodology is based on a repeated linear admittance matrix solution technique ($YV=I$), where the bus injection currents are computed from the component models and the bus voltages are directly solved using a complex number admittance matrix. Functionally, this technique provides faster solution times in comparison to the Jacobian-based Newton method solution. Additionally, this method provides a direct (non-iterative) solution for cases without multiple PV buses and for all loads represented as impedance loads. This is a highly desirable feature for cases that may be difficult to solve using Newton type methods.

Non-linearity in the domain of the $YV=I$ solution space is handled by updating the current injections based on the component models, and by resolving the $YV=I$ equations. Non-linearity arises from the modeling of PQ loads and PV buses.

PQ loads are treated as constant current injections, and the current is computed from the specified power (a function of voltage) and the latest terminal voltages. Non-impedance loads are transformed into current injections using Equation 5.1 and the voltage from the previous iteration (or initial conditions on the first iteration). The algorithm iterates until the convergence criteria are met or the maximum iteration is reached.

Convergence is defined as the maximum P, Q, and V (voltage magnitude for a PV bus) being met. Additionally, no control actions (such as tap moves) should be pending. Tap control movements are handled outside the $YV=I$ solution between iterations.

The $YV=I$ matrix solution is performed using the usual sparse linear algebra LU decomposition techniques. A minimum degree ordering technique is used to minimize the number of non-zero fill-ins. Updates to the LU factors are achieved with fast factor update methods.

A matrix diagonal block is allocated for each bus within the solution. Each block contains one equation for each non-ground phase within the bus. Thus, a system with N three-phase buses contains N blocks with three equations, for a total of 3N equations.

Incorporating shunt impedance to ground into the admittance matrix is straightforward, because the terms only go on the true diagonal. There is no mutual coupling for shunt impedance components. Line components are similarly easy to handle. Transformers require the use of a connectivity matrix, and the actual matrix terms are built up using a formula. The formula is required so that tap adjustments can be made and reflected in the Y matrix. The connectivity matrix depends on the transformers and transformer types. Generator internal buses are also modeled in the matrix, although they are equivalenced to a single phase.

The core losses are voltage-dependent (second degree or higher) and therefore are treated as non-linear elements in DPF. DPF handles non-linearity during the iterative process.

The angle reference bus is modeled in the matrix with a special term to force its update to be zero (or to the specified reference angle value, which we assume is zero). The regulation equation can be thought of as reduced into the bus structure by having a very small (but not zero) term on its diagonal. This small term is referred to as the inverse of "big", which is a solution tuning parameter. The angle reference is always at a PVbus.

5.1.2. Load Representation

The different types of loads are handled in the DNAF using the load-to-voltage coefficients given in the model. The load-to-voltage dependencies for real power and reactive power are modeled as second order polynomials; the coefficients of the polynomial can be either positive or negative floating point numbers in the range from -99 to +99. Additionally, the sum of these polynomial coefficients must be equal to 1.0; otherwise, an internal routine enforces this condition by normalizing them before the DPF algorithm is executed. These coefficients are used to adjust the P and Q values of the loads inside the power flow.

For example, given a constant power load, the power is held constant (assuming a constant power profile is given in the load category), whereas in a constant current load, the current is held constant. The load to voltage dependencies are represented by the following equations:

$$P = P_{nom} \cdot \left[A \left(\frac{V}{V_{nom}} \right)^2 + B \left(\frac{V}{V_{nom}} \right) + C \right] \quad (\text{Equation 5.2})$$

$$Q = Q_{nom} \cdot \left[D \left(\frac{V}{V_{nom}} \right)^2 + E \left(\frac{V}{V_{nom}} \right) + F \right] \quad (\text{Equation 5.3})$$

Where:

$A+B+C = 1.0$ and $D+E+F = 1.0$

Constant power loads have $C = F = 1.0$ and $A = B = D = E = 0.0$.

Constant current loads have $B = E = 1.0$ and $A = C = D = F = 0.0$.

Constant impedance loads have $A = D = 1.0$ and $B = C = E = F = 0.0$.

5.1.3. PQ to PV Bus Conversion

Another feature of DPF is the automatic conversion of PQ to PV buses (and vice versa), which is triggered when the computed kVAR demand from a PV bus is found to be outside its pre-specified kVAR limits. In this procedure, the PV bus is marked as a PQ bus with its Q generation pegged to this pre-specified limit for further iterations. If it is determined, during subsequent iterations, that the computed kVAR demand is within the generator's kVAR limits, its status as a PV bus is restored

and the kVAR generation is set to the computed kVAR demand. This procedure is handled outside the DPF iterations. For more information on this issue, see [Generator PV/PQ Bus Conversion](#).

Each iteration of DPF includes two voltage solutions: $Y1 \cdot V1 = I1$ and $Y2 \cdot V2 = I2$. In the first solution, buses are modeled as fixed voltage magnitude and angle (in $Y1$), while in the second solution, buses are modeled as constant current injection (in $Y2$). The transition from modeling a bus as fixed V to fixed PQ modeled as a fixed current injection is handled using fast rank one-factor update techniques, which vastly reduces processing time for computations requiring matrix inversion. Additionally, tap changes that modify the Y matrix are handled using partial re-factorization techniques.

Each time convergence is achieved to some degree (possibly not the final user convergence), the tap adjustment logic is activated. If the tap adjustment logic results in a tap change in the system, the admittance matrices ($Y1$ and $Y2$) are recalculated and DPF continues to run until convergence is achieved.

5.1.4. Voltage Regulation of Generator Buses

DPF provides the capability to incorporate three-phase generator models into the distribution network. These models are generally small customer co-generator plants that contribute with both active and reactive support.

The generator remote voltage regulation is handled by outer loop adjustments between iterations. The regulation is "remote" in the sense that the internal generator bus voltage is not held constant between iterations. There are no matrix terms regarding the location of the remotely regulated bus. Sensitivity helps solve cases where the terminal bus voltage is being dragged down since, as the internal voltage grows during iterations, the algorithm realizes that the sensitivity is very small.

For cases where the internal voltage is near one, this change has little effect on convergence. The consideration of non-linear loads, remote voltage regulation, and very stressed network conditions increases the number of iterations, and can cause voltage deviations of up to +/- 0.07 p.u.

5.1.5. Representation of Distributed Energy Resource (DER)

Energy resource objects, such as solar, wind, and batteries, are modeled as negative constant power loads. The real and reactive loads are set to be the negative of the scheduled real and reactive output. The real power, reactive power, and voltage target (in the case that a DER is used as the slack bus) could come from dynamic schedules (from an external system) or static schedules (defined in model file). However, if manually entered values or associated SCADA analog values are available, they are used instead.

In cases of islanded operation or when no external generation is given, the DG or DER with the largest size (kVA) is automatically chosen as the slack bus.

5.1.6. Arc Suppression Coil Grounding Impedance

For power transformers with a regulated Arc Suppression Coil (ASC) connected, DPF recalculates the transformer secondary side grounding Impedance ($R+jX$). The grounding impedance value depends on the capacitance of downstream lines. More lines connected to a transformer result in larger capacitance and smaller grounding impedance.

For more details about calculating the ASC grounding impedance, see [Regulating ASCs](#).

5.1.7. Unsolvability DPF Cases

The solution of unbalanced, multiple-phase power flow is similar to that of the more typical balanced power flow, with some important exceptions. The differences generally make the unbalanced solution more difficult to model realistically and to solve.

The power flow equations use the Y matrix, instead of the typical Newton-Raphson method. This method solves extremely stressed networks when the loads are modeled as pure impedance and multiple PV generators are not modeled. This means it can solve short-circuit cases (either intentionally or due to data errors), and it can obtain voltage and power flow results that reflect the input data. Even though they are not realistic, such solutions can often provide insight toward identifying specific modeling errors.

Non-linear magnetizing models can create unsolvable cases. DPF has a set of parameters that can be relaxed to allow convergence in difficult cases.

If two generators attempt to regulate a bus to different values, this may cause the solution to diverge. If the generators have the same effective voltage target (considering line drop compensation), there should be no problems.

Unfortunately, when using non-impedance loads and non-linear transformer magnetizing models, mathematical solutions may not be available. There are many variations of this over-specified problem, but they typically result in DPF not being able to converge and reach maximum iterations. Note that the above case can have a perfectly balanced load, but an imbalance somewhere else in the problem makes the case unsolvable. Changing the loads to impedance loads, or grounding the loads, makes the case solvable.

A limited number of PV buses (1–2 per distribution circuit) may substantially increase the number of iterations, but should not diverge the iteration process. Nevertheless, the possibility of divergence exists due to unsolvable problem formulations.

To help address diverged cases, when DPF solves (iterative procedure), a set of tests is conducted to check convergence (for example, tests for unrealistic voltages). If any bus voltage violates a threshold set by the `DeadLoadTol` parameter in the `alg.dat` configuration file and a load is involved, DPF can switch the non-linear load model for the offending phase load to constant impedance, allowing for a solution. The load model can be changed to normal if the voltage is higher than the threshold set by the `DeadLoadTolRestartTol` parameter in the `alg.dat` file. Model conversion is an automatic DPF feature that users can enable or disable via the `SkipDeadLoads` parameter in `alg.dat`. If DPF switches to a constant impedance load to avoid an unrealistic solution, this fact is reported to

the user on the Base Application Status grid tabular display as the suspect DPF status (if the "Suspect Load" option is enabled in DCE) and the load involved is added to the suspect loads list. The suspect loads list can be queried from the Station DPF Results Summary display and is recorded in the station DPF log file.

5.2. Key Output Summary and Manual Input Displays

One of the main motivations for performing power flow on any electric network is to compute nodal voltages and branch flows (kW, kVAR, and Amps) and thereby identify any existing voltage and flow violations. ADMS provides many ways to view these violations in multiple levels of detail, including grid tabular displays and component-specific pop-up displays. For example, the Station Exception display, which can be accessed by selecting "Station Exception" menu item under the RTDMS drop-down menu, provides a quick glance at flow and voltage violations categorized by substation. It also provides links to additional tabular displays that list specific violations such as High Voltage Violations, Low Voltage Violations, Flow Imbalance Violations, Overload Violations, and Capacity Margin Violations.

To access all DPF results, a suitable starting point is the DPF Results tabular display, which can be accessed from the DNAF pop-up display and by selecting the DPF Results link for the desired station in the Station Output Summary display. The link for this display is provided in the RTDMS drop-down menu. The bottom pane of the DPF Results tabular display shows kW, kVAR, Amp, and power factor data for each feeder in the station. The top pane contains links to additional displays such as the Transformer Summary, Capacitor Summary, Load Summary, Generator Summary, and Distribution Transformer Summary for the station.

ADMS also provides manual entry pop-up displays to override the statically modeled parameters of components such as power transformers, breakers, switches, fuses, etc. To access these pop-ups, select the desired component and click the "Enter Data" toolbar button in the ADMS toolbar. Note that the toolbar is context-specific: different buttons appear based on the selected component. ADMS provides other tabular and pop-up displays, including the Application Input Summary, Manual Input Parameters Entry, and the DNAF System Configuration display, where users can override or specify station-wide or feeder-wide parameters such as load scaling factors or settings that enable or disable all the network applications.

5.3. Voltage Regulation

Voltage regulation is available for phase-to-neutral and phase-to-phase quantities. It is valid for generators, capacitors, and transformers.

When flow reverses (for reversing regulators), the regulating sensing point reverses and a new set point value is used. The sign of the real power determines the direction.

Transformer tap control regulates the voltage magnitude at the specified bus; both line-to-line and line-to-ground voltage are supported. The logic assumes unity (a parameter) for the sensitivity of the transformer tap ratio to the regulated bus voltage magnitude. The sign of the tap sensitivity is

negated when the controlling tap is on the opposite end from the regulated bus.

The logic moves to within one tap of the solution. No attempt is made to move to the closest tap when the high and low voltages are within one tap of each other. Tap movements are only allowed after convergence is reached to some degree (that is, only when the maximum of the voltage, reactive, and real power mismatch computed during each DPF iteration is less than a pre-set tolerance specified in DCE, which takes priority over equivalent parameters in alg.dat). Increasing the tap ratio at an end increases voltage at that same end (or decreases voltage at the opposite end).

Line drop compensation is modeled as a series impedance between the terminals of the transformer and the regulated bus. The terminal of the transformer corresponds to the terminal of the tap. For example, if the tap is on the primary winding, the current is taken as the current out of the transformer at the primary terminals. The voltage held at the regulated bus is $V_{\text{target}} + I \cdot Z$, where I is the current out of the transformer at the end where the tap is modeled. The compensating term is only applied on phases that exist at the regulated bus and at the end of the transformer where the tap is modeled.

When voltage reduction is enabled, the regulator's effective voltage target is calculated based on the active band-center setpoint. If the voltage setpoint measurement (type VoltSetpoint) is not available, the effective voltage target is calculated based on the nominal voltage target specified in the static model. For example: if the bandcenter measurement value is 125 V, voltage reduction is enabled, and the value is 2%, the target value will be calculated as $125 * (1 - .02) = 122.5$ V.

Generally, the regulated bus is assumed to be on the secondary side of the transformer. However, with optional regulation control file, advanced regulation modes with different regulated bus location and methods are also supported. The regulation control definition and regulation mode defined in the optional file overwrite the regular regulation parameters in the station internal/external file. For modeling details, refer to the section Regulation Control File in the *ADMS Modeling Overview and Converter User Guide*.

If advanced regulation mode exists, the current regulation mode is determined from the valid SCADA mode measurement (measurement type AnalogActiveRegulationMode) if it exists, or from the default mode defined in the model. The status measurement (measurement type StatusFlowDirection) drives regulation running in forward or reverse mode. If the status measurement is not available, regulation mode depends on the modeling parameter (field REACTFLOWSENSE_REGMODE) and uses the sign of the active or reactive power to determine the sign of the current. Then, the current value is compared to the pre-defined forward or reverse threshold to determine if the regulation runs in forward or reverse mode. In each mode, the regulated bus can be defined on the primary side or the secondary side of the transformer. The tap movement behavior can also be defined among three choices:

- **Regulate:** Moves the tap according to the regulated bus voltage.
- **Idle:** Freezes the tap at its current position.
- **RunToNeutral:** Moves the tap to its nominal position.

5.3.1. Influence of Measurements on Effective Regulation Status

This section describes how the real-time DPF determines if it can internally control taps on a regulator or transformer LTC depending on certain SCADA measurements associated with the control status of the device. This only applies to devices that have internal voltage regulation capability.

Possible SCADA measurements considered include the following:

- Tap position measurement (analog)
- Remote/Local measurement (status)
- Auto/Manual measurement (status)
- Blocked measurement (status)

If a valid tap position measurement (item 1) is present, this always overrides all other cases, and results in the regulation device being set to a fixed tap position. Even if the device has regulation capabilities, they are ignored for the current DPF analysis and the specified tap position is used instead.

If there is no valid tap position measurement, then various combinations of the other 3 measurements (items 2,3,4) determine whether the DPF analysis treats the device as having a fixed tap position (using the most recently set value) or whether the device self-regulates according to its internal control settings (voltage setpoint, voltage reduction, deadband, tap increment, etc.). The following table shows the different possibilities.

Table 4. Effective Tap Regulation Status in DPF

Remote/Local Measurement Value	Auto/Manual Measurement Value	Blocked Measurement Value	Resulting Tap Control
Remote Control	Automatic Control	Regulation Blocked	Fixed
Remote Control	Automatic Control	Not Blocked	Fixed
Remote Control	Automatic Control	Missing / Not Valid	Fixed
Remote Control	Manual Control	Regulation Blocked	Fixed
Remote Control	Manual Control	Not Blocked	Fixed
Remote Control	Manual Control	Missing / Not Valid	Fixed
Remote Control	Missing / Not Valid	Regulation Blocked	Fixed
Remote Control	Missing / Not Valid	Not Blocked	Fixed
Remote Control	Missing / Not Valid	Missing / Not Valid	Fixed
Local Control	Automatic Control	Regulation Blocked	Fixed
Local Control	Automatic Control	Not Blocked	Regulated
Local Control	Automatic Control	Missing / Not Valid	Regulated
Local Control	Manual Control	Regulation Blocked	Fixed

Remote/Local Measurement Value	Auto/Manual Measurement Value	Blocked Measurement Value	Resulting Tap Control
Local Control	Manual Control	Not Blocked	Fixed
Local Control	Manual Control	Missing / Not Valid	Fixed
Local Control	Missing / Not Valid	Regulation Blocked	Fixed
Local Control	Missing / Not Valid	Not Blocked	Regulated
Local Control	Missing / Not Valid	Missing / Not Valid	Regulated
Missing / Not Valid	Automatic Control	Regulation Blocked	Fixed
Missing / Not Valid	Automatic Control	Not Blocked	Regulated
Missing / Not Valid	Automatic Control	Missing / Not Valid	Regulated
Missing / Not Valid	Manual Control	Regulation Blocked	Fixed
Missing / Not Valid	Manual Control	Not Blocked	Fixed
Missing / Not Valid	Manual Control	Missing / Not Valid	Fixed
Missing / Not Valid	Missing / Not Valid	Regulation Blocked	Fixed
Missing / Not Valid	Missing / Not Valid	Not Blocked	Regulated
Missing / Not Valid	Missing / Not Valid	Missing / Not Valid	Regulated

5.3.2. Influence of Measurements on Effective DPF Capacitor Regulation Status

This section describes how DPF determines if it can internally regulate switched capacitors depending on whether certain SCADA measurements associated with the capacitor are available or not. The SCADA measurements that are considered include the following:

- Capacitor Remote/Local measurement (status)
- Capacitor Auto/Manual measurement (status)
- Capacitor Step measurement (status)

The expected DPF behavior depending on various combinations of the above-mentioned measurement types is tabulated below. Note that for capacitors with phase-independent switching, the evaluation is performed separately for each phase. Therefore, it is possible that DPF may be able to regulate, for example, Phase A of a capacitor, but treat Phases B and C as fixed.

Table 5. Effective Capacitor Regulation Status in DPF

SCADA Measurement Quality			Effective DPF Regulation Status
Step	Remote/Local	Auto/Manual	
Valid	Valid	Valid	Fixed
Valid	Valid	Invalid or not mapped	Fixed
Valid	Invalid or not mapped	Valid	Fixed
Valid	Invalid or not mapped	Invalid or not mapped	Fixed

SCADA Measurement Quality			Effective DPF Regulation Status
Invalid or not mapped	Valid	Valid	Regulating only if in "Local" and "Auto"; fixed otherwise.
Invalid or not mapped	Valid	Invalid or not mapped	Regulating if in "Local" mode; fixed otherwise.
Invalid or not mapped	Invalid or not mapped	Valid	Regulating if it has local control capability and regulation is on; fixed otherwise.
Invalid or not mapped	Invalid or not mapped	Invalid or not mapped	Regulating if it has local control capability; fixed otherwise.

5.3.3. Voltage Regulator Rating

The rating of a voltage regulator is set in transformer model. If it is the regulated kVA, the Units filed in the transformer model should be set to 3. It is defined in IEEE Std C57.15 IEEE Standard Requirements, Terminology, and Test Code for Step-Voltage Regulators as follows:

- **Rating in kVA of a voltage regulator: (A)** The rating that is the product of the rated load amperes and the rated "raise" or "lower" range of regulation in kilovolts (kV). If the rated raise and lower range of regulation are unequal, the larger shall be used in determining the rating in kVA. **(B)** The rating in kVA of a three-phase voltage regulator is the product of the rated load amperes and the rated range of regulation in kilovolts multiplied by 1.732.

Effective rating of a voltage regulator is used to calculate the impedance and losses of the regulated, and it is calculated from the rating of a voltage regulator:

$$kVA_{effective} = \frac{100.0}{Range} \cdot kVA_{rated} \quad \text{(Equation 5.4)}$$

$kVA_{effective}$ is the effective rating in kVA

kVA_{rated} is the rating in kVA

Range is the larger of the rated raise and lower range of regulation in percent

5.4. Service Drop Voltage and Voltage Drop Schedule

In some station external models, the secondary network ends at the secondary node of the distribution (secondary) transformers. For a more complete model, the analyst can optionally model a voltage drop schedule to estimate the voltage drop for the connecting service drop lines. The voltage drop schedule for the load is specified in the corresponding load category in the global.csv file.

Figure below shows a sample voltage drop schedule. To specify this schedule in the global file:

- The enumerated schedule type (SCHEDTYPE_SCHED) is set to 4.
- The x-axis (XAXISTYPE_SCHED) of the schedule is the transformer loading percentage, and the x-axis unit (XAXISUNITS_SCHED) is %Loading.

- The corresponding y1-axis (YAXISTYPE_SCHED) is Voltage, and the y1-axis unit (Y1AXISUNITS_SCHED) is %Volt Drop.

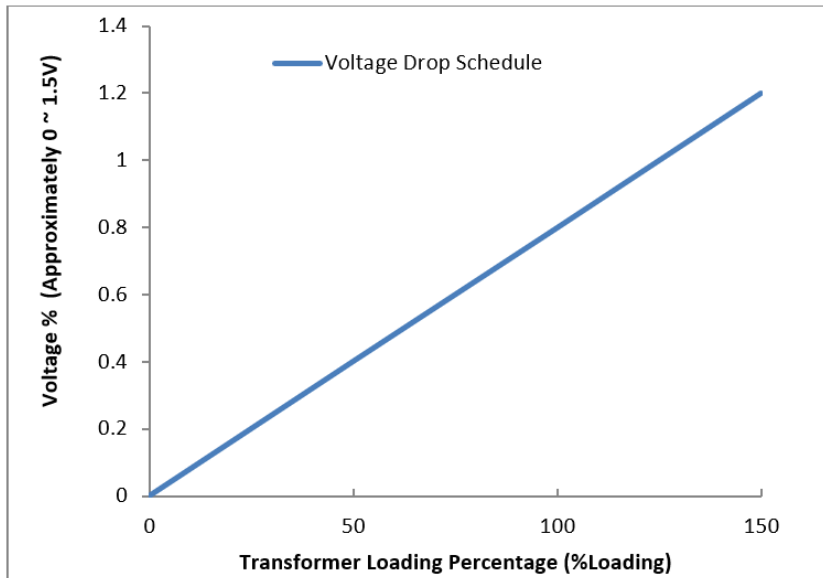


Figure 26. Sample Voltage Drop Schedule

In figure above, the 100% transformer loading level corresponds to a 0.8% voltage drop, which is approximately 1V on a 120.0 Volt base. If the transformer is overloaded to 150%, the voltage drop is approximately 1.5V on a 120.0 Volt base.

The schedule points of a secondary voltage drop schedule are discrete values. If SHAPE_SCHED = 1 (Linear), linear interpolation is performed if transformer loading falls into a bracket. If SHAPE_SCHED = 2 (Step), the first point of the bracket is selected. If the transformer loading is out of the scheduled range, the closest point is selected.

If a voltage drop schedule is provided, the customer voltage is slightly lower than the DPF calculated secondary voltage of the distribution transformer. Otherwise, the customer voltage is identical to the secondary voltage of the distribution transformer.

5.5. Generator PV/PQ Bus Conversion and Capability Curves

As described in [PQ to PV Bus Conversion](#), a voltage-controlled bus is converted to a load bus between DPF iterations. This is done to replicate the conditions occurring in the real world when the generator exceeds its maximum kVAR limits or falls below the minimum kVAR limits, as follows:

$$Q_{gen} = \begin{cases} Q_{min} & \text{if } Q_{gen} < Q_{min} \text{ and bus status is set to PQ bus} \\ Q_{max} & \text{if } Q_{gen} > Q_{max} \text{ and bus status is set to PQ bus} \end{cases}$$

During subsequent DPF iterations, if it is determined that the computed kVAR falls between the limits, its status is changed back to a PV bus.

These kVAR limits are usually inferred from the Generator's Capability Curve (GCC). There are currently two ways of modeling a generator capability curve:

- Using a "box" curve that is represented by minimum and maximum real power and reactive power output, which is required for a station to be DNAF-capable.
- Using a piecewise linear generation capability curve, which is optional for DNAF capability.

The generator kVAR limits are compared to the total positive sequence kVAR output at the terminal. The generator's internal voltage is adjusted, similarly to how voltage is regulated, to keep the reactive power within the limits. This is achieved via PV/PQ switching.

5.5.1. Box Generator Capability Curve

The "box" generator capability curve, shown on the left side of figure below, is modeled by the minimum and maximum real power and reactive power output values that are entered in the generator model in the global.csv file (and/or global_internals.xml file from the Substation Editor).

5.5.2. Piecewise Linear Generator Capability Curve

The piecewise linear generator capability curve, shown on the right side of figure below, describes the real and reactive power capability of a generator more accurately. Real power is plotted on the horizontal axis, while reactive power is plotted on the vertical axis. The three individual curves that make up the capability curve represent different physical limitations of the generator (for example, continuous stator current limit, continuous rotor current limit, and continuous field winding current limit).

The first segment is an arc of a circle centered on the Q axis. The arc joins the positive Q axis with the constant MVA portion of the curve, and defines the upper boundary of reactive power out of the generator. It is the arc of a circle because it also reflects current-based heating. The limitation described here is that of the rotor winding.

The second segment is the arc of a circle centered at the origin of the reactive capability curve. Because the radius of that circle is the apparent power, S (in MVA), it is based on the thermal heating limitations inherent in the stator winding and reflects the fact that the stator limitation is based on current alone.

The third segment joins the negative Q axis (representing reactive power into the machine) with the constant MVA portion of the curve. This segment reflects end-ring heating while in under-excited operation. The following graph shows a typical three-segment capability. It also gives a visual relationship between generator operating point (P, Q) and angle.

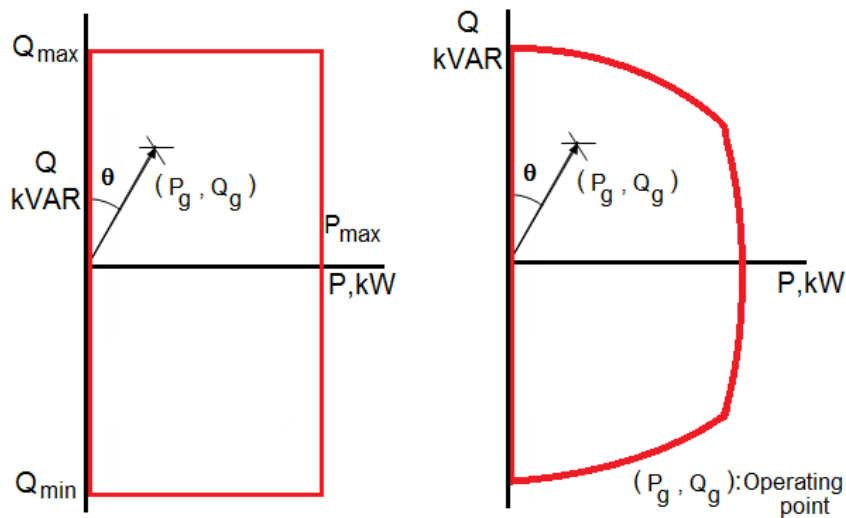


Figure 27. Box and Piecewise Generator Capability Curves

Note

Users can set the generator bus type to be either PV or PQ by setting the GenType field in the station externals file to true or false, respectively. If the field is not set, generators default to PV.

The real power generator schedule and reactive power generator schedule support sub hours on the X-axis. The application uses linear interpolation to calculate real power and reactive power for a given time. If users only define a schedule based on hours and still want the application to use the value of Y-axis on hour of X-axis (not sub hours), the "shape" value must be set as 2 (step, not linear) for the generation schedule specified in the global.csv file.

5.6. Inverter Autonomous Modes

If a DER has an associated inverter or set of inverters, and there are no higher priority DER power input sources, the DER's output may be controlled by an inverter autonomous mode. A DER's real and reactive power input sources are decided in the following priority:

1. Manual entry
2. SCADA measurement
3. 2030.5 Gateway measurement
4. Dynamic schedule
5. Autonomous mode
6. Static schedule (profile)

There can be more than one non-conflicting autonomous mode, but only one active control curve is allowed in one active control mode. For each supported mode, a DER will use the autonomous mode based on status and analog measurements from SCADA. For example, StatusInverterActiveModeFrequencyWatt and AnalogActiveInverterFrequencyWattCurve are the

measurements used for a Frequency-Watt control mode. The AnalogActiveInverterFrequencyWattCurve measurement indicates the active Frequency-Watt curve when the Frequency-Watt mode is active (determined by measurement type StatusInverterActiveModeFrequencyWatt). The SCADA analog value for each Frequency-Watt curve is defined in the static model. If a Frequency-Watt control curve's index matches the analog measurement value, it will be used for the Frequency-Watt control mode. Table below lists the available SCADA measurements used for modeling autonomous curves.

Table 6. Status and Analog Measurements Used for Inverter Autonomous Modes

Measurement Type	Description
AnalogActiveInverterFrequencyWattCurve	Index of active Frequency-Watt curve.
StatusInverterActiveModeFrequencyWatt	Frequency-Watt control mode active.
AnalogActiveInverterVVcurve	Index of active Volt-VAR curve.
StatusInverterActiveModeVoltVar	Volt-VAR control mode active.
AnalogActiveInverterVWcurve	Index of active Volt-Watt curve.
StatusInverterActiveModeVoltWatt	Volt-Watt control mode active.

If there is no valid SCADA measurement, the DER will try to find the autonomous mode based on the time of evaluation. If the schedule of multiple curves matches the evaluation time, the first curve found will be chosen. If none of the curves meets the requirement, the DER will use the static schedule (profile). The study mode follows the same procedure as the real time mode when determining the DER power output.

The DNAF applications (DPF, LVM, and the reconfiguration applications including FISR, AFR, and PLOS) support five DER inverter autonomous modes: Volt-VAR mode, Volt-Watt mode, Fixed Power Factor Mode, Maximum Power Limiting Mode, and Frequency-Watt Mode. These modes are described in the following sections.

For modes using voltage parameters (Volt-VAR mode, Volt-Watt mode), a reference voltage and an offset voltage are defined as listed in table below. All inverters behind one Point of Common Coupling (PCC) have a common reference voltage but may differ in the voltage between their own Electrical Connection Point (ECP) and the PCC due to instrumentation errors or voltage shifts within a plant. These differences can be corrected by the parameter VRefOfs that is to be applied by each inverter.

Table 7. Voltages Used for Inverter Autonomous Modes

Name	Description
VRef	The normal operating voltage for this DER site / service connection, in Volts.
VRefOfs	An offset voltage that represents an adjustment for a DER, relative to VRef, in Volts. VRefOfs is defined as the voltage at the ECP, relative to the PCC. For example, if the PCC VRef is 120 V, and the nominal voltage at the DER's ECP is 122V, then VRefOfs = +2V.

5.6.1. Volt-VAR Mode

In this mode, the DER's VAR output is set based on the measured voltage from SCADA or 2030.5 Gateway, or a calculated voltage if valid voltage measurements are not available at the DER terminal. The "configurable-curve" approach is used for this mode. This mechanism allows the inverter to be configured using an array of points, where the points define a piece-wise linear "curve" that establishes VAR output as a function of the local voltage.

In this configuration, the voltages are to be represented in the form of "Percent of VRef". Each DER will compute an "Effective Percent Voltage" based on its real-time local voltage measurement, nominal voltage setting, and offset voltage setting, as follows:

$$V_{effective} = \frac{V_{meas} - V_{RefOfs}}{V_{Ref}} \cdot 100 \quad (\text{Equation 5.5})$$

Where:

$V_{effective}$ is the effective voltage in percent

V_{meas} is the local voltage measurement at the DER in volts

V_{RefOfs} is an offset voltage that represents an adjustment for this DER, relative to VRef, in volts

V_{Ref} is the normal operating voltage for this DER site / service connection, in volts

The inverter compares this "Effective Percent Voltage" value to the voltages (X-Values) in the curve, which has been represented also in the form of "Percent of VRef", as follows:

$$V_{X,percent} = \frac{V_X}{V_{Ref}} \cdot 100 \quad (\text{Equation 5.6})$$

Where:

$V_{X,percent}$ is the X-Value voltage in percent

V_X is the X-Value voltage from the curve, in volts

V_{Ref} is the normal operating voltage for this DER site / service connection, in volts

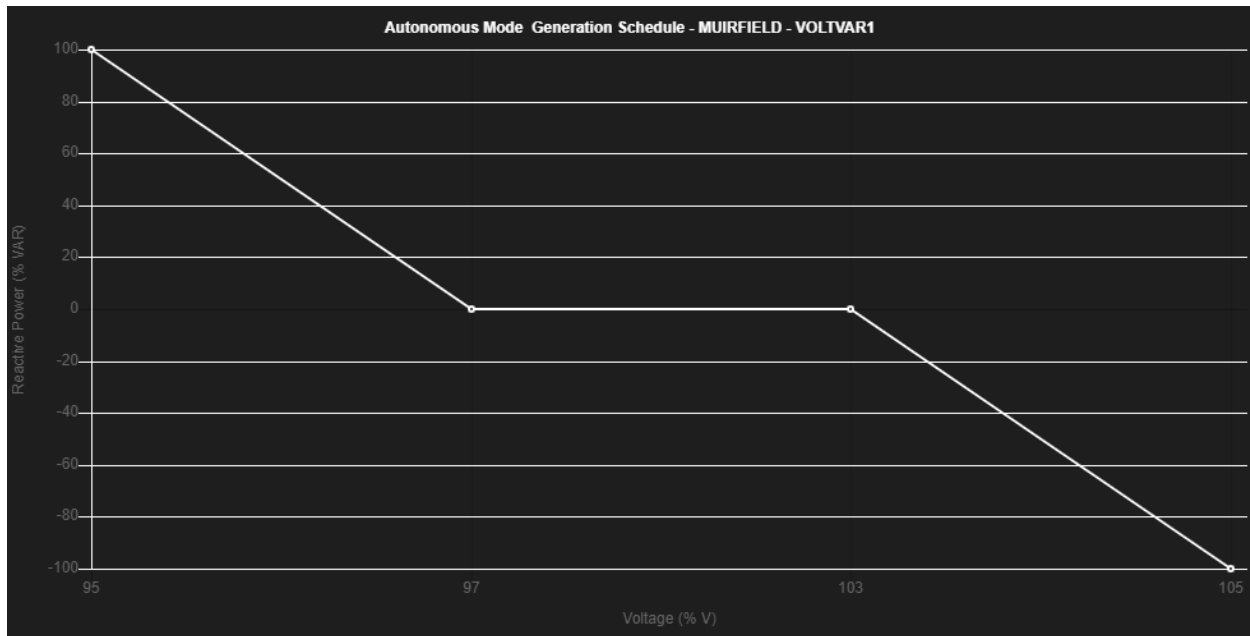


Figure 28. Volt-VAR Curve Example

If a Volt-VAR mode is active and there are no valid VAR measurements, the reactive power is decided from this mode. Inside the power flow solution engine, after each iteration when the voltages are calculated, adjustments are made to the DER reactive power output based on the calculated voltage.

5.6.2. Volt-Watt Mode

In this mode, the DER's Watt output is set based on the measured voltage from SCADA or 2030.5 Gateway, or a calculated voltage if valid voltage measurements are not available at the DER terminal. The "configurable-curve" approach is used for this mode. This mechanism allows the inverter to be configured using an array of points, where the points define a piece-wise linear "curve" that establishes watt output as a function of the local voltage.

In this configuration, the voltages used are the same as Volt-VAR mode, which are computed in equations 5.5 and 5.6.

If a Volt-Watt mode is active and there are no valid Watt measurements, the real power is decided from this mode. Inside the power flow solution engine, after each iteration when the voltages are calculated, adjustments are made to the DER real power output based on the calculated voltage.

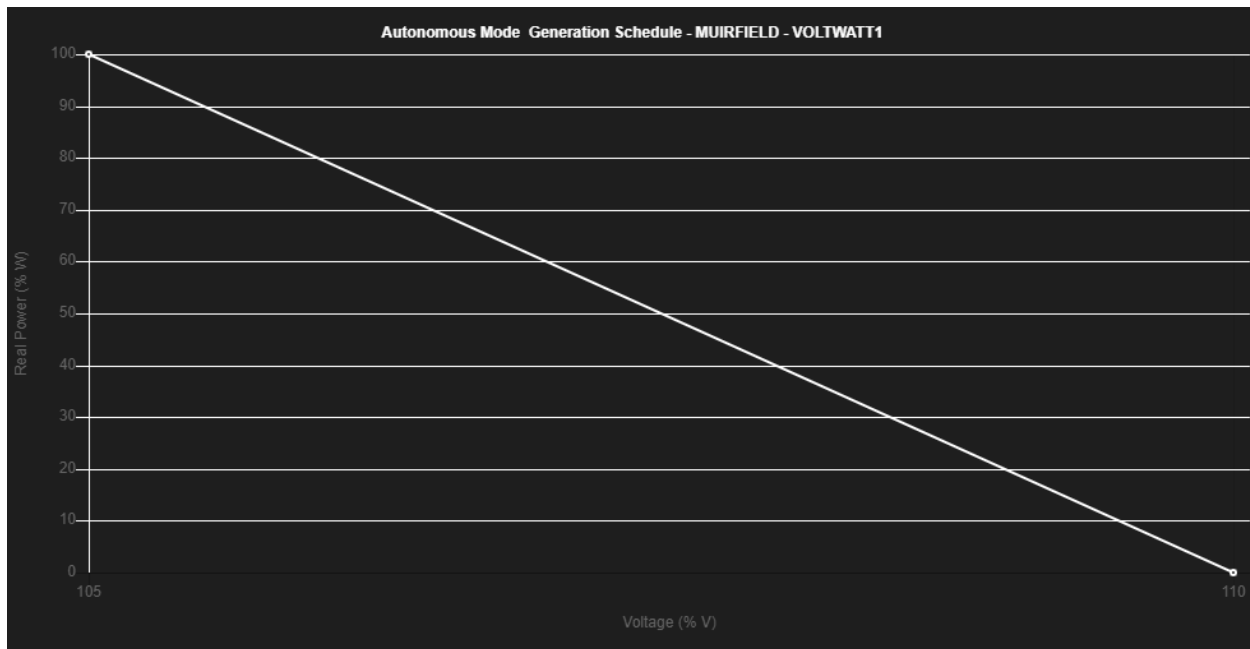


Figure 29. Volt-Watt Curve Example

5.6.3. Fixed Power Factor Mode

In this mode, the DER's power factor is set to a fixed value. The following parameters are involved:

- **Power Factor Target:** The power factor that the DER can deliver to the grid.
- **Power Factor Type:** The sign convention of the set power factor target. The IEC and IEEE conventions are supported.

For IEC Convention, supplying or generating active power is positive and demanding active power is negative (quadrants 2 and 3 have a negative power factor; quadrants 1 and 4 have a positive power factor). For IEEE Convention, a leading (capacitive) power factor is positive, and a lagging (inductive) power factor is negative (quadrants 1 and 3 have a negative power factor; quadrants 2 and 4 have a positive power factor).

If the Fixed Power Factor mode is active and there is no valid real power measurement, but there are valid reactive power measurements, the real power value is determined from 1) the reactive power value from a measurement from SCADA or 2030.5 Gateway, or a value from its dynamic schedule or static schedule (if a dynamic schedule is not available) and 2) the power factor target. When setting the real power value, a check is performed to make sure the inverter's VA limit is not exceeded.

If the Fixed Power Factor mode is active and there is no valid reactive power measurement, the reactive power value is determined from 1) the real power value from either a measurement from SCADA or 2030.5 Gateway, or a value from its dynamic schedule or static schedule (if a dynamic schedule is not available) and 2) the power factor target. When setting the reactive power value, a check is performed to make sure the inverter's VA limit is not exceeded.

5.6.4. Maximum Power Limiting Mode

In this mode, the DER's real power output is limited to a percentage of the maximum real power that the DER can deliver to the grid at its ECP. The following parameters are involved:

- WMax: The maximum real power (in Watts) that the DER can deliver to the grid.
- Upper Watt Limit: The upper Watt limit the DER can deliver to the grid, expressed as a percentage of WMax.

If Maximum Power Limiting mode is active and there are no valid Watt measurements from SCADA or 2030.5 Gateway, the real power is decided from this mode. If the real power from a schedule is larger than the maximum allowable real power, the real power value is set to the maximum allowable real power value. The maximum allowable real power value is calculated as follows:

$$W_{max_allowed} = \frac{W_{upper_limit}}{W_{max}} \cdot 100 \quad (\text{Equation 5.7})$$

Where:

$W_{max_allowed}$ is the maximum allowable real power

W_{upper_limit} is the upper real power limit the DER can deliver to the grid, in percent

W_{max} is the maximum real power that the DER can deliver to the grid

5.6.5. Frequency-Watt Mode

In this mode, the DER's watt output is set based on the frequency taken from a user manual entry or a measurement from SCADA or 2030.5 Gateway. The "configurable-curve" approach is used for this mode. This mechanism allows the inverter to be configured using an array of points, where the points define a piece-wise linear "curve" that establishes watt output as a function of the frequency.

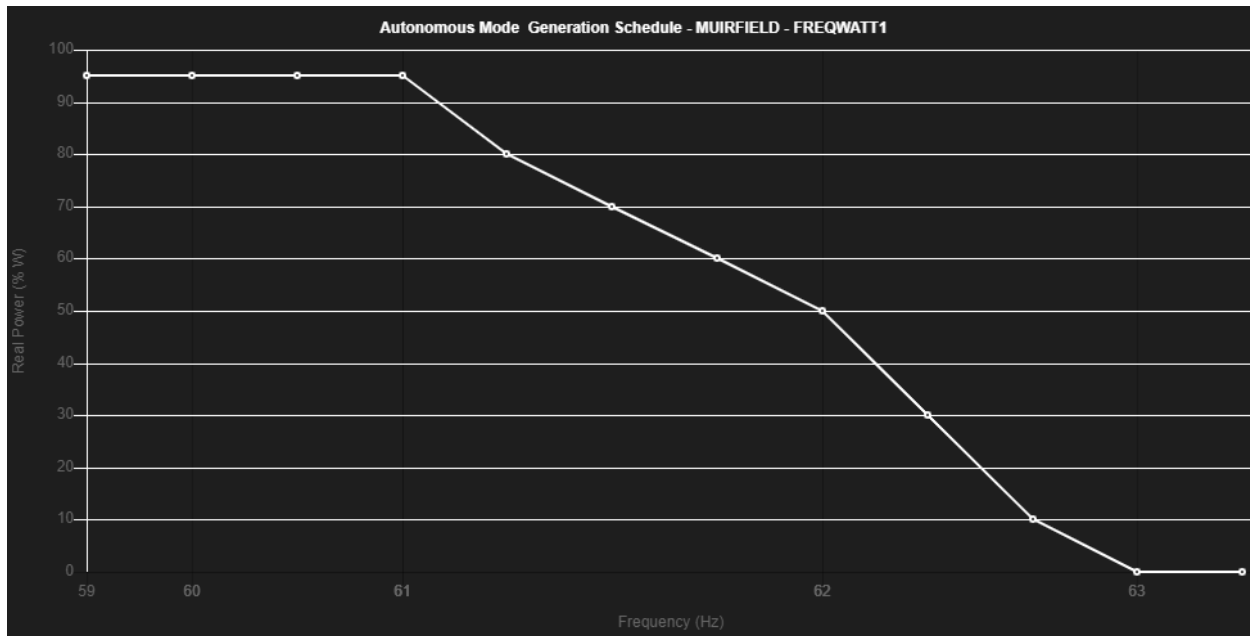


Figure 30. Frequency-Watt Curve Example

If a Frequency-Watt mode is active and there are no valid real power measurements, the real power is decided from this mode. In this mode, the real power output is determined based on the input frequency.

5.7. Locally Controlled Capacitor Switching Algorithm

DNAF applications support six types of locally controlled capacitors, which are described in the following sections. Figure below illustrates a locally controlled capacitor. Note that the process of selecting the state of the capacitor for various ranges has been modified. The process of selecting the operating ranges via the station input files using the variables OnLevel and OffLevel (ONLEVEL_SHIMP and OFFLEVEL_SHIMP for .csv file) is retained in the current version with slight variations. However, in the previous version, the device states in these ranges were fixed and pre-determined. While this was a valid approach, these variables and their labels introduced considerable confusion with regards to when the device turns on and off. To eliminate this confusion, the current approach allows the user to select the state of the device in these operating ranges using the DNAF Configuration Editor as described in the following subsections.

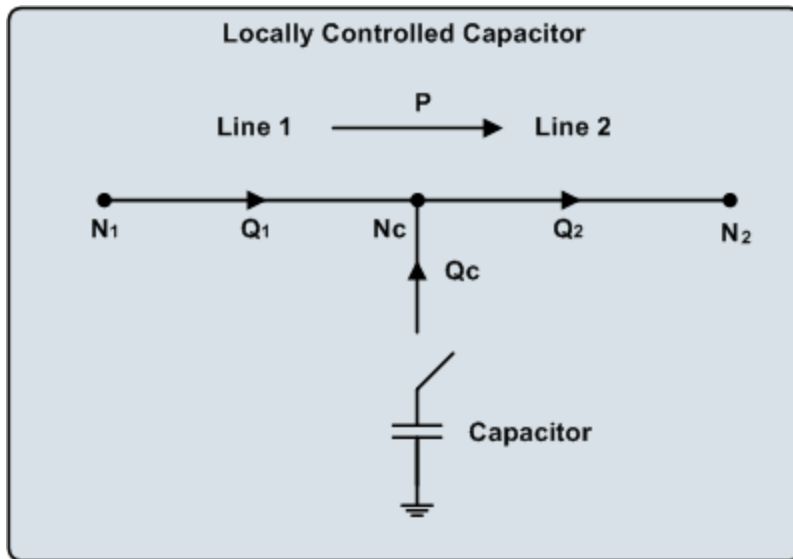


Figure 31. Locally Controlled Capacitor

5.7.1. Voltage-Controlled Capacitor

The following parameters must be set for a voltage-controlled capacitor:

- The regulated node is set in the Regnode field in the XML file or in the REGNODE_SHIMP field in the CSV file.
- The monitored phases are set in the PhaseSense field in the XML file or in the PHASESENSE_SHIMP field in the CSV file.
- The Control Level 1 (CL1) voltage is set in the OnLevel field in the XML file or in the ONLEVEL_SHIMP field in the CSV file.
- The Control Level 2 (CL2) voltage is set in the OffLevel field in the XML file or in the OFFLEVEL_SHIMP field in the CSV file.

In some cases, the regulation node of a station capacitor is the unregulated node between the transformer secondary node and the tap changer. This node is not explicitly modeled in the system but can be determined from the transformer model.

The fields ReguseUnRegulatedNode and RegTransformer in the XML file are used to indicate whether the capacitor is monitoring the unregulated node and the name of the monitored transformer. The unregulated node voltage is calculated by the transformer secondary side voltage and current transformer tap ratio. If the tap is in its nominal position, the unregulated node voltage should be equal to the transformer secondary node voltage.

Based on CL1 and CL2, two new parameters (CS1 and CS2) are defined as follows: CS1 is the smaller (lower voltage magnitude) of CL1 and CL2, and CS2 is the larger (higher voltage magnitude) of CL1 and CL2. These two parameters, in turn, define the three operating ranges of the capacitor as follows:

- The device is in Range 1 when the controlled variable (voltage) is less than CS1.

- The device is in Range 2 when the controlled variable (voltage) is greater than CS1 and smaller than CS2.
- The device is in Range 3 when the controlled variable (voltage) is greater than CS2.

The user sets the state of the capacitor switch for the three ranges via the DNAF Configuration Editor's three combo box controls, as shown in figure below. Each combo box has the following three options: Turn on, No Status Change, and Turn Off. Figure below illustrates the three available operating ranges for the capacitor (where "Voltage" is the average voltage magnitude of the monitored phases at the regulated node).

Locally Controlled Capacitors

Controller Type	Below Lower	Between	Above Upper	Control Target Adj. (%)
Voltage	Turn On	No Status Change	Turn Off	0
VAR	Turn Off	No Status Change	Turn On	0
Amp	Turn Off	No Status Change	Turn On	0
Power Factor	Turn On	No Status Change	Turn Off	0
Time	Turn Off	Turn On	Turn Off	
Temperature	Turn Off	No Status Change	Turn On	

Figure 32. Parameters for Locally Voltage-Controlled Capacitors

Figure below illustrates the states of the capacitor for the above user-selected configuration for the three ranges. (The combo box settings above are shown for illustrative purposes only.) The gray area indicates that the device remains in its previous position, that is, its status is unchanged.

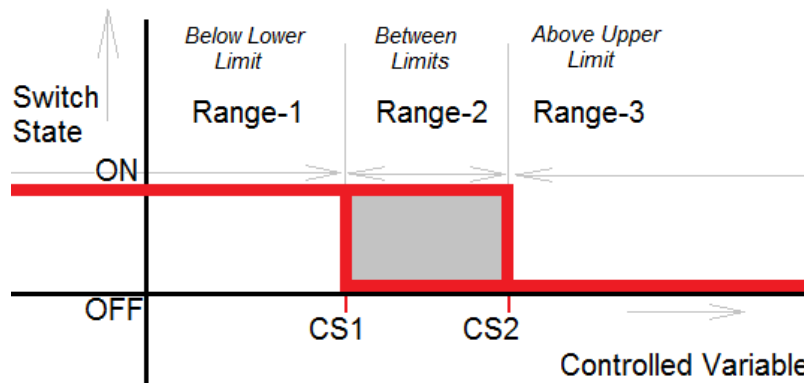


Figure 33. Locally Voltage-Controlled Capacitor States and Operating Ranges

5.7.2. VAR-Controlled Capacitor

The following parameters must be set for a VAR-controlled capacitor:

- The regulated branch (line, switch, transformer, etc.) is set in the field RegBranch in the XML file or in the field REGBRANCH_SHIMP in the CSV file. The VAR flow in the regulated branch is monitored. In the locally controlled capacitor example shown in [figure](#), **Line 1** is the regulated

branch.

- The monitored phases are set in the field PhaseSense in the XML file or in the field PHASESENSE_SHIMP in the CSV file.
- The regulated end of the branch is set in the field RegBrEnd in the XML file and in the field REGBRANCHEND_SHIMP in the CSV file.

The field has the following values: 0=FROM side, 1=TO side, and 2=TERTIARY side (only if the branch is a three-winding transformer). In [figure](#), assume that the FROM side is N1, the TO side is Nc for Line 1, and the end with node Nc is regulated, so the value is 1. If the end with the node N1 is regulated, the value is 0.

- The regulated flow direction is set in the field RegBrDirect in the XML file and in the field REGBRANCHDIRECT_SHIMP in the CSV file.

The field has the following values: 1=OUT of the regulated branch, 2=INTO the regulated branch. In [figure](#), if the direction is OUT then the value is 1 and if the direction is INTO then the value is 2. This parameter is chosen regardless of the regulation end defined above.

- Control Level 1 (CL1) kVAR is set in the OnLevel field in the XML file or in the ONLEVEL_SHIMP field in the CSV file.
- Control Level 2 (CL2) kVAR is set in the OffLevel field in the XML file or in the OFFLEVEL_SHIMP field in the CSV file.

Based on CL1 and CL2, two new parameters (CS1 and CS2) are defined as follows: CS1 is the smaller (lower kVAR) of CL1 and CL2, and CS2 is the larger (higher kVAR) of CL1 and CL2. These two parameters, in turn, define the three operating ranges of the capacitor as follows:

- The device is in Range 1 when the controlled variable (kVAR) is less than CS1.
- The device is in Range 2 when the controlled variable (kVAR) is greater than CS1 and smaller than CS2.
- The device is in Range 3 when the controlled variable (kVAR) is greater than CS2.

The user sets the state of the capacitor switch for the three ranges via the DNAF Configuration Editor's three combo box controls, as shown in [figure](#) below. Each combo box has the following three options: Turn on, No Status Change, and Turn Off. The corresponding device states are shown in following [figure](#) (where "VAR" is the sum of the reactive power in the monitored phases in the regulated branch end and direction).

Locally Controlled Capacitors

Controller Type	Below Lower	Between	Above Upper
Voltage	Turn On	No Status Change	Turn Off
VAR	Turn Off	No Status Change	Turn On
Amp	Turn Off	No Status Change	Turn On
Power Factor	Turn On	No Status Change	Turn Off
Time	Turn Off	Turn On	Turn Off
Temperature	Turn Off	No Status Change	Turn On

Figure 34. Parameters for Locally VAR-Controlled Capacitors

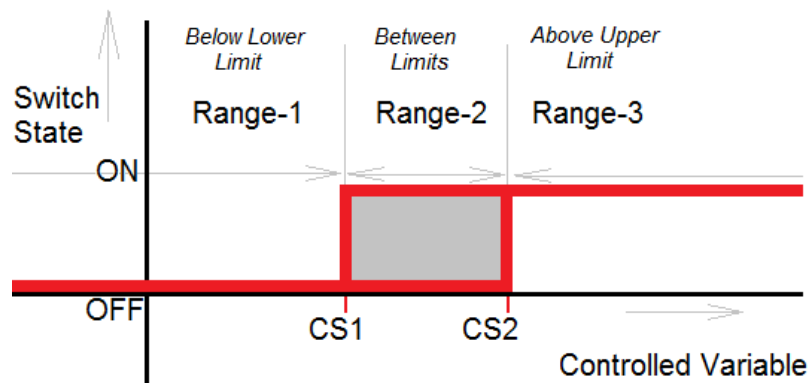


Figure 35. Locally VAR Controlled Capacitor States and Operating Ranges

5.7.3. Amp-Controlled Capacitor

The following parameters must be set for an Amp-controlled capacitor:

- The regulated branch is set in the field RegBranch in the XML file or in the field REGBRANCH_SHIMP in the CSV file. The Amp flow in this branch is monitored. In [figure](#), **Line 1** is the regulated branch.
- The monitored phases are set in the field PhaseSense in the XML file or in the field PHASESENSE_SHIMP in the CSV file.
- The regulated end of the branch is set in the field RegBrEnd in the XML file and in the field REGBRANCHEND_SHIMP in the CSV file.

The field has the following values: 0=FROM side, 1=TO side, and 2=TERTIARY side (only if the branch is a three-winding transformer). In [figure](#), assume the FROM side is N1, the TO side is Nc for Line 1, and the end with the node Nc is regulated, so the value is 1. If the end with the node N1 is regulated, the value is 0.

- The regulated flow direction is set in the field RegBrDirect in the XML file and field REGBRANCHDIRECT_SHIMP in the CSV file.

The field has the following values: 1=OUT of the regulated branch, 2=INTO the regulated branch. In [figure](#), since Nc is the regulated end, the direction is out of the regulated branch and

the value is 1. If N1 is the regulated end, the direction is into the regulated branch, and the value is 2.

- Control Level 1 is set in the OnLevel field in the XML file or in the ONLEVEL_SHIMP field in the CSV file.
- Control Level 2 is set in the OffLevel field in the XML file or in the OFFLEVEL_SHIMP field in the CSV file.

Based on CL1 and CL2, two new parameters (CS1 and CS2) are defined as follows: CS1 is the smaller (lower Amp) of CL1 and CL2, and CS2 is the larger (higher Amp) of CL1 and CL2. These two parameters, in turn, define the three operating ranges of the capacitor as follows:

- The device is in Range 1 when the controlled variable (Amp) is less than CS1.
- The device is in Range 2 when the controlled variable (Amp) is greater than CS1 and smaller than CS2.
- The device is in Range 3 when the controlled variable (Amp) is greater than CS2.

The user sets the state of the capacitor switch for the three ranges via the DNAF Configuration Editor's three combo box controls, as shown in figure below. Each combo box has the following three options: Turn on, No Status Change, and Turn Off. The corresponding device states are shown in following figure (where "Amp" is the sum of the current in monitored phases in the regulated branch end and direction).

Locally Controlled Capacitors

Controller Type	Below Lower	Between	Above Upper
Voltage	Turn On	No Status Change	Turn Off
VAR	Turn Off	No Status Change	Turn On
Amp	Turn Off	No Status Change	Turn On
Power Factor	Turn On	No Status Change	Turn Off
Time	Turn Off	Turn On	Turn Off
Temperature	Turn Off	No Status Change	Turn On

Figure 36. Parameters for Locally Amp-Controlled Capacitors

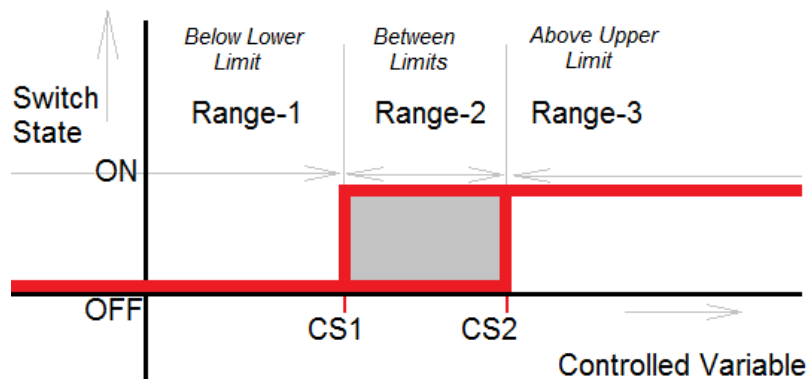


Figure 37. Locally Amp Controlled Capacitor States and Operating Ranges

5.7.4. Power Factor-Controlled Capacitor

The following parameters must be set for a power factor-controlled capacitor:

- The regulated branch is set in the field RegBranch in the XML file or in the field REGBRANCH_SHIMP in the CSV file. The Watt and VAR quantities in this branch are monitored. In [figure](#), **Line 1** is the regulated branch.
- The monitored phases are set in the field PhaseSense in the XML file or in the field PHASESENSE_SHIMP in the CSV file.
- The end of the regulated branch is set in the field RegBrEnd in the XML file and in the field REGBRANCHEND_SHIMP in the CSV file.

The field has the following values: 0=FROM side, 1=TO side, and 2=TERTIARY side (only if the branch is a three-winding transformer). In [figure](#), assume the FROM side is N1 and TO side is N2 for Line 1, and the end with node N2 is regulated, so the value is 1. If the end with node N1 is regulated, the value is 0.

- The regulated flow direction is set in the field RegBrDirect in the XML file and in the field REGBRANCHDIRECT_SHIMP in the CSV file.

The field has the following values: 1=OUT of the regulated branch and 2=INTO the regulated branch. In [figure](#), if the direction is OUT then the value is 1 and if the direction is INTO then the value is 2. This parameter is chosen regardless of the regulation end defined above.

- Control Level 1 is set in the OnLevel field in the XML file or in the ONLEVEL_SHIMP field in the CSV file.
- Control Level 2 is set in the OffLevel field in the XML file or in the OFFLEVEL_SHIMP field in the CSV file.

Based on CL1 and CL2, two new parameters (CS1 and CS2) are defined as follows: CS1 is the smaller (more lagging) of CL1 and CL2, and CS2 is the larger (more leading) of CL1 and CL2. These two parameters, in turn, define the three operating ranges of the capacitor as follows:

- The device is in Range 1 when the controlled variable (power factor) is less than CS1.
- The device is in Range 2 when the controlled variable (power factor) is greater than CS1 and smaller than CS2.
- The device is in Range 3 when the controlled variable (power factor) is greater than CS2.

The user sets the state of the capacitor switch for the three described ranges via the DNAF Configuration Editor's three combo box controls, as shown in [figure](#) below. Each combo box has the following three options: Turn on, No Status Change, and Turn Off. The corresponding device states are shown in [following figure](#).

Locally Controlled Capacitors

Controller Type	Below Lower	Between	Above Upper
Voltage	Turn On	No Status Change	Turn Off
VAR	Turn Off	No Status Change	Turn On
Amp	Turn Off	No Status Change	Turn On
Power Factor	Turn On	No Status Change	Turn Off
Time	Turn Off	Turn On	Turn Off
Temperature	Turn Off	No Status Change	Turn On

Figure 38. Parameters for Locally Power Factor-Controlled Capacitors

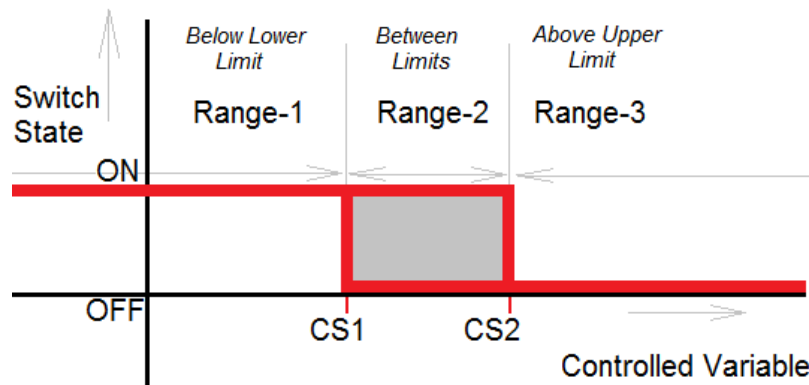


Figure 39. Locally Power Factor Controlled Capacitor States and Operating Ranges

In figure above, "Controlled Variable" is the power factor of the real and reactive power sum in the monitored phases at the regulated branch end. Assuming all three phases are monitored, the power factor is $\pm (PA + PB + PC) / \sqrt{(PA + PB + PC)^2 + (QA + QB + QC)^2}$, where the sign is determined by whether the power factor is leading (-) or lagging (+).

5.7.5. Time-Controlled Capacitor

The following parameters must be set for a time-controlled capacitor:

- Control Level 1 is set in the OnLevel field in the XML file or in the ONLEVEL_SHIMP field in the CSV file.
- Control Level 2 is set in the OffLevel field in the XML file or in the OFFLEVEL_SHIMP field in the CSV file.

Based on CL1 and CL2, two new parameters (CS1 and CS2) are defined as follows: CS1 is the smaller (earlier time) of CL1 and CL2, and CS2 is the larger (later time) of CL1 and CL2. These two parameters, in turn, define the three operating ranges of the capacitor as follows:

- The device is in Range 1 when the controlled variable (time) is less than CS1.
- The device is in Range 2 when the controlled variable (time) is greater than CS1 and smaller than CS2.

- The device is in Range 3 when the controlled variable (time) is greater than CS2.

The user sets the state of the capacitor switch for the three described ranges via the DNAF Configuration Editor's three combo box controls, as shown in figure below. Each combo box has the following three options: Turn on, No Status Change, and Turn Off. The corresponding device states are shown in following figure.

Locally Controlled Capacitors

Controller Type	Below Lower	Between	Above Upper
Voltage	Turn On	No Status Change	Turn Off
VAR	Turn Off	No Status Change	Turn On
Amp	Turn Off	No Status Change	Turn On
Power Factor	Turn On	No Status Change	Turn Off
Time	Turn Off	Turn On	Turn Off
Temperature	Turn Off	No Status Change	Turn On

Figure 40. Parameters for Locally Time-Controlled Capacitors

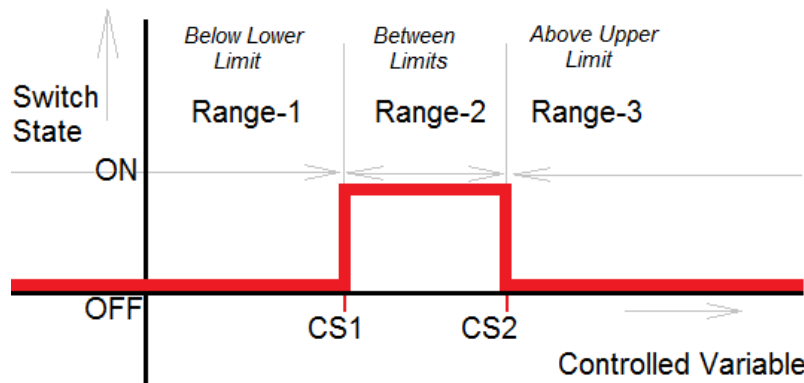


Figure 41. Locally Time Controlled Capacitor States in Operating Ranges

In figure above "Controlled Variable" is the current time. For devices that respect more than one time range, the above scheme can be easily achieved by creating a new schedule and associated schedule points (in the globals.csv file). The X value is the time, and the Y1 value can be 0, 1, or 2 indicating TURN_OFF, TURN_ON, or NO_STATUS_CHANGE, respectively. Using a schedule, an arbitrary number of ranges and actions for device/controller types can be encoded generically. This schedule is given priority over the DNAF Configuration Editor scheme, which is used as a default (that is, if this schedule is not entered, the DNAF Configuration Editor scheme is used).

When different time schedules are needed for a given day type and/or season, multiple time control schedules can optionally be added to the capacitor's associated shunt impedance model. The server automatically determines the appropriate schedule to use based on the current day type and season. The schedules defined in the shunt impedance model take priority over a single schedule, and the single-schedule takes priority over the single pair of on and off level values.

5.7.6. Temperature-Controlled Capacitor

The following parameters must be set for a temperature-controlled capacitor:

- Control Level 1 is set in the OnLevel field in the XML file or in the ONLEVEL_SHIMP field in the CSV file.
- Control Level 2 is set in the OffLevel field in the XML file or in the OFFLEVEL_SHIMP field in the CSV file.

Based on CL1 and CL2, two new parameters (CS1 and CS2) are defined as follows: CS1 is the smaller (lower temperature) of CL1 and CL2, and CS2 is the larger (higher temperature) of CL1 and CL2. These two parameters, in turn, define the three operating ranges of the capacitor as follows:

- The device is in Range 1 when the controlled variable (temperature) is less than CS1.
- The device is in Range 2 when the controlled variable (temperature) is greater than CS1 and smaller than CS2.
- The device is in Range 3 when the controlled variable (temperature) is greater than CS2.

The user sets the state of the capacitor switch for the three described ranges via the DNAF Configuration Editor's three combo box controls, as shown in figure below. Each combo box has the following three options: Turn on, No Status Change, and Turn Off. The corresponding device states are shown in following figure.

Locally Controlled Capacitors

Controller Type	Below Lower	Between	Above Upper
Voltage	Turn On	No Status Change	Turn Off
VAR	Turn Off	No Status Change	Turn On
Amp	Turn Off	No Status Change	Turn On
Power Factor	Turn On	No Status Change	Turn Off
Time	Turn Off	Turn On	Turn Off
Temperature	Turn Off	No Status Change	Turn On

Figure 42. Parameters for Locally Temperature Controlled Capacitors

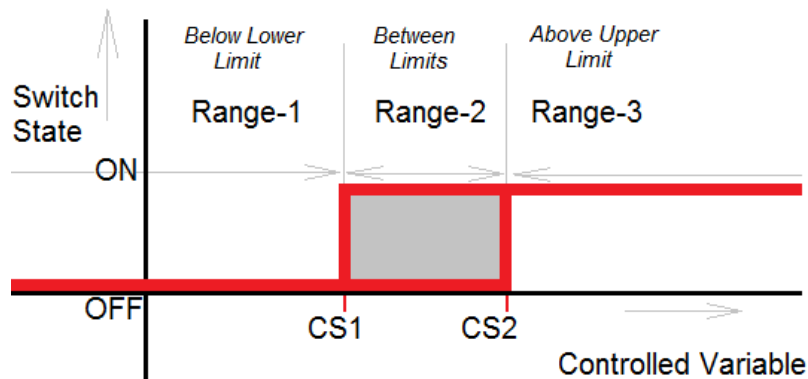


Figure 43. Locally Temperature Controlled Capacitor States and Operating Ranges

For devices requiring more than one temperature range, the scheme can be easily achieved by creating a new schedule and associated schedule points (in the `globals.csv` file). The X value is the temperature, and the Y1 value can be 0, 1, or 2 indicating `TURN_OFF`, `TURN_ON`, or `NO_STATUS_CHANGE`, respectively. Using the schedule, an arbitrary number of ranges and actions for device/controller types can be encoded in a generic way. This schedule is given priority over the DNAF Configuration Editor scheme, which is used as a default (that is, if this schedule is not entered, the DNAF Configuration Editor scheme is used).

When different temperature schedules are needed for a given day type and/or season, multiple temperature control schedules can optionally be added to the capacitor's associated shunt impedance model. The server automatically determines the appropriate schedule to use based on the current day type and season. The schedules defined in the shunt impedance model take priority over a single schedule, and the single-schedule takes priority over the single pair of on and off level values.

For manually entering temperature, use the Manual Entry Parameters pop-up display, which can be accessed via the RTDMS (or STDMS for study mode) drop-down menu. You can enter temperature for the entire system or at the station and division level.

5.7.7. Temperature and Voltage Override

In addition to the above control strategies, ADMS also supports an override capability, which prioritizes temperature and voltage control strategies over the other four (VAR, power factor, time, and Amp). This can be illustrated with an example of VAR control with voltage override (VAR-Voltage), involving the device settings. For these settings, the control is relinquished to the VAR-based strategy only when the DPF computed voltage falls between the lower and upper limit (that is, Range-2, which has the option No Status Change). The device is turned on in Range-1, and turned off in Range-3 (that is, it follows a volt-based strategy, regardless of the VAR-based strategy's settings). Currently, voltage and temperature overrides are supported, both of which can also be implemented via their respective schedules (temperature vs status and voltage vs status schedules). Information on the necessary parameters for applying an override can be found in the "Shunt Impedance" section in the *Modeling Reference Tables*.

5.7.8. Permissive Strategies

In permissive strategies, the primary control strategy is only allowed when the permissive control strategy permits it. An example is Temperature-Time, where the primary control strategy is time and the permissive control strategy is temperature. The capacitor switches based on temperature settings, however, it is only allowed to do so during the time the schedule is in the "No Status Change" range. Outside of this range, the device is OFF. A user can enter the required parameters for handling these strategies in the capacitor record in the station externals and internals file:

- The permissive control enumeration indicates whether the capacitor has temperature or time permissive control (`LOCALPERM_SHIMP` in CSV file format and `LocalPerm` in XML file format). The enumeration has the following values: 0 = none, 1 = time, and 2 = temperature.

- Permissive control level 1 (PERMCTRLLEVEL1_SHIMP in CSV file format and PermCtrlLevel1 in XML file format)
- Permissive control level 2 (PERMCTRLLEVEL2_SHIMP in CSV file format and PermCtrlLevel2 in XML file format)
- Permissive control schedule (optional) if the schedule is to be used (PERMSCHED_SHIMP in CSV file format and PermSchedule in XML file format). If the permissive control variable is in the "No Status Change" range, the primary control is allowed; otherwise, the capacitor is OFF. If available, the permissive control schedule takes priority over the parameters defined above (PERMCTRLLEVEL1_SHIMP and PERMCTRLLEVEL2_SHIMP).

5.7.9. Capacitor Bank

A locally controlled capacitor bank is a group of coordinated and sequentially operated capacitors that can be operated via any type of local control (Volt, Amp, PF, or VAR). The local control scheme for all capacitors inside the capacitor bank must have the same type.

For each capacitor inside the capacitor bank, two user-configured properties are available: OnSequence and OffSequence. The DPF engine uses the OnSequence property to turn on the capacitors sequentially and the OffSequence property to turn off the capacitors sequentially. After each DPF calculation, only one capacitor in an individual capacitor bank can be turned on or off.

Depending on how the capacitors are modeled within the capacitor bank, there may be a switching device, such as a circuit breaker, connecting each individual capacitor. When using such optional switching devices, they should be added for all capacitors in the bank, for consistency.

Figure below shows two capacitor bank configurations: with and without individual capacitor switching devices.

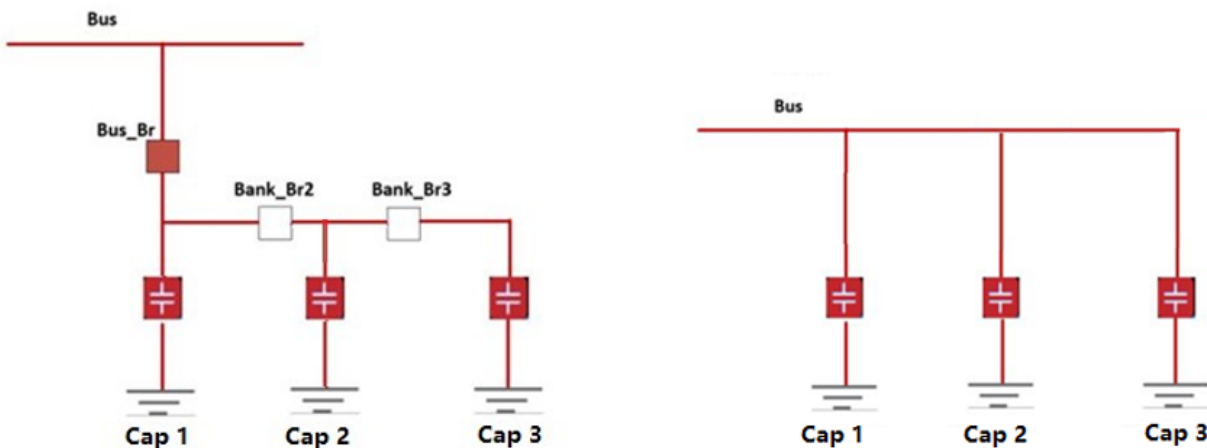


Figure 44. Capacitor Bank Configurations: With and Without Associated Switching Devices

5.7.10. Control Mode Switching

For a capacitor that has both remote and local control capability, it can be considered solved in local control mode when communications are lost. In local mode, the BLA/DPF determines the open/closed status based on the local controller on/off settings. The control mode switching of a capacitor when communications are lost is enabled/disabled by a user configured option set in the DNAF Configuration Editor's "Solve with Capacitor in Local Mode if Communications are Lost" option, located in the Solution Formulation section of the Distribution Power Flow (DPF) tab.

5.8. Suspect Controls and Hunting Prevention Algorithm

For locally controlled (non-telemetered) devices such as capacitors, tap changing transformers (regulators and LTC), generators, and energy resources, DPF adjusts the status according to a preset local control strategy. In some cases, these locally controlled devices may have incorrect on/off level settings, or the settings may conflict with other locally controlled devices, resulting in frequent capacitor status changes (open/closed) or tap changer tap step changes and a diverged DPF solution. The solution is truly not diverged in a mathematical sense, but due to the frequent capacitor status changes, the power flow runs until it reaches the maximum number of iterations.

To prevent these excessive switching cases from occurring, DPF employs the methodology described below, which is used for controlled capacitors, tap changers, generators, and energy resources. There are, however, minor differences in the processing of capacitors and tap changers. These variations mostly concern the priority in which the devices are processed and subsequently marked as suspect or locally-controlled during the DPF iterations. When DPF detects excessive switching, each locally controlled capacitor in the network selection is added to a list and each tap changer device is added to another list. The tap-changers list is processed before the locally controlled capacitor list. In the case of tap changers, the tap changer that is located downstream on a feeder is given priority over a tap changer that is located upstream on a feeder and is thus added to the list first; there is no particular priority in processing the capacitors.

The methodology for determining if a device is considered locally-controlled or suspect is as follows:

- DPF resets all the devices' positions to their previous position before DPF was executed, and then blocks all these devices except for the first device in the list.
- If DPF reaches the maximum iteration count (specified in alg.dat; default is 30) because a device is hunting, this device is marked as a suspect control and blocked at its previous position before DPF was executed. Otherwise, the first device is left as a locally controllable device.
- The next step is to change the second device in the list to local control along with the first device.
- If DPF reaches the maximum iteration count because any locally controlled device is hunting, the second device is identified as a suspect control and blocked at its previous position.
-

This step is repeated until all capacitors and tap changing transformers in the list are processed.

The devices marked as suspect can be queried from the Station DPF Results Summary display and are recorded in the station DPF result file.

5.9. Modes of Operation

DPF and BLA support two modes of operation for the distribution analysis functions, and the same Distribution Power Flow algorithm is applied for each mode.

Real-Time Mode: Executes periodically and upon pre-specified changes in status or analog points. It provides the operators and other DNAF applications with estimates of kW, kVAR, kV, Amps, and power losses in the actual distribution and adjacent transmission/sub-transmission systems in real time.

The real-time DPF function executes under the following conditions:

- Periodically based on user-defined parameters: maximum and minimum periodic time intervals, specified by the DNAF_DPF_MAX_PERIOD and DNAF_DPF_MIN_PERIOD parameters in the DNAF server configuration file, and the maximum load change percentage, specified by the Maximum Loading Difference for a Periodic DPF Execution (%) parameter on the BLA page in DCE.

At each periodic interval, if the station load has changed by more than the maximum load change percentage from the previous solution, another solution is executed for the station. If a solution has not been performed within the maximum periodic time interval, a forced solution is executed for the station.

To avoid excessive solutions, periodic triggering can be skipped for stations where DPF was recently solved as part of another station's network selection. The time period in seconds, within which periodic application triggering for a station is skipped, is configured by the DNAF_PERIODIC_DEADBAND parameter in the DNAF server configuration file. The time period counts after an application is solved for a station or after an application is triggered for an adjacent station in the network selection. This feature applies to manual or event-triggered solutions.

- Upon certain events that affect the distribution system. These triggering events include the following:
 - Alteration in feeder main or lateral network topology (change in switching device status or the insertion/removal of a temporary modification). These event triggers are configured by the DNAF_EVENT_TRIGGER_TOPOLOGY_BACKBONE and DNAF_EVENT_TRIGGER_TOPOLOGY_LATERAL parameters in the DNAF server configuration file.
 - SCADA sudden measurement changes in a station internal object (reference bus interface, station transformer, breaker, or feeder head). The sudden measurement change threshold values (Amps and Volts) are set by the Current Measurement Delta Value (A) and Voltage

Measurement Delta Value (%) parameters on the DPF page in the DNAF Configuration Editor. This event trigger is configured by the DNAF_EVENT_TRIGGER_MEAS_CHANGE parameter in the DNAF server configuration file.

- Tap estimation or tap position change from SCADA. This event trigger is configured by the DPF_EVENT_TRIGGER_TAP_CHANGE parameter in the DNAF server configuration file.
- Capacitor status change. This event trigger is configured by the DPF_EVENT_TRIGGER_CAP_STEP_CHANGE parameter in the DNAF server configuration file.
- Generator and energy resource status change. This event trigger is configured by the DPF_EVENT_TRIGGER_ENERGYSOURCE_STATUS_CHANGE parameter in the DNAF server configuration file.
- When any triggering event occurs, a short delay, specified by the DNAF_EVENT_DELAY parameter in the DNAF server configuration file, is imposed before the DPF function is initiated. This delay is to allow the power system to reach steady state so that a snapshot of the real-time data can be provided as input to the DPF application. DPF execution initiated by a triggering event includes all components energized from the same primary substation bus.
- Upon operator manual request. In this case, the application is executed immediately, without a delay. The operator can select the stations to be analyzed.

Study Mode: The study DPF is initiated on demand. It provides operators and engineers with estimates of kW, kVAR, kV, Amps, and power losses in the distribution and adjacent transmission/subtransmission systems under hypothetical conditions represented in the study case. A "study case" is typically a snapshot of the real-time system, with user-defined changes representing "what-if" conditions.

Study mode has the following capabilities:

- The operator can run the study DPF for one or more selected substations and their connected feeders.
- The operator can designate an individual load node (distribution transformer) as the subject for the study mode execution, in which case DPF executes for the station(s) in which the device is electrically connected.
- The operator can start from the last DPF real-time run or from a saved case.
- The operator can initiate "what if" studies to determine the loading, voltage profile, and losses on a specific feeder given the following changes:
 - Sections of an adjacent feeder were switched to it or from it
 - A capacitor was switched on/off
 - A tie-switch was closed leading to looped circuits
 - Loading levels were changed
 - The voltage of a reference bus was changed
 - Other connectivity changes were made
 - The capacitor or regulator/LTC controller settings were changed

- The operator can study "what if" scenarios in different control modes for capacitors and tap changing transformers. The studying scenarios include:
 - Respect remote/local measurements for all devices
 - Solve with all devices in local control mode
 - Solve with all devices in remote control mode
- Network loop conditions can be created by energizing a network section from multiple points belonging to a single source (transformers at one station).
- Parallel loop conditions can be created by energizing a network section from multiple points belonging to multiple sources (transformers at different stations).
- DPF results can be calculated and stored for later review and analysis by creating a savecase or retrieving the LST file located in the study server's dynamics folder.

5.10. Voltage Representation

To facilitate quicker understanding of system conditions, DNAF allows the user to choose the units in which the different primary network nodal voltage values can be displayed in the device pop-up displays and grid-tabular displays. This is done by selecting the desired voltage units in the DPF tab of the DNAF Configuration Editor, as shown in figure below. In addition to the typical per unit and S.I unit system, the user can also specify other base voltage systems, such as the 120V base system which is common for many U.S. utilities. To specify other base systems, select the "Use Voltage Base (V)" option and enter the voltage base value in the adjacent text box.

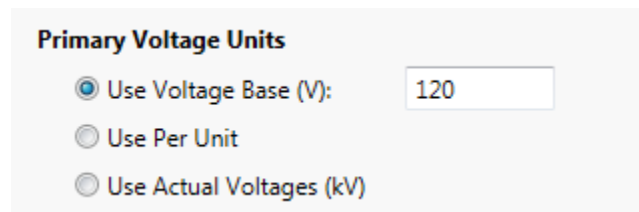


Figure 45. Selection of Primary Voltage Units using the DNAF Configuration Editor

To eliminate confusion, the voltages of nodes in the secondary (low voltage or LV) network are always displayed in actual Volts regardless of this input selection.

5.11. Tap Estimation Methodology

5.11.1. Core Tap Estimation Algorithm

When the tap-position measurements for a regulating transformer (LTC or regulator) are not available, the DNAF server uses the following formula to estimate the tap position.

Suppose:

$V_{1p.u.}$ = measured primary node voltage in per unit.

$V_{2p.u.}$ = measured secondary node voltage in per unit.

kV_{1rated} = rated primary voltage (kV) for the transformer.

kV_{1nom} = nominal voltage (kV) at the transformer primary node.

kV_{2rated} = rated secondary voltage (kV) for the transformer.

kV_{2nom} = nominal voltage (kV) at the transformer secondary node.

First, the voltage ratio is determined as follows:

If the tap position is to be estimated for a primary side tap changer, then the voltage ratio is calculated as follows:

$$\text{Voltage ratio, } x = \left(\frac{V_{1p.u.}}{V_{2p.u.}} \right) \cdot \left(\frac{kV_{1nom}}{kV_{1rated}} \right) \cdot \left(\frac{kV_{2rated}}{kV_{2nom}} \right) \cdot \text{FixedTapRatio} \quad (\text{Equation 5.5})$$

Where:

- If a tap changer exists on the secondary side:

$$\text{FixedTapRatio} = (1 + \text{PrimaryTapStepIncrement} \cdot \text{PrimaryTapPositionFromNominal} \cdot 0.01)$$

- Otherwise:

$$\text{FixedTapRatio} = 1$$

If the tap position is to be estimated for a secondary side tap changer, then the voltage ratio is calculated as follows:

$$\text{Voltage ratio, } x = \left(\frac{V_{2p.u.}}{V_{1p.u.}} \right) \cdot \left(\frac{kV_{1rated}}{kV_{1nom}} \right) \cdot \left(\frac{kV_{2nom}}{kV_{2rated}} \right) \cdot \text{FixedTapRatio} \quad (\text{Equation 5.6})$$

Where:

- If a tap changer exists on the primary side:

$$\text{FixedTapRatio} = (1 + \text{SecondaryTapStepIncrement} \cdot \text{SecondaryTapPositionFromNominal} \cdot 0.01)$$

- Otherwise:

$$\text{FixedTapRatio} = 1$$

Since the ratio of these voltages, x , is dependent on the tap position, the tap position can be estimated if x is known. Hence, the estimated tap position, ESTIMTAPPOSN, rounded off to the nearest integer, is given by the following equation:

$$ESTIMTAPPOSN = Round \left(NOMINALTAPMOD + \frac{(x - 1) \cdot 100}{TAPINCRTAPMOD} \right) \quad (\text{Equation 5.7})$$

Where:

NOMINALTAPMOD is the nominal tap setting

TAPINCRTAPMOD is the tap increment in percentage Volts.

In addition, the tap estimation procedure ensures that a physically available tap position is estimated:

$$ESTIMTAPPOSN = \begin{cases} MAXTAP_TAPMOD & \text{if } ESTIMTAPPOSN = MAXTAP_TAPMOD + 1 \\ MINTAP_TAPMOD & \text{if } ESTIMTAPPOSN = MINTAP_TAPMOD - 1 \end{cases}$$

Where:

MAXTAP_TAPMOD is the maximum tap step

MINTAP_TAPMOD is the minimum tap step

If the primary or secondary current measurement is available, the voltage drop across the transformer can also be taken into account for a more accurate tap estimation. Assume the current (in per unit) is represented by I_{pu} , and the impedance of the transformer is represented by Z , then the magnitude of voltage drop can be calculated by the following equation:

$$V_{drop} = |I_{p.u.} \cdot Z| \quad (\text{Equation 5.8})$$

Where:

Vdrop is voltage drop

In this case, the voltage drop is subtracted from the primary voltage measurement before calculating the voltage ratio:

$$V_{1p.u.effective} = V_{1p.u} - |I_{p.u.} \cdot Z| \quad (\text{Equation 5.9})$$

In this case, the difference of the magnitude of the primary voltage and the voltage drop directly is used as an approximation because the phase angle between the voltage and current is commonly not available.

5.11.2. Tap Estimation without Source Voltage Measurement

For tap changing transformers and regulators, if a tap position measurement is not available and a source side voltage measurement is not available, then a source voltage is assumed. Using this

assumed source voltage, a tap position is determined using the procedure described in the previous section. The estimated tap position is compared to the parameter "Furthest Tap Position for Re-estimation" from the DPF page of the DNAF Configuration Editor. If the estimated tap is outside of the range specified by this parameter, then the assumed source voltage is adjusted, and the tap position is re-estimated. This procedure is illustrated in the flowchart in figure below.

The relevant parameters for tap estimation without source voltage measurement are described below:

- Source voltage magnitude: This parameter specifies the initial primary voltage to be used for determining the tap position for a given LTC. This parameter is set for the corresponding reference bus interface object via the Substation Editor. If this parameter is not set via the Substation Editor, then the default value from the Converter is used. Furthermore, the interface voltage magnitude can be overridden by the operator via a manual entry.
- Source voltage magnitude adjustment range: This parameter specifies the range within which the assumed source voltage can be adjusted to find an estimated tap position that meets the criteria set by the option "Furthest Tap Position for Re-estimation". The source voltage magnitude adjustment range can be specified for each interface object via the Substation Editor. If, for an interface object, the source voltage magnitude adjustment range is not specified via the Substation Editor, then the default value from the "Default Source Voltage Magnitude Range" field on the DPF page of the DNAF Configuration Editor is used.
- Furthest Tap Position for Re-estimation: This parameter specifies the furthest tap position from nominal for re-executing tap position estimation with an adjusted source voltage. This parameter is specified via the field "Furthest Tap Position for Re-estimation" on the DPF page of the DNAF Configuration Editor. Setting this parameter to the highest tap position from nominal, disables the source voltage magnitude re-adjustment process.

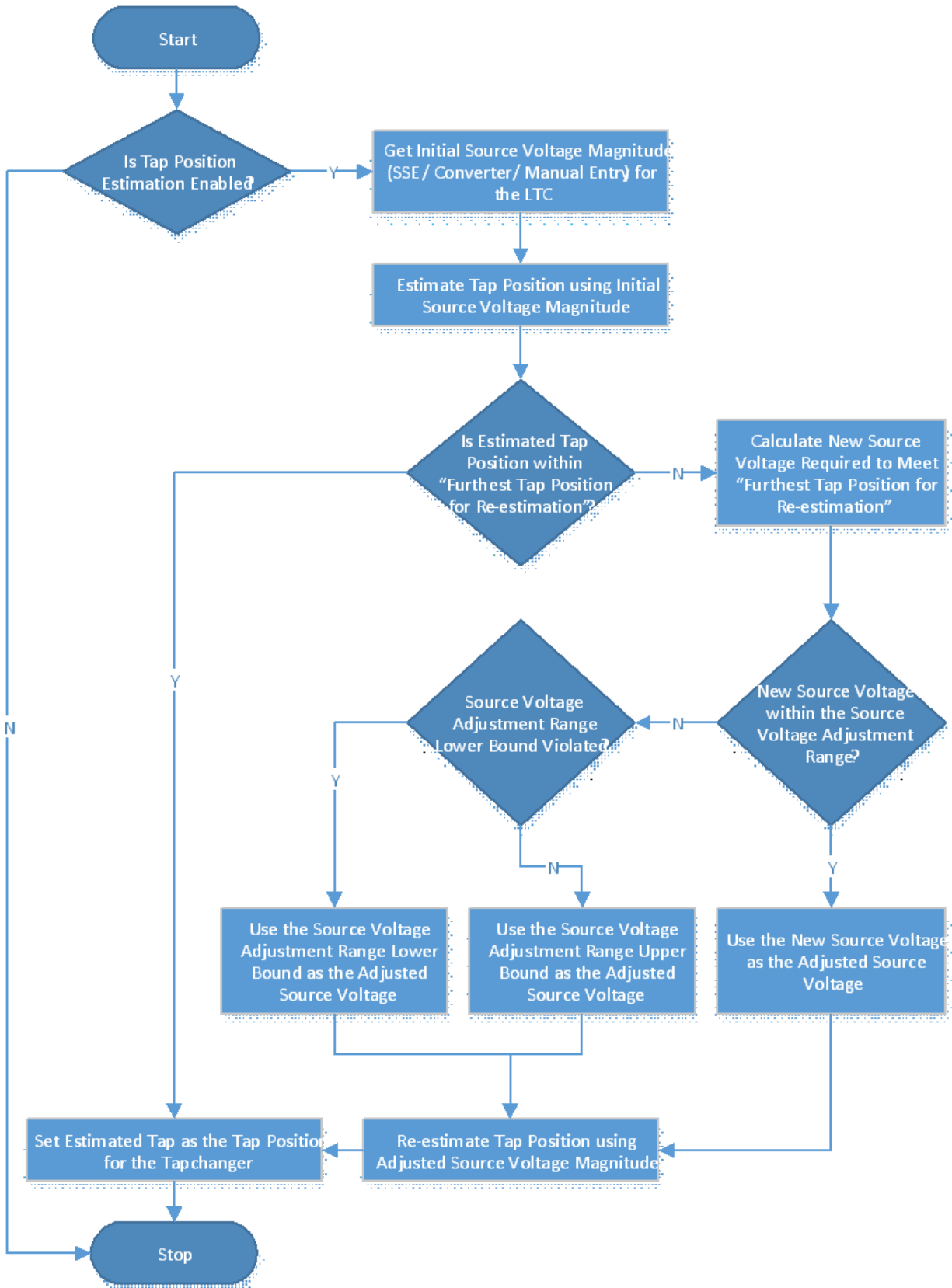


Figure 46. Modified Tap Estimation Algorithm

5.11.3. Tap Estimation Triggers

To estimate tap position, the "Perform Tap Estimation" option on the DPF page of the DNAF Configuration Editor must be enabled. Tap position estimation is triggered under the following conditions:

- On server startup: Immediately after the MAIN server or the DNAF server starts and has received the necessary voltage measurements for estimating tap position for LTCs/regulators, tap position is estimated for LTCs/regulators that do not have an available tap position analog measurement, but do have valid load-side voltage measurement(s) and a valid source-side voltage measurement or interface voltage magnitude.
- On SCADA updates: For any LTC/regulator without a tap position analog measurement, if updated voltage measurements are received from SCADA, then tap position is estimated for that LTC/regulator taking into account the new voltage measurements. Almost instantaneously, an updated tap position can be viewed from the Analyst Output pop-up display for the device, or from the LVM Device Eligibility grid tabular display.
- On DNAF Configuration Editor updates: If the "Default Source Voltage Magnitude Range" and "Furthest Tap Position for Re-estimation" parameters (on the DPF page of the DNAF Configuration Editor are changed) and the new configuration is saved online, then tap position estimation is triggered again to pick up the new values of these parameters.

5.12. Back Calculation of Interface Voltages

Within the distribution substation, the Reference Bus Interface (voltage source) is commonly connected to the bus connected at the source side of the station power transformer as shown in figure below. This voltage (both magnitude and phase angle) is the starting point for the distribution power flow and impacts all the calculated voltages both inside the substation and along the feeders.

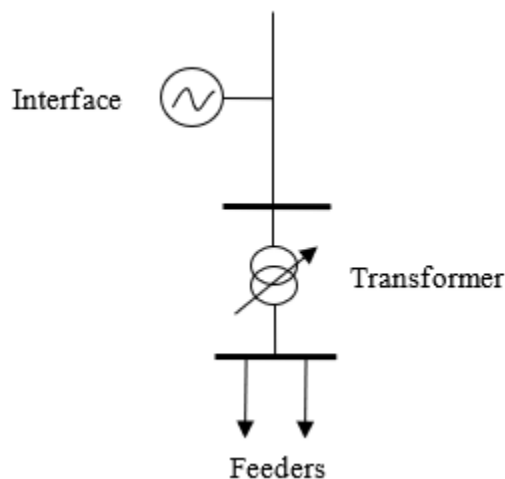


Figure 47. Single Interface Configuration

For a given reference bus interface group, which is a set of interface objects connected by zero impedance, the voltage magnitude and phase are determined via the following hierarchy: manually entered values, measured values, back calculation values, and default values. Additionally, the magnitude and phase values can come from two separate interface objects within the same group. Back calculation only works on real interfaces (study mode only interfaces are never used).

In some cases, there is no direct measurement of the interface voltage. Since the Energy Management System (EMS) State Estimator is generally the source of the interface voltage magnitude and angle values, there is only one value for all three phases (EMS model solves a positive sequence network). If there are enough measurements on the load side of the station power transformer, the interface voltage magnitude (three-phase or per phase depending on the available measurements) can be back-calculated from the load side measurements and downstream power transformer impedance and loss parameters. If these measurements are not available, the interface voltage cannot be calculated and the default voltage magnitude and phase values are used. Interface voltage magnitude and phase default values are set for each interface bus object in the substation internals file.

5.12.1. Measurement Sets and Calculation Methods

Calculating a three-phase unbalanced interface voltage correctly requires several measurements on the load side of the power transformer. Either three-phase or single-phase measurements are acceptable. However, if only three-phase values are available in the measurement set, the calculation assumes that the corresponding measurements (voltages, currents, or power factors) on three different phases are balanced and applies the same value for all three phases. This yields a balanced reference bus interface voltage magnitude if all measurements available are three-phase values. If an actual reference voltage measurement exists, it must be removed from service before the back-calculation function works, as it has priority over the value determined by the back-calculation function. If "Ignore Reference Bus Interface Voltage Measurements" on the DPF page of the DNAF Configuration Editor is selected, the actual reference voltage measurements are ignored and the back-calculation function determines the reference bus voltage.

The required measurements to perform back calculation include the following:

- Phase voltage at the load-side bus

Either a three-phase or single-phase measurements can be considered. If both are available, single-phase measurements receive higher priority.

- Measured transformer tap position / fixed or measured regulator phase tap position
- Phase current at load side of transformer

Either three-phase or single-phase measurements can be considered. If both are available, single-phase measurements receive higher priority).

- Real and reactive power at load side of transformer

These values are used to calculate the phase angle between the voltage and the current. The load side real and reactive power analog flow sign must be consistent with DMS flow

conversion. When the flow is flowing out of the transformer or regulator, the load side flow values are negative. When there is reverse flow, the load side flow values are positive.

5.12.2. Different Station Configurations

If the three-phase power transformer is connected directly to a regulator and there are measurements on the load side of a regulator, the voltage and current values can be transformed to the high side of the regulator first to satisfy the back-calculation requirement. The regulator may not be gang operated, so the tap position on each phase could be different.

The back-calculation procedure is applicable to the following scenarios:

- **Transformer with tap changer**

The measurements are all on the load side of the three-phase transformer after the tap changer. This configuration was introduced in [Measurement Sets and Calculation Methods](#).

- **Transformer (fixed tap) with station regulator**

Figure below shows this configuration. The measurements are all on the load side of the regulator. The voltages and currents on the load side of the three-phase power transformer can be calculated from measurements on the load side of the station regulator with the current regulation tap ratio. After the calculation, this scenario is identical to scenario 1.

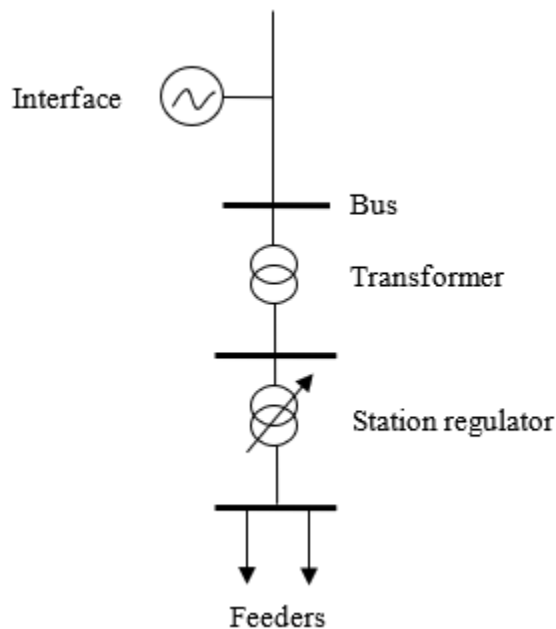


Figure 48. Transformer with Station Regulator Configuration

- **Transformer (fixed tap) with individual feeder regulators**

The measurements are all on the load side of the feeder regulators. Figure below shows this configuration. Again, the voltages and currents on the load side of the three-phase power

transformer can be calculated from measurements on the load side of the feeder regulators with the current regulation tap ratio. After the calculation, the scenario is identical to scenario 1 with a fixed turns ratio.

However, if multiple sets of measurements exist on two or more feeders, the voltages on the feeder head, which is also the voltage on the low side of the power transformer, can be calculated from the set of measurements on either feeder. Theoretically it does not matter which feeder is selected. However, due to the measurement accuracy, it is possible that the voltage values calculated from different feeders are not exactly the same, in which case the calculated voltages from each feeder are averaged.

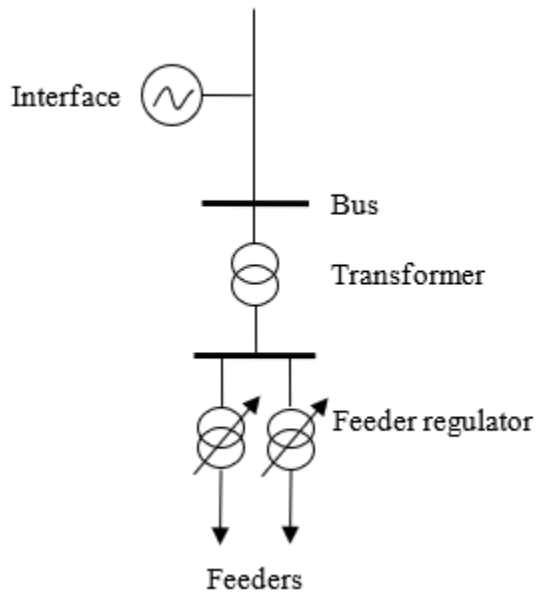


Figure 49. Transformer with Individual Feeder Regulators Configuration

- **Transformer (fixed tap)**

The measurements are all on the load side of the three-phase power transformer. Figure below shows this configuration. This scenario can be treated as the same as scenario 1 with a fixed turns ratio number.

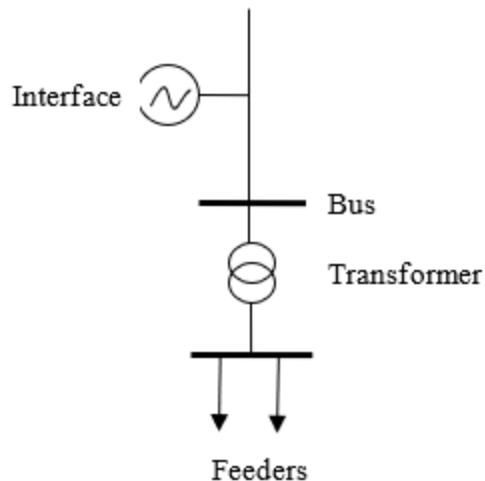


Figure 50. Transformer with Fixed Tap Configuration

- **Transformer (fixed tap) with different feeder regulators**

In this scenario, some of the feeders have no regulators, and the other feeders have individual feeder regulators. The measurements are on the load side of both the transformer and the feeder regulators. Figure below shows this configuration. The measurements on the load side of the transformer can be used directly in the back calculation. In this case, it is the same as scenario 1.

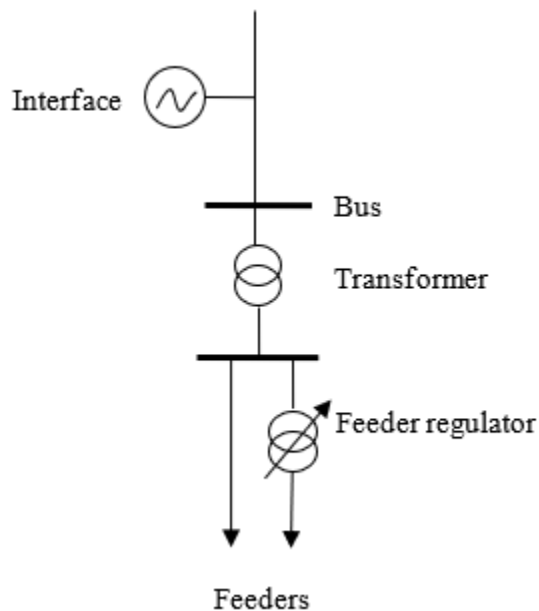


Figure 51. Transformer with Different Feeder Regulators Configuration

- **Regulators connected to Reference Interface Bus**

There is no power transformer connected to the interface bus, but a three-phase regulator or three single-phase regulators are connected to the interface bus directly. The voltage

measurements are all on the load side of the regulators. The interface bus voltage could be calculated easily from regulator voltage measurements.

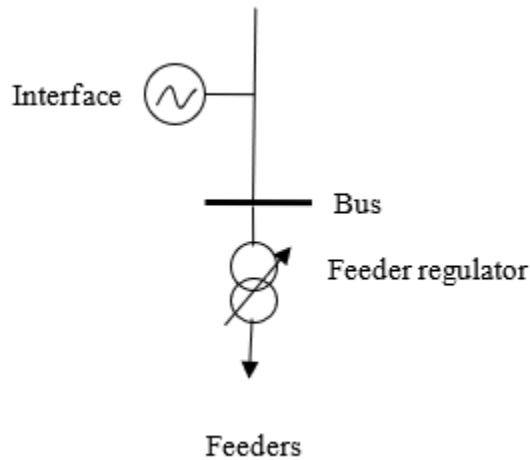


Figure 52. Regulators Connected to Reference Interface Bus

- **Station with Multiple Interfaces**

If the substation has multiple power transformers, it is possible to have multiple interfaces at the source side of each power transformer as shown in Figure below. These interfaces have the same voltage if the high side buses are tied together. However, in some cases when the bus breaker is open, the buses are not connected and each interface may have different voltage values.

Each interface voltage is calculated separately if all the required measurements exist on the corresponding power transformer and the high side buses are not connected. A topology analysis is performed to determine if the interfaces are connected. If the interfaces are not connected, each interface uses its own back-calculated value. If they are connected, theoretically the value should be the same. However, due to measurement and numerical calculation error, the voltages could be different. In this case, the average of the calculated source voltages is used in the network applications.

The power transformer also has a back-calculation priority option to indicate if this transformer should be used in the back-calculation function. All power transformers that have been chosen for back calculation are solved separately and the average value is used for the source voltage.

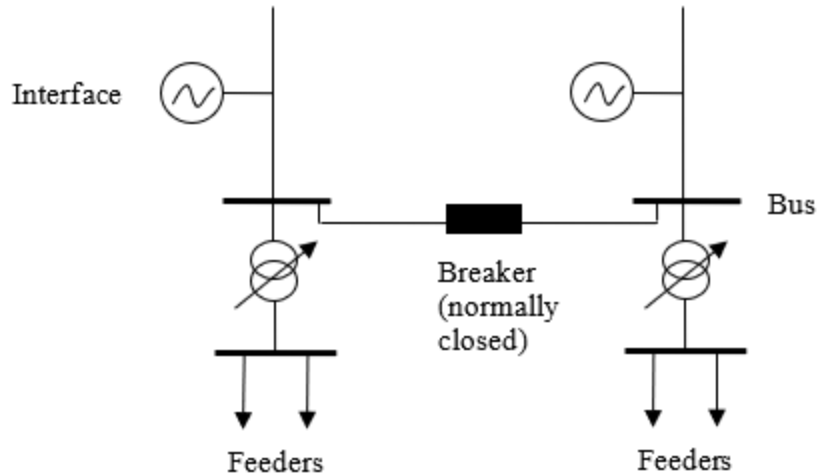


Figure 53. Multiple Interface Configurations

5.13. Limit Monitor

The Limit Monitor (LIM) function identifies all limit violations after an execution of the DPF. The LIM function performs an algebraic comparison between sets of real values. All load and bus voltages and branch flows calculated by DPF are checked against their limits. If values are outside of an acceptable range, then a limit violation has occurred. Limit violations are displayed to the operator. In real time, flow and voltage violations messages are optionally sent to the centralized alarm facility based on the results from LIM. Optionally, reserved capacity can be taken into account to de-rate branch flow limits. This is described in a section below.

The following limits are checked by the LIM function:

- Voltage limits:
- Normal and emergency voltage deviation limit at a distribution bus.
- Normal and emergency voltage deviation limit at the equivalent customer terminal considering the secondary voltage drop schedule.
- Voltage imbalance limit of a three-phase node at the three-phase load terminal, which is based on the Voltage Imbalance Calculation Option on the Limit Monitor page in the DNAF Configuration Editor.

If the Maximum Voltage Difference from the Average Voltage option is selected, the voltage imbalance factor is calculated as shown below:

$$V_{ave} = \frac{V_a + V_b + V_c}{3} \quad (\text{Equation 5.10})$$

$$V_{imb} = \max_{i \in (a,b,c)} |V_i - V_{ave}| \cdot 100 \quad (\text{Equation 5.11})$$

Where:

V_a , V_b , and V_c are the phase A, phase B, and phase C voltage magnitude value in per unit, respectively.

V_{ave} is the average of the three single-phase voltage magnitude value in per unit.

V_{imb} is the largest voltage magnitude difference between the phase voltage magnitude and the average of the three single-phase voltage magnitudes, in percent.

If the Negative Sequence Voltage Divided by the Positive Sequence Voltage option is selected, the voltage imbalance factor is calculated as shown below:

$$V_{imb} = \frac{V_{neg}}{V_{pos}} \cdot 100 \quad (\text{Equation 5.12})$$

Where:

V_{neg} is the negative sequence voltage magnitude in per unit.

V_{pos} is the positive sequence voltage magnitude in per unit.

V_{imb} is the ratio of the negative sequence voltage magnitude to the positive sequence voltage magnitude, in percent.

- Flow limits:
 - Flow limits on lines, switching devices, transformers, and series reactors (in Amps, kVA, or MVA). For transformers, flow limits are per-phase values specified in Amps while limits are total values specified in kVA or MVA.
 - Flow imbalance limit on three-phase switching devices, which is the largest difference between the single-phase flows and the average flow of all three phases.

Multiple ampacity limits are supported.

There are two types of limits: static limits and dynamic limits. Static limits are assigned based on season (time range defined in the season record in the global.csv file. If no season is specified then it is assumed the limit is active for the entire year), while dynamic limits are calculated using linear interpolation applied to the temperature ampacity schedule, which is also defined in the global.csv file.

In the real-time and study mode network applications, the LIM application looks for the current temperature value (a SCADA measurement or a manual entry), and then looks up the appropriate ampacity value in the temperature schedule. If no temperature value is found, the static limit value is evaluated.

5.13.1. Multiple Circuits in a Duct Bank

When multiple circuits are placed in a duct bank, the heating effects (multiple conductors in close proximity) may dramatically reduce the cable ampacity. To model this heating effect and its impact on cable ampacity, a conductor model has an ampacity reduction schedule (optional) that specifies

the ampacity reduction (in percent) as a function of the number of circuits in the duct bank figure below shows a sample ampacity reduction schedule.

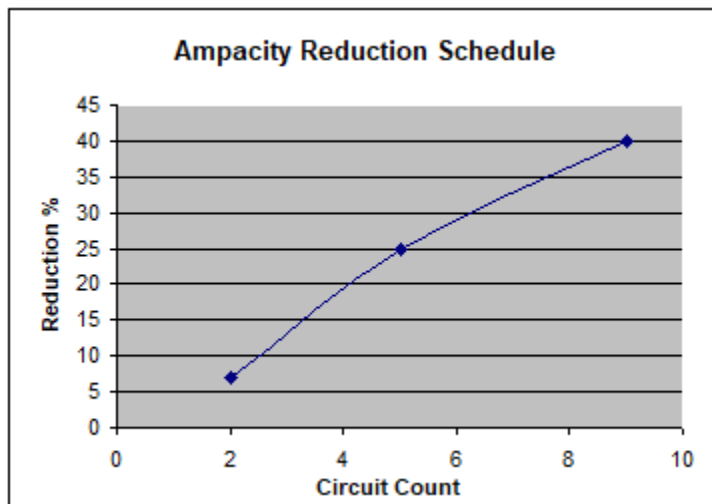


Figure 54. Sample Ampacity Reduction Schedule

5.13.2. Regulator

For regulating transformers, the single static limit and "dynamic" ampacity limits at specific steps are checked.

The following equation describes how the dynamic limits are calculated:

$$I_{dyn} = I_{step,3} + (I_{step,4} - I_{step,3}) \cdot (T - T_{step,3}) / (T_{step,4} - T_{step,3}) \quad (\text{Equation 5.13})$$

Where:

I_{dyn} is the calculated dynamic amp limit at tap T

T is current tap position

$T_{step, n}$ is the effective tap position)

$V_{step, n}$ is the voltage step

$I_{step, n}$ is the Amp limit at given voltage step (the voltage step can be translated to a tap position)

It is common for regulators to have multiple dynamic voltage "steps" for the ampacity limit. Table below illustrates these limits. This table is defined for each regulator transformer model in the global.csv file.

To calculate the effective tap position, the voltage step (in percent) is divided by the tap increment. For a 32-step regulator (with $\pm 10\%$ raise/lower), the tap increment is 0.625%.

Table 8. Dynamic Voltage Steps for the Ampacity Limit

Step	V _{step,n}	Effective Tap Position (T~step, n~)	I~step, n~
1	±5.00	±8	160.0
2	±6.25	±10	135.0
3	±7.50	±12	120.0
4	±8.75	±14	110.0
5	±10.00	±16	100.0

The effective limit at tap position 7 is 160.0amps (tap position is less than 8). The effective limit at tap position 13 is calculated below (linear interpolation between correct boundary points):

$$I_{dyn} = I_{step,3} + (I_{step,4} - I_{step,3}) \cdot (T - T_{step,3}) / (T_{step,4} - T_{step,3})$$

$$I_{dyn} = 120.0 + (110.0 - 120.0) (13 - 12) / (14 - 12)$$

$$I_{dyn} = 120.0 - 10.0 / 2.0 = 115.0$$

5.13.3. Limit De-Rating due to Reserved Capacity

Reserved Capacity is a contractual agreement between a distribution customer and the power supply provider. Through this contract, the customer is guaranteed a stipulated flow capacity or bandwidth on the supply paths feeding the customer loads.

The reserved capacity can be specified in kVA or in kW. Internally in the power flow engine, the specified reserved capacity is converted to kVA, if necessary, using the power factor derived from the nominal loads modeled in the reserved capacity object.

Limit monitor can be configured to take into account reserved capacity into limit calculations. This is configured by the Adjust Limits Based on Reserved Capacity option in the DNAF Configuration Editor as shown in figure below. If this option is enabled, Limit Monitor calculates the impact of enabled reserved capacity objects on upstream branch devices, such as lines, switches, inline transformers (regulators, power transformers, step up/down transformers, and auto-transformers), breakers, reclosers, regulators, and power transformers. The reserved capacity objects can be individually enabled or disabled by the operator from the device toolbars. If the Only Consider Firm Commitments is selected, only those reserved capacity objects with the Firm Commitment flag set to True are considered.

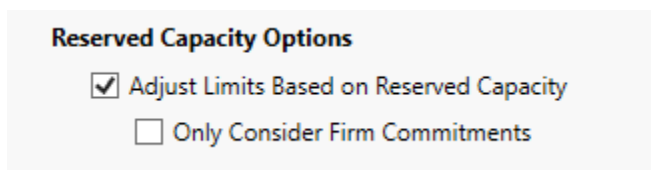


Figure 55. Reserved Capacity Options

The impact of a given reserved capacity object on any branch device is determined after converged

power flow results are obtained. Otherwise, reserved capacity calculations are not meaningful. The following steps are executed for determining the impact of reserved capacity objects on a given branch device.

1. The incremental reserved capacity on the branch device is calculated.
 - Incremental reserved capacity is equal to the specified reserved capacity for the feed upstream of which the branch device is located minus the load consumed at the supply point. For example, if the specified reserved capacity is 500 kVA, while the customer is already drawing 300 kVA through a feed point, then only 200 kVA needs to be reserved on the upstream branch devices.
 - Additionally, let us say the alternate feed specified in the reserved capacity object stipulates a reservation of 400 kVA and currently, the customer is being supplied through the primary feed only. Then, for branch devices upstream of the secondary feed point, the full 400 kVA is reserved.
 - If a branch device is on a path common to more than one feed for the given reserved capacity object, then the attribute "Add Reserves on Common Path" associated with the reserved capacity object is used to decide the effective reserved capacity. If this attribute is set to "true", the incremental reserved capacities from the impacting feeds are added. In the ongoing example, $200 + 400 = 600$ kVA is reserved. If this attribute is set to "false", only the maximum reserved capacity is reserved on that branch device. In the example above, 400 kVA is reserved.
 - If there are parallel branches upstream of any feed, then the option "Apply Full Reserve on Parallel Path" is used to determine the effective reserved capacity on those paralleled branches. If this option is set to true, the incremental reserved capacity calculated in the preceding steps is applied to all branches in the parallel paths. If this option is set to false, only half the incremental reserved capacity calculated above is applied to the branches in the parallel paths.
 - If a branch device is impacted by several reserved capacity objects, then the kVA reserved due to each reserved capacity object is calculated as described in the preceding steps. Then, the reserved kVA values are added to obtain the final effective reserved capacity on the branch device.
2. The final effective reserved capacity on any branch device, as calculated above, is converted into appropriate units for derating purposes. The modeled limit units for the branch device are used. If the branch device has limits modeled in kVA or MVA, no conversion is necessary. If the branch device has limits modeled in per phase Amps, the per-phase final effective reserved capacity in Amps is obtained using the nominal line-ground voltage level of the branch device. This value is called the "limit de-rating value".
3. The "limit de-rating value" for any branch device is used to de-rate the modeled limit for that branch device.

Device loading and overload levels are calculated based on the de-rated limit. The de-rating is applied after any multipliers such as a flow limit multiplier are applied.

5.14. Loss Analysis

The DPF is an unbalanced power flow algorithm, supporting the modeling and analysis of a three-phase distribution power system consisting of three-phase, two-phase, and single-phase facilities and loads. The losses are available for every device involved in the DPF network selection.

The Loss Analysis (LA) function uses the DPF results to aggregate real and reactive losses for the following devices:

- Station
- All station power transformers

Power transformer core and copper losses. Core losses are mainly related to heat. Copper losses are losses due to resistance in the conductors of the windings (proportional to the resistance of the wire and the square of the current).

- All station regulators

Regulator core and copper losses.

- All station series reactors
- All feeders (energized by the station)
- Distribution feeders, including feeder regulators and distribution transformers, as well as the total feeder loss

The LA function executes after power flow and it accumulates technical losses to provide a report on demand. The function uses the latest DPF results and calculates instantaneous losses for different components of the distribution circuit.

Line and Cable Loss (per phase):

$$P_{Loss,line} = I_{line}^2 R_{line} \quad \text{(Equation 5.14)}$$

$$Q_{Loss,line} = I_{line}^2 X_{line} \quad \text{(Equation 5.15)}$$

Where:

R_{line} is the impedance of the line (ohms).

X_{line} is the reactance of the line (ohms).

I_{line} is the current in the line in (Amps).

$P_{Loss,line}$ is the real power line losses (Watts).

$Q_{Loss,line}$ is the reactive power line losses (VARs).

Transformer Copper Loss:

$$P_{Loss,copper} + jQ_{Loss,copper} = \sum_{All\ Windings} I_{tf}^2 \cdot (R_{winding} + jX_{winding}) \quad (Equation\ 5.16)$$

$$R_{winding} = R \cdot Z_{base} \quad (Equation\ 5.17)$$

$$X_{winding} = X \cdot Z_{base} \quad (Equation\ 5.18)$$

$$Z_{base} = \frac{V_{base}^2}{S_{base}} \quad (Equation\ 5.19)$$

Where:

I_{tf} is the current flow magnitude in a transformer winding (Amps).

$Z_{winding}$ is the impedance (R +jX) of a transformer winding (ohms).

R is the per-unit impedance of the winding on rated transformer power base at nominal voltage.

X is the per-unit reactance of the winding on rated transformer power base at nominal voltage.

V_{base} is the nominal transformer winding voltage.

S_{base} is the transformer rated power (kVA).

Transformer Core Loss:

$$Loss_{core} = |V_{tf}|^2 \cdot (P_{nom-core} + jQ_{nom-core}) \quad (Equation\ 5.20)$$

Where:

V_{tf} is the primary winding average voltage (per unit).

$P_{nom-core}$ is the rated core real power loss (kW).

$Q_{nom-core}$ is the rated core reactive power loss (kVAR).

+ The nominal reactive core loss is calculated using the following equation:

$$Q_{nom-core} = \sqrt{(I_{excit} \cdot S)^2 - P_{nom-core}^2} \quad (Equation\ 5.21)$$

Where:

S is the transformer rating (kVA)

I_{excit} is the excitation current percentage (per unit)

Shunt Impedance Real Power Loss (per phase):

$$Loss_{shimp} = P_{nom} \cdot V^2 \quad (Equation\ 5.22)$$

Where:

P_{nom} is the nominal real power consumed by the shunt impedance.

V is the solved capacitor voltage (per unit).

5.15. Power Quality Index Evaluation

The Power Quality Index Evaluation (PQI) function is responsible for estimating the quality of service. The PQI function compares the calculated load voltage values with the associated voltage limits and determines the extent of voltage violations. Three power quality indices are computed:

- **High-Voltage Power Quality Index:** The sum over all loads of the product of the load (kW) and the amount by which the voltage at the load exceeds the normal high-quality limit (kW*high voltage deviation)
- **Low-Voltage Power Quality Index:** The sum over all loads of the product of the load (kW) and the amount by which the voltage at the load violates the normal low-quality limit (kW*low voltage deviation)
- **Imbalance Voltage Power Quality Index:** The sum over all loads of the product of the three-phase load (kW) and quality limit (kW*imbalance voltage deviation)

For each load in the model, the load high power quality index (in %kW) is calculated using the following equation:

$$PQI_h = \sum_{all\ phases} kW_{ph} \cdot (V_{ph} - Vlim_{h,ph}) \cdot 100 \quad if\ V > Vlim_h \quad (Equation\ 5.23)$$

Where:

kW_{ph} is the single-phase nodal load (kW)

V_{ph} is the measured or calculated phase load nodal voltage (per unit)

$Vlim_{h,ph}$ is the normal high voltage limit (per unit)

For each load in the model, the load low power quality index (in %kW) is calculated using the following equation:

$$PQI_l = \sum_{all\ phases} kW_{ph} \cdot (V_{ph} - Vlim_{l,ph}) \cdot 100 \quad if\ V < Vlim_l \quad (Equation\ 5.24)$$

Where:

kW_{ph} is the single-phase nodal load (kW)

V_{ph} is the measured or calculated phase nodal voltage (per unit)

$Vlim_{l,ph}$ is the normal low voltage limit (per unit)

For each load in the model, the imbalance voltage power quality index, PQI_{imbal} (in % kW) is calculated as shown below. Note that this index is only calculated for three-phase loads.

The imbalance voltage power quality index, which is calculated only if the V_{imb} is higher than the specified imbalance limit ($V_{imbalance\ limit}$), is calculated using the following equation:

$$PQI_{imbal} = kW_{3-ph} \cdot (V_{imb} - V_{imbalance\ limit}) \quad (\text{Equation 5.25})$$

Where:

kW_{3-ph} is the total three-phase load (real power) in kW.

5.16. DNAF System Aggregation

An aggregation of calculated and measured values for a variety of system parameters is displayed using the following hierarchical levels (sorted from lowest to highest level):

- Feeder level
- Station power transformer level
- Station level
- Optimization group level/Division level
- System level

At each hierarchical level, the aggregation values of the components in the level below it are summed along with any additional values to produce a single set of aggregated values for that component. For example, the aggregation of values for a power transformer includes summing the aggregated values for each feeder that is dynamically fed by the power transformer as well as any contributions that may exist between the power transformer and feeder heads.

The following sections describe the calculation of these values in more detail.

Calculated Primary/Secondary Real Power Demand

At the feeder level, the Primary real power is the aggregation of real power at the source side. The real power demand is the DPF-calculated real power flow on the upstream side of the nominal head branch.

At the power transformer level, the real power demand from each downstream feeder is aggregated along with contributions from any loads, DERs, DGs, and capacitors that are between the feeder heads and the power transformer. The DPF-calculated real power flow from loads and capacitors is added to the demand while the values from DERs and DGs are subtracted from the demand.

At the substation level, aggregation is performed in the same fashion as the power transformer level with the inclusion of any loads, DERs, DGs, and capacitors that are not fed by power transformers. This could include components that are upstream of the power transformers or are islanded.

At the optimization group, division, and system level, aggregation is performed by summing the

real power supply and demand of each substation in the collection.

Measured Secondary Real Power Demand

At the feeder level, the real power demand is the three-phase measured real power flow at the feeder head branch.

At the power transformer level, the real power demand is the three-phase measured real power flow at the load side (secondary side) of the power transformer.

At the substation level, the real power demand is the sum of the measured real power demand for all power transformers in the station. In addition, the DPF-calculated real power flows from loads and capacitors are added to the demand while the values from DERs and DGs are subtracted from the demand. These are included so that a direct comparison between station calculated real power demand and station measured real power demand is more accurate.

At the optimization group, division, and system level, aggregation is performed by summing the measured real power demand of each substation in the collection.

Secondary Real Power Demand Difference

At the power transformer level, the secondary real power demand difference is the difference between the calculated and measured secondary real power at the load side of the power transformer.

At the substation level, the secondary real power demand difference is the difference between the calculated and measured secondary real power for all power transformers in the station.

At the optimization group, division, and system level, aggregation is performed by calculating the real power demand difference of each substation in the collection.

Calculated Native Real Power Demand

At the feeder level, the native real power demand is the sum of the DPF-calculated real power flow for each load on the feeder. Embedded load generation is considered when calculating the native real power demand.

At the power transformer level, the native real power demand from each downstream feeder is aggregated along with the sum of the DPF-calculated real power flow for any loads that are downstream of the power transformer and upstream of the feeder head.

At the substation level, the native real power demand from each power transformer in the station is aggregated. The DPF-calculated real power flow from any load upstream of the power transformer or islanded is also added to the demand.

At the optimization group, division, and system level, aggregation is performed by summing the native real power demand of each substation in the collection.

Calculated DER Real Power Injection

At the feeder level, the DER real power injection is the sum of the DPF-calculated real power flow for each DER and DG on the feeder. Embedded load generation is considered when calculating the DER

real power injection.

At the power transformer level, the DER real power injection from each downstream feeder is aggregated along with contributions from any DERs, DGs, and capacitors upstream of the feeder head and downstream of the power transformer.

At the substation level, the DER real power injection from each power transformer is aggregated along with contributions from any DERs, DGs, and capacitors that are upstream of the power transformer or islanded.

At the optimization group, division, and system level, aggregation is performed by summing the DER real power injection of each substation in the collection.

Real Power Loss

At the feeder level, the real power loss is the sum of the DPF-calculated real power loss for the feeder.

At the power transformer level, the real power loss from each downstream feeder is aggregated along with the DPF-calculated loss for the power transformer and each capacitor and series reactor that is upstream of the feeder head and downstream of the power transformer.

At the substation level, the real power loss from each power transformer is aggregated along with the DPF-calculated loss for each capacitor and series reactor that is upstream of the power transformers or is islanded.

At the optimization group, division, and system level, aggregation is performed by summing the real power loss of each station in the collection.

Calculated Primary/Secondary Reactive Power Demand

The aggregation of calculated reactive power at the source side and at the load side follows the same process as the aggregation of calculated real power demand.

Measured Secondary Reactive Power Demand

The aggregation of measured secondary reactive power demand follows the same process as the aggregation of measured secondary real power demand.

Secondary Reactive Power Demand Difference

At the power transformer level, the secondary reactive power demand difference is the difference between the calculated and measured secondary reactive power at the load side of the power transformer.

At the substation level, the secondary reactive power demand difference is the difference between the calculated and measured secondary reactive power for all power transformers in the station.

At the optimization group, division, and system level, aggregation is performed by calculating the reactive power demand difference of each substation in the collection.

Calculated Native Reactive Power Demand

The aggregation of calculated native reactive power demand follows the same process as the aggregation of calculated native real power demand.

Calculated DER Reactive Power Injection

The aggregation of calculated DER reactive power injection follows the same process as the aggregation of calculated DER real power injection.

Reactive Power Loss

The aggregation of calculated reactive power loss follows the same process as the aggregation of calculated real power loss.

Calculated Primary/Secondary Power Factor

The calculated primary/secondary power factor is calculated using the calculated real power and reactive power demand at the source side and the load side for each aggregation object.

Measured Secondary Power Factor

The measured power factor is calculated using the measured real and reactive power demand at the load side for each aggregation object.

Calculated Average Voltage, Station Internal Regulated Bus Only

The calculated average voltage on station buses averages the voltage magnitude of each phase on all power transformers downstream of and including the given aggregation object. Only station internal regulated MV buses are included.

Measured Average Voltage, Station Internal Regulated Bus Only

The measured average voltage on station buses averages the measured voltage magnitude of each phase on the load side of all power transformers downstream of and including the given aggregation object. Only station internal regulated MV buses are included.

Calculated Weighted Average Voltage, Station Internal Regulated Bus Only

The calculated weighted average voltage on station buses averages the calculated voltage magnitude of each phase on all power transformers downstream of and including the given aggregation object. However, when averaging the voltages, each voltage is weighted by the real power flow through that phase of the power transformer.

This weighting is performed by multiplying each voltage by its accompanying real power flow. These values are then summed and divided by the total real power flow from all calculated voltage locations. Only station internal regulated MV buses are included.

Measured Weighted Average Voltage, Station Internal Regulated Bus Only

The measured weighted average voltage on station buses averages the measured voltage magnitude of each phase on all power transformers downstream of and including the given aggregation object. These values are weighted using the same process as the calculated weighted average voltage on station buses. Only station internal regulated MV buses are included.

Calculated Average Voltage, All Regulated Buses

The calculated average voltage on all regulated buses uses the same process as the calculated average voltage on station buses with the inclusion of voltages on all regulators downstream of and including the given aggregation object.

Measured Average Voltage, All Regulated Buses

The measured average voltage on all regulated buses uses the same process as the measured average voltage on station buses with the inclusion of voltages on all regulators downstream of and including the given aggregation object.

Calculated Weighted Average Voltage, All Regulated Buses

The calculated weighted average voltage on all regulated buses uses the same process as the calculated weighted average voltage on station buses with the inclusion of voltages on all regulators downstream of and including the given aggregation object.

Measured Weighted Average Voltage, All Regulated Buses

The measured weighted average voltage on all regulated buses uses the same process as the measured weighted average voltage on station buses with the inclusion of voltages on all regulators downstream of and including the given aggregation object.

Calculated Online Capacitor Reactive Power

The calculated online capacitor reactive power is an aggregation of all online capacitors downstream of the given aggregation object. This includes connected fixed and remote capacitors as well as all local capacitors. The reactive power capacity is calculated as $(V_{Mag} * V_{Mag} * \text{NomKVar})$ where V_{Mag} is the magnitude of the phase voltage at the capacitor terminal and NomKVar is the nominal KVAR capacity defined in the static model of the capacitor. The local capacitors are always included in the online capacitor reactive power calculation, even if disconnected at the moment, as this could change to connected automatically according to the local control strategy.

Calculated Offline Capacitor Reactive Power

The calculated offline capacitor reactive power is an aggregation of all offline capacitors downstream of the given aggregation object. As opposed to the calculation for online capacitor reactive power, the offline calculation only includes disconnected fixed and remote capacitors. The reactive power capacity is calculated using the same equation seen in calculated online capacitor reactive power. Note that disconnected local control capacitors are counted in the online calculation.

Calculated Total Capacitor Reactive Power

The total capacitor reactive power is simply the sum of the online and offline capacitor reactive power for each aggregation object.

Calculated Real Power Generation Capacity

The calculated real power generation capacity is an aggregation of the reserve capacity for all DERs

and DGs downstream of the given aggregation object. The real power generation capacity for each device is the difference between the maximum real power output and the calculated real power output for the device.

Calculated Reactive Power Generation Capacity

The calculated reactive power generation capacity is an aggregation of the reserve capacity for all DERs and DGs downstream of the given aggregation object. The reactive power generation capacity for each device is the difference between the maximum reactive power output and the calculated reactive power output for the device.

5.17. DNAF Violation Counts

The aggregated count of various violations based on different components (switches, lines, distribution transformers and loads) is displayed using the following hierarchical levels (sorted from lowest to highest level):

- Feeder level
- Station power transformer level
- Station level
- Optimization group level/Division level
- System level

At each hierarchical level, the aggregated violation counts from the components in the lower level are summed along with any additional counts to produce a consolidated set of violation counts for that component. For example, the aggregated violation count for a power transformer includes the summed violation counts from each feeder dynamically fed by the transformer, as well as any contributions that may exist between the power transformer and the feeder heads. The following violations are displayed based on the LIM function (refer to section “Limit Monitor”) which identifies all limit violations after each DPF execution:

Number of Switches with Flow Imbalance Violation: Number of three phase switching devices with a flow imbalance limit violation. **Number of Loads with Voltage Imbalance violation:** Number of three phase loads with a voltage imbalance limit violation. **Number of Switches with Flow Limit Violation:** Number of three phase switching devices with a flow imbalance limit violation. **Number of Lines with Flow Limit Violation:** Number of lines with a flow limit violation. **Number of Distribution Transformers with Flow Limit Violation:** Number of distribution transformers with a flow limit violation. **Number of Loads with Voltage Limit Violation:** Number of loads with a high or low voltage limit violation.

5.18. DPF Alarms

Table 9 lists the supported DPF alarms. Note that the alarms can be enabled or disabled using the DNAF System Configuration display. For more information, see [DNAF Alarm Format](#).

Table 9. DPF Alarms

Alarm	Description
Failed to Complete	If DPF failed to complete for a station (for example, due to diverged power flow or model data errors), this alarm is issued for the station.
Failed to Complete (Return to Normal)	After DPF is successfully solved for a station and the "DPF Failed to Complete" alarm was issued for the last solution, this alarm is issued for the station.
Station Flow Violation	After DPF is successfully solved for a station, the flows of all station internal branches (lines, switches, etc. without feeder assignment) are checked against their limits. If the DPF estimated flow exceeds any of the specified limits, this alarm is issued for the station. There are up to five levels for this alarm. When an alarm level changes, it is noted in the alarm text.
Station Flow Violation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Station Flow Violation" alarm was issued for the station, and the flows of all station internal branches were found to be within limits, this alarm is issued for the station.
Feeder Flow Violation	After DPF is successfully solved for a station, the flows of all branches that have feeder assignment (lines, switches, etc.) are checked against their limits. If the DPF estimated flow exceeds any specified limit, this alarm is issued for the associated feeder.
Feeder Flow Violation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Feeder Flow Violation" alarm was issued for a feeder, and the flows of all branches on that feeder are found to be within limits, this alarm is issued for the feeder.
Feeder Low Voltage Violation	After DPF is successfully solved for a station, the voltages of load and bus nodes on feeders in the network selection are checked against their lower limits. If the DPF estimated node voltage is less than any of the specified limits, this alarm is issued for the associated feeder.
 Note This alarm is not applicable to station internal buses.	
Feeder Low Voltage Violation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Feeder Low Voltage Violation" alarm was issued for a feeder, and the DPF estimated node voltages on the feeder are found to be within limits, this alarm is issued for the feeder.
Feeder High Voltage Violation	After DPF is successfully solved for a station, the voltages of all load and bus nodes on feeders in the network selection are checked against their upper limits. If the DPF estimated node voltage is greater than any of the specified limits, this alarm is issued for the associated feeder.
Feeder High Voltage Violation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Feeder High Voltage Violation" alarm was issued for a feeder, and the DPF estimated node voltages on the feeder are found to be within limits, this alarm is issued for the feeder.
Feeder Flow Imbalance Violation	After DPF is successfully solved for a station, the flows for three-phase switching devices are checked. If there is a flow imbalance in any of these three-phase switching devices, this alarm is issued for the associated feeder. Flow imbalance is described in Limit Monitor .
Feeder Flow Imbalance Violation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Feeder Flow Imbalance Violation" alarm was issued for a feeder, and the flow imbalance is no longer found in any of the three-phase switching devices of the feeder, this alarm is issued for the feeder.

Alarm	Description
Feeder Voltage Imbalance Violation	After DPF is successfully solved for a station, the voltages of all three-phase load nodes are checked. If there is a voltage imbalance in any of the three-phase load nodes, this alarm is issued for the associated feeder. Voltage imbalance is described in Limit Monitor .
Feeder Voltage Imbalance Validation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Feeder Voltage Imbalance Validation" alarm was issued for a feeder, and the voltage imbalance is no longer found in any of three-phase load nodes of the feeder, this alarm is issued for the feeder.
Feeder Capacitor Position Validation	After DPF is successfully solved for a station, if the DPF estimated state of any telemetered locally controlled capacitor on a feeder does not match with the measurement status, this alarm is issued for the associated feeder.
Feeder Capacitor Position Validation (Return to Normal)	After DPF is successfully solved for a station after the "DPF Capacitor Position Validation" alarm was issued for a feeder, and the state of all telemetered locally controlled capacitors of the feeder matches with the measurement status, this alarm is issued for the feeder.
Tap Position Estimation Source Voltage Adjusted	If DPF is solved with the "Perform Tap Estimation" option from the DPF page of the DNAF Configuration Editor enabled, and if source voltage adjustment was performed for tap estimation purposes, then this alarm is issued for the associated station power transformer for which the tap position was estimated.
Feeder Back-Feed State Caused by Distribution Generation Event	After DPF is successfully solved for a station, the active power flow on all feeders that have connected and working distribution generators or energy sources is checked against its backfeed threshold limit (set in DNAF Configuration Editor). If the DPF estimated active power flow is greater than the specified threshold, this alarm is issued for the associated feeder.
Feeder Back-Feed State Caused by Distribution Generation Event (Return to Normal)	After DPF is successfully solved for a station after the "Feeder Back-Feed State Caused by Distribution Generation Event" alarm was issued for a feeder, and the DPF estimated active power flow is found to be within its backfeed threshold limit for the feeder, this alarm is issued for the feeder.
Large Mismatch between Calculation and AMI Measurement Event	After DPF is successfully solved for a station, the maximum deviation between a measured load AMI voltage and the DPF calculated voltage is checked against the threshold (set in the DNAF Configuration Editor). If the mismatch is greater than the specified threshold, this alarm is issued for the associated substation.
Large Mismatch between Calculation and AMI Measurement Event (Return to Normal)	After DPF is successfully solved for a station after the "Large Mismatch between Calculation and AMI Measurement Event" alarm was issued for a station, and the maximum deviation between a measured load AMI voltage and the DPF calculated voltage is found to be within its threshold limit for the substation, this alarm is issued for the substation.

5.19. DPF Check Before Operate

The DPF Check Before Operate (CBO) functionality can show results from a Distribution Power Flow analysis of the post operation conditions. The following checks are considered before and after operation, on either side of the device, if applicable:

- Minimum and maximum voltages and violation state on the feeder.
- Maximum loading and violation state on the feeder.
- Real power, reactive power, and current flows through the device and violation state.

The DPF analysis of the post operation conditions is supported for switching operations, tap changing transformer tap position changes, and capacitor status changes.

The before-operation power flow and limit monitoring results are retrieved from the latest DPF results. Therefore, to have valid before-operation results, a recent converged DPF solution should be completed.

DPF CBO performs a power flow and limit monitor analysis for the post-operation topology using the internally simulated new status of the device. The DPF CBO study does not impact the system topology or DPF results.

The loading of the system is retrieved from the current system loading using the DPF_CBO_LOOK_AHEAD_DURATION parameter in the DNAF server configuration file. This parameter specifies the look-ahead duration in hours to find the peak loading time used in this check. DPF CBO is solved at the peak loading time. If DPF_CBO_LOOK_AHEAD_DURATION is set to 0, DPF CBO uses the current system loading values. For more information about the parameters used in DPF CBO, refer to the *ADMS Software Installation and Maintenance Guide*.

5.20. DPF Configuration

This section describes the Distribution Power Flow configuration parameters.

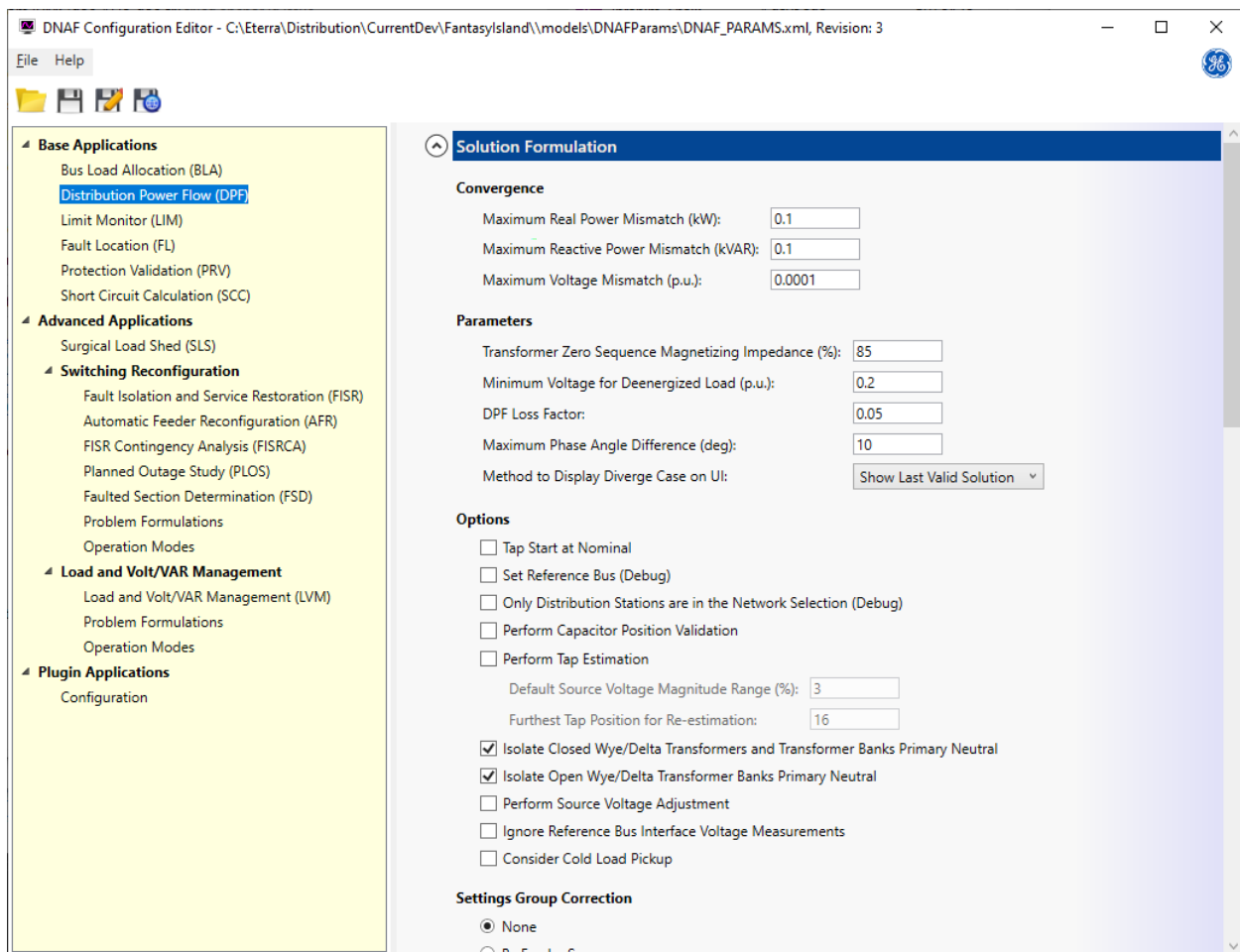


Figure 56. DPF Configuration

DPF configuration parameters are provided in table below. All listed parameters are contained by the <Params xsi:type = "DPFParams"> element in the DNAF_PARAMS.xml configuration file.

Table 10. DPF Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Convergence			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Real Power Mismatch (kW)	PMISMATCH_DPF <PMismatch>	Text box	This value specifies the internal power flow's real power convergence criterion in kW for the DPF and LVM applications. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The power flow solution status is set to Diverged if any load's real power difference between DPF iterations or any generator's real power mismatch between its target and output is larger than this specified value. A typical value is 0.1 kW. The allowable range is 0-10.
Maximum Reactive Power Mismatch (kVAR)	QMISMATCH_DPF <QMismatch>	Text box	This value specifies the internal power flow's reactive power convergence criterion in kVAR for the DPF and LVM applications. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The power flow solution status is set to Diverged if any load's reactive power difference between DPF iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. A typical value is 0.1 kVAR. The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Voltage Mismatch (p.u.)	VMISMATCH_DPF <VMismatch>	Text box	This value specifies the internal power flow's voltage convergence criterion in per unit for the DPF and LVM applications. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. You can find the name and ID of the device connected to the largest voltage mismatch node in the DPF SOLUTION SUMMARY section of the DPF result file and in the DPF Station Results display. The power flow solution status is set to Diverged if the voltage mismatches are larger than this specified value. A typical value is 0.0001 p.u. The allowable range is 0-0.1.
Parameters			
Transformer Zero Sequence Impedance (%)	ZEROSEQTFZ_DPF <TransformerZ0>	Text box	This value specifies the default transformer zero sequence impedance as a fraction of the transformer positive sequence impedance. It is used when the zero sequence impedance is not specified in the transformer model. • To enter a zero sequence magnetizing shunt susceptance, data must be entered in the field MagB0 or MAGB0_TFMODE (defined in the Transformer Model in the global file) for xml and csv fields, respectively. • To enter a zero sequence magnetizing shunt conductance, data must be entered in the field MagG0 or MAGG0_TFMODE (defined in the Transformer Model in the global file) for xml and csv fields, respectively. A typical value is 85.0 (%). The allowable range is 0.0-100.0.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Minimum Voltage for Deenergized Load (p.u.)	VDEENERGIZE_DPF <VDeenergizeLoad>	Text box	This value specifies the voltage threshold for detecting a de-energized load in per unit. If the voltage is smaller than this value, the load is considered de-energized. This value is also used in computing the number of de-energized customers (in FISR, FISRCA, AFR, and PLOS) and used in identifying voltage violations for de-energized loads (voltage violation is detected if a de-energized load has voltage greater than this value). The allowable range is 0.01-0.4.
DPF Loss Factor	LOSSFACTOR_DPF <LossFactor>	Text box	This value specifies the ratio of losses to total demand. When a feeder is not modeled in detail, it is modeled as an equivalent load (constant impedance) at the feeder head node. This value is used to estimate the equivalent load at the feeder head with the following equation: Demand $\times$ (1 + DPF Loss Factor). This factor is not used if real and reactive power values are provided by SCADA. The allowable range is 0-1.
Maximum Phase Angle Difference (deg)	<MaxPhaseAngleDiff>	Text box	This value specifies the maximum phase angle difference in degrees across energized, three-phase currently open switching devices. It is used by the reconfiguration application (FISR, AFR, and PLOS) switch eligibility checks. If the voltage phase angle difference between the From and To nodes of a feeder main switching device exceeds this value, this switch is not eligible in the FISR, AFR, and PLOS applications. This value is also used to identify dangerous switching conditions and show the associated bitmap near the switching device. The allowable range is 0-20.
Method to Display Diverge Case on UI	SAVESHOWDIVOUTPUT_DPF <MethodDisplayDivCase>	Combo box	This value specifies how the DPF results are stored and displayed if DPF diverges. There are three options: • Store and display the latest DPF results no matter if DPF has diverged or not. • Store and display only the latest converged DPF results, and discard the diverged results. This option is used by default. • Store only the converged DPF results, the diverged results are discarded, and display "--" for all DPF results.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Options			
Tap Start at Nominal	TAPNOMPOSSTART_DPF <TapNominal>	Check box	For locally controlled tap changing transformers (both regulators and LTCs), DPF determines tap positions that correspond to the voltage within the acceptable range determined by voltage regulation parameters, including the voltage target, voltage bandwidth, and line drop compensation R and X settings. After each DPF solution, the system adjusts tap positions for these transformers based on the calculated tap positions. If this option is selected, the system starts from the nominal tap position, as defined in the static model, when adjusting the tap position for a locally controlled tap changing transformer after a DPF solution. For example, if the nominal tap position is 0 and the acceptable tap positions are 0-2, the system sets tap position to 0. If this option is cleared, the system starts from the latest tap position when adjusting the tap positions for a locally controlled tap changing transformer after a DPF solution. For example, if the latest tap position is 6 and the acceptable tap positions are 0-2, the system sets tap position to 2. This option is cleared by default.
Set Reference Bus (Debug)	CREATEDDEBUGREFBUSOUTPUT_DPF <DebugSetRefBuses>	Check box	This indicates whether to set the reference bus strings from NTP and DPF. These strings can be placed in DPF logs or data queries used for tabular and other displays. This is a debugging option for performing a system-wide network selection comparison. If the option is selected, the reference bus strings are set. This option is cleared by default.
Distribution Stations are in the Network Selection (Debug)	DISTSUBNETSELECT_DPF <DistSubNetSelect>	Check box	This indicates whether to include only distribution substations in the network selection. This is a debug only option for debugging DPF/BLA problems. If this option is selected, only distribution substations are included, and no sub-transmission stations are included. This option is cleared by default.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Perform Capacitor Position Validation	<CapacitorPositionValidation>	Check box	This indicates whether to check if the DPF estimated state of any telemetered locally controlled capacitor matches the SCADA measurement status. If this option is selected, DPF compares the internally estimated status for capacitors in local control mode with their respective telemetered status measurement. If there is a mismatch, the "DPF Feeder Capacitor Position Validation" alarm, if enabled, is issued. This option is cleared by default.
Perform Tap Estimation	<PerformTapEstimation>	Check box	This indicates whether tap position should be estimated for LTCs or regulators for which no tap position analog measurement is available, if the corresponding voltage measurement exists. There may be cases where no tap position measurement is available for some LTCs (and possibly regulators). In such a case, if this option is selected and if both the source and load side voltage measurements on the transformers are available, the tap position is estimated using the measurements and the transformer model parameters. This option is cleared by default.
Default Source Voltage Magnitude Range (%)	<SourceVoltageMagnitudeRange>	Text box	This value specifies the default source voltage magnitude range, in percent voltage, for source voltage adjustment in tap position estimation for a given LTC. For a given interface, if a source voltage magnitude range is not defined via the Substation Editor, the Default Source Voltage Magnitude Range value is used as the source voltage magnitude range for that interface object. If the source voltage is set to 102%, and the source voltage range is set to 3% (either via the Substation Editor, or via the DCE), then the tap estimation logic assumes that the source voltage is in the 99% - 105% range. The source voltage magnitude range is used only if a valid source side voltage analog measurement could not be found for the LTC. Although tap estimation is performed on both LTCs and regulators, this parameter applies only to LTCs. The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Furthest Tap Position for Re-Estimation	<FurthestTapPositionForReEstimation>	Text box	This value specifies the furthest tap position from nominal before tap position estimation is re-executed using an adjusted source voltage for a given LTC. For example, if the furthest tap position is specified as 8, and tap position is estimated at 9 using the interface source voltage (specified via the Substation Editor, or via a manual entry), then the source voltage is adjusted within the allowed range defined by the source voltage magnitude range (via the Substation Editor, or via the Default Source Voltage Magnitude Range parameter) and tap position is recalculated using the adjusted source voltage. This parameter is used only if a valid source side voltage analog measurement could not be found for the LTC. Although tap estimation is performed on both LTCs and regulators, this parameter applies only to LTCs. The allowable range is 0-16.
Isolate Closed Wye/Delta Transformers and Transformer Banks Primary Neutral	FLOATTFNEUTRALCLOSED_DPF <FloatTFNeutralClosed>	Check box	This indicates whether to isolate the transformer primary side neutral for closed wye – delta transformer banks and three-phase transformers for all DNAF applications. If the option is selected, the transformer primary side neutral is isolated (not connected to the system neutral wire). This is internally implemented using a fictitious switch that opens to disconnect the neutral. This option is selected by default.
Isolate Open Wye/Delta Transformer Banks Primary Neutral	FLOATTFNEUTRALOPEN_DPF <FloatTFNeutralOpen>	Check box	This indicates whether to isolate the transformer primary side neutral of open wye – delta transformer banks for all DNAF applications. If the option is selected, the primary side neutral for each transformer in the bank is isolated (not connected to the system neutral wire). This is internally implemented using a fictitious switch that opens to disconnect the neutral. This option is selected by default.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Perform Source Voltage Adjustment	SOURCEVOLTADJ_DPF <SourceVoltageAdjustment>	Check box	This indicates whether to adjust the source voltage according to the downstream measurements if there is no voltage measurement at the source. If the option is selected, the source voltage is adjusted. (For more information, see Back Calculation of Interface Voltages). This option is cleared by default.
Ignore Reference Bus Interface Voltage Measurement	<IgnoreEmsVoltFlag>	Check box	This specifies whether to ignore an EMS (interface) reference bus voltage measurement during DPF even if it is available. If this option is selected, the EMS voltage measurement is not used. The interface voltage could come from a manual entry, back calculation, or a nominal value defined in the static model. (For more information, see Measurement Sets and Calculation Methods). This option is cleared by default.
Consider Cold Load Pickup	<ConsiderColdLoadPickup>	Check box	This specifies whether cold load pickup effect is considered in DPF. This option is cleared by default.
Settings Group Correction			
Settings Group Correction	<EnableGroupSettingsCheck>	Radio buttons	These buttons determine how the DMS will check for incorrect post-DPF values of active protection settings groups. None – No check will be performed. Default value. By Feeder Source – The feeder source adaptive relaying option will be used to ensure the active settings group matches the setting associated with the feeder currently supplying the protection device. By Nominal Setting – The active settings group will be set to nominal if the topology of the parent feeder is normal (contains no off-nominal switching).
Triggering			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Measurement Delta Value (%)	VOLTMEASDELTA_DPF <VoltDeltaRate>	Text box	This value specifies the voltage measurement change threshold in percent for the SCADA Sudden Measurement Change (SMC) trigger. If a station internal voltage change is larger than this value, and the DPF SMC trigger is enabled, DPF is triggered. However, this event trigger is conditional on the block timer value defined in the DNAF_SCADA_ROC_BLOCK_TIME field in the NetServerDnafConfig_localhost.txt file. For example, if DNAF_SCADA_ROC_BLOCK_TIME =5, each successive ROC-triggered DPF is blocked for 5 minutes. The allowable range is 0-20.  Note The sudden measurement change trigger currently applies only to voltage and current measurements that are attached to the station internal objects (such as interface, station transformer, breaker, and feeder head). The trigger does not apply to the measurements of any device outside the station fence (such as reclosers and line sensors).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Current Measurement Delta Value (A)	AMPMEASDELTA_DPF <CurrentDeltaRate>	Text box	This value specifies the current measurement change threshold (in Amps) for the Sudden Measurement Change trigger. If a station internal current change is above this value, and the DPF SMC trigger is enabled, DPF is triggered. However, this is conditional on the block timer value defined in the DNAF_SCADA_ROC_BLOCK_TIME field in the NetServerDnafConfig_localhost.txt file. For example, if DNAF_SCADA_ROC_BLOCK_TIME =5, each successive SMC triggered DPF is blocked for 5 minutes. The allowable range is 1-200.  Note The sudden measurement change trigger currently applies only to voltage and current measurements that are attached to the station internal objects (such as interface, station transformer, breaker, and feeder head). The trigger does not apply to the measurements of any device outside the station fence (such as reclosers and line sensors).
Primary Voltage Units			
Use Voltage Base	VOLTAGERESULTUNIT_DPF <ResultTabularDisplay>	Option button	This setting specifies that the primary network voltage values are displayed via the base voltage specified using the Base Voltage parameter. The node voltages of secondary networks are always displayed in S.I. units.
Base Voltage (V)	VBASEVALUE_DPF <VoltBase>	Text box	This text box, located next to the Use Voltage Base setting, specifies the base voltage to be used in displaying primary network voltage values. The text box is enabled only if the Use Voltage Base (V) option is selected. The allowable range is 1-1000.
Use Per Unit (p.u.)	VOLTAGERESULTUNIT_DPF <ResultTabularDisplay>	Option button	This setting specifies that the primary network voltage values are displayed as per unit values. The node voltages of secondary networks are always displayed in S.I. units, regardless of the user's selection.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Actual Voltages (kV)	VOLTAGERESULTUNIT_DPF <ResultTabularDisplay>	Option button	This setting specifies that the primary network voltage values are displayed as actual voltage values. The node voltages of secondary networks are always displayed in S.I. units, regardless of the user's selection.
DPF Suspect Solution Conditions			
BLA Not Converged	DPFSUSPECTBLANOTCONVERGED_DPF <DPFSuspectBLANotConverged>	Check box	This indicates whether to set the station DPF status to Suspect when BLA convergence is not met for all measurement islands in the network selection. • If this option is selected, the DPF status is set to Suspect for cases when BLA convergence is not met. • If the option is cleared, the DPF Suspect status is not set based on the BLA convergence status.
Suspect Local Control	DPFSUSPECTLOCALCONTROL_DPF <DPFSuspectLocalControl>	Check box	This indicates whether to set the station DPF status to Suspect when there are any suspect locally controlled capacitors, regulating transformers (regulators and LTCs), generators, or energy resources in the station. • If this option is selected, the DPF status is set to Suspect for suspect locally controlled devices. • If the option is cleared, the DPF Suspect status is not set based on the presence of suspect locally controlled devices.
Suspect Load	DPFSUSPECTLOAD_DPF <DPFSuspectLoad>	Check box	This indicates whether the station DPF status is set to Suspect when there are any suspect loads in this station. See Unsolvable DPF Cases for more information on suspect loads. • If the option is selected, the DPF status is set to Suspect if there are suspect loads in the solution. • If the option is cleared, the DPF Suspect status is not set based on the presence of suspect loads.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Interface Voltage Not Set	DPFSUSPECTINTERFACEVOLTNOTSET_DPF <DPFSuspectInterfaceVoltNotSet>	Check box	This check box indicates whether the station DPF status is set to Suspect when the interface voltage magnitude or angle measurement is invalid or a voltage magnitude or angle manual entry is missing. • If the option is selected, the DPF status is set to Suspect if there are missing interface voltage values. • If the option is cleared, the DPF Suspect status is not set based on the presence and/or SCADA quality of the interface voltage data.
Locally Controlled Capacitors			
Voltage: Below Lower	COMBOVOLTAGERANGE1INDEX_DPF <VoltageRange1Action>	Combo box	This value specifies the action that is taken if the DPF calculated voltage value at the regulation node of a locally controlled capacitor is less than the minimum of the two on/off level control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Voltage: Between	COMBOVOLTAGERANGE2INDEX_DPF <VoltageRange2Action>	Combo box	This value specifies the action that is taken if the DPF calculated voltage value at the regulation node of a locally controlled capacitor is between the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Voltage: Above Upper	COMBOVOLTAGERANGE3INDEX_DPF <VoltageRange3Action>	Combo box	This value specifies the action that is taken if the DPF calculated voltage value at the regulation node of a locally controlled capacitor is greater than the maximum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage: Control Target Adjust Factor	LOCALCAPVTARGETADJ_DPF <LocalCapVTargetAdj>	Text box	This value is used to increase voltage control value 2 and decrease voltage control value 1 by a specified percentage. The affected fields are OnLevel and OffLevel in the xml file (or ONLEVEL_SHIMP and OFFLEVEL_SHIMP for a csv file). The allowable range is 0-10.
VAR: Below Lower	COMBOVARRANGE1INDEX_DPF <VARRange1Action>	Combo box	This value specifies the action that is taken if the DPF calculated VAR value at the regulation branch end of a locally controlled capacitor is less than the minimum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
VAR: Between	COMBOVARRANGE2INDEX_DPF <VARRange2Action>	Combo box	This value specifies the action that is taken if the DPF calculated VAR value at the regulation branch end of a locally controlled capacitor is between the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
VAR: Above Upper	COMBOVARRANGE3INDEX_DPF <VARRange3Action>	Combo box	This value specifies the action that is taken if the DPF calculated VAR value at the regulation branch end of a locally controlled capacitor is greater than the maximum of the two on/off control settings. The enumeration has the following values: • 0: Turn Off • 1: Turn On • 2: No Status Change
VAR: Control Target Adjust Factor	VARADJUSTFACTOR_DPF <VARAdjustFactor>	Text box	This value is used to increase VAR control value 2 and decrease VAR control value 1 by a specified percentage. The affected fields are OnLevel and OffLevel in the xml file (or ONLEVEL_SHIMP and OFFLEVEL_SHIMP for a csv file). The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Amp: Below Lower	COMBOAMPRANGE1INDEX_DPF <AmpRange1Action>	Combo box	This value specifies the action that is taken if the DPF calculated Amp value at the regulation branch end of a locally controlled capacitor is less than the minimum of the two on/off control settings. The enumeration has the following values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Amp: Between	COMBOAMPRANGE2INDEX_DPF <AmpRange2Action>	Combo box	This value specifies the action that is taken if the DPF calculated Amp value at the regulation branch end of a locally controlled capacitor is between the two on/off control settings. The enumeration has the following values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Amp: Above Upper	COMBOAMPRANGE3INDEX_DPF <AmpRange3Action>	Combo box	This value specifies the action that is taken if the DPF calculated Amp value at the regulation branch end of a locally controlled capacitor is greater than the maximum of the two on/off control settings. The enumeration has the following values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Amp: Control Target Adjust Factor	AMPADJUSTFACTOR_DPF <AmpAdjustFactor>	Text box	This value is used to increase Amp control value 2 and decrease Amp control value 1 by a specified percentage. The affected fields are OnLevel and OffLevel in the xml file (or ONLEVEL_SHIMP and OFFLEVEL_SHIMP for a csv file). The allowable range is 0-10.
Power Factor: Below Lower	COMBOPOWERFACTORRANGE1INDEX_DPF <PowerFactorRange1Action>	Combo box	This value specifies the action that is taken if the DPF calculated Power Factor value at the regulation branch end of a locally controlled capacitor is less than the minimum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Power Factor: Between	COMBOPOWERFACTORRANGE2INDEX_DPF <PowerFactorRange2Action>	Combo box	This value specifies the action that is taken if the DPF calculated Power Factor value at the regulation branch end of a locally controlled capacitor is between the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Power Factor: Above Upper	COMBOPOWERFACTORRANGE3INDEX_DPF <PowerFactorRange3Action>	Combo box	This value specifies the action that is taken if the DPF calculated Power Factor at the regulation branch end of a locally controlled capacitor value is greater than the maximum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Power Factor: Control Target Adjust Factor	POWERFACTORADJUSTFACTOR_DPF <PowerFactorAdjustFactor	Text box	This value is used to increase Power Factor control value 2 and decrease Power Factor control value 1 by a specified percentage. The affected fields are OnLevel and OffLevel in the xml file (or ONLEVEL_SHIMP and OFFLEVEL_SHIMP for a csv file). The allowable range is 0-10.
Time: Below Lower	COMBOTIMERANGE1INDEX_DPF <TimeRange1Action>	Combo box	This value specifies the action that is taken if the operation Time value of a locally controlled capacitor is earlier than the minimum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Time: Between	COMBOTIMERANGE2INDEX_DPF <TimeRange2Action>	Combo box	This value specifies the action that is taken if the operation Time value of a locally controlled capacitor is between the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Time: Above Upper	COMBOTIMERANGE3INDEX_DPF <TimeRange3Action>	Combo box	This value specifies the action that is taken if the operation Time value of a locally controlled capacitor is later than the maximum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Temperature: Below Lower	COMBOTEMPERATURERANGE1INDEX_DPF <TemperatureRange1Action>	Combo box	This value specifies the action that is taken if the operation Temperature value of a locally controlled capacitor is less than the minimum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Temperature: Between	COMBOTEMPERATURERANGE2INDEX_DPF <TemperatureRange2Action>	Combo box	This value specifies the action that is taken if the operation Temperature value of a locally controlled capacitor is between the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Temperature: Above Upper	COMBOTEMPERATURERANGE3INDEX_DPF <TemperatureRange3Action>	Combo box	This combo box specifies the action that is taken if the operation Temperature value of a locally controlled capacitor is greater than the maximum of the two on/off control settings. The enumeration has these values: • 0: Turn Off • 1: Turn On • 2: No Status Change
Solve with Capacitor in Local Mode if Communications are Lost	<CapSolveInLocalMode>	Check box	This specifies whether a capacitor that has both remote and local control capability is treated as a locally controlled device when communications are lost, which is determined by a "Communications Established" measurement. If one does not exist, communications are deemed to be lost if the status measurement is invalid. In local control mode, the DPF determines the open/closed status based on the local controller on/off settings.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Display and Failsafe LVM Options			
Maximum Voltage Mismatch at Station Internal Buses (%)	<MaxVMeasMismatchInternal>	Text box	This value specifies, as a percentage, the maximum deviation between a measured station internal bus voltage and the DPF calculated bus voltage. If the voltage mismatch exceeds this deviation, the mismatch is shown in red on the Bus Voltage tabular display. Additionally, if the mismatch is greater than the specified amount and LVM is run using an optimization-based problem formulation with HLVM failsafe enabled, then the failsafe heuristic problem formulation is used. The allowable range is 0-10.
Maximum Voltage Mismatch at Station External Buses (%)	<MaxVMeasMismatchExternal>	Text box	This value specifies, as a percentage, the maximum deviation between a measured station external bus voltage and the DPF calculated bus voltage. If the voltage mismatch exceeds this deviation, the mismatch is shown in red on the Bus Voltage tabular display. Additionally, if the mismatch is greater than the specified amount and LVM is run using an optimization-based problem formulation with HLVM failsafe enabled, then the failsafe heuristic problem formulation is used. The allowable range is 0-10.
Maximum Voltage Mismatch at Load Point (%)	<MaxVMeasMismatchLoad>	Text box	This value specifies, as a percentage, the maximum deviation between a measured load AMI voltage and the DPF calculated voltage. If the voltage mismatch exceeds this deviation, the mismatch is shown in red on the Meter Measurement Data tabular display. Additionally, if the mismatch is greater than the specified amount and the Discard Load Voltage Measurement with Large Deviations option on the LVM Problem Formulations page is enabled, this voltage measurement will not be used in the LVM voltage limit adjustment process. The allowable range is 0-10.
Solutions Results			
Create DPF Result File	PRINTOUT_DPF <PrintDPFResult>	Check box	This specifies whether DPF result files are created. If the option is selected, DPF result files are created. The file name contains the station name with the prefix "DPF Results_" (for example: DPF Results_BAKER.dat).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Create Full DPF Result File	FULLDPFREPORT_DPF <FullDPFReport>	Check box	This specifies whether to add all detailed flow and voltage results in the DPF result files. These detailed results are written to the same file as the basic DPF results. The suboptions provide for user to configure equipment flows (like line, load etc.) they want to add in the DPF result file as per their need.
Include Line	FULLDPFLINEREPORT_DPF <FullDPFLineReport>	Check box	This specifies whether to add line flows in the full DPF result file. These detailed results will be written to the same file as the basic DPF results.
Include Load	FULLDPFLOADREPORT_DPF <FullDPFLoadReport>	Check box	This specifies whether to add load voltages in the full DPF result file. These detailed results will be written to the same file as the basic DPF results.
Include Switch	FULLDPFSWITCHREPORT_DPF <FullDPFSwitchReport>	Check box	This specifies whether to add switch flows in the full DPF result file. These detailed results will be written to the same file as the basic DPF results.
Include Distribution Transformer	FULLDPFDISTTFREPORT_DPF <FullDPFDistTfReport>	Check box	This specifies whether to add distribution transformer flows in the full DPF result file. These detailed results will be written to the same file as the basic DPF results.
Include UPF Iteration Data	PRINTUPFINFO_DPF <PrintUPFInfo>	Check box	This specifies whether to include the DPF Iteration Information section in the DPF result files.
Use Unique File Name	UNIQUELOG_DPF <UniqueDPFLog>	Check box	This specifies whether to create unique DPF result files that are not overwritten. These files are created in the same directory as the basic DPF files. Each unique file contains the date and time stamp when it was created. GE recommends against using this option, to avoid filling up the hard drive.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Number of DPF Result Files Created for Any One Substation	<MaxNumDpfLogFile>	Text box	This value specifies the maximum number of accumulated DPF result files for a single station. This option is effective only if the Use Unique File Name option is enabled. Otherwise, the file is overwritten each time DPF is run on a particular station. The allowable range is 0-200. DNAF server will only perform periodic cleanup of those over-limit result files for each station based on creation time. So, you may still see more than MaxNumDpfLogFile number of files between the cleanup period in the file system. The period can be configured from DNAF server configuration file via option (DNAF_LOG_CLEANUP_PERIOD_MINUTES), its default value is 2 hours (120 minutes), the minimum of this option is 10 minutes.
Create Detailed BLA Result Report	<DetailBLAResult>	Check box	This specifies whether to generate a detailed BLA result report within the DPF result file. If it is selected, the BLA measurement island load data is output during each BLA/DPF iteration. This check box is cleared by default, to avoid having too many IO operations (thus negatively impacting application performance).
Only BLA Not Converged Islands	<OnlyBlaNotConvergedIslandResult>	Check box	This specifies whether only the results from BLA not converged measurement islands are included in the detailed BLA section in the DPF result file. This setting applies only if the Create Detailed BLA Result Report option is enabled.
Include DER/DG Injection Data	<IncludeDERDGInjectionData>	Check box	This specifies whether the DOWNSTREAM DER/DG INJECTIONS AT SWITCHES section with DER/DG real and reactive power injection is included the DPF result file.
Include DER Measurement Used in BLA/DPF	<IncludeDERMeasurementInjection>	Check box	This indicates whether DER measurement injection data is included in the DPF results file. If this option is selected, the DER Gateway and Scada DER measurements that are used by BLA and DPF are included in the DPF results file.
Create Debug Files for Diverged Case	<PrintDivCaseLog>	Check box	This specifies whether to create DPF debug files if DPF diverges.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Number of Sets of Debug Files:	MAXUPFOUTPUTFILE_DPF <MaxNumUpfOutFile>	Text box	This value specifies the maximum number of sets of DPF debug files that can be generated for a given station. When DPF does not converge, a set of debug files (including two debug files) is created. Debug files are as follows: • The first file contains the prefix "upfin_export" along with the station name, time, and date. • The second file contains the prefix "alg_export" along with station name, time, and date. These debug files contain the system status at the DPF execution time and the parameters used in the solution, including those from alg.dat, optalg.dat, system connectivity data, and system device data. The allowable range is 0-50.
Other Result Files			
Create Historical Load Data File	WRITESTUDYHISTLDFILE_DPF <PrintOutHistLdFile>	Check box	This specifies whether to create a historical load data file in real time mode. If the option is selected, the historical load data file is created, with the name HistLoad.dat. This file holds information on each load category, the day type, and each individual load for every substation.
Create DPF Statistics File	STATS_DPF <CreateDPFStatsFile>	Check box	This specifies whether to create a DPF statistics file. If the option is selected, a DPF statistics file named DPF_STAT.dat is created. This file has DPF and BLA statistics for a given substation, which includes convergence data, mismatch data, and station device counts.
Manual Execution			
Allow in Real Time	REALTIME_DPF <RealTime>	Check box	This specifies whether the DPF application can be invoked through the RT DNAF Solution pop-up display (selected means the application is shown). Note that DPF could still be executed via periodic or event triggers in RT by setting the triggering parameters in the appropriate server configuration file.
Allow in Study Mode	STUDYMODE_DPF <Study>	Check box	This specifies whether the DPF application can be invoked through the ST DNAF Solution pop-up display (selected means the application is shown).
Old and Deprecated Parameters			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Name	NAME_DPF <Name>	Text box	This value specifies the function name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_DPF	Text box	This text box represents the network analysis function Index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCENUM_DPF <FuncEnum>	Text box	This text box indicates the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISIR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

5.20.1. LIM Configuration

This section describes the Limit Monitor configuration parameters.

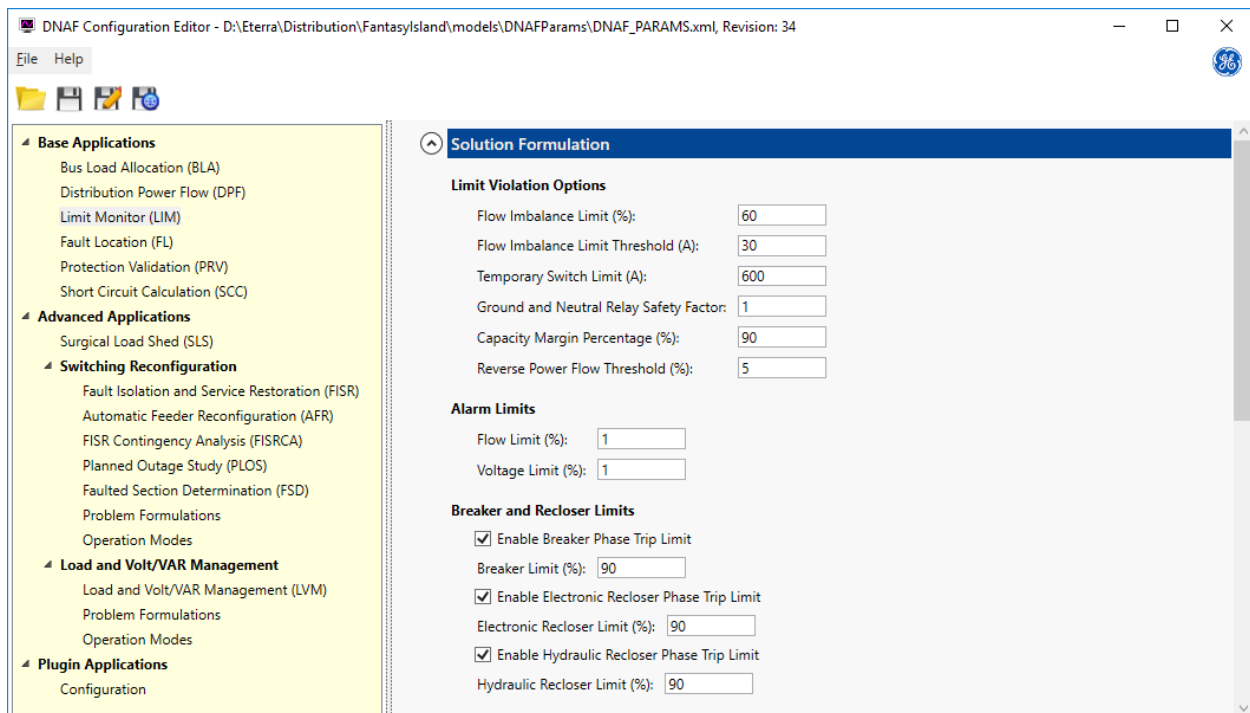


Figure 57. Limit Monitor Configuration

The Limit Monitor configuration parameters are provided in table below. All listed parameters are contained by the <Params xsi:type = "LIMParams"> element in the DNAF_PARAMS.xml configuration file.

Table 11. LIM Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Limit Violation Options			
Flow Imbalance Limit	FLOWIMBLIM_LIM <FlowImbalLimit>	Text box	This value specifies the flow imbalance limit (in percentage) for three-phase switching devices. The allowable range is 0-100.
Flow Imbalance Limit Threshold	<FlowImbalViolationThreshold>	Text box	This value specifies the threshold for determining if a flow imbalance condition exists for a three-phase switching device. If the average current (in Amps) is less than this value, no flow imbalance is reported. The allowable range is 0-200.
Temp Switch Limit	TEMPSWITCHLIMIT_LIM <TempSwitchLimit>	Text box	This value specifies the flow limit (in Amps) for temporary jumpers. The allowable range is 0-1200.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Ground and Neutral Relay Safety Factor	GRNDRTNFACTOR_LIM <GroundRelaySafetyFactor>	Text box	This value specifies a safety factor used when comparing the phase imbalance flow (the vector sum of currents in the three phases) for breakers and reclosers to the minimum ground relay pick-up setting and neutral relay current setting. The ground return flow is compared to the ground relay pick-up current setting multiplied by this safety factor to make sure the ground relay will not be tripped. The neutral flow is compared to the neutral relay current setting multiplied by this safety factor to make sure the neutral relay will not be tripped. The allowable range is 0-2.
Capacity Margin Percentage	<CapaMarginPercentage>	Text box	This value specifies the capacity margin threshold as a percentage of the flow limit rating for branch devices. If the calculated flow in any one phase of a branch device exceeds this threshold but the flow is lower than the level 1 flow limit, a capacity margin violation exists for the device. The allowable range is 0-99.9.
Reverse Power Flow Threshold	<BackFeedThreshold>	Text box	This value specifies the real power threshold for feeder (as a percentage of the feeder breaker phase trip limit) to initiate a backfeed alarm. The allowable range is 0.1-15.
Alarm Limits			
Flow Limit (%)	ALARMFLOWSCALER_DPF <AlarmLimitFlow>	Text box	This value specifies the percentage increase in the flow limit as the alarm flow limit. The alarm flow limit is used to determine if a flow violation event is issued in the Alarm subsystem. An alarm is issued if the flow is greater than the original flow limit multiplied by (100 + Flow Limit)/100. All available flow limit levels are affected. The allowable range is 0-1000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Limit (%)	ALARMVOLTSCALER_DPF <AlarmLimitVolt>	Text box	This value specifies the percentage increase of the high voltage limit or decrease of the low voltage limit as the alarm voltage limit. The alarm voltage limit is used to issue a high or low voltage violation event in the Alarm subsystem. All available voltage limits are affected. - For a high voltage violation, an alarm is issued if the voltage is greater than the high voltage limit plus Voltage Limit/100 in per unit. - For a low voltage violation, an alarm is issued if the voltage is less than the low voltage limit minus Voltage Limit/100 in per unit. The allowable range is 0-100.
Breaker and Recloser Limits			
Enable Breaker Phase Trip Limit	<EnableBreakerPhaseTripLimit>	Check box	This indicates whether to compare the flow results against the percentage of phase trip limit for breakers for DPF, LVM, and switching reconfiguration application. If this option is selected, the regular flow limits defined in the static model in the switch model are not checked.
Breaker Limit (%)	CBPHTRIPLIMPERCENT_LIM <CBLimitPercent>	Text box	This value specifies the percentage of the phase trip setting for breakers (%). For example, if this value is set to 90% for a breaker with a 600 Amp phase trip setting, the breaker flow is compared to the phase trip setting of the breaker multiplied by this breaker limit percentage ($0.90 * 600 = 540.0$ Amps). This option is used only if the Enable Breaker Phase Trip Limit check box is selected. The allowable range is 0-200.
Enable Electronic Recloser Phase Trip Limit	<EnableElectronicRecloserPhaseTri pLimit>	Check box	This indicates whether to compare the flow results against the percentage of phase trip limit for electronic reclosers for DPF, LVM, and switching reconfiguration application. If this option is selected, the regular limits defined in the static model in the switch model are not checked.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Electronic Recloser Limit (%)	<ElectronicRecloserLimitPercent>	Text box	This value specifies the percentage of the phase trip setting for electronic reclosers (%). For example, if this value is set to 90% for an electronic recloser with a 600 Amp phase trip setting, the recloser flow is compared to the phase trip setting of the recloser multiplied by this recloser limit percentage ($0.90 * 600 = 540.0$ Amps). This option is used only if the Enable Electronic Recloser Phase Trip Limit check box is selected. The allowable range is 0-200.
Enable Hydraulic Recloser Phase Trip Limit	<EnableHydraulicRecloserPhaseTripLimit>	Check box	This indicates whether to compare the flow results against the percentage of phase trip limit for hydraulic reclosers for DPF, LVM, and switching reconfiguration application. If this option is selected, the regular limits defined in the static model in the switch model are not checked.
Hydraulic Recloser Limit (%)	<HydraulicRecloserLimitPercent>	Text box	This value specifies the percentage of phase trip limit for hydraulic reclosers (%). For example, if this value is set to 90% for a hydraulic recloser with a 600 Amp phase trip setting, the recloser flow is compared to the phase trip setting of the recloser multiplied by this recloser limit percentage ($0.90 * 600 = 540.0$ Amps). This option is used only if the Enable Hydraulic Recloser Phase Trip Limit check box is selected. The allowable range is 0-200.
Voltage Change Limits			
Line Voltage Change Limit	<LineVoltDropLimit>	Text box	This value specifies the voltage change threshold (%) for determining whether an excessive line voltage drop or rise has occurred. This check is performed on a per phase basis. Lines with an excessive voltage drop or rise are shown on the DPF Analyst/Debug tabular display. The allowable range is 0-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Auto Transformer Voltage Change Limit	<AutoTfVoltDropLimit>	Text box	This value specifies the voltage change threshold (%) for determining whether an excessive auto transformer voltage drop or rise has occurred. The voltage change is calculated as the difference between the average primary side voltage and the average secondary side voltage. • For three-phase auto transformers, the primary side average line to line voltage and secondary side average line to line voltage values are calculated. • For a single-phase auto transformer, the primary line to ground and secondary line to ground voltage values are used. Auto transformers with an excessive voltage drop/rise are shown on the DPF Analyst/Debug tabular display. The allowable range is 0-20.
Regulator Voltage Change Limit	<RegulatorVoltDropLimit>	Text box	This value specifies the voltage change threshold (%) for determining whether an excessive regulator voltage drop or rise has occurred. For three-phase regulators, the primary side average line to line voltage and secondary side average line to line voltage values are calculated. For a single-phase regulator, the primary line to ground and secondary line to ground voltage values are used. The voltage change is calculated as the difference between the average primary side voltage and the average secondary side voltage. Regulators with an excessive voltage drop/rise are shown on the DPF Analyst/Debug tabular display. The allowable range is 0-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Distribution Transformer Voltage Change Limit	<DistTfVoltDropLimit>	Text box	This value specifies the voltage change threshold (%) for determining whether an excessive distribution transformer voltage drop or rise has occurred. For two-phase and three-phase distribution transformers (banked or three-phase devices), the primary side average line to line voltage and secondary side average line to line voltage values are calculated. For a single-phase distribution transformer, the primary line to ground and secondary line to ground voltage values are used. The voltage change is calculated as the difference between the average primary side voltage and the average secondary side voltage. Distribution transformers with an excessive voltage drop/rise are shown on the DPF Analyst/Debug tabular display. The allowable range is 0-20.
Step Transformer Voltage Change Limit	<StepTfVoltDropLimit>	Text box	This value specifies the voltage change threshold (%) for determining whether an excessive step transformer voltage drop or rise has occurred. For three-phase step transformers, the primary side average line to line voltage and secondary side average line to line voltage values are calculated. For a single-phase step transformer, the primary line to ground and secondary line to ground voltage values are used. The voltage change is calculated as the difference between the average primary side voltage and the average secondary side voltage. Step transformers with an excessive voltage drop/rise are shown on the DPF Analyst/Debug tabular display. The allowable range is 0-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Power Transformer Voltage Change Limit	<PowerTfVoltDropLimit>	Text box	This value specifies the voltage change threshold (%) for determining whether an excessive power transformer voltage drop or rise has occurred. For three-phase power transformers, the primary side average line to line voltage and secondary side average line to line voltage values are calculated. For a single-phase power transformer, the primary line to ground and secondary line to ground voltage values are used. The voltage change is calculated as the difference between the average primary side voltage and the average secondary side voltage. Power transformers with an excessive voltage drop/rise are shown on the DPF Analyst/Debug tabular display. The allowable range is 0-20.
Reserved Capacity Options			
Adjust Limits Based on Reserved Capacity	<AdjustLimForReservedCapacity>	Check Box	This specifies whether reserved capacity commitments are used to de-rate limits used by the limit monitor function.
Only Consider Firm Commitments	<ReservedCapacityConsiderOnlyFirmCommitments>	Check Box	This specifies whether only firm reserved capacity commitments are considered for adjusting limits used by the limit monitor function. This option is effective only if the Adjust Limits Based on Reserved Capacity option is selected.
Voltage Imbalance Calculation Option			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Imbalance Calculation Option	<VoltageImbalanceCalculationOption>	Radio buttons	These buttons determine the calculation method for the three-phase load imbalance. • Maximum Voltage Difference from the Average Voltage - This radio button indicates that the three-phase load imbalance (in %) is calculated as the maximum voltage difference between each single-phase voltage and the average of the single-phase voltages. • Negative Sequence Voltage Divided by the Positive Sequence Voltage - This radio button indicates that the three-phase load imbalance (in %) is calculated as the negative sequence voltage divided by the positive sequence voltage. By default, the "Maximum Voltage Difference from the Average Voltage" method is used.
Check Before Operate			
Ferro-Resonance Check - Transformer Resistance Limit (p.u.)	<SwitchingValidationResonanceTfResistance>	Text box	This value specifies the transformer resistance limit (in per unit, on the transformer power base) for the "Ferro-resonance switching conditions" Check Before Operate (CBO) switching validation check. The "Ferro-resonance switching conditions" CBO violation will appear when changing status of a switching device upstream of a transformer with the resistance value lower than the specified resistance limit. The allowable range is 0-22.
Ferro-Resonance Check - Transformer Loading Percentage (%)	<SwitchingValidationResonanceTfLoadingPercentage>	Text box	This value specifies the transformer loading limit (in percent, relative to the full load) for the "Ferro-resonance switching conditions" Check Before Operate (CBO) switching validation check. The "Ferro-resonance switching conditions" CBO violation will appear when changing status of a switching device upstream of a transformer with the loading lower than the specified loading limit. The allowable range is 0-200.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Ferro-Resonance Check - Deviation of Line Capacitance to Transformer Magnetizing Reactance (%)	<SwitchingValidationResonanceLCRatio>	Text box	This value specifies a threshold (in percent) for the ratio of the total line capacitance between the switch and the transformer to the transformer magnetizing reactance for the "Ferro-resonance switching conditions" Check Before Operate (CBO) switching validation check. The "Ferro-resonance switching conditions" CBO violation will appear when changing status of a switching device upstream of a transformer with the absolute difference of the calculated ratio and 1.0 lower than the threshold value divided by 100. The allowable range is 0-50.
Old and Deprecated Parameters			
Function Name	NAME_LIM <Name>	Text box	This value specifies the function name in the DNAF Solution pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_LIM	Text box	This value represents the network analysis function Index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCENUM_LIM <FuncEnum>	Text box	This value indicates the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Enable Phase Trip Limit	ENABLEPHTRIP_LIM <EnablePhaseTripLimit>	Check box	This indicates whether to compare the DPF flow results against the percentage of phase trip limit for breakers and reclosers. The regular limits defined in the static mode in the switch model are not checked.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Recloser Limit	RECLOSERPHTRIPLIMITPERCENT_LIM <RecloserLimitPercent>	Text box	Percentage of phase trip limit for reclosers (for example, 90%). If phase trip limit checking is enabled, the flow of the recloser is compared to the phase trip setting of the recloser multiplied by this recloser limit percentage. This option is used only if the "Enable Phase Trip Limit" option is selected.

5.20.2. LA Configuration

Currently there are no user-configurable parameters for the Loss Analysis function. The LA configuration parameters used in the old DNAF Configurator are provided in table below.

Table 11. LA Configuration Parameters

Configurator Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Name	NAME_LA <Name>	Text box	This value specifies the function name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_LA	Text box	This value represents the network analysis function Index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCENUM_LA <FuncEnum>	Text box	This value indicates the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

5.20.3. PQI Configuration

Currently there are no user configurable parameters for the Power Quality Index Evaluation function. The PQI configuration parameters used in the old DNAF Configurator are provided in table below.

Table 12. PQI Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Name	NAME_PQI <Name>	Text box	This value specifies the function name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_PQI	Text box	This value represents the network analysis function Index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCENUM_PQI <FuncEnum>	Text box	This value indicates the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISIR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

6. Protection Validation

The Protection Validation (PRV) application analyzes distribution systems under fault conditions and provides basic information regarding system protection provided by circuit breakers, reclosers, and fuses. It analyzes whether these protection devices will operate correctly to protect the system under a full range of short circuit conditions. It should be noted that the sectionalizer is not considered as a protection device in the Protection Validation application.

PRV functionality includes verifying that protection devices will trip for all types of faults and verifying that device fault ratings are not exceeded. Options are provided to check backup protection for a PRV zone, check fuses, and check upstream coordination between adjacent devices. The impact of Distributed Energy Resources (DERs) with the capability to contribute fault current is also considered in the analysis.

6.1. Modeling and PRV Concepts

6.1.1. Model Input Data

The key data for breaker relays and reclosers includes the phase and ground overcurrent trip settings and the maximum fault interruption rating of the device. Note that the current values for overcurrent trip settings should represent the long-time inverse trip current and not instantaneous values.

The key data for fuses used in performing study mode only analyses is the nominal fuse rating. Additional data includes the largest fuse that will coordinate size for breakers and reclosers.

Note

The Protection Validation application performs ground fault calculations only if both the Ground Fault Capable and Ground Fault Enabled flags are true.

6.1.2. PRV Zone

Protection Validation (PRV) uses the unbalanced power system network model and divides the system into 'PRV zones'. A PRV zone represents the area of the system downstream of a single protective device (for example, breaker, recloser or fuse – for study mode) which does not contain any other protective devices. PRV considers the range of short-circuit currents that will occur for faults in the PRV zone and determines whether the protective device will trip for these conditions.

The protective devices used to connect generators and Distributed Energy Resources (DERs) to the grid, such as circuit breakers, are considered during the formulation of PRV zones. For the protective device connecting a generator, the corresponding PRV zone does not contain any other downstream protective devices. For the protective device connecting a DER (such as PV or wind), if the DER can contribute fault current, which means the DER remains connected during a fault, the

PRV zone of the protective device is evaluated in the same pattern as the generator; if the DER is disconnected from the grid during a fault, the protective device is not evaluated.

Protective devices such as reclosers which are currently set to act in a non-protection mode such as Switch Mode or Sectionalizer Mode are not used to define the PRV Zone boundaries.

For the protective device energized from two directions, for example a circuit breaker that has remote current infeed from other power sources (such as generators, distributed energy resources, or other feeders), its PRV zone is analyzed as a radial topology by opening the switches/breakers of the other power sources.

For a power transformer, there is a Transformer Self-Protection attribute (enumeration) that identifies alternative methods of protection, such as differential protection. These protection methods/devices may or may not be explicitly modeled in the ADMS. This self-protection attribute determines whether a power transformer's upstream PRV zone is extended downstream through the power transformer:

- If the power transformer has an alternative method of protection, the upstream PRV zone will not be extended downstream through the power transformer. In this case, the upstream protective device will not protect the power transformer nor the network downstream from the power transformer.
- If the transformer does not have an alternative method of protection, the upstream PRV zone will be extended downstream through the power transformer and stop when the next protective device (breaker, recloser, or fuse) is encountered. In this case, the upstream protective device will protect the power transformer and the network downstream from the power transformer.

Figure below shows a sample distribution substation. Station B is energized (sub-fed) from Station A. Recloser RCL in station A is upstream of power transformer XF in station B. Power transformer XF in sections (a) and (b) has self-protection; XF in section (c) does not have self-protection. Breaker CB is the feeder head breaker downstream of XF. In section (a), a breaker BKR is modeled upstream of power transformer XF in station B; it can be opened by the self-protection scheme of XF. In section (b), a switch SW is modeled upstream of power transformer XF in station B; it can be opened by the self-protection scheme of XF.

- Because power transformer XF in section (a) has an alternate method of protection, the PRV zone of BKR will not be extended downstream through XF. In this case, BKR will protect the network between BKR and XF, but not including XF. Three PRV zones will be formed for the PRV evaluation: one for RCL, one for BKR, and the third for CB.
- Because power transformer XF in section (b) has an alternate method of protection, the PRV zone of RCL will not be extended downstream through XF. In this case, RCL will protect the network between RCL and XF, but not including XF. Two PRV zones will be formed for the PRV evaluation: one for RCL and the other for CB. Switch SW is inside the PRV zone of RCL. If SW is not explicitly modeled, the same two PRV zones will be formed: one for RCL and the other for CB.
- Because power transformer XF in section (c) does not have an alternate method of protection, the PRV zone of RCL will be extended downstream through XF. In this case, RCL will protect the

network between RCL and CB, including XF.

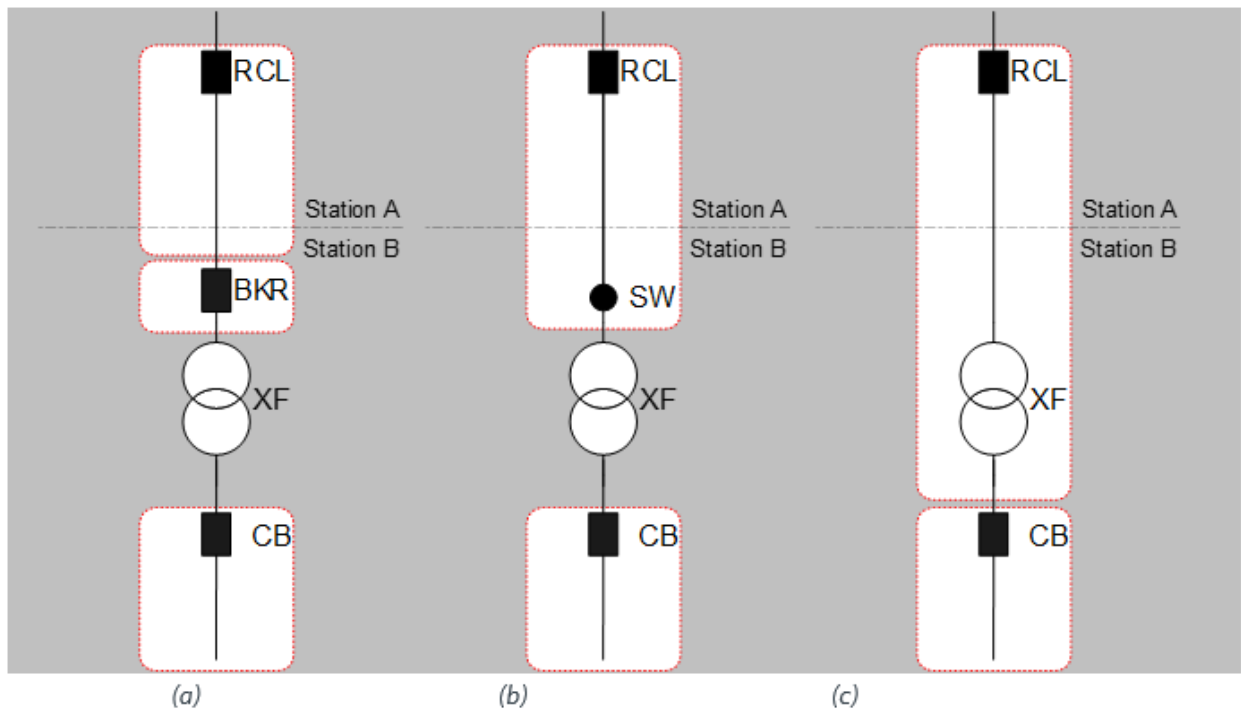


Figure 58. Network for Demonstrating Power Transformer Self-Protection PRV Zones

6.1.3. PRV Analysis

The PRV application calculates the fault current seen at the protective device, and not at a specific faulted location. Figure below shows a sample distribution substation. The normally open bus tie breaker, BUSTIE, is closed and the two power transformers, XF1 and XF2, are paralleled. Breakers CBF1, CBF2, CBF3, and CBF4 are the four feeder head breakers.

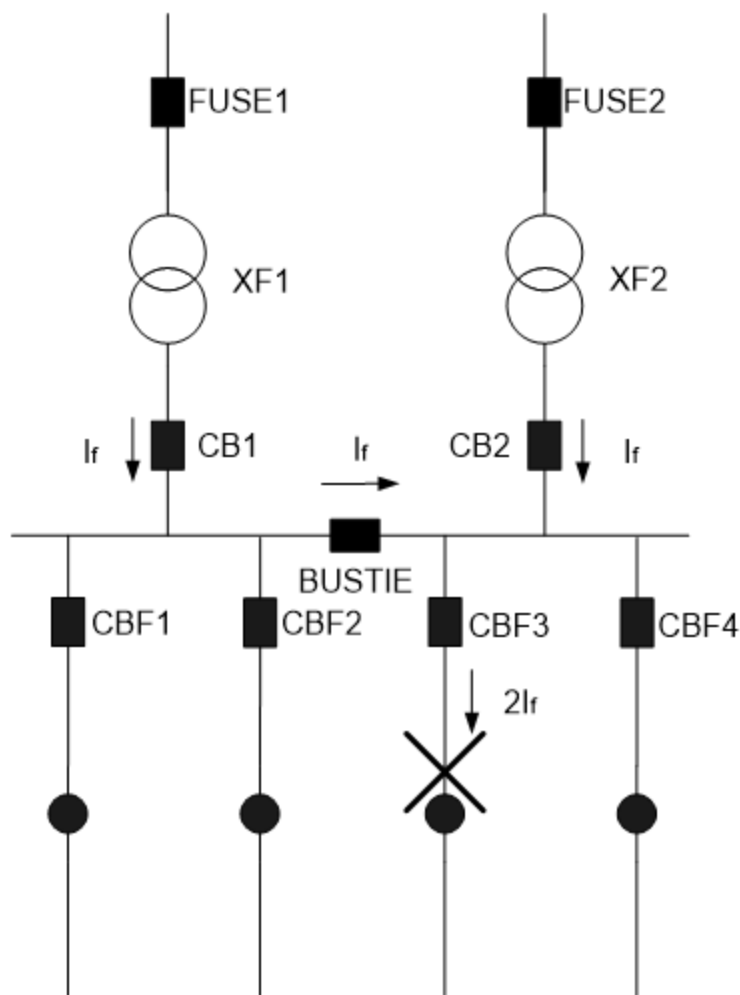


Figure 59. Fault Current with Bus Tie Closed

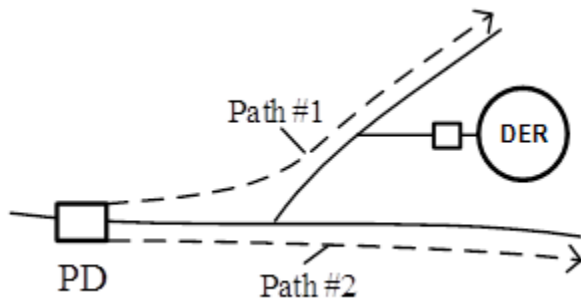
For a fault downstream of CBF3, as illustrated in figure above, the fault current flowing through CBF3 will be approximately double the fault current in CB1, CB2, and the BUSTIE breaker; the actual current seen by the protection device will be used during validation for breaker CBF3.

6.1.4. Impact of DER on a PRV Protection Zone

A DER that can provide substantial current during a short circuit fault can change the fault current pattern, and thus may change the maximum interruption current and minimum pickup current for a protective device. The impact on the current levels for a given protective device is determined by the DER's characteristics (such as the type of energy resource, such as solar or wind), electrical parameters (such as capacity and equivalent impedance), and location.

For a protective device, the highest possible fault current is obtained when a solid three-phase fault takes place at one of the terminals of the protective device. A DER may change the magnitude of the fault current at the protective device, and thus change the maximum interruption current. The lowest possible fault current is found at the point with largest equivalent impedance from the protective device. A DER in the protection zone may change the equivalent impedance as seen from

the protective device. For example, in the figure below, without the DER the equivalent impedance of the Path #1 for the PD (Protective Device) is slightly greater than that of Path #2. Considering the impact of the DER, the Path #2 end may become the location with the minimum fault current.



To consider the impact of the DERs on finding the minimum fault current point, the following steps are performed:

1. Find all the equivalent impedances starting from the protection zone at the protective device by assuming that all DERs are disconnected.
2. Modify the equivalent impedances using the equation below:

$$Z^{new} = Z^{old} \cdot 1 + \left(\sum_{i=1}^n \frac{Z_{s,g_i} \cdot Z_{f,g_i}}{Z_{g_i} \cdot (Z_{s,g_i} + Z_{f,g_i})} \right) \quad (\text{Equation 6.1})$$

Where:

Z^{new} and Z^{old} are the modified and original equivalent impedances.

i is the i^{th} DER while n is the total number of DERs.

Z_{s,g_i} represents the equivalent impedance from the i^{th} DER to the protective device while Z_{f,g_i} represents the equivalent impedance from the i^{th} DER to the candidate node, which is any node in this protection zone.

Z_{g_i} is the impedance of i^{th} DER.

3. Sort the modified equivalent impedance. The point with the minimum fault current is located at the node with the greatest equivalent impedance.

The impact of DER objects is not considered for fuses.

6.2. Basic Validation

Protection Validation considers a full range of fault conditions to detect unprotected or improperly protected zones in the network. The largest possible fault levels for a PRV zone are determined at the protective device at the start of the zone. The lowest possible fault levels are determined for the point in the PRV zone with the highest impedance (that is, furthest electrically) from the protective device.

For the maximum interruption validation, 11 different fault types (ABC, ABCG, ABG, BCG, ACG, AG, BG, CG, AB, BC, AC) are considered for three-phase devices. For non-three-phase devices, the appropriate subset of faults is simulated in the analysis. For example, for an A phase device, only an AG fault is considered; for a device with AB phases, the AG, BG, AB, and ABG fault types are evaluated.

For the minimum phase pickup validation, the AB, AC, and BC fault types are valid for three-phase devices. For two-phase devices, one L-L fault is considered. For example, the AB fault type is considered for AB phase devices.

For the minimum ground pickup validation, only ground fault types are considered. For example, the AB, AC, BC, and ABC fault types are not considered for three-phase devices. Likewise, an AB fault type is not considered for AB phase devices.

To allow improved performance, a configuration option exists on the PRV page of the DNAF Configuration Editor to "Simulate Minimum Number of Faults". When this option is selected, instead of running a full set of fault simulations (eleven fault types for three-phase devices and four fault types for two-phase devices), only a single type of fault is considered for each device (except for the minimum phase pickup calculation, which simulates all three LL faults):

- For three-phase devices a single ABC fault is simulated.
- For non-three-phase devices a single phase-ground fault is simulated.

Protection Validation performs five basic checks: maximum interruption, minimum phase pick-up, minimum ground pick-up, phase fault sensitivity, and ground fault sensitivity.

6.2.1. Maximum Interruption

If the largest fault current in the PRV zone exceeds the maximum interruption rating of the breaker or recloser (in Amps), a maximum interruption violation is reported. Note that a direct comparison is performed for this check - no safety factor is considered.

6.2.2. Minimum Phase/Ground Pickup

The smallest fault current in a PRV zone is determined for the device with the largest impedance (that is, furthest electrically from the protection device).

If the smallest fault current is less than the minimum pickup setting for the recloser or breaker (multiplied by a phase or ground trip safety factor), a Minimum Phase Pickup or a Minimum Ground Pickup violation is reported.

For phase fault checks a violation exists if:

$$I_{fmin_ph} < I_{ph_pickup} \cdot Safety_Factor_{ph} \quad (Equation 6.2)$$

For ground fault checks, a violation exists if:

$$I_{fmin_grnd} < I_{grnd_pickup} \cdot Safety_Factor_{grnd} \quad (Equation 6.3)$$

Where:

I_{fmin_ph} and I_{fmin_grnd} represent the minimum phase and ground fault currents within the PRV protection zone, respectively.

I_{ph_pickup} and I_{grnd_pickup} represent the phase and ground overcurrent pickup settings for the breaker or recloser in Amps, respectively.

$Safety_Factor_{ph}$ and $Safety_Factor_{grnd}$ represent the "Phase Trip Safety Factor" and "Ground Trip Safety Factor" configuration options for PRV, respectively, and are typically greater than 1.0 (for example, 1.5).

i Note

If no ground trip setting is provided in the static model or the ground trip capability is disabled, the function uses the switch size.

6.2.3. Phase/Ground Fault Sensitivity

Fault sensitivity values in a PRV zone are calculated for phase and ground faults. The phase fault sensitivity is the inverse of the minimum phase pick-up ratio. The ground fault sensitivity is the inverse of the minimum ground pick-up ratio.

Fault sensitivity is determined by calculating the ratio of minimum fault current to the minimum pick-up current (multiplied by a safety factor).

$$Minimum_Phase_Pick_Up_Ratio = 100\% \cdot \frac{I_{ph_pickup} \cdot Safety_Factor_{ph}}{I_{fmin_ph}} \quad (Equation 6.4)$$

$$Phase_Fault_Sensitivity = 100\% \cdot \frac{I_{fmin_ph}}{I_{ph_pickup} \cdot Safety_Factor_{ph}} \quad (Equation 6.5)$$

$$Minimum_Ground_Pick_Up_Ratio = 100\% \cdot \frac{I_{grnd_pickup} \cdot Safety_Factor_{grnd}}{I_{fmin_grnd}} \quad (Equation 6.6)$$

$$Phase_Fault_Sensitivity = 100\% \cdot \frac{I_{fmin_ph}}{I_{ph_pickup} \cdot Safety_Factor_{grnd}} \quad (Equation 6.7)$$

If a phase or ground fault sensitivity is less than or equal to 100.0%, a Phase Fault Sensitivity or Ground Fault Sensitivity violation is reported.

6.3. Real Time Options

6.3.1. Bound Protection Area by Fuses

The Bound Protection Area by Fuses option on the PRV page of the DNAF Configuration Editor

specifies whether the Protection Validation (PRV) zone is bounded by fuses. This option applies to the real-time analysis only.

If the Bound Protection Area by Fuses option is selected, during PRV calculations of the lowest fault current, the solution finds and calculates the lowest fault current utilizing a network location within the breaker or recloser PRV zone and does not evaluate locations downstream of a fuse.

If this option is not selected, the lowest fault current network location could be downstream of a fuse.

For a three-phase protective device, the minimum phase fault location (for calculating a line to line fault) is always on the three-phase network component. If a PRV zone has no three-phase fused lateral taps and contains only one or two phase laterals protected by fuses, the minimum phase fault location is on the feeder main backbone upstream of the lateral tap fuses no matter whether this option is selected or not. If a PRV zone has a three-phase lateral (with a tapped fuse), the minimum phase fault location could be along the lateral if this option is disabled.

6.4. Study Mode Options

6.4.1. Check Fuses

Fuses can be included in the protection validation in Study Mode by selecting the "Check Fuses" configuration option on the PRV page of the DNAF Configuration Editor. Fuses are never considered during the Real Time analysis.

A Minimum Pickup violation is created for a fuse if:

$$I_{min} < I_{rating} * 2 * Safety_Factor_{grnd}$$

I_{min} is the minimum ground fault current (Amps).

I_{rating} is the nominal rating of the fuse (Amps).

$Safety_Factor_{grnd}$ is the same Protection Validation configuration option described in the previous section.

6.4.2. Backup Protection Option

The "Check Backup Protection" option on the PRV page of the DNAF Configuration Editor applies to Study Mode analysis only. It provides an option to consider upstream (that is, backup) protection when determining whether a PRV zone is protected.

If the protective device at the head of a PRV zone does not provide adequate protection (that is, a violation is found when performing the basic checks described in [Basic Validation](#)), the algorithm traces upstream and finds any other protection devices between the fault and the source. Each device is checked in turn to determine if it will protect the zone. If these devices also cannot protect

the PRV zone, a violation is reported.

6.4.3. Upstream Device Coordination

The "Check Upstream Device Coordination" option on the PRV page of the DNAF Configuration Editor applies to Study Mode analysis only. It provides an additional check to ensure that the settings for a protection device coordinate with those of another adjacent protection device further upstream.

To validate the coordination of protective devices, the Protection Validation performs a comparison of fuse minimum trip ratings, protective device minimum trip points, and the Largest Fuse That Will Coordinate value. The fuse minimum trip rating is set as twice the fuse size.

Protection coordination starts from the furthest downstream PRV zone. Each protection device's minimum trip point (or rating for fuses in study mode in Amps) is compared to the next upstream (toward the feeder breaker) protection device's minimum trip point or rating. If the downstream device's minimum trip point or rating is not less than the upstream protection island supply-side protection device's minimum trip point or rating by a configurable "Coordination Factor", the two protective devices are considered to be uncoordinated.

If the downstream protection device is a fuse, and the upstream device has an adjustable trip point (recloser, breaker, etc.), the fuse rating must be less than or equal to the Largest Fuse That Will Coordinate value set in the static model or the two protection devices are considered to be uncoordinated.

When comparing devices with adjustable settings (breakers and reclosers), if the downstream device has a ground trip setting and the upstream device does not, the ground trip setting for the downstream device is compared against the upstream device's phase trip setting multiplied by the coordination factor (set in the DNAF Configuration Editor, PRV page, Coordination Factor field).

If the downstream device does not have a ground trip setting, but the upstream device does have one, then the downstream device's phase trip setting is compared to the upstream device's ground trip setting (multiplied by the coordination factor).

Figure below represents an example of a feeder with a breaker, a three-phase recloser, a two-phase recloser, a single-phase recloser with no ground trip enabled, and 3 fuses. The data for the breakers and reclosers includes the phase trip current, the ground trip current, and the largest fuse that will coordinate. The PRV zones are shown as dotted red lines.

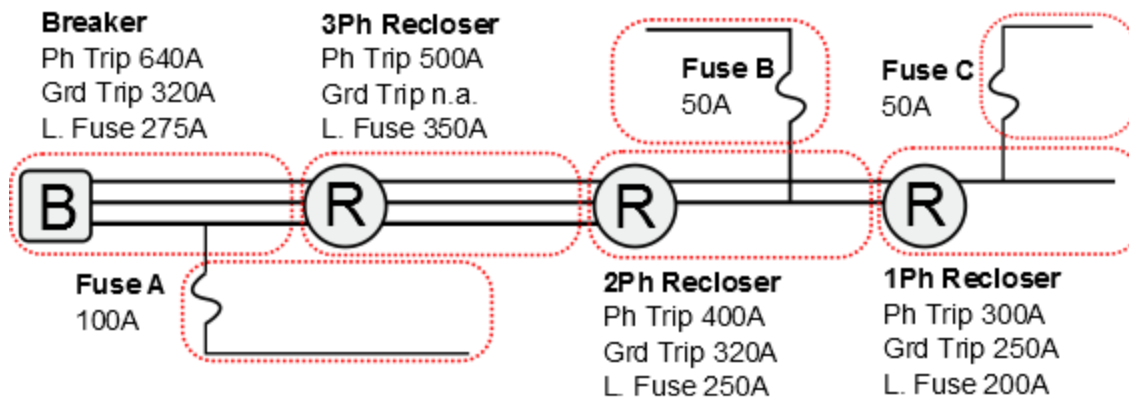


Figure 60. PRV Zones

The calculations in the table below are based on the sample feeder in figure above and use a Coordination Factor of 0.8.

Table 13. PRV Coordination

Device	Upstream Device	Comparison Type	Comparison	Result
Fuse C	1Ph Recloser	Rating / L Fuse	$2 * 50 < 200$	Ok
Fuse B	2Ph Recloser	Rating / L Fuse	$2 * 50 < 250$	Ok
Fuse A	Breaker	Rating / L Fuse	$2 * 100 < 275$	Ok
1Ph Rec	2Ph Recloser	Gr Trip / Gr Trip	$250 < 320 * 0.8$	Ok
2Ph Rec	3Ph Recloser	Gr Trip / Ph Trip	$320 < 500 * 0.8$	Ok
3Ph Rec	Breaker	Ph Trip / Gr Trip	$500 < 320 * 0.8$	Fail

6.5. Fault Impedance

The fault impedance configurations allow setting the value of the short circuit impedance to be applied to the fault. For PRV short circuit calculations, the fault impedance is applied when calculating both the minimum and maximum short-circuit current for phase-to-phase and phase-to-ground faults (Minimum Phase/Ground Pickup and Maximum Phase/Ground Pickup).

For all seven fault types (ABC, ABG, BCG, ACG, AG, BG, and CG), the fault impedance is inserted for each phase separately, between the phase and ground, or between the faulted phases. The examples of how fault impedance is inserted for different types of faults are shown in figure below.

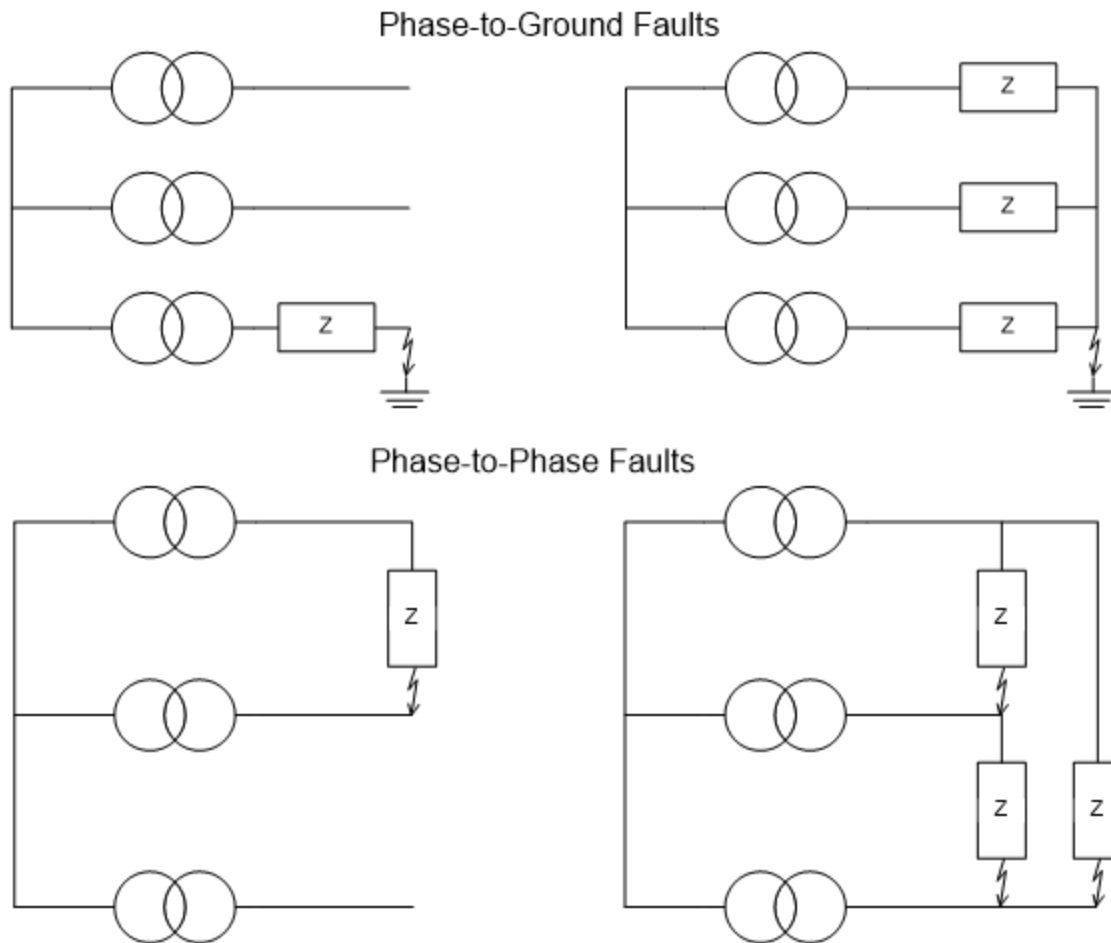


Figure 61. Fault Impedance

The PRV application is able to calculate fault currents using a bolted fault and/or a user-configured fault impedance. PRV violations are determined as follows:

- When checking for a maximum interruption violation which is based on the calculated maximum capacity ratio, the largest fault current calculated from either the bolted fault or the user-configured fault impedance in the PRV zone is used.
- When checking for a minimum phase pick-up violation which is based on the calculated minimum phase pick-up ratio (the phase fault sensitivity is the inverse of this ratio), the smallest phase-to-phase fault current calculated from either the bolted fault or the user-configured fault impedance in the PRV zone is used.
- When checking for a minimum ground pick-up violation which is based on the calculated minimum ground pick-up ratio (ground fault sensitivity is the inverse of this ratio), the smallest phase-to-ground fault current calculated from either the bolted fault or the user-configured fault impedance in the PRV zone is used.

6.6. Fault Current Contributions

For information about the fault current contribution characteristics of different energy sources, see [Fault Current Contributions](#).

6.7. PRV Check Before Operate

The Check Before Operate (CBO) functionality issues notifications related to potential protection settings violations when the operator attempts to issue a switching action.

If the Enable Protection Validation option is enabled in the Viewer Configuration Editor, then the ADMS client sends a request to the ADMS server to solve the PRV application for the network configuration resulting from the proposed switching action. The ADMS server solves PRV and notifies the operator if a relay setting violation could occur because of the switching action. The CBO solution also reports the following details:

- All protective devices impacted due to the switching action.
- For each impacted protective device, whether the following violations were detected:
 - Maximum interruption violation and associated fault current.
 - Minimum phase pick-up violation and associated fault current.
 - Minimum ground pick-up violation and associated fault current.

6.8. PRV Alarms

Table below lists the supported PRV alarms. The alarms can be enabled or disabled using the DNAF System Configuration display. For more information on Alarm naming, see [DNAF Alarm Format](#).

Table 14. PRV Alarms

Alarm	Description
PRV Failed to Complete	If PRV fails to solve for a station (due to a base case power flow that is not able to converge), this alarm is issued for the station. Note that this alarm is issued for the first time PRV diverges after a successful PRV solution was obtained for the station. Subsequent diverged PRV solutions do not generate this alarm until a valid PRV solution is obtained and the next PRV fails to solve for that station.
PRV Failed to Complete (Return to Normal)	When PRV solves successfully after a diverged solution for a station, this alarm is issued for that station.
Protection Issue Detected	If PRV detects a PRV zone protection issue for a station, this alarm is issued for that station. Note that this alarm is issued for the first time a PRV zone protection issue is detected after a PRV solution without any violation or protection zone issue. Subsequent PRV zone protection issue situations for the station do not generate an alarm until a PRV solution without any PRV zone protection issues is found for that station and the next PRV solution detects a PRV zone protection issue.

Alarm	Description
Protection Issue Detected Return to Normal	When PRV solves successfully and there is no longer a PRV zone protection issue detected, this alarm is issued for the protection device (breaker or recloser).

6.9. PRV Configuration

The Protection Validation function calculates phase-to-ground, double-phase-to-ground, and three-phase fault currents for each application protection zone, where an application protection zone is defined by feeder circuit breakers and field reclosers.

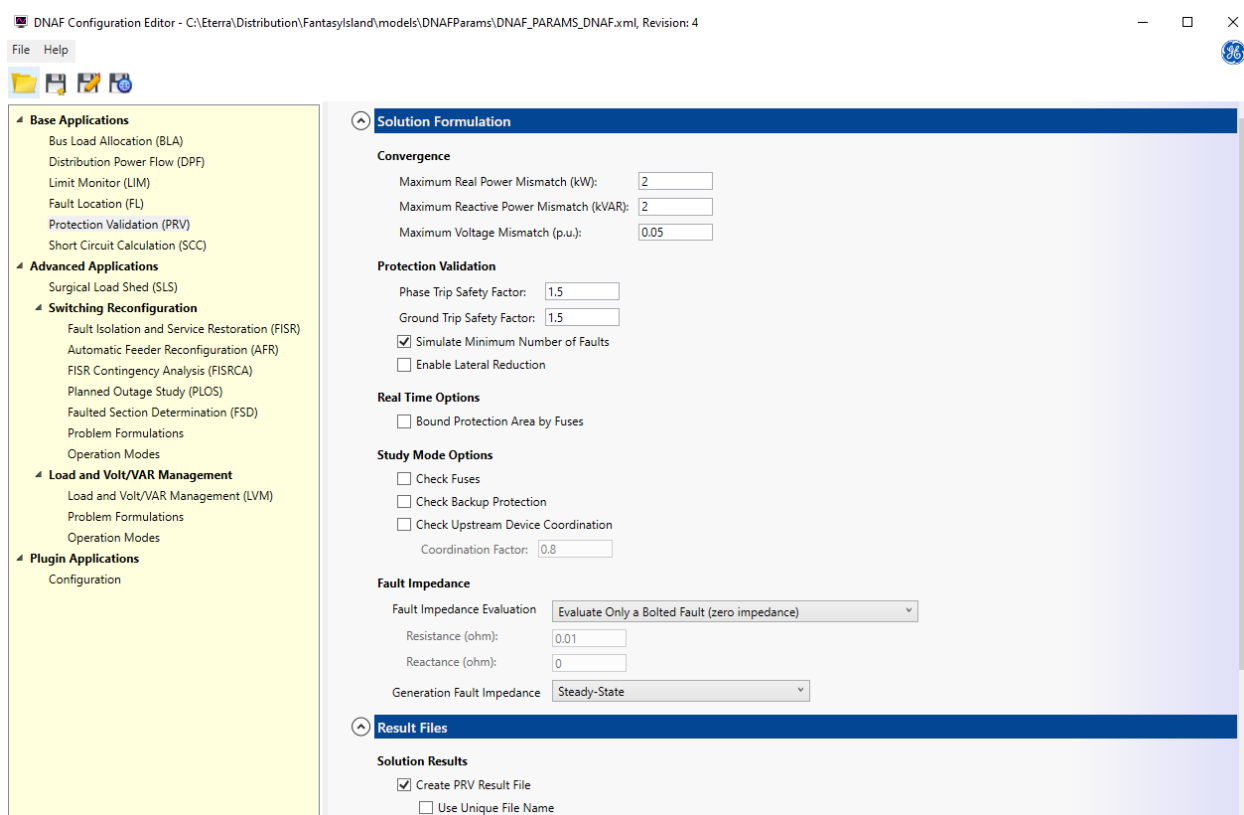


Figure 62. Protection Validation Configuration

Table below lists the PRV configuration attributes. All listed parameters are contained by the <Params xsi:type = "PRVParams"> element in the DNAF_PARAMS.xml configuration file.

Table 15. PRV Configuration Attributes

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Convergence			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Real Power Mismatch	PMISMATCH_SC <PMismatch>	Text box	This value specifies the internal power flow's real power convergence criterion in kW. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The PRV solution status is set to Diverged if any load's real power difference between DPF iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Reactive Power Mismatch	QMISMATCH_SC <QMismatch>	Text box	This value specifies the internal power flow's reactive power convergence criterion in kVAR. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The PRV solution status is set to Diverged if any load's reactive power difference between DPF iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Voltage Mismatch	VMATCH_SC <VMismatch>	Text box	This value specifies the internal power flow's voltage convergence criterion in per unit. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. The PRV solution status is set to Diverged if the voltage mismatches are larger than this specified value. The allowable range is 0-0.1.

Protection Validation

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Phase Trip Safety Factor	<PhaseTripSafetyFactor>	Text box	This value specifies a safety factor used to calculate the minimum phase fault pick-up limit. The larger the factor, the lower a device setting must be to pick up for a given fault. A value of 1.0 would provide no safety factor. A value of 1.5 is typical. The allowable range is 0.1-10.
Ground Trip Safety Factor	<GroundTripSafetyFactor>	Text box	This value specifies a safety factor used to calculate the minimum ground fault pick-up limit. The larger the factor, the lower a device setting must be to pick up for a given fault. A value of 1.0 would provide no safety factor. A value of 1.5 is typical. The allowable range is 0.1-10.
Simulate Minimum Number of Faults	SIMMINFAULTS_SC <SimMinFault>	Check box	This indicates whether the minimum number of faults should be simulated. Selecting this option improves the performance by reducing the number of fault simulations that must be performed.
Real Time Options			
Bound Protection Area by Fuses	BOUNDPROTECTION_SC <BoundProtectionAreaFuses >	Check box	This indicates whether the Protection Validation (PRV) zone is bounded by fuses. If this option is checked, the PRV analysis calculates the lowest fault current utilizing a network location within the breaker or recloser PRV zone; it does not evaluate locations downstream of a fuse. If this option is not checked, the lowest fault current network location could be downstream of a fuse.
Study Mode Options			
Check Fuses	INCLUDEFUSE_SC <IncludeFuse>	Check box	This indicates whether fuses should be included in the protection validation checks (in study mode analyses only) The fuses are also included in the protection coordination check if the Check Upstream Device Coordination option is selected.
Check Backup Protection	CHECKUPSTREAM_SC <CheckUpstream>	Check box	This indicates whether backup protection devices upstream of the current PRV application protection zone should be considered in addition to the device at the head of the PRV application protection zone. A violation is raised only if no upstream protection device can provide adequate protection for the PRV application protection zone.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Check Upstream Device Coordination	PROTECTIONCOORD_SC <ProtectionCoord>	Check box	This indicates whether coordination between protection devices should be checked. This provides an evaluation of whether a device (recloser, breaker, or optionally a fuse in study mode) coordinates with the next upstream protective device.
Coordination Factor	COORDFACTOR_SC <CoordFactor>	Text box	This value specifies the safety factor used when checking protection coordination between devices. The smaller the factor, the larger the required difference between device set points. A value of 1.0 would provide no current difference between adjacent devices. A typical value is 0.8. The allowable range is 0.1-5.
Fault Impedance			
Fault Impedance Evaluation	<FaultImpEvalType>	Drop-down list	This drop-down allows setting the fault impedance evaluation values that are evaluated in the PRV solution. There are three options: • Evaluate Only a Bolted Fault (zero impedance). If this option is selected, the PRV application solves for the short circuit fault current values using a bolted fault impedance (resistance and reactance values that are approximately zero ohm each). • Evaluate Only the User-Configured Fault Impedance. If this option is selected, the PRV application solves for short circuit fault current values using the user-configured fault impedance that are specified in the Resistance (ohm) and Reactance (ohm) parameters. • Evaluate a Bolted Fault and the User-Configured Fault Impedance. If this option is selected, the PRV application solves for the short circuit fault current using both a bolted fault impedance (resistance and reactance values that are approximately zero ohm each) and a user-configured fault impedance that is specified by the Resistance (ohm) and Reactance (ohm) parameters.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Resistance	<FaultImpResistance>	Text box	Short circuit fault resistance in Ohms. This text box is disabled when the Fault Impedance Evaluation option is set to "Evaluate only a bolted fault (zero impedance)". The allowable range is 0.0001-200.0.
Reactance	<FaultImpReactance>	Text box	Short circuit fault reactance in Ohms. This text box is disabled when the Fault Impedance Evaluation option is set to "Evaluate only a bolted fault (zero impedance)". The allowable range is 0.0-20.0.
Generation Fault Impedance	<GenerationFaultImpType>	Drop-down list	This parameter specifies generation fault impedance values that are used by the PRV solution. • 0: Steady-state: Synchronous Reactance (Xd) determines the current flow after steady-state condition is reached. • 1: Transient: Transient Reactance (X'd) determines the current flow after several cycles. This condition lasts from 0.5 to 2 seconds. • 2: Sub-transient: Sub-transient Reactance (X''d) determines the current flow during the first cycles after a fault occurs. This condition lasts for approximately 0.1 seconds. The generation fault impedance values are defined in the static model.
Solution Results			
Create PRV Result File	PRINTOUT_SC <PrintSCResult>	Check box	This indicates whether PRV results should be written to a result file. There is a separate file for each station. Only one file is kept for each station (that is, it is overwritten after each solution) unless the Create Unique Log option is selected.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Unique File Name	UNIQUELOG_SC <UniqueLog>	Check box	This indicates whether a unique PRV result file is generated for each PRV solution (that is, whether to overwrite the previously created file). If this option is selected, each result file includes a time stamp as part of the filename. If this option is left on for an extended period, a large number of files can be generated, potentially causing disk space issues.
Debug Files			
Maximum Number of Debug Files	MAXUPFOUTPUTFILE_SC <MaxNumUpfOutFile>	Integer	This value specifies the maximum number of PRV debug files that can be generated for a given station. When PRV does not converge, two debug files are created. The first file contains the prefix "upfin_export" along with the station name, time, and date. The second file contains the prefix "alg_export" along with station name, time, and date. These debug files contain the system status at the PRV execution time and the parameters used in the solution. The allowable range is 0-50.
Manual Execution			
Allow in Real Time	REALTIME_SC <RealTime>	Check box	This indicates whether the application is shown in the RT DNAF Solution pop-up display. (PRV can be periodically triggered and manually triggered in RT.)
Allow in Study Mode	STUDYMODE_SC <Study>	Check box	This indicates whether the PRV application can be invoked through the ST DNAF Solution pop-up display (selected means the application is shown).
Old and Deprecated Parameters			
Function Name	NAME_SC <Name>	String	Function display name – does not appear in the UI. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_SC	String	Function index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Enumeration	FUNCTENUM_SC <FuncEnum>	Integer	This value specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISIR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

7. Short Circuit Calculation

It is important to study the distribution system under fault conditions for system protection, distribution planning, electric device design, and so on. The Short Circuit Calculation (SCC) application simulates different kinds of fault types for selected locations and determines the resulting short-circuit current values. The SCC application can be executed in real-time mode or study mode.

The application is triggered by manual requests.

7.1. Methodology

The Short Circuit Calculation application uses the unbalanced power system network model to calculate the fault currents. It simulates all types of fault conditions at a user-selected location based on the phasing configuration.

The generators and distributed energy resources, such as solar, wind, and batteries, are supported in the short-circuit calculation. The generator is modeled as a synchronous machine. The energy resources that can contribute to fault currents are modeled in the calculation; the other energy resources, which cannot contribute to fault current, are not modeled and an open circuit is placed in their place. The calculation results show the phase currents flowing into the faulted location.

The fault impedance configurations in the DNAF Configuration Editor allow setting the short circuit resistance and reactance values to be applied to the fault during the short circuit calculation. These values are applied to any type of fault.

7.1.1. Fault Simulation

Short Circuit Calculation simulates bolted three-phase, phase-to-phase-to-ground, line to line, and phase-to-ground faults for each selected location. For a three-phase switch, fuse, recloser, sectionalizer, breaker, transformer (power, regulator, step, and auto), and line, the following eleven fault types are simulated:

- ABCG
- ABC
- ABG
- BCG
- ACG
- AG
- BG
- CG
- AB

- BC
- AC

For two-phase devices (for example, lines), phase to ground, phase-to-phase, and two phase-to-ground faults are simulated. For a single-phase line, a single phase-to-ground fault is simulated.

7.2. Fault Current Contributions

This section provides a high-level summary of the fault current contribution characteristics of different energy sources.

7.2.1. Synchronous Generator

If a short circuit is applied at the terminal of a synchronous machine, the current starts out very high and decays to a steady-state value. Synchronous machines generally deliver about six times the rated current for several cycles before decaying to between 200 - 400% of rated current. In a synchronous machine, the field current is supplied by an external DC source. This external source continues to supply voltage to the field windings of the generator. The prime mover continues to drive the rotor that produces the required induced voltage in the stator winding, which in turn supplies a continuous fault current. The steady-state short-circuit current value persists unless interrupted by a switching device, such as a recloser or a circuit breaker.

As short circuit current continues flowing in the circuit, the machine's impedance increases due to the increase in winding temperature. This helps the AC envelope to decay faster. The industry has established three reactance variables:

- **Sub-transient Reactance (X''_d):** Determines the current flow during the first cycles after a fault occurs. This condition lasts for approximately 0.1 seconds.
- **Transient Reactance (X'_d):** Determines the current flow after several cycles. This condition lasts from 0.5 to 2 seconds.
- **Synchronous Reactance (X_d):** Determines the current flow after steady-state condition is reached.

A global configuration option "Generation Fault Impedance" in the DNAF Configuration Editor indicates which impedance values (sub-transient, transient, or steady-state) are used in the SCC application.

7.2.2. Inverter Coupled DERs

Inverters dynamically behave differently from synchronous machines. Inverters do not have a rotating mass component; therefore, they do not develop inertia to carry fault current based on an electro-magnetic characteristic. Power electronic inverters have a much faster decaying envelope

for fault currents because the devices lack predominately inductive characteristics that are associated with rotating machines. These characteristics dictate the time constants involved with the circuit. Inverters also can be controlled in a manner unlike rotating machines because they can be programmed to vary the length of time it takes them to respond to fault conditions. This also impacts the fault current characteristics of the inverter.

Typically, inverter-based DERs are designed to act as ideal current sources. Therefore, they provide minimal fault current contributions and have little effect on overcurrent protection and coordination strategies for fuses and circuit breakers at low penetration levels. However, this is not always true with increased DER penetration (10% or more of system capacity).

The following are typical characteristics of inverter-coupled DER devices:

- The inverter current contribution during a fault is not zero and varies by design.
 - For most fault conditions, inverters can supply current to the feeder, subsequent to a fault for a period ranging from four to ten cycles.
 - The current contribution level is a function of the voltage at the terminal of the PV inverter during a fault, which is determined by the fault type and location.
 - For most vendor-specific inverter models, the inverter current during the fault is above the rated inverter current and may reach 120% of the inverter-rated current.
 - For a few inverter models, the inverter current during faults is maintained at the pre-fault inverter current.
 - In one inverter model, the inverter current drops to zero and the inverter is disconnected in less than 0.5 cycles for a fault case in which the terminal voltage reached below 50% on any phase.

In the static model, a field defined in the Energy Resource Model determines if an energy resource can contribute fault current. This field is respected in the SCC application. For inverters that have a fast disconnect capability (for example, inverters that disconnect in less than a cycle), this value should be set to false. For large inverters that produce infeed during the sub-transient period (1-6 cycles), this field should be set to true.

The SCC accounts for the inverter-coupled DERs based on their specified short circuit output. This is done by modeling inverter fault behavior depending on the voltage. A set of voltage ranges and their associated fault current levels are defined for an Energy Resource Model in the global.csv file. The following repeating fields have been added to the static model:

- NUMSCVOLTAGERANGES_ENERGYRESMOD – The number of voltage ranges.
- HIGHVOLTRANGEn_ENERGYRESMOD – High voltage value per unit V (for example, 0.90 per unit).
- LOWVOLTRANGEn_ENERGYRESMOD – Low voltage value per unit V (for example, 0.80 per unit).
- SCCURRENTLEVELn_ENERGYRESMOD – Fault current as a rated multiplier of the rated output (for example, 1.2 times rated output).

Based on the voltage that is calculated when simulating a fault at a selected location, the short

circuit contribution from the DERs is added to the original short circuit value.

7.3. Arc Suppression Coils

Arc suppression coils provide a power transformer grounding reactance; they are used to control the magnitude of ground fault current. ASCs can be connected directly to the secondary neutral of transformers with a secondary Wye or Zig-Zag winding. They can also be connected to the neutral or secondary of a dedicated grounding transformer (Delta-Wye or Zig-Zag respectively). ASCs are typically used on 3-wire systems where ground fault currents can be very small.

7.3.1. Fixed ASCs

Fixed grounding coils are modeled implicitly as the reactive component of the transformer secondary grounding impedance. This impedance is specified as part of the static transformer model.

7.3.2. Regulating ASCs

Regulating ASCs are designed to allow the coil reactance to vary to keep fault currents at a minimum value as system topology changes. Specifically, the ideal configuration is to set the impedance so that reactive power through the coil will cancel out the capacitive power generated by lines attached to the power transformer.

$$Y_t = \sum_{Lines} Y_{line} \text{ (Siemens)} \quad \text{(Equation 7.1)}$$

The total capacitive admittance is the sum of the individual line-ground reactive admittances of each line (both underground and overhead are included). Y_{line} represents the average admittance of the 3 individual phases.

$$X_{ASC} = \frac{-1}{3Y_t} \text{ (Ohms)} \quad \text{(Equation 7.2)}$$

The target reactance of the coil is determined from the total capacitive admittance.

To ensure that the reactance of the coil always matches the capacitance of the attached lines, the reactance of regulating ASCs is updated every time DPF runs. This ensures that topology changes, which trigger power flow solutions, will always be captured and the ASC reactance will always match the current topology.

The regulating state of an ASC (on/off) is set in the static model and can be overridden manually.

The regulation respects the maximum and minimum allowed values for ASC reactance (Ohms),

which are specified in the static model for the transformer. For example, if the calculated grounding impedance is greater than the ASC default maximum grounding impedance, the default maximum grounding impedance is applied.

7.4. Short Circuit Calculation Results

Short Circuit Calculation can only be triggered manually. The solution results can be viewed on the SCC Summary tabular display, which can be accessed from the Station Output Summary display and from the DNAF Solution display.

An SCC result file is generated whenever SCC is triggered. This file provides a solution summary, application configurations, and short circuit calculation results. For more information, refer to [SCC Result File](#).

7.5. SCC Configuration

This section describes the Short Circuit Calculation configuration parameters.

The screenshot shows the DNAF Configuration Editor window. The title bar indicates the file path: D:\Eterra\Distribution\FantasyIsland\models\DNAFParams\DNAF_PARAMS.xml, Revision: 30. The left sidebar contains a tree view with the following categories:

- Base Applications
 - Bus Load Allocation (BLA)
 - Distribution Power Flow (DPF)
 - Limit Monitor (LIM)
 - Fault Location (FL)
 - Protection Validation (PRV)
 - Short Circuit Calculation (SCC)
- Advanced Applications
 - Surgical Load Shed (SLS)
 - Switching Reconfiguration
 - Fault Isolation and Service Restoration (FISR)
 - Automatic Feeder Reconfiguration (AFR)
 - FISR Contingency Analysis (FISCA)
 - Planned Outage Study (PLOS)
 - Problem Formulations
 - Operation Modes
 - Load and Volt/VAR Management
 - Load and Volt/VAR Management (LVM)
 - Problem Formulations
 - Operation Modes
 - Plugin Applications
 - Configuration

The main panel displays the configuration for the Short Circuit Calculation (SCC) application, organized into sections:

- Solution Formulation**
 - Convergence**
 - Maximum Real Power Mismatch (kW): 2
 - Maximum Reactive Power Mismatch (kVAR): 2
 - Maximum Voltage Mismatch (p.u.): 0.05
 - Fault Impedance**
 - Fault Impedance Evaluation: Evaluate Only a Bolted Fault (zero impedance)
 - Resistance (ohm): 0.01
 - Reactance (ohm): 0
 - Generation Fault Impedance: Steady-State
- Result Files**
 - Solution Results**
 - ☒ Create SCC Result File
 - ☐ Use Unique File Name
- Miscellaneous**
 - Manual Execution**
 - ☒ Allow in Real Time
 - ☒ Allow in Study Mode

Figure 63. Short Circuit Calculation Configuration

Table below lists the SCC configuration attributes. All listed parameters are contained by the <Params xsi:type = "SCCParams"> element in the DNAF_PARAMS.xml configuration file.

Table 16. SCC Configuration Attributes

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Convergence			
Maximum Real Power Mismatch	PMISMATCH_SCC <PMismatch>	Text box	This value specifies the internal power flow's real power convergence criterion in kW. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The SCC solution status is set to Diverged if any load's real power difference between DPF iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Reactive Power Mismatch	QMISMATCH_SCC <QMismatch>	Text box	This value specifies the internal power flow's reactive power convergence criterion in kVAR. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The SCC solution status is set to Diverged if any load's reactive power difference between DPF iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Voltage Mismatch	VMATCH_SCC <VMismatch>	Text box	This value specifies the internal power flow's voltage convergence criterion in per unit. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and interface. The SCC solution status is set to Diverged if the voltage mismatches are larger than this specified value. The allowable range is 0-0.1.
Fault Impedance			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Fault Impedance Evaluation	<FaultImpEvalType>	Drop-down list	This drop-down allows setting the fault impedance evaluation values that are evaluated in the SCC solution. There are three options: • Evaluate Only a Bolted Fault (zero impedance). If this option is selected, the SCC application solves for the short circuit fault current values using a bolted fault impedance (resistance and reactance values that are approximately zero ohm each). • Evaluate Only the User-Configured Fault Impedance. If this option is selected, the SCC application solves for short circuit fault current values using the user-configured fault impedance that are specified in the Resistance (ohm) and Reactance (ohm) parameters. • Evaluate a Bolted Fault and the User-Configured Fault Impedance. If this option is selected, the SCC application solves for the short circuit fault current using both a bolted fault impedance (resistance and reactance values that are approximately zero ohm each) and a user-configured fault impedance that is specified by the Resistance (ohm) and Reactance (ohm) parameters.
Resistance	<FaultImpResistance>	Text box	Short circuit fault resistance in Ohms. The allowable range is 0.0001-200.
Reactance	<FaultImpReactance>	Text box	Short circuit fault reactance in Ohms. The allowable range is 0.0001-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Generation Fault Impedance	<GenerationFaultImpType>	Drop-down list	This parameter specifies generation fault impedance values that are used by the SCC solution. • 0: Steady-state: Synchronous Reactance (Xd) determines the current flow after steady-state condition is reached. • 1: Transient: Transient Reactance (X'd) determines the current flow after several cycles. This condition lasts from 0.5 to 2 seconds. • 2: Sub-transient: Sub-transient Reactance (X''d) determines the current flow during the first cycles after a fault occurs. This condition lasts for approximately 0.1 seconds. The generation fault impedance values are defined in the static model.

Solution Results

Create SCC Result File	PRINTOUT_SCC <PrintSCCResult>	Check box	This indicates whether SCC results should be written to a result file. There is a separate file for each station. Only one file is kept for each station (that is, it is overwritten after each solution) unless the Create Unique Log option is checked.
Use Unique File Name	UNIQUELOG_SCC <UniqueLog>	Check box	This indicates whether a unique SCC result file is generated for each SCC solution (that is, whether to overwrite the previously created file). If this option is selected, each result file includes a time stamp as part of the filename. If this option is left on for an extended period, a large number of files can be generated, potentially causing disk space issues.

Manual Execution

Allow in Real Time	REALTIMEMODE_SCC <RealTime>	Check box	This indicates whether the SCC application can be invoked through the RT DNAF Solution pop-up display (selected means the application is shown).
Allow in Study Mode	STUDYMODE_SCC <Study>	Check box	This indicates whether the SCC application can be invoked through the ST DNAF Solution pop-up display (selected means the application is shown).

Old and Deprecated Parameters

Function Name	NAME_SCC	String	Function display name. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
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Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Index	INDEX_SCC	Integer	Function Index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCTENUM_SCC	Integer	This value specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISIR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

8. Fault Location

The Fault Location (FL) function utilizes fault Amp measurements and faulted phase status measurements on devices to determine potential fault locations along distribution feeders. Any device that can provide fault current Amp measurements and faulted phase information can be a triggering device and provide input for the FL application.

There are two different algorithms that can be configured to solve Fault Location. The Radial algorithm is optimized for radial circuits and solves more quickly. The Meshed algorithm takes longer to solve but solves for cases with distributed generation and looped topology.

8.1. Radial FL Algorithm

Historically, short-circuit current calculations for unbalanced faults in a normally balanced three-phase system have been performed using symmetrical components analysis. Due to the unbalanced nature of a distribution network this method generally does not work very well.

To overcome the issues with using symmetrical components, a Thevenin equivalent phase-frame circuit method is employed. For more information about this calculation method, see *Distribution System Modeling and Analysis* by William H. Kersting (CRC Press, 2002). The Radial FL algorithm was written based on this Thevenin equivalent approach.

Note that generators and distributed energy resources may contribute to fault currents. Looped and paralleled circuits will also have a significant effect on fault currents and fault location. The impact of these is not considered in the Radial algorithm. This is a known limitation addressed by the Meshed algorithm described below.

The short-circuit currents immediately downstream of the system source voltage are often determined from a transmission system short-circuit study. With the results from the transmission analysis in terms of three-phase and single-phase short-circuit MVA values, the positive and zero sequence impedances of the equivalent system can be determined as follows (kVLL is the nominal line-to-line voltage in kV of the transmission system):

$$Z_+ = \frac{kVLL^2}{MVA_{3-phase}} \Omega \quad (\text{Equation 8.1})$$

$$Z_0 = \frac{3 \cdot kVLL^2}{(MVA_{1-phase})^*} - 2Z_+ + \Omega \quad (\text{Equation 8.2})$$

The positive and zero sequence impedances can then be translated into the phase impedance matrix by using this equation:

$$Z_{phase} \cong [A_s] \cdot [Z_{seq}] \cdot [A_s]^{-1} \quad (\text{Equation 8.3})$$

Where:

$$Z_{seq} = \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_+ & 0 \\ 0 & 0 & Z_+ \end{bmatrix} \quad A_s = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \quad \text{(Equation 8.4)}$$

The short-circuit calculations at nodes further downstream in the distribution network are calculated using the Thevenin equivalent three-phase circuit at the short-circuit location.

Figure below shows the Thevenin equivalent circuit.

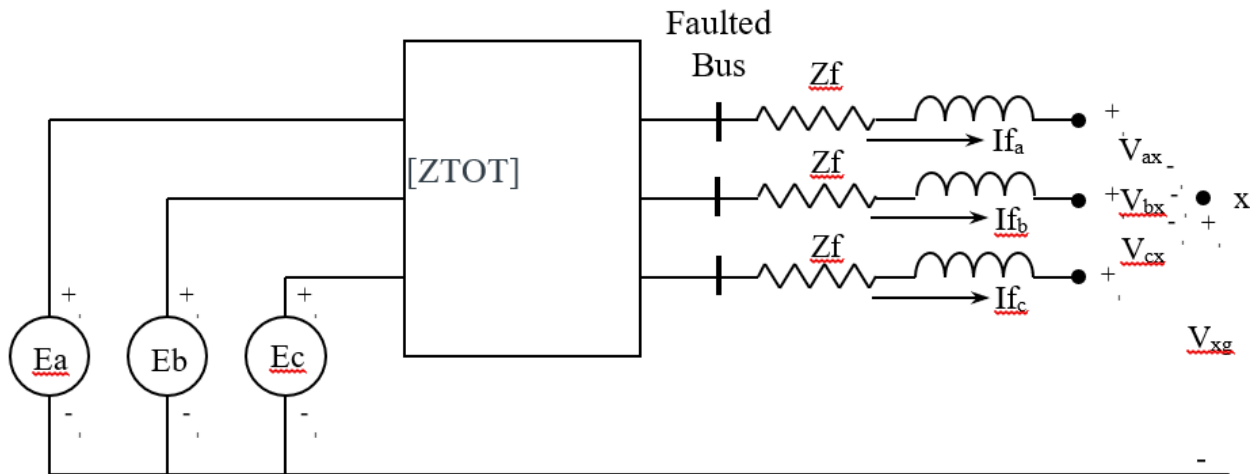


Figure 64. Thevenin Equivalent Circuit

The Thevenin equivalent voltages are the nominal line-to-ground voltages with their associated angles. The Thevenin equivalent phase impedance matrices are the sum of the phase impedance matrices of each device between the system voltage source and the fault location. If a transformer is encountered (a typical winding configuration for a distribution substation transformer), the total phase impedance matrix on the source side of the transformer is referred to the secondary side. Any grounding impedances of the transformer are also accounted for.

Source impedances are also included in the Thevenin equivalent impedance matrices. These values represent impedance upstream of the power transformers and downstream of the voltage sources; they are provided by either the substation or interface objects. If nonzero resistance and reactance values are provided for each sequence for every interface object in the station, then they are used. Otherwise, the values defined for the substation object are used.

The voltage solution is recorded from the latest power flow execution. Therefore, if the breaker is open at the time of FL execution, the latest power flow calculated pre-fault voltages are used. If no pre-fault voltages are available (that is, power flow has not solved or has not converged), the pre-fault voltage is taken to be 1.0 per unit.

The result from the FL application is a list of potential fault locations. These locations are summarized in a tabular display, and they are also shown graphically on the geographic user interface.

8.1.1. Trigger Type

The radial FL application can be triggered and solved on feeder breakers or other devices with fault current analog and fault indication status measurements, which is configured in the checkbox **Only Use Feeder Breaker Fault Current Measurements** in the DNAF Configuration Editor (DCE). If checked, the radial FL application only considers a feeder breaker as the triggering device and uses its measurements for the FL calculation. If there are no valid feeder breaker measurements, the application is not triggered and no potential fault locations are found. If unchecked, a different triggering device is determined. The application searches for the nearest upstream protection device with valid fault measurements that is closest to the fault location as determined by the Network Topology Processor (NTP). FL iterates through all possible triggering devices and evaluates each device. Devices without fault current and fault indication status measurements are also skipped. After that, FL checks if this triggering device is the downstream-most device in the faulted network section. If it is, the device is marked as a candidate triggering device. If the feeder breaker is the only potential triggering device, the application still uses the feeder breaker fault current and/or faulted phase measurements if they are available.

8.1.2. Ground Fault Flag

The Radial FL algorithm has a configurable option to determine if the fault type is a ground fault based on the fault indicator associated with the triggering device alone or the ground fault flag from fault indicators in the faulted area. This feature can help to identify a ground fault when the ground fault indicator is not set correctly or not modeled. For example, an L-L-L-G fault has occurred downstream of recloser R2 in figure below whose ground fault flag is not set correctly. If FL is configured to trigger by the nearest upstream triggering device that is closest to the fault location, R2 will be the triggering device and the fault type will be identified as ABC due to R2's ground fault flag issue. If this feature is enabled, other fault indicators in the fault area will also be used to determine the fault type. If any of the fault indicators show that there is a ground flag in the faulted area, the fault will be identified as a ground fault when this feature is enabled.

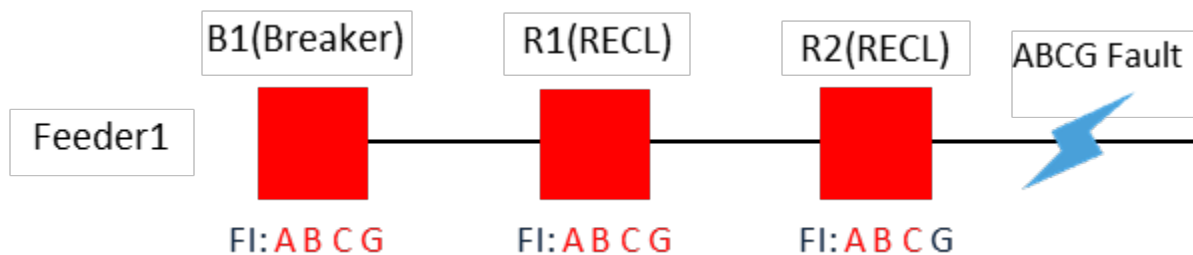


Figure 65. Example of Ground Fault Flag

To enable this feature, the Radial FL application needs to be configured with the "Only Use Feeder Breaker Fault Current Measurements" option unchecked and the "Consider the Ground Fault Flag of any Fault Indicator in the Faulted Section" option checked in the FL tab of the DCE. If the "Only Use Feeder Breaker Fault Current Measurements" option is unchecked and the "Consider the Ground Fault Flag of any Fault Indicator in the Faulted Section" option is checked, the fault will be

identified as a ground fault (L-G, L-L-G, or L-L-L-G) if any of the fault indicators show a ground flag in the faulted network area. If FL is configured to be triggered by the feeder breaker (with the "Only Use Feeder Breaker Fault Current Measurements" option checked) or the "Consider the Ground Fault Flag of any Fault Indicator in the Faulted Section" option is unchecked, the ground fault type is determined only by the fault indicators associated with the triggering device.

8.2. Meshed FL Algorithm

Generators and distributed energy resources connected to the electrical distribution network may contribute to fault currents. The Meshed Fault Location (FL) algorithm considers the impact of loops and parallels along with distributed energy resources on the fault currents, and therefore on the predicted fault location. These are not considered in the Radial FL application algorithm ([Radial FL Algorithm](#)).

Specifically, the Meshed FL algorithm considers the following conditions when determining fault locations: parallel conductors and with multiple sources and injection points, networked primary distribution systems, network loops, parallel underground cables, meshed secondary networks, distributed generation, energy storage devices, and temporary system conditions, such as jumpers, cuts, and phase swaps.

The Meshed FL algorithm works by breaking the faulted network topology up into different sections, either radial sections, loops or sections fed by multiple fault infeeds. Each of the sections is efficiently searched by evaluating if a fault can exist on a section and, if so, where on that section the fault is located.

To evaluate for potential locations in a section, the Meshed algorithm will run a Short Circuit (SCC) analysis multiple times for different locations to iteratively narrow down potential locations until one or more valid fault locations are found, or it is confirmed that none exist. Each time a SCC analysis is run, the algorithm will compare the calculated currents through triggering devices to the current values (SCADA or manual) used to trigger the FL analysis, and use the results of this comparison to narrow down potential locations. If multiple fault current devices are triggered for an analysis, the Meshed FL algorithm will use the trigger device with the highest average faulted currents for comparison.

Due to the nature of the algorithm, the Meshed FL must run multiple SCC calculations, and each of these is based on a load flow solution under faulted conditions. As such, the Meshed FL solution is technically very accurate, but will take longer to solve than the Radial FL algorithm. The time to solve varies on conditions but will increase as the following items change:

- Larger station model size or number of meshed stations.
- Lower numbers of fault indicators (which are used to narrow down the search area).
- Larger numbers of potential locations (often seen for phase-ground faults on systems with many single-phase laterals).

The ALLOWCONTRIBUTETOFAULT_ENERGYRESMOD field (in CSV format) or the AllowContributeToFault field (in XML format) in the energy resource model determines if an energy

resource can contribute to fault current. This field is respected in the PRV, SCC, and FL applications. For inverters that have a fast disconnect capability (for example, inverters that disconnect in less than a cycle), this value should be set to false. For large inverters that produce infeed during the sub-transient period (1-6 cycles), this field should be set to true.

The meshed FL algorithm accounts for the inverter-coupled DERs based on their specified short circuit output (steady state analysis is still performed, and the DER is considered as a generator in the FL analysis). This is done by modeling inverter fault behavior depending on the voltage. A set of voltage ranges and their associated fault current levels are defined for an Energy Resource model in the global.csv file.

8.3. Algorithm Configuration

To enable the Meshed FL algorithm, run the Distribution Network Applications Scheduler (DNAScheduler) application and set the DNA_SCHEDULER_HOST and DNA_SCHEDULER_PORT parameters in both the NetServerDnafConfig.txt and the NetServerDNAConfig.txt files. Select the "Meshed" option for the Algorithm in DCE for both the DNAF_PARAMS_DNAF.xml and DNAF_PARAMS_SIM.xml (for a DOTS system) under the FL tab. By default, the "Radial" button is selected to use the radial FL algorithm.

Note that the DNAF Configuration Editor for FL contains different sets of options for either the Radial or the Meshed algorithm. See [FL Configuration](#) for more details.

8.4. Non-directional and Directional Fault Indicators

A non-directional fault indicator is set when it senses a fault current, but it cannot determine which side has the fault. Non-direction fault indicators work fine if there is only one source in the system as illustrated in the figure below (set fault indicators are red, and unset fault indicators are green).

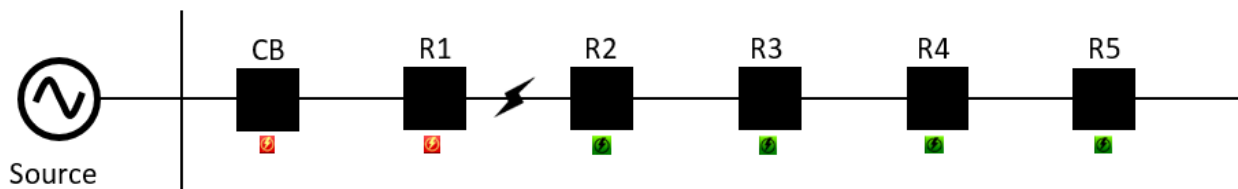


Figure 66. Fault on Single Source Feeder with Only Non-directional Fault Indicators

Figure above shown a feeder with one source located upstream of feeder head breaker CB. There is a non-directional fault indicator on every downstream breaker and recloser. If a fault occurs between reclosers R1 and R2, the fault indicators on CB and R1 are set, and all of the fault indicators downstream of R1 remain unset. In this case, even though no directional information is known, FISR can correctly predict that the faulted section is between R1 and R2.

However, when there is more than one source in the network, either by a switching operation or a large generator connected along the feeder that can contribute fault current, the fault indicators on

all devices that are in series between the fault and the energy sources will be set no matter if the fault is located upstream or downstream of the fault indicating device. This condition is illustrated in the figure below with a large generator connected along the feeder that can contribute fault current.

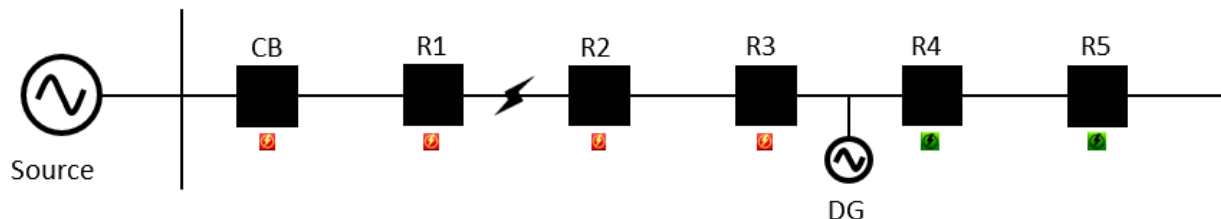


Figure 67. Fault on Feeder with a Generator with only Non-directional Fault Indicators

Figure above shows a large generator, DG, that is connected to the feeder between reclosers R3 and R4. This generator is capable of contributing fault current. If a fault occurs between R1 and R2, the fault indicators on CB and R1 are set. The fault indicators on R2 and R3 are also set because the generator DG can also contribute to the overall fault current. Due to all of the fault indicators along the feeder being non-directional and the existence of multiple fault current sources, the exact location of the faulted section cannot be determined. This means that FISR will incorrectly predict that the faulted location is between R3 and R4 because R3 has the last set fault indicator status and R4 has the first unset fault indicator status.

Figure below shows the same system as figure above, but the fault indicators are directional. The fault indicators on CB and R1 are set in forward direction and the fault indicators on R2 and R3 are set in the reverse direction.

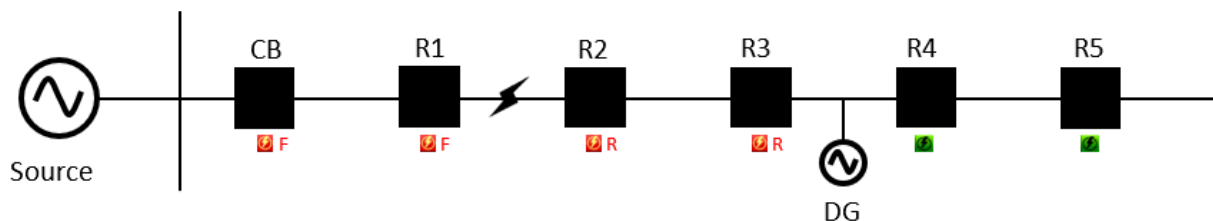


Figure 68. Fault on Feeder with a Generator with only Directional Fault Indicators

The meshed FL algorithm uses the device with the highest average measured phase fault currents to compare the measured and calculated fault current values. In figure above, the breaker CB has the highest average measured fault currents, so this device is used in the FL analysis.

The two options "Only Consider Faults Downstream of Set Fault Indicators" and "Only Consider Faults Upstream of Unset Fault Indicators" in the Fault Location (FL) tab in the DNAF Configuration Editor impact the selection of the potential fault locations for the internal short circuit calculation analysis and determination of the fault current at the telemetered fault current device.

If the "Only Consider Faults Downstream of Set Fault Indicators" option is selected, the meshed FL algorithm will first check if there is meshed topology or any generator capable of contributing fault current along the feeder. If meshed topology or one or more generators that can contribute fault current are found, the faults downstream of the fault current sensing device with the highest

average measured phase fault currents are considered. If there is no meshed topology or generator capable of contributing fault current along the feeder and only non-directional fault indicators are available, only faults downstream of set non-directional fault indicators are considered. If there is no meshed topology or generator capable of contributing fault current along the feeder and only directional fault indicators are available, only faults downstream of forward direction set fault indicators are considered. In this scenario, FL detects a DG capable of contributing fault current along the feeder, and FL can find the correct fault location because it considers faults downstream of the CB regardless of the fault indicator type (directional or non-directional) and their status.

If the "Only Consider Faults Upstream of Unset Fault Indicators" option is selected, only faults upstream of unset non-directional fault indicators are considered in the meshed FL algorithm. In this scenario in figure above, FL can find the correct fault location because it only considers faults upstream of R4. However, since the fault is not in the area between R2 and R4, the search area is larger which may impact the FL performance. For directional fault indicators, only faults upstream of set reverse or unset fault indicators are considered. In this scenario in figure above, FL can find the correct fault location because it only considers faults upstream of set reverse fault indicator R2.

If neither option is selected ("Only Consider Faults Downstream of Set Fault Indicators" and "Only Consider Faults Upstream of Unset Fault Indicators"), which is the default setting, the meshed FL algorithm will not use fault indicators to determine the faults to be considered no matter if there is meshed topology or a generator capable of contributing fault current along the feeder, so the faults downstream of the fault current sensing device with the highest average measured phase fault currents are considered. In this scenario, FL can find the correct fault location because it considers faults downstream of CB.

If both options are selected ("Only Consider Faults Downstream of Set Fault Indicators" and "Only Consider Faults Upstream of Unset Fault Indicators"), only the faults that meet the requirement of both options are considered in the meshed FL algorithm. If meshed topology or one or more generators that can contribute fault current are found, and only non-directional fault indicators are available, the faults downstream of CB and upstream of R4 are considered. If meshed topology or one or more generators that can contribute fault current are found, and only directional fault indicators are available, the faults downstream of CB and upstream of R2 are considered. If there is no meshed topology or a generator capable of contributing fault current along the feeder and only non-directional fault indicators are available, the faults downstream of R1 and upstream of R2 are considered. If there is no meshed topology or generator capable of contributing fault current along the feeder and only directional fault indicators are available, the faults downstream of R1 and upstream of R2 are considered. In this scenario, FL can always find the correct fault location.

If all of the fault indicators along this feeder are directional, the faulted section can be determined. FISIR will be able to determine that the fault is downstream of both CB and R1, and upstream of R2 and R3, allowing FISIR to correctly predict that the fault is located between R1 and R2.

8.5. Modes of Operation

The FL application can be executed in real-time mode or study mode.

8.5.1. Real-Time Mode

In real-time mode, the FL application can be triggered upon receiving a valid device fault current (Amps) analog measurement. To enable the event trigger, the `DNAF_FL_EVENT_TRIGGER` parameter must be set to a value of 1. Once the fault current measurement has been received, the FL application will be executed.

When `DNAF_FL_CHECK_MEAS_CHANGE` is also set to **1**, two additional validation checks are performed before the FL event trigger is issued:

1. **Value change check:** The difference between the current measurement value and the previous scan value must be greater than a threshold (0.1 A). If the measurement value has not changed significantly between scans, the FL event trigger is skipped.
2. **Measurement age check:** The measurement timestamp must be recent, i.e., within the maximum age defined by the DCE setting "Maximum Age of Fault Inputs". If the measurement is outdated, the FL event trigger is skipped.

These checks prevent redundant FL evaluations caused by unchanged or stale SCADA current measurements.

The FL application can also be configured to block the triggering for a specified time period after the last execution, to prevent repetitive executions if different analog fault current measurements are received for a protective device within a short time period. This time period, specified in minutes, is configured by the `DNAF_FL_BLOCK_SOLVE_TIMER` parameter in the DNAF server configuration file.

The existing FL results (data on the Fault Location tabular display and halos on the geographic display) are cleared every 30 minutes on the half hour (:00 and :30) if their lifetime has exceeded the minimum value, which is set by the "Minimum Fault Location Halo Life Time (min)" parameter in the DCE (default is 5 minutes). If the option "Only Remove Fault Location Result When Energized" is enabled, then the fault location results will only be removed if both the lifetime minimum has been met and the location is energized. Newly triggered execution of a fault analysis for the same feeder also clears previous fault location results on that feeder.

The fault indication halos are set by the client ADMS Network Topology Processor (NTP) application according to the fault indicator status. The halos are cleared after the fault indicators are reset. The reset time is defined by the `NTP_FAULTIND_RESET_TIME` parameter, which is configured in the Simulator server configuration file.

If the manual trigger is enabled, the FL application can also be manually invoked after setting the appropriate breaker fault current and faulted phase values. These values are set using the Fault Location Input display. To enable the manual trigger, the parameter `DNAF_MANUAL_TRIGGER` must be set to 1 in the DNAF server configuration file. Manually triggering the fault analysis clears previous fault location halos and results for all station feeders (Radial FL) or the faulted feeder only (Meshed FL).

8.5.2. Study Mode

The FL application can be manually invoked in study mode after setting the appropriate breaker fault current and faulted phase values. These values are set using the Fault Location Input display. To enable the manual trigger, the DNAF_MANUAL_TRIGGER parameter must be set to 1 in the study server configuration file.

8.6. Fault Location Result Parameters

The FL output can be most easily seen by using the fault location halos shown on the geographic UI displays. To change the width or color of the potential fault location halos, use the Lines tab in the Viewer Configuration Editor.

The FL application can support up to three equally spaced potential fault location ranges. The mismatch used to determine the range is calculated by averaging the mismatch from each faulted phase. For example, three ranges at a 2% mismatch each for a fault current target of 5000.0 amps result in the following ranges:

Range 1: 4900 – 5100.0 amps ($5000 \pm 2\%$)

Range 2: 4800 – 4900.0 amps and 5100 – 5200.0 amps ($5000 \pm 2 - 4\%$)

Range 3: 4700 – 4800.0 amps and 5200 – 5300.0 amps ($5000 \pm 4 - 6\%$)

The number of fault location ranges (Number of Ranges) and the accuracy (Fault Location Mismatch %) are specified in the FL page of the DNAF Configuration Editor.

8.7. Fault Location Input and Output

Input data for running FL can be obtained either from SCADA data or manual inputs in real-time mode, and from manual inputs in study mode.

Outputs from FL include up to 3 different colored halos representing different likelihoods of fault location. Fault indication halos are also associated with faults but result from topology processing rather than from the Fault Location analysis itself.

8.7.1. Fault Location Manual Input

Input fault levels and targets can be entered using the Fault Location Input tabular display, which is accessed via the Application Input Summary tabular display as shown in figure below. The Fault Location Input tabular display is used for manual triggers in both real-time and study mode and allows the user to enter fault currents and phases for each potential trigger device in the selected station. This display lists potential trigger devices that are either feeder breakers or other switching devices that have a mapped MeasFaultCurrent measurement type.

Name	ID	Locate	Device Type	Phase A Fault	Phase B Fault	Phase C Fault	Ground Fault	Fault Type	Phase A Fault Current (Amps)	Phase B Fault Current (Amps)	Phase C Fault Current (Amps)	Maximum Fault Current (Amps)
5631F	A1AB561...		Breaker	☒	☐	☒	☒	ACG	0.00	0.00	0.00	3200.00
5632F	2A18CB4...		Breaker	☐	☐	☐	☐	--	0.00	0.00	0.00	0.00
5633F	0D055B32...		Breaker	☐	☐	☐	☐	--	0.00	0.00	0.00	0.00
5634F	D0416F4C...		Breaker	☒	☒	☐	☐	AB	3200.00	3000.00	0.00	0.00
5635F	25C35EBF...		Breaker	☐	☐	☐	☐	--	0.00	0.00	0.00	0.00
5636F	CC79EA64...		Breaker	☐	☐	☐	☐	--	0.00	0.00	0.00	0.00
5637F	B8D742B9...		Breaker	☐	☐	☐	☐	--	0.00	0.00	0.00	0.00
5638F	DD1CC49...		Breaker	☐	☐	☐	☐	--	0.00	0.00	0.00	0.00

Figure 69. Fault Location Input Tabular Display

The Fault Location Input display supports entering either individual phase fault currents or a peak fault current for a device:

- When the operator enters the peak fault current value among the faulted phases (via the Maximum Fault Current column), the FL application determines the most probable fault locations based on the difference between the measured fault current value and the highest calculated fault current values of the faulted phases (A, B, or C). If the difference is within the user-configured range (specified in the Fault Location Mismatch (%) field in the DNAF Configuration Editor), it is considered as a potential fault location.

The Fault Location application uses the same method to identify fault locations for devices where the MeasFaultCurrent measurement type is mapped to the peak fault current measurement.

- If individual phase fault current values are provided via the Phase A/B/C Fault Current columns, the FL application determines the most probable fault locations based on the difference between measured and calculated fault current values for each faulted phase (A, B, or C). If the difference for a phase is within the user-configured range (specified in the Fault Location Mismatch (%) field in the DNAF Configuration Editor), it is considered as a potential fault location.

The Fault Location application uses the same method to identify fault locations when individual phase fault current values are provided via a set of single SCADA fault current measurements (for devices where the MeasFaultCurrent measurement type is mapped to the individual phase fault current measurements).

When running using Manual Inputs, the non-radial FL algorithm will ignore the Fault Location Limits checkboxes in the DNAF Configuration Editor. As fault indicators are typically not set when running manually, fault indicators will not be used to constrain the search area.

8.7.2. Fault Location SCADA Inputs

There are two main sources of Fault Location input data from SCADA: fault current analogs and

fault indicator status. This section describes the behavior the Fault Location algorithms expect from these measurements.

Fault current measurements should be set in SCADA when the fault occurs for at least one of the upstream protection devices sensing the fault. Setting currents for all upstream devices is preferred and will result in a smaller search area for the Meshed FL algorithm and a faster solution. When using the Radial FL algorithm, currents should not be reported for non-faulted phases. The Meshed FL algorithm is agnostic whether non-faulted phases are reported.

Fault indicator status measurements should be set for all faulted phases on all fault indicators upstream of the fault. If there is a chance that one or more fault indicators may not be set (for example, due to device configuration or potential bad communications), the "Only Consider Faults Upstream of Unset Fault Indicators" must not be set in the DNAF Configuration Editor. Providing a fault indicator for the Ground phase during ground faults is not required but providing such a point on at least one of the trigger devices will improve the ability to distinguish between LL and LLG faults.

Measurements that are older than a configurable limit ("Maximum Age of Fault Inputs" in the DNAF Configuration Editor) will be ignored. This can help avoid bad inputs when a subsequent fault follows an initial fault but with different phasing.

8.7.3. Fault Halos

There are two types of line halos used in conjunction with fault location: fault location halos and fault indication halos.

8.7.3.1. Fault Location Halos

Graphically, the Fault Location results are shown by halos that are drawn over the line sections. These halos are called "fault location halos". The user can configure from one to three levels of fault location halos, where level1 represents the most likely locations and level3 represents the less-probable locations (for more details, refer to the *Viewer Configuration Editor User Guide*).

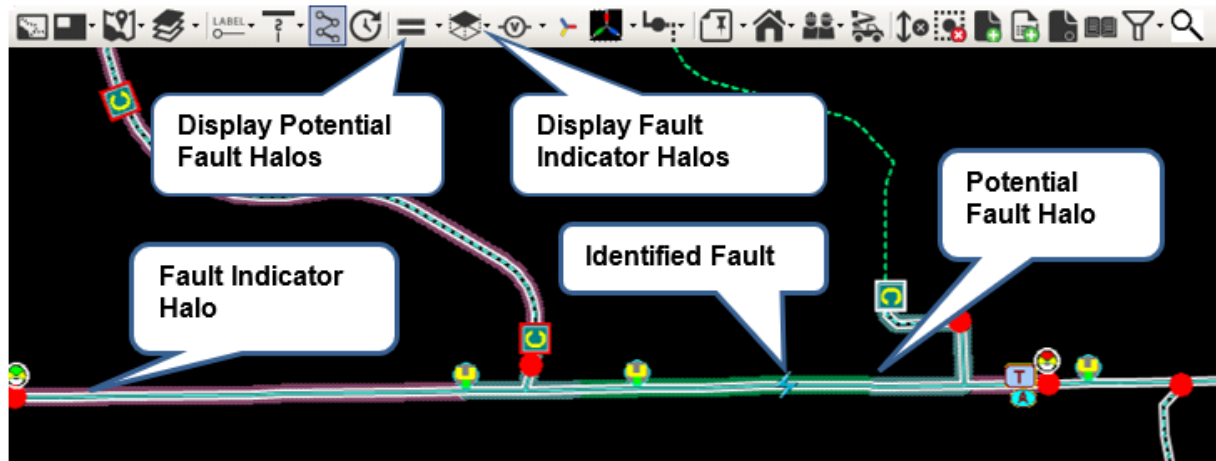


Figure 70. Fault Location Halos on a Geographic Display

Figure above shows typical halos associated with a fault analysis, the fault location halo toolbar button, and the halos (green and cyan). This figure demonstrates two fault indicator halo levels (a maximum of three levels is available). The display of potential fault halos is controlled by the Display Potential Fault Halos  button on the ADMS geographic (common) toolbar.

8.7.3.2. Fault Indicator Halos

When a telemetered or manual fault indicator is set, the client ADMS Network Topology Processor (NTP) application puts halos around all lines between the furthest downstream set fault indicator and the next set of fault indicators that are unset. These fault indicator halos, used in conjunction with the fault location halos, can reduce the number of potential fault locations.

To enable the fault indicator halos on the geographic display, click the Display Fault Indicator Halos  button on the ADMS geographic (common) toolbar. After the fault indicator halos have been enabled, the halos are drawn on the geographic UI using the fault indicator halo colors defined in the Viewer Configuration Editor. Figure above shows an example of the fault indicator halos (in purple).

8.7.4. Fault Location Tabular Results

Results in the Fault Locations tabular display, as shown in figure below, provide another view of the same data shown by the fault location halos.

3LIONS - Fault Locations (Fault Current in Amps: Maximum: 3200.0 A: -- B: -- C: -- G: --, Fault Type: ABG)									
Station Name	Branch Device Id	Phase A Calculated Fault Current	DifferenceA (%)	Phase B Calculated Fault Current	DifferenceB (%)	Phase C Calculated Fault Current	DifferenceC (%)	Fault Location	Locate in URD
3LIONS	136098574	3316.50	3.64	3006.60	6.05	0.00	--		
3LIONS	181162827	3229.90	0.93	2904.20	9.24	0.00	--		
3LIONS	22496786	3199.80	0.01	2874.50	10.17	0.00	--		
3LIONS	8238341	3229.90	0.93	2904.20	9.24	0.00	--		
3LIONS	8238351	3229.90	0.93	2904.20	9.24	0.00	--		
3LIONS	8238488	3229.90	0.93	2904.20	9.24	0.00	--		
3LIONS	188580149	3227.70	0.86	2902.20	9.31	0.00	--		
3LIONS	222629868	3239.40	1.23	2912.70	8.98	0.00	--		
3LIONS	9977765	3301.90	3.18	2973.70	7.07	0.00	--		
3LIONS	9927640	3301.90	3.18	2973.70	7.07	0.00	--		
3LIONS	222630741	3322.30	3.82	2992.80	6.48	0.00	--		
3LIONS	181162139	3181.40	0.58	2857.50	10.70	0.00	--		
3LIONS	8237140	3181.40	0.58	2857.50	10.70	0.00	--		
3LIONS	8237150	3181.40	0.58	2857.50	10.70	0.00	--		
3LIONS	8237284	3181.40	0.58	2857.50	10.70	0.00	--		
3LIONS	8237294	3181.40	0.58	2857.50	10.70	0.00	--		
3LIONS	222370345	3292.80	2.90	2922.60	8.67	0.00	--		
3LIONS	173400271	3160.80	1.23	2837.70	11.32	0.00	--		
3LIONS	8238112	3160.80	1.23	2837.70	11.32	0.00	--		
3LIONS	8238244	3160.80	1.23	2837.70	11.32	0.00	--		

Figure 71. Fault Location Results Tabular Display

8.7.5. Comparison of Short Circuit and Fault Location Results

The FL algorithm accepts fault currents at triggering devices as an input and provides locations as outputs. The SCC application accepts a location as input and provides total fault currents at that point as outputs. It is important to recognize that there are differences in these two analysis functions.

One key difference is that the FL application works from fault currents as seen at the feeder breaker or triggering device, while SCC provides total fault current at the location of the fault. These are often similar; however, where voltage regulation devices such as regulators or step-down transformers lie between the breaker and the fault location, the FL input and SCC output currents may be very different (by the ratio of the voltage change through the inline transformer or regulator).

Cases where there are multiple fault infeeds will also cause differences between the FL and SCC values. The fault location only compares values seen at the main triggering device, whereas SCC will report the total fault current at the fault, including all infeeds.

The other key difference is the set of factors considered by the applications. SCC is an exact solution and considers load contribution to total current, mutual coupling, etc. under a single specific fault condition. The FL application is more general and reports on likely (usually multiple) locations for faults. It is unable to consider the same detailed factors, which means that FL outputs may not correspond exactly to the values calculated by the SCC application.

8.8. Fault Current Contributions

For information about the fault current contribution characteristics of different energy sources, see [Fault Current Contributions](#).

8.9. FL Alarms

Table below lists the supported FL alarms. The alarms can be enabled or disabled using the DNAF System Configuration display. For more information on Alarm naming, see [DNAF Alarm Format](#).

Table 17. Fault Location Alarms

Alarm	Description
FL Started Event	Alarm to indicate that the FL calculation started for a substation.
FL Success Event	Alarm to indicate that the FL calculation completed successfully for a substation.
FL Failed Event	Alarm to indicate that the FL calculation failed to complete for a substation.

8.10. FL Configuration

The Fault Location function uses fault Amp measurements and faulted phase status measurements to determine potential fault locations along distribution feeders.

This section describes the Fault Location configuration parameters.

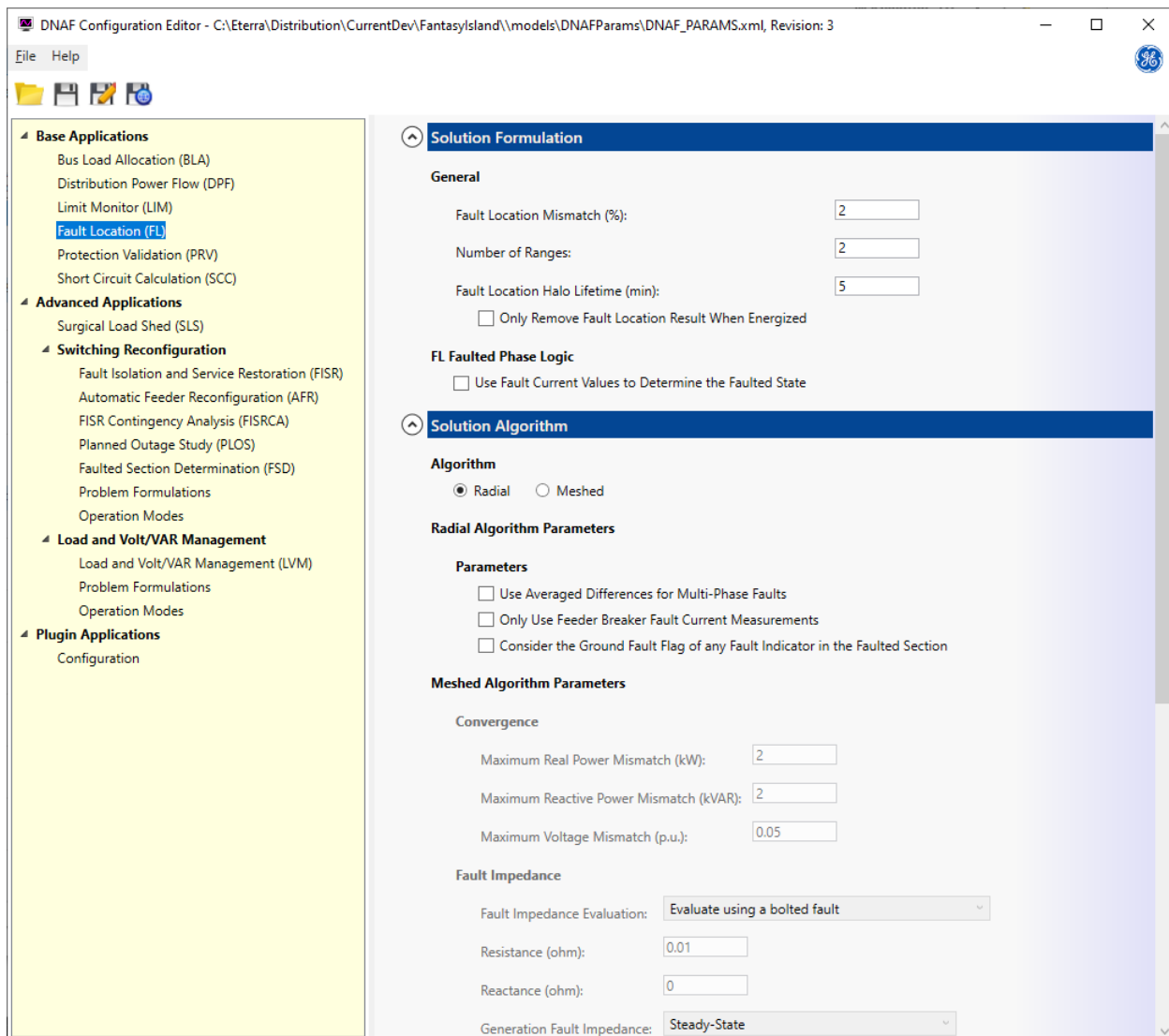


Figure 72. Fault Location Configuration

Table below lists the FL configuration attributes. All listed parameters are contained by the <Params xsi:type = "FLParams"> element in the DNAF_PARAMS.xml configuration file.

Table 18. FL Configuration Attributes

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
General			
Fault Location Mismatch (%)	FAULTLOCTOL_FL <Tolerance>	Text box	This value specifies the fault current mismatch (%). This value is the percentage difference between the measured and calculated fault current values for each fault location results range. The allowable range is 0.01-10.0.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Number of Ranges	NUMRANGES_FL <NumToleranceRanges>	Text box	This value specifies the number of equally sized tolerance ranges (3maximum). If this parameter is set to 3 and the Fault Location Mismatch tolerance above is 2.5, then the tolerance ranges are +/-2.5%, +/-2.5 - 5.0%, and +/- 5.0 - 7.5%. The allowable range is 1-3.
Fault Location Halo Lifetime (min)	<MinHaloTime>	Text box	This value specifies the minimum number of minutes that the fault location halos and tabular display solution results are shown. The server removes the fault location results every 30 minutes on the hour (at :00 and :30) unless the minimum life time is not met. This feature only applies to halos displayed in RT. The allowable range is 1-120.
Only Remove Fault Location Result When Energized	<OnlyRemoveFLResultWhenEnergized>	Check box	This indicates whether the energization status of a line is checked when removing the fault location results. If this option is selected, the fault location results are removed only after the section is energized and the results life time is met.

FL Faulted Phase Logic

Use Fault Current Values to Determine the Faulted State	<UseFaultCurrentMeas>	Check box	This specifies whether the faulted phase is only determined using relay and fault current measurements. If selected, the faulted phase is determined by analyzing the phase current measurement. The FL faulted phase logic tries to use the ground trip current of the triggering device as the threshold first. If ground trip current is not set, FL tries to use the phase trip current of the triggering device. If not selected, the faulted phase is determined by the fault indicator state.
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Algorithm

Algorithm	<FLUseRadialAlgorithm>	Radio buttons	These buttons determine the selected Fault Location (FL) algorithm. Radial: When this option is selected, the radial FL algorithm is used. Meshed: When this option is selected, the meshed FL algorithm is used.
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Radial Algorithm Parameters

Use Averaged Differences for Multi-Phase Faults	<CompareAverageCurrent>	Check box	When this check box is checked, FL will compare calculated currents to target currents by averaging the difference across all faulted phases. This produces a simpler single predicted fault location when there are differences between phases. If this box is unchecked, FL will compare calculated currents to target currents on a phase-by-phase basis. This may show multiple predicated locations associated with the locations where individual phases each match best.
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Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Only Use Feeder Breaker Fault Current Measurements	<UseFeederBreakerFaultAmps>	Checkbox	This checkbox determines the triggering device for the radial Fault Location (FL). If checked, the radial FL application only considers a feeder breaker as the triggering device and uses its measurements for the FL calculation. If there are no valid feeder breaker measurements, the application is not triggered, and no potential fault locations are found. If unchecked, the application finds the nearest upstream protection device with valid fault measurements that is closest to the fault location determined by the Network Topology Processor (NTP) as the triggering device. If the feeder breaker is the only potential triggering device, the application still uses the feeder breaker fault current and/or faulted phase measurements if they are available.
Consider the Ground Fault Flag of any Fault Indicator in the Faulted Section	<CheckFisGroundFlag>	Checkbox	This checkbox indicates whether the ground fault type is determined by the fault indicator associated with the triggering device alone or by the ground fault flag from fault indicators in the faulted area. For radial-only FL applications, if the "Only Use Feeder Breaker Fault Current Measurements" option is unchecked and this option is checked, the fault will be identified as a ground fault (L-G, L-L-G, or L-L-L-G) if any of the fault indicators show a ground flag in the faulted network area. If FL is configured to trigger by the feeder breaker (with the "Only Use Feeder Breaker Fault Current Measurements" option checked) or this option is unchecked, the ground fault type is determined only by the fault indicators associated with the triggering device.
Maximum Age of Fault Inputs			
Maximum Real Power Mismatch (kW)	<PMismatch>	Text box	This value specifies the internal power flow's real power convergence criterion in kW for the FL application. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The FL solution status is set to Diverged if any load's real power difference between DPF iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Reactive Power Mismatch (kVAR)	<QMismatch>	Text box	This value specifies the internal power flow's reactive power convergence criterion in kVAR for the FL application. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The FL solution status is set to Diverged if any load's reactive power difference between DPF iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Voltage Mismatch (p.u.)	<VMismatch>	Text box	This value specifies the internal power flow's voltage convergence criterion in per unit for the FL applications. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. The FL solution status is set to Diverged if the voltage mismatches are larger than this specified value. The allowable range is 0-0.1.
Fault Impedance			
Fault Impedance Evaluation	<FaultImpEvalType>	Drop-down list	This parameter specifies the fault impedance evaluation type that is applied by the FL solution. 0: Evaluate only a bolted fault. If this option is selected, the FL solution solves for short circuit fault resistance and reactance values that are close to zero. 1: Evaluate only the user-configured fault impedance. If this option is selected, the FL solution solves for short circuit fault resistance and reactance values that are specified by the Resistance and Reactance parameters.
Resistance (ohm)	<FaultImpResistance>	Text box	Short circuit fault resistance in Ohms. The allowable range is 0-20.
Reactance (ohm)	<FaultImpReactance>	Text box	Short circuit fault reactance in Ohms. The allowable range is 0-20.
Generation Fault Impedance	<GenerationFaultImpType>	Drop-down list	This parameter specifies generation fault impedance values that are used by the FL solution. 0: Steady-state: Synchronous Reactance (Xd) determines the current flow after steady-state condition is reached. 1: Transient: Transient Reactance (X'd) determines the current flow after several cycles. This condition lasts from 0.5 to 2 seconds. 2: Sub-transient: Sub-transient Reactance (X''d) determines the current flow during the first cycles after a fault occurs. This condition lasts for approximately 0.1 seconds. The generation fault impedance values are defined in the static model.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Only Consider Faults Downstream of Set Fault Indicators	<RespectSetFIs>	Check box	This specifies whether faults are only considered downstream of set fault indicators. Select this option to minimize the upstream fault location area. Deselect this option if you are concerned that fault indicators may be set incorrectly during a fault. This option only applies to the advanced (Meshed) FL application.
Only Consider Faults Upstream of Unset Fault Indicators	<RespectUnsetFIs>	Check box	This specifies whether faults are only considered upstream of unset fault indicators. Select this option to minimize the downstream fault location area. Deselect this option if you are concerned that fault indicators may not be set during a fault. This option applies only to the advanced (Meshed) FL application.
Maximum Age of Fault Inputs	<MaxAgeFaultInput>	Text box	This value specifies the maximum age of fault input data (fault current measurements) for them to be considered part of the current event. Data older than this age in seconds will be ignored.
Solution Results			
Create FL Result File	PRINTOUT_FL <PrintFLResult>	Check box	This indicates whether FL results should be written to a result file. There is a separate file for each feeder. Only one file is kept for each feeder (that is, it is overwritten after each solution) unless the Create Unique Log option is selected.
Use Unique File Name	UNIQUELOG_FL <UniqueLog>	Check box	This indicates whether a unique FL result file is generated for each FL solution (that is, do not overwrite the previously created file). If this option is selected, each result file includes a time stamp as part of the filename. If this option is left on for an extended period, a large number of files can be generated, potentially causing disk space issues.
Use Detailed Result Reporting	DETAILEDLOG_FL <DetailedLog>	Check box	This indicates whether the FL result file includes detailed results providing additional information, including the impedance data for multiple feeder locations.
Manual Execution			
Allow in Real Time	REALTIME_FL <RealTime>	Check Box	This indicates whether the FL application can be invoked through the RT DNAF Solution pop-up display (selected means the application is shown).
Allow in Study Mode	STUDYMODE_FL <Study>	Check Box	This indicates whether the FL application can be invoked through the ST DNAF Solution pop-up display (selected means the application is shown).
Old and Deprecated Parameters			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Name	NAME_FL <Name>	String	Function display name – does not appear in the UI. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_FL	String	Function index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Radial Algorithm Parameters			
Function Enumeration	FUNCTENUM_FL <FuncEnum>	Integer	This value specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Switching 8 = FISIR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

9. Parameter Files

This chapter describes all the parameters defined in the network application data files located in the server directory. There are five data files: `alg.dat`, `optalg.dat`, `msgdef.dat`, `phid.dat`, and `tftype.dat`.

9.1. Alg.dat

The `alg.dat` file contains parameters used inside the power flow solution engine. Most of the time, these parameters do not need to be modified unless unusual cases are being solved.

Attributes	Default Value	Description
IterMax	30	Maximum number of iterations.
IterMin	2	Minimum number of iterations. If the solution converged with fewer iterations than this value, a new iteration is run until the number of iterations reaches this value.
CapMoveMax	10	Maximum number of moves for one capacitor in one solution. If any capacitor moves more than this number, the solution exits and it is considered diverged.
StopOnV1	1	During DPF, there are two steps in each iteration. V1 is the voltage magnitude from the first step; V2 is the voltage magnitude from the second step. When the solution hits maximum iterations, the loop can optionally stop with the V1 or V2 solution results. If this value is 1, the V1 results are the final results; otherwise, the V2 results are the final results.
VhighMax	100	The per-unit voltage at which the solution is considered invalid. If the highest voltage magnitude in the network reaches this value, the solution is considered invalid.
NumValInit	200	Initial number of values used for transformer variables (and transformer tap variables) and other purposes. This is the initial allocation block of CSMatValueC objects.
NumValMax	-1	Maximum number of CSMatValueC objects that can be used by the program. This is an upper limit on allocations. Minus one is interpreted as INT_MAX and is the default.
NumValInc	1000	Incremental allocation size for CSMatValueC variables.
FillFactor	1.6	Initially allocated ODs = $iNumBr * FillFactor + FillOffset$
FillOffset	100	
FillIncFactor	0.2	Incremental allocations of ODs = $iNumBr * FillIncFactor + FillOffset$

Attributes	Default Value	Description
RankHighestFactor	1.0	The RankHighestFactor is used to allocate storage for the highest expected rank during factorization. The highest expected rank is computed as: $\text{RankHighestFactor} * \text{iNumBus} + 1.$ If the highest expected rank is exceeded, extra processing time may occur, but rigorous minimum degree ordering is still performed. Memory is allocated for an array sized to the highest expected rank. A value of 1.0 is for maximum CPU speed and minimum memory. A value of 0.2 is reasonable for most power system models.
InvertTol	1e-17	Matrix inversion tolerance for detecting singular or ill-conditioned matrices. This is used by both the sparse matrix and dense matrix classes. This number is in terms of internal p.u. admittance. If singular matrices messages are being issued, and one does not believe the matrix is singular, try decreasing this tolerance. Decreasing this tolerance too much may cause ill conditioned cases to go undetected and the results may be unpredictable (and probably do not converge or have unreasonably high voltage solutions).
InvertTolJac	1e-8	Jacobian matrix inversion tolerance for detecting singular or ill-conditioned Jacobian matrices. If the Jacobian matrix is detected as singular during calculation, the DNAF application will stop and return an error. This value is different from the non-Jacobian matrix inversion tolerance by default.
TapMovePmmTol	10	Tap movement real power mismatch tolerance. To prevent unnecessary tap movement, it is beneficial to disable tap movement logic before power flow convergence. The tolerances (TapMovePmmTol, TapMoveQmmTol, and TapMoveVmmTol) are used to determine when the tap movement logic is enabled. The P, Q, and V tolerance values are multiplied by the P, Q, and V mismatch DPF convergence criteria, respectively. If the largest DPF P, Q, and V mismatches (differences between the target and calculated values) are within the tolerance multiplication results, the tap movement is enabled. To disable tap movement completely, set any one of these tolerance values to zero or a negative number. Normally these numbers should be larger than one, or you can get convergence without moving taps that should have moved. To get tap movement only after total power flow convergence, set all tolerances to 1.0. In that case, the tap will not move unless the power flow converges. Note that these parameters also control shunt impedance regulation as well as tap switching. These parameters are unitless and all default to 10.
TapMoveQmmTol	10	Tap movement reactive power mismatch tolerance.
TapMoveVmmTol	10	Tap movement voltage mismatch tolerance.

Attributes	Default Value	Description
VregUpdateScaler	1.0	Used to scale the voltage magnitude update of Ea, the internal voltage from center point to internal phase voltage Va, to better hold the regulated bus voltage magnitude. Normally set to 1.0. Setting to less than one produces slower but possibly more reliable convergence.
GenAngleUpdateScaler	1.0	PV generator angle update scaler. Used during voltage magnitude updates of generator voltage angles during the second step of each DPF iteration.
TapSensitivity	1.0	The assumed partial derivative of regulated voltage with respect to tap ratio. Normally, set this number to 1.0. To "damp" tap movements, set this number to a value greater than one. To accelerate taps (dangerous), set this number to less than one. Accelerating taps may cause oscillating tap movements. A value of zero causes infinite tap movements (divide by zero).
TapMoveMaxUp	4	Maximum number of tap positions that can be moved up per iteration. This only applies to the master of a group. The slaves move to follow the master and may actually move more than the maximum number of taps. This number is positive or zero. Zero freezes tap movement in the specified direction.
TapMoveMaxDown	4	Maximum number of tap positions that can be moved down per iteration. This only applies to the master of a group. The slaves move to follow the master and may actually move more than the maximum number of taps. This number is positive or zero. Zero freezes tap movement in the specified direction.
BigVRef	1000000.0	Large number used for folding voltage reference equations. The folded equation is the voltage regulation equation with $-1/\text{BigVRef}$ on its diagonal. The folded variable is the generator phase current at terminal. If voltage convergence for generators is slow, but otherwise the solution numerically looks good, one can increase this number. If one starts to see numerical difficulties in later iterations of the solution (with PV buses), one should try decreasing this number. Normally this number should be about as large as or slightly larger than the largest expected admittance value. The number is in terms of internal per-unit admittance.
BigZLink	1000000.0	The large number used for zero impedance link equations. If flows are computed on the links, this number is used as an incremental impedance value. It should have no impact upon the final solution, but if the number is too small, excessive iterations with voltage mismatch may occur. If the number is too large, numerical error may result. The value is used as the reactive part of the link admittance for incremental updates. The small voltage difference caused by this non-zero impedance is iterated away during the solution.
GroundVRef	0.0001	Number used for grounding ungrounded phase bus groups. It is treated like capacitor admittance. This number is in terms of internal per-unit admittance. Normally this number should be quite small compared to regular admittances in the network. If internal base is 100MVA, then 0.000001 is tenth of kW, which should prevent singularity, but not skew regular charging much if present.

Attributes	Default Value	Description
VnoiseTol	0.0000000001	Tolerance for smallest per-unit voltage that is not noise. Any per-unit voltage with magnitude less than VnoiseTol is not rotated in the second step voltage magnitude solution. The setting should be about 1.0e-10 on 64-bit precision. A value of zero removes all effects, but may first appear as numerical problems in neutral voltages (extremely large). A value too large may lead to cases which struggle with latter stages of convergence due to neutral angles that do not change.
MVABaseInternal	100.0	All internal calculations are performed in the MVABaseInternal MVA (power) base. This parameter is independent of the MVA base specified in the input format. Values in the input file are converted from their MVA base (specified in the input file) to the MVA base specified here as MVABaseInternal. The internal MVA base can have some impact on numerical error but typically a number around 1 or 100 gives good results. Note that the number is divided by 3 before being stored as the internal MVA base. The division by 3 allows exact specification of an internal MVA base such as 100/3.
LoadImpedanceFraction	1.0	If a load is not an impedance load (that is, Pexp and Qexp not 2.0), a small impedance load is placed in parallel with the load phase branch. This parallel impedance has the value of the load's impedance (as if the load were pure impedance at nominal voltage) times LoadImpedanceFraction. The current through this parallel impedance load is compensated for in the computed current injections of the load so it has no effect on the final result. The purpose of introducing this small impedance term is to avoid singular matrices for cases such as an ungrounded star load that is modeled as constant current. A number such as 0.1 usually prevents singularity and has a negligible impact on convergence strength. Making this number too small causes singularity of the initial Y1 matrix. Making this number too big slows convergence for PQ or constant I loads. A value of zero removes all effects.
LoadGroundFraction	0.0	This is the same as the LoadImpedanceFraction, but is for adding phase to ground terms at both terminals of a phase load. The admittance to ground is computed as the nominal impedance multiplied by LoadGroundFraction. (The admittance value added is totally independent of the value of the LoadImpedanceFraction parameter.) Very large values cause the network to move more slowly to the actual PQ solution of a load. A value of zero removes all effects.
TfCouplingTol	0.2	This parameter is used by topology analysis to determine energization between windings of a transformer. Terms of the Zprimitive matrix are compared to determine if the coupling is sufficient to consider winding j energized if winding i is energized. If magnitude (Z_{ij}/Z_{ii}) is greater than TfCouplingTol, a magnetic energization coupling is considered to exist between the two windings.
LineCouplingTol	0.0	Similar to TfCouplingTol but applied to lines with mutual coupling between the phases. This is used to test coupling in the Z series of the line. If magnitude (Z_{ij}/Z_{ii}) is greater than LineCouplingTol, a magnetic energization coupling is considered to exist between phase i and phase j wires of the line.

Attributes	Default Value	Description
SkipDeadLoads	1	If set, then automatically convert any PhLoad with per-unit voltage lower than DeadLoadTol to a pure impedance PhLoad. This option only takes place if LoadImpedanceFraction is 1 and LoadGroundFraction is 0 such that the matrix representation of the PhLoad does not have to change.
DeadLoadTol	0.5	Tolerance for low per-unit voltage used to automatically convert load to pure impedance.
DeadLoadRestartTol	0.75	Tolerance for low per-unit voltage used to automatically convert converted impedance back to load.
CommonSlackLoops	1	Consider connections between slack generators to form a loop. If it is 0, you may have multiple generators slack generators feeding a part of the network without creating loops.
DetermineLoops	1	Determine the loops in the network during the solution. Loops are not required for a valid flow and voltage solution, but sometimes useful as additional information. A non-zero value causes the loops to be computed during the solution. Loops are determined in conjunction with energization paths. A breadth first search is made starting from generators. When the search closes on an already searched portion of network, a loop is formed.
FilterAmpViolations	1	If True, filter out amp violations to the largest amp violation per phase in a radial section of network.
FilterVoltHiViolations	1	If True, filter out high voltage violations to the largest high voltage violation per phase in a radial section of network.
FilterVoltLoViolations	1	If True, filter out low voltage violations to the largest low voltage violation per phase in a radial section of network.
ForceImpedanceLoads	0	Force all loads to pure impedance model. If set to 1, then all loads are treated as if they were pure impedance. If set to zero, then the model specified on each individual load object is used.
ComputeLossesByIIR	0	Compute losses by special I^2R calculation instead of sum at terminals.

9.2. Optalg.dat

The optalg.dat file contains parameters used inside the optimization solution engine. Most of these parameters do not need to be modified unless unusual cases are being solved.

Attributes	Default Value	Description
OptControlDamping	1.0	Damping for implementing control moves in non-linear solution. The control update from the Linear Programming (LP) solution is scaled by this before updating the power flow solution input. A value of less than one (assuming pure linear system) causes multiple iterations, but a solution that approaches the optimal feasible solution. A value larger than one causes fewer iterations (possibly only one) but may terminate with a solution that is feasible but not optimal. A value of less than 1.0 can cause funny results in the feasibility solution where the cheaper controls do not get fully utilized before more expensive controls get used.
OptConstraintDamping	0.9	Damping of recommended constraint moves for linear solution. A value of 1.0 solves constraints to their limits. Undersolve constraints by setting to less than 1.0. Oversolve constraints by setting to greater than 1.0 (not recommended). Setting this parameter to less than 1.0 does <u>not</u> cause the final constraint values to be undersolved, but does help reduce the amount of oversolved constraints in a pure feasibility solution.
MaxSwitchAttempts	160	Maximum number of switching plans to be generated. It is used by tree switching mode reconfiguration functions in the uopf solution.
MarginPITol	50.0	Tolerance used to calculate Amps margin PI.
DnstrOfBrViol	1	Allows switch controls downstream of worst flow violation to be used. It is used by tree switching mode reconfiguration functions.
UpstrOfBrViol	0	Allows switch controls upstream of worst flow violation to be used. It is used by tree switching mode reconfiguration functions.
DnstrOfVloViol	1	Allows switch controls downstream of worst low voltage violation to be used. It is used by tree switching mode reconfiguration functions.
UpstrOfVloViol	0	Allows switch controls upstream of worst low voltage violation to be used. It is used by tree switching mode reconfiguration functions.
DnstrOfVhiViol	0	Allows switch controls downstream of worst high voltage violation to be used. It is used by tree switching mode reconfiguration functions.
UpstrOfVhiViol	1	Allows switch controls upstream of worst high voltage violation to be used. It is used by tree switching mode reconfiguration functions.
ImproveDiscreteMoves	0	Forces discrete control moves in a single state solution to be moved one at a time to discrete positions.
TfSensTolJac	0.0001	Tolerance used to adjust the step size in computing incremental sensitivity of transformer controls. Only used when using the Jacobian. If using transformer models with very small primary or secondary positive sequence series impedances values, it may be required to enter a number that is very small and on the order of these impedances in per-unit. If the number is too small, some numerical accuracy may be lost in the sensitivity calculations.
BreakLoops	1	Activates switch controls that break open loops but otherwise is not activated. It is used by tree switching mode reconfiguration functions.

Attributes	Default Value	Description
Allow2ndRatingViolations	1	Allows building upon solutions with 2 nd (emergency) rating violated. If this parameter is zero, then do not allow building plans upon a previous plan that caused the emergency rating to be violated in an intermediate step. It is used by tree switching mode reconfiguration functions.
MaxSwitchingsPerControl	1	Maximum switchings per control in one plan. It is used by tree switching mode reconfiguration functions.
MaxPlansPerDepth	5	Maximum number of switching plans per depth of switch control moves. It is used by tree switching mode reconfiguration functions.
MaxPlansPerDepth2	4	This number multiplied by depth is added to MaxPlansPerDepth for the actual test. This allows solving more plans at deeper depths. It is used by tree switching mode reconfiguration functions.
FlatStartIter1	1	Use a flat start on first powerflow solution within optimization solution
FlatStartIterNot1	0	Use flat start logic on non-first iterations of plans.
ReuseMatrix	1	Reuse the power flow matrix when possible.
LpEngineType	1	Type of LP engine to use. 0=SLP, 1=LpSolve.
MaxPI	9999999.0	Maximum intermediate Performance Index to build new plans upon. Negative value means ignore this logic. It is used by tree switching mode reconfiguration functions.
SaveViolationsOnPlans	1	Save violation results on plans.
SaveDroppedLoadsOnPlans	1	Save dropped loads on plans.
MaxPfPerState	20	The maximum number of power flow solutions per switching state. This is the maximum number times the LP/power flow loop can go within the solution of one switching state.
MaxExtraConstraints	6	The maximum number of constraints to be added after violated constraints are attempted to be reduced but cannot be reduced further. The count is applied only within a single switching state. To have an effect, MaxExtraConstraints must be less than MaxPfPerState.

Attributes	Default Value	Description
FeasibleTol kVA	0	Limit parameters
OverSatTol kVA	0.005	FeasibleTol is the amount over the limit allowed while still determining the constraint to be feasible.
FeasibleTol kVARhi	0	OverSatTol is the amount under the limit to be the lower range of the oversatisfied constraint region.
OverSatTol kVARhi	0.5	
FeasibleTol kVARlo	0	The possible limit types are:
OverSatTol kVARlo	0.5	• kVARhi (upper reactive power limit in terms of kVAR)
FeasibleTol kW	0.005	• kVARlo (lower reactive power limit in terms of kVAR)
OverSatTol kW	0	• kW (upper real power limit in terms of kW)
FeasibleTol Vhi	0.005	• Vhi (upper voltage limit in terms of p.u. voltage)
OverSatTol Vhi	0	• Vlo (lower voltage limit in terms of p.u. voltage)
FeasibleTol Vlo	0.0005	• Amp (upper ampere limit in terms of p.u. current)
OverSatTol Vlo	0.0005	
FeasibleTol Amp		
OverSatTol Amp		
ControlMoveTol VOLT	0.0001	Control move tolerances by control type. These are the minimum moves to cause further optimization LP/UPF iterations. VOLT is in units of p.u. voltage. Note that this should be larger than the parameter VmmTol in alg.dat file.
ControlMoveTol TAP	0.6	TAP is in units of tap positions.
ControlMoveTol SWSH	0.6	SWSH (switched shunt) is in units of integer steps.
ControlMoveTol SWOC	0.5	SWOC (switch open/close) is open=1 and closed=0.
ControlMoveTol SWPH	0.5	SWPH (switch phase rotation) is in integer rotations of phases.
ControlMoveTol LDGP	0.001	LDGP (load drop group) is in units of scaling times full load.
KeepMoves	0	Keep the optimization control moves in the power system state solution. This is only useful for scenarios with a single optimization plan. Thus, it is incompatible with switching controls being active. The previous default was to assume this value was false.

9.3. Msgdef.dat

The msgdef.dat file is the message definition file used within the power flow/optimization engine. Specific messages can be enabled/disabled through this file. Most of the time, these messages are only seen while debugging, so there is little need to alter this file.

Message Type	Message Text	Status
Node Energization	A node is de-energized.	Disabled
Informative	Generic informational text.	Enabled

Message Type	Message Text	Status
Suspect Conditions	Generic warning text.	Enabled
Debug Output	Generic debug text.	Enabled
Loop Conditions	A loop is detected.	Enabled
Suspect Loads Energized	An AutoExp2 load (a load that has been modeled as constant impedance due to low voltage) is energized.	Disabled
Suspect Loads De-energized	An AutoExp2 load is de-energized.	Disabled
Multiple Slack Generators	Multiple generator slacks in an island.	Disabled
Error Conditions	Generic error text.	Enabled
Switch Flow Limits	Zlink monitored without flow computed.	Disabled
Test Message	From file debug test message.	Enabled

9.4. Phid.dat

The phid.dat file contains the phase identifier information, including phase identifier, bitmask representation, flat start voltage magnitude and angle, and description. Currently, the phase identifiers are specified as follows:

ID	Bitmask Value	Voltage Magnitude	Voltage Angle	Description
a	1	1.0	0.0	"a (red)"
b	2	1.0	-120.0	"b (yellow)"
c	4	1.0	120.0	"c (blue)"
n	8	0.0	0.0	"neutral (n)"

9.5. Tftype.dat

The tftype.dat file contains a connection matrix© and a formula for computing the primitive branch impedance matrix(Zprimitive), from variables specific to the transformer instance for each DMS transformer type. A transformer is assigned to a transformer type to avoid storing data that is duplicated for many transformers. Therefore, the formula specified on the transformer type must provide for a way to make the matrix non-singular, and the transformer instance must provide proper variable values to actually make the instance of the Zprimitive matrix evaluate to a non-singular matrix.

The Zprimitive matrix can typically be expressed in terms of available sequence impedance data from the test results of a transformer instance. The Zprimitive matrix is always symmetrical. The transformer type stores the formulas in a simple "language". The formulas used for the transformer may contain the following operations:

^.	Raise the left operand to the power of the right operand (the right operand is treated as an integer)
*.	Multiply the left and right operands

/.	Divide the left operand by the right (non-zero) operand
+	Add the left and right operands
-.	Subtract the right operand from the left operand

The typical rules of operator precedence apply, with ^ being highest precedence, * and / being next-highest precedence, and + and - being lowest precedence. The operands can be either constants specified in the transformer type, or variables obtained from the transformer instance.

- Constants: The unary minus operator is accepted. The unary plus operator is not accepted. Scientific notation is not accepted.
- Variables (a-z): These are lowercase letters. The values come from a transformer instance.
- Tap ratio variables (R-Z): These are uppercase letters in the range R-Z. The tap ratio variables come from the tap ratio calculations of a transformer instance. Note that only valid entries of R-Z for the transformer are accepted (for example, a transformer type with two tanks cannot use V, W, or Z. A transformer type with no secondary2 windings cannot use X, Y, or Z). The following list gives the tank and "end" for the variable:
 - R – tank 1 primary
 - S – tank 1 secondary1
 - T – tank 2 primary
 - U – tank 2 secondary1
 - V – tank 3 primary
 - W – tank 3 secondary1
 - X – tank 1 secondary2
 - Y – tank 2 secondary2
 - Z – tank 3 secondary2

The tap ratio values are computed from a tap (if modeled at that end for that tank) multiplied by the fixed portion of the tap ratio (modeled for all modeled tanks and modeled ends).

The transformer instance contains a rotation field, which has the effect of rotating the phases of the transformer type when applied to that transformer instance.

The connectivity matrix[©] consists of rows corresponding to branches (windings or grounding branches) and columns corresponding to phase connections. Note that the rows of C are the same as the rows/columns of Zprimitive. The C matrix is a real-number matrix, although internally it is sometimes treated as complex with only the real part of the numbers being non-zero.

A transformer instance need not use all grounding branches or neutral connections; however, the transformer instance must supply sufficient variables to create a non-singular primitive branch admittance matrix (for example, non-zero ground impedance for unused grounds must still be supplied, even though they are not reflected in the final solution).

Each row of C for a transformer instance has one or two non-zero terms. Terms between -0.01 and

0.01 are considered zero. If two terms are present, they must be of opposite sign. If only one term is present, the connection is considered to ground. Positive numbers correspond to the "from" end of branches, and negative numbers correspond to the "to" end of branches.

The initial C matrix may sometimes contain more than two terms in a grounding branch row when optional neutral connection points for the secondary1 tank are modeled. The transformer instance specifies which of the possible secondary1 neutral connections is actually being connected externally. In this case, the non-zero terms in both the secondary1 grounding branch row and the columns of "unconnected neutrals" are treated as zeros. After zeroing the terms in the secondary1 grounding row at the columns of "unconnected neutrals", the same branch connection rules apply. (Specifically, the rules require that there be only one or two non-zero terms per row; if two terms are present, they must be of opposite sign. Zero non-zero terms per row or column are not allowed. Any positive number of non-zero terms per column is valid.)

The "unconnected neutrals" are assumed to only occur at the center tap position of windings. Thus, physically a branch connection exists from the two extreme ends of the winding with the center tap.

"Unconnected neutrals" are center tap points modeled in the transformer types, but not externally connected for the transformer instance. These "unconnected neutral" voltages are not solved. The solution method "reduces" or "equivalences away" these points, but the proper impedance is accurately reflected in the final solution results.

It is not supported to specify a grounded star transformer that has no neutral (of the star) external connection. Instead, use two buses and put a zero-impedance link between all phases except the neutral.

Additionally, the ground branch row for an "end" might not exist for a transformer instance. In this case, the row and column are ignored in the Zprimitive matrix, and the row is ignored in the C matrix. Existence of the grounding branch is explicitly specified in the transformer instance for the primary, secondary1, and secondary2 ends. The existence of the ground branch has a relation to whether the neutrals are connected (however, you cannot have a ground on an unconnected neutral, because that causes three connection points on an unconnected neutral).

Similar to the unconnected neutral, the zigzag type of connection causes an unconnected phase bus, which is explicitly modeled as a column of the Cmatrix of the transformer type. The unconnected "elbow" on the zigzag winding reduces away using the same logic as that of the unconnected neutral. The software "knows about" the unconnected winding because the winding type is specified as zigzag; therefore, there are really two sets of phase buses for the zigzag end of the transformer. It is always assumed that the external connection is to the first phase bus column of C for a particular phase identifier at a zigzag end, and that the "elbow" phase bus column of C is reduced away because it is not externally connected.

The Ynetwork admittance used by the solution is $\text{transpose}(Y_{\text{primitive}} * C)$, where $Y_{\text{primitive}}$ is matrix inverse of Zprimitive, and Y and C have been manipulated according to the optional grounding branches and neutral connections, as specified above.

For a three-phase two-winding transformer (neglecting any grounding branches), the primitive impedance matrix is given below:

	V ₁	V ₂	V ₃	V ₄	V ₅	V ₆
V ₁	A	B	C	C	C	C
V ₂	B	D	C	C	C	C
V ₃	C	C	A	B	C	C
V ₄	C	C	B	D	C	C
V ₅	C	C	C	C	A	B
V ₆	C	C	C	C	B	D

The voltages V₁ to V₆ represent the six winding voltages.

In a typical wye-wye connection, the primary phase A connects winding1 from phase "A" to ground, winding2 from secondary phase "a" to ground, winding3 from primary phase "B" to ground, winding4 from secondary phase "b" to ground, winding5 from primary phase "C" to ground, and winding6 from secondary phase "c" to ground.

In the provided tftype.dat file, the matrix above is represented with transformer(TF) instance impedance variables as follows (neglecting tap ratios for the moment):

$$A = a + b$$

$$B = b$$

$$C = c$$

$$D = d + b$$

Where:

$a = Z_{p1}$ = primary positive sequence impedance in p.u.

$$b = (2Z_{m1} + Z_{m0})/3$$

$$c = (Z_{m0} - Z_{m1})/3$$

$d = Z_{s1}$ = secondary positive sequence impedance in p.u.

Where:

Z_{m1} = positive sequence magnetizing impedance in p.u.

Z_{m0} = zero sequence magnetizing impedance in p.u.

$$Z_{m1} = 1/Y_{m1}$$

$$Z_{m0} = 1/Y_{m0}$$

$$Y_{m1} = G_{m1} + jB_{m1}$$

$$Y_{m0} = G_{m0} + jB_{m0}$$

G_{m1} = iron loss in p.u.

B_{m1} = reactive part of magnetizing losses in p.u.

B_{m1} is sometimes computed from power factor and magnetizing current magnitude.

Z_{m0} may be assumed to be the same as Z_{m1} if mutuals are not important. Otherwise, Z_{m0} may be computed as a fraction of Z_{m1} based on the transformers of a similar construction type. Z_{m0} is less than Z_{m1} for core type transformers.

The grounding impedances are typically the terms alone on the diagonal at the end of the transformer type's impedance matrix. The grounding impedances normally have no mutual coupling to other windings, so grounding impedances are straightforward to include. However, each transformer type may need a different number of grounding impedances.

For example, a delta-delta transformer needs none, while a wye-wye-wye needs three. Thus, there are slight differences in the variable names used for grounding depending on how many windings and grounds are present.

For example, a wye-wye transformer is given below:

	V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8
V_1	A	B	C	C	C	C	0	0
V_2	B	D	C	C	C	C	0	0
V_3	C	C	A	B	C	C	0	0
V_4	C	C	B	D	C	C	0	0
V_5	C	C	C	C	A	B	0	0
V_6	C	C	C	C	B	D	0	0
V_7	0	0	0	0	0	0	Z_{GP}	0
V_8	0	0	0	0	0	0	0	Z_{GS}

where the Z_{GP} term is the primary grounding impedance connected from the primary neutral to ground, and the Z_{GS} term is the secondary grounding impedance connected from the secondary neutral to ground.

If the transformer was a wye-delta connection, the primitive Zmatrix looks exactly the same, except that the column and row for winding8 are omitted. Of course, the connection matrices for the wye-wye and wye-delta are also different.

	V_1	V_2	V_3	V_4	V_5	V_6	V_7
V_1	A	B	C	C	C	C	0
V_2	B	D	C	C	C	C	0
V_3	C	C	A	B	C	C	0
V_4	C	C	B	D	C	C	0
V_5	C	C	C	C	A	B	0
V_6	C	C	C	C	B	D	0
V_7	0	0	0	0	0	0	Z_{GP}

If the connection was delta-wye, the matrix looks very similar, but the Z_{GP} term is replaced with the

Z_{GS} term, and the seventh winding represents the secondary grounding branch.

	V_1	V_2	V_3	V_4	V_5	V_6	V_7
V_1	A	B	C	C	C	C	0
V_2	B	D	C	C	C	C	0
V_3	C	C	A	B	C	C	0
V_4	C	C	B	D	C	C	0
V_5	C	C	C	C	A	B	0
V_6	C	C	C	C	B	D	0
V_7	0	0	0	0	0	0	Z_{GS}

Adding Tap Ratios

Adding taps to the primitive impedance matrix is most easily understood by taking the simple winding pair given previously.

0	0	0	C	C	0	0	0
0	0	0	C	C	0	0	0
C	C	C	A	B	C	C	C
C	C	C	B	D	C	C	C
0	0	0	C	C	0	0	0
0	0	0	C	C	0	0	0

Now, A, B, C, and D are modeled exactly as before, except that each row and column is multiplied by the tap ratio of that winding. Diagonals are multiplied by the square of the tap ratio of that column. The matrix is given below, where R is the primary tap ratio and S is the secondary tap ratio.

0	0	0	R^*C	S^*C	0	0	0
0	0	0	R^*C	S^*C	0	0	0
R^*C	C	C	R^*R^*A	R^*S^*B	R^*C	R^*C	R^*C
S^*C	S^*C	S^*C	R^*S^*B	S^*S^*D	S^*C	S^*C	S^*C
0	0	0	R^*C	S^*C	0	0	0
0	0	0	R^*C	S^*C	0	0	0

Tap ratios are used in the following sense. Assume a transformer with no load, and a primary voltage of one p.u. voltage. A secondary tap ratio of $S=1.0$ gives nominal voltage on the secondary. With $S=1.05$, one expects to see a secondary voltage of 105% or 1.05p.u. Similarly, a primary tap ratio of $R=1.0$ gives nominal voltages. With $R=1.05$ and $S=1.0$, a primary voltage of 1.05p.u. is needed to achieve nominal voltage on the secondary. So, both R and S are the ratio of turns to the nominal turns at that end of the transformer. Note that in this implementation, all wye-delta p.u. conversions are handled in the connection matrices and are not assumed to occur in the tap ratios, so the tap ratios are expected to be near one.

Figure below shows the simplified transformer model for a typical transformer winding pair

(neglecting phasing).

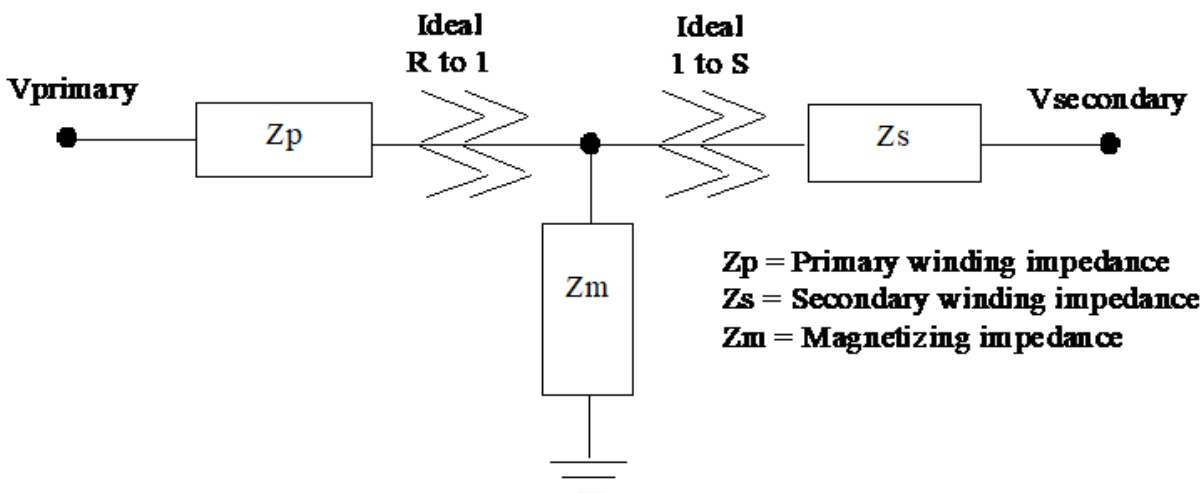


Figure 73. Single-Phase Wye Connection

The model above is a phase model representation for a single-phase wye-wye connection with the neutral grounded. Of course, this winding pair shows no information about interphase mutual couplings. However, this simplified view of a winding pair is the basis for the more-complex phase-oriented model.

Connection Matrices

Connection matrices include the p.u. transformations for line-to-line vs. line-to-ground voltages. In the phase p.u. domain, the p.u. voltage is a line-to-ground voltage. Thus, any windings that connect line-to-line must be multiplied by the factor one over the square root of three.

Similarly, a center tap winding must use a factor of 0.5 for each winding when the windings are connected in series from line to ground (or neutral). A connection from neutral to ground also gets a multiplier of 1, and the p.u. voltages from neutral to ground are normally computed to be near zero (for nearly balanced networks).

An example of a connection matrix for a wye-delta is given below. Note that this shifts the phase voltages by 30degrees from primary to secondary on the same phase. Other connections of delta-wye are possible to shift by 150degrees.

Note also that uppercase phase letters represent primary and lowercase letters represent secondary.

In this matrix, the letter **A** represents one divided by the square root of three.

	V_A	V_a	V_B	V_b	V_C	V_c	V_n
V_1	A	0	0	0	A	0	0
V_2	0	1	0	0	0	0	-1
V_3	-A	0	A	0	0	0	0
V_4	0	0	0	1	0	0	-1

V_5	0	0	-A	0	A	0	0
V_6	0	0	0	0	0	1	-1
V_7	0	0	0	0	0	0	1

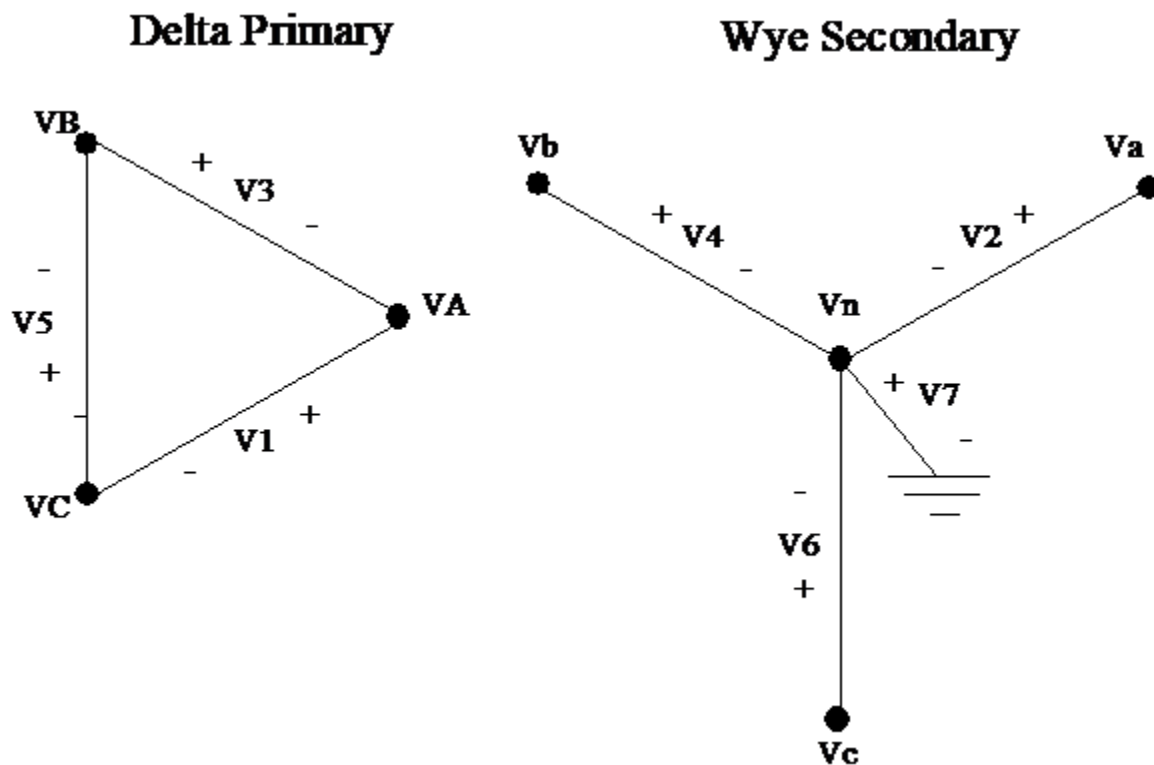


Figure 74. Single-Phase Delta Wye Connection

10. Line Impedance and Admittance Calculations

This chapter gives the equations that are used to compute the series impedance matrix and the shunt admittance matrix for various overhead line and underground cable configurations.

The following configurations are supported in ADMS:

- Overhead Lines
- Tape Shielded Cables: 1 cable per phase (optional bare wire neutral)
- Tape Shielded Cable: 1 three-phase cable (optional bare wire neutral)
- Concentric Neutral Cables: 1 cable per phase (optional bare wire neutral)
- Concentric Neutral Cable: 1 three-phase cable (optional bare wire neutral)
- Secondary Cable: The series impedance matrix and the shunt admittance matrix are calculated from positive, negative, and zero sequence data

The equations given in this chapter were extracted from *Distribution System Modeling and Analysis* by William H. Kersting (CRC Press).

The relative permittivity values used in calculating the shunt admittances for all cables are listed in table below.

Table 19. Values of Relative Permittivity (ϵ_r)

Material	Range of Values for ϵ_r	Value Used
Polyvinyl Chloride(PVC)	3.4 – 8.0	3.4
Ethylene-Propylene Rubber (EPR)	2.5 – 3.5	2.5
Polyethylene (PE)	2.5 – 2.6	2.5
Cross-Linked Polyethylene (XLPE)	2.3 – 6.0	2.3
Steel in Paper	Not listed	2.3
High Molecular Weight Polyethylene (HMWPE)	Not listed	2.3
Tree-Retardant Cross-Linked Polyethylene (TR_XLPE)	Not listed	2.3
Paper Insulated Lead Covered (PILC)	Not listed	2.3

10.1. Overhead Lines

Self impedance (modified Carson's equation):

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_i} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B1.1})$$

Where:

z_{ii} is the self impedance of conductor i (Ω/kft)

r_i is the conductor per-unit length resistance (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_i is the geometric mean radius of conductor i (feet)

Mutual impedance:

$$Z_{ij} = r_d + j0.022981 \cdot \left(\ln \frac{1}{D_{ij}} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B1.2})$$

Where:

z_{ij} is the mutual impedance between conductors i and j (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

D_{ij} is the distance between conductors i and j (feet)

Shunt admittance:

$$P_{ii} = 11.17689 \cdot 5.28 \cdot \ln \left(\frac{S_{ii}}{RD_i} \right) \text{ kft}/\mu\text{f} \quad (\text{Equation B1.3})$$

$$P_{ij} = 11.17689 \cdot 5.28 \cdot \ln \left(\frac{S_{ij}}{D_{ij}} \right) \text{ kft}/\mu\text{f} \quad (\text{Equation B1.4})$$

Where:

P_{ii} is the self potential coefficient ($\text{kft}/\mu\text{F}$)

P_{ij} is the mutual potential coefficient ($\text{kft}/\mu\text{F}$)

S_{ii} is the distance from conductor i to its image i' (feet)

RD_i is the radius of conductor i (feet)

S_{ij} is the distance from conductor i to the image of conductor j' (feet)

D_{ij} is the distance from conductor i to conductor j (feet)

$$Y_{abc} = j2 \cdot \pi \cdot f \cdot [P_{abc}]^{-1} \mu\text{S/kf} \quad (\text{Equation B1.5})$$

Where:

Y_{abc} is the shunt admittance matrix ($\mu\text{S/kft}$)

P_{abc} is the phase potential coefficient matrix calculated using equations (3) and (4)

f is the system frequency

10.2. Tape Shielded Cables - One Cable per Phase

Self impedance of phase conductor (modified Carson's equation):

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_i} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B2.1})$$

Where:

Z_{ii} is the self impedance of conductor i (Ω/kft)

r_i is the conductor per-unit length resistance (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_i is the geometric mean radius of conductor i (feet)

The resistance and GMR of the tape shield conductor are given below:

$$r_{shield} = \frac{0.003565559}{d_s T_s} \text{ } (\Omega/\text{kft}) \quad (\text{Equation B2.2})$$

$$GMR_{shield} = \frac{d_s - T_s}{24} \text{ (ft)} \quad (\text{Equation B2.3})$$

Where:

r_{shield} is the mutual tape shield resistance, assuming earth resistivity is equal to $2.3715\text{e-}08$ ohm-meter (ohms/kft)

GMR_{shield} is the tape shield geometric mean radius (feet)

d_s is the outside diameter over the tape shield (inches)

T_s is the thickness of the tape shield (inches)

Mutual impedance (modified Carson's equation):

$$Z_{ij} = r_d + j0.022981 \cdot \left(\ln \frac{1}{D_{ij}} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B2.4})$$

Where:

Z_{ij} is the mutual impedance between conductors i and j (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

D_{ij} is the distance between conductors i and j (feet)

The tape shield conductor is removed through Kron reduction.

Shunt admittance:

$$Y_{ii} = 6.37038 \cdot \varepsilon \cdot \ln \left(\frac{R_b}{RD_i} \right) \mu\text{S/kft} \quad (\text{Equation B2.5})$$

$$Y_{ij} = 0.0 \mu\text{S/kft} \quad (\text{Equation B2.6})$$

Where:

Y_{ii} is the self shunt admittance of a tape shielded cable

Y_{ij} is the mutual shunt admittance of a tape shielded cable

ε is the relative permittivity value (ε_r)

R_b is the radius of circle passing through the center of the tape shield (inches), which equals the difference of the diameter over the tape shield and the thickness of the tape shield divided by 2

RD_i is the radius of phase conductor i (inches)

10.3. Tape Shielded Cables - One Three-Phase Cable

Self impedance of phase conductor (modified Carson's equation):

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_i} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B3.1})$$

Where:

z_{ii} is the self impedance of conductor i (Ω/kft)

r_i is the conductor per-unit length resistance (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_i is the geometric mean radius of conductor i (feet)

The resistance and GMR of the tape shield conductor are given below:

$$r_{shield} = \frac{0.003565559}{d_s T_s} (\Omega/\text{kft}) \quad (\text{Equation B3.2})$$

$$GMR_{shield} = \frac{d_s - T_s}{24} (\text{ft}) \quad (\text{Equation B3.3})$$

Where:

r_{shield} is the mutual tape shield resistance, assuming earth resistivity is equal to $2.3715\text{e-}08$ ohm-meter (ohms/kft)

GMR_{shield} is the tape shield geometric mean radius (feet)

d_s is the outside diameter over the tape shield (inches)

T_s is the thickness of the tape shield (inches)

Mutual impedance (modified Carson's equation):

$$Z_{ij} = r_d + j0.022981 \cdot \left(\ln \frac{1}{D_{ij}} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B3.4})$$

Where:

Z_{ij} is the mutual impedance between conductors i and j (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

D_{ij} is the distance between conductors i and j (feet)

The tape shield conductors are removed through Kron reduction.

Shunt admittance:

$$Y_{ii} = 6.37038 \cdot \epsilon \cdot \ln \left(\frac{R_b}{RD_i} \right) \mu\text{S/kft} \quad (\text{Equation B3.5})$$

$$Y_{ij} = 0.0 \mu\text{S/kft} \quad (\text{Equation B3.6})$$

Where:

Y_{ii} is the self shunt admittance of a tape shielded cable

Y_{ij} is the mutual shunt admittance of a tape shielded cable

ϵ is the relative permittivity value (ϵ_r)

R_b is the radius of circle passing through the center of the tape shield (inches), which equals the difference of the diameter over the tape shield and the thickness of the tape shield divided by 2

RD_i is the radius of phase conductor i (inches)

10.4. Concentric Neutral Cables - One Cable per Phase

Self impedance of phase conductor (modified Carson's equation):

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_i} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B4.1})$$

Where:

Z_{ii} is the self impedance of conductor i (Ω/kft)

r_i is the conductor per-unit length resistance (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_i is the geometric mean radius of conductor i (feet)

Mutual impedance (modified Carson's equation):

$$Z_{ij} = r_d + j0.022981 \cdot \left(\ln \frac{1}{D_{ij}} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B4.2})$$

Where:

Z_{ij} is the mutual impedance between conductors i and j (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

D_{ij} is the distance between conductors i and j (feet)

Self impedance of the concentric neutral "bundle":

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_{cn}} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B4.3})$$

Where:

Z_{ii} is the self impedance of concentric neutral wire i (Ω/kft)

r_i is the conductor per-unit length resistance for one strand (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_{cn} is the geometric mean radius of conductor concentric neutral bundle, which is represented by the following equation (feet):

$$GMR_{cn} = \sqrt[k]{GMR_s \cdot k \cdot R^{k-1}} \quad (\text{Equation B4.4})$$

Where:

k is the number of concentric neutral strands

GMR_s is the geometric mean radius of a single neutral strand (feet)

R is the radius of a circle passing through the center of the concentric neutral strands (feet)

$$Y_{ii} = \frac{6.37038 \cdot \epsilon}{\ln \left(\frac{R_b}{RD_c} \right) - \frac{1}{k} \ln \frac{k \cdot RD_s}{R_b}} \quad (\mu\text{S/kft}) \quad (\text{Equation B4.5})$$

$$Y_{ij} = 0.0 \mu\text{S/kft} \quad (\text{Equation B4.6})$$

Where:

Y_{ii} is the self shunt admittance of a tape shielded cable

Y_{ij} is the mutual shunt admittance of a tape shielded cable

ϵ is the relative permittivity value (ϵ_r)

R_b is the radius of circle passing through the center of concentric neutral (inches)

RD_c is the radius of phase conductor i (inches)

RD_s = radius of concentric neutral strand (inches)

10.5. Concentric Neutral Cables - One Three-Phase Cable

Self impedance of phase conductor (modified Carson's equation):

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_i} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B5.1})$$

Where:

Z_{ii} is the self impedance of conductor i (Ω/kft)

r_i is the conductor per-unit length resistance (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_i is the geometric mean radius of conductor i (feet)

Mutual impedance between phase conductors (modified Carson's equation):

$$Z_{ij} = r_d + j0.022981 \cdot \left(\ln \frac{1}{D_{ij}} + 7.93402 \right) \text{ ohms/kft} \quad (\text{Equation B5.2})$$

Where:

Z_{ij} is the mutual impedance between conductors i and j (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

D_{ij} is the distance between conductors i and j (feet)

Mutual impedance between phase conductor and concentric neutrals (modified Carson's equation):

$$Z_{ij} = r_d + j0.022981 \cdot (7.93402 - \ln (GMR)) \text{ ohms/kft} \quad (\text{Equation B5.3})$$

Where:

Z_{ij} is the mutual impedance between conductors i and j (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR is the effective GMR given by the following equation:

$$GMR = \sqrt[k]{Radius^k - \left(\frac{PhaseDist}{\sqrt{3}}\right)^k} \quad (\text{Equation B5.4})$$

Where:

k is the number of concentric neutral strands

$Radius$ is the distance between phase conductor and concentric neutral strand (inches)

$PhaseDist$ is the phase conductor spacing (inches)

Self impedance of the concentric neutral "bundle":

$$Z_{ii} = (r_i + r_d) + j0.022981 \cdot \left(\ln \frac{1}{GMR_{cn}} + 7.93402\right) \text{ ohms/kft} \quad (\text{Equation B5.5})$$

Where:

z_{ii} is the self impedance of concentric neutral wire i (Ω/kft)

r_i is the conductor per-unit length resistance for one strand (Ω/kft)

r_d is the earth resistance (Ω/kft), which equals $0.00030083 \cdot \text{system frequency (in Hz)}$

GMR_{cn} is the geometric mean radius of conductor concentric neutral bundle, which is represented by the following equation (feet):

$$GMR_{cn} = \sqrt[k]{GMR_s \cdot k \cdot R^{k-1}} \quad (\text{Equation B5.6})$$

Where:

k is the number of concentric neutral strands

GMR_s is the geometric mean radius of a single neutral strand (feet)

R is the radius of a circle passing through the center of the concentric neutral strands (feet)

$$Y_{ii} = \frac{6.37038 \cdot \epsilon}{\ln \left(2.0 \cdot \frac{R_b \cdot \frac{PhaseDist}{\sqrt{3}}}{RD_c} \right) - \frac{1}{k} \ln \frac{k \cdot RD_s}{R_b}} \quad (\mu\text{S/kft}) \quad (\text{Equation B5.7})$$

$$Y_{ij} = 0.0 \mu\text{S/kft} \quad (\text{Equation B5.8})$$

Where:

Y_{ii} is the self shunt admittance of a tape shielded cable

Y_{ij} is the mutual shunt admittance of a tape shielded cable

ϵ is the relative permittivity value (ϵ_r)

R_b is the radius of circle passing through the center of the concentric neutral strand (inches)

$PhaseDist$ is the phase conductor spacing (inches)

RD_c is the radius of phase conductor i (inches)

RD_s is the radius of concentric neutral strand (inches)

10.6. Secondary Cable

The Secondary Cable impedance is calculated from Sequence Impedance (positive, negative, and zero) values. The positive, negative, and zero sequence impedances can then be translated into the phase impedance matrix by using the equation below:

$$Z_{phase} \cong [A_s] \cdot [Z_{seq}] \cdot [A_s]^{-1} \quad (Equation B6.1)$$

Where:

$$Z_{seq} = \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_+ & 0 \\ 0 & 0 & Z_- \end{bmatrix} \quad A_s = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \quad (Equation B6.2)$$

X_1 , X_0 is the positive sequence impedance and zero sequence impedance value, respectively, given at 1 ft spacing and must be converted to the real value according to the construction model.

$$X = X_{std} \cdot \ln(Distance/GMR) \quad (Equation B6.3)$$

The final value of X is calculated as follows:

$$X = X(\text{at 1 ft spacing}) \cdot \ln(Distance/GMR)/\ln(1 \text{ feet}/GMR) \quad (Equation B6.4)$$

where *Distance* is the geometric mean of the phase conductor spacing (ft) distance given in the construction model.

For example, for a three phase cable,

$$Distance = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} \quad (Equation B6.5)$$

11. Manual Entry Files

Manual_Entry_Init.dat and Manual_Entry.dat use the same set of commands to set/unset the manual entries. These files have the following properties:

- Every line is one command. A line starting with "#" is a comment line. Any word with underscore(s) should be replaced with an actual substation/feeder/device name or ID. For example, "SUB_NAME" should be replaced by the name of a substation, such as "RAVEN".
- The last word of most manual entry commands is ENABLE_DISABLE. If the last word is ENABLE, this manual entry is enabled; otherwise, this manual entry is disabled. There are a few commands that do not end with ENABLE_DISABLE. In these cases, manual entry is always enabled.

All the manual entry commands are listed and explained below.

1. SUBSTATION

- Set the substation load scaling factor:

SUB,SUB_NAME,LSCALER,LOAD_SCALER,ENABLE_DISABLE

For example:

SUB,RAVEN,LSCALER,1.2,ENABLE

- Block/unblock all station regulators:

SUB,SUB_NAME,BLKREG,TRUE_FALSE

For example:

SUB,RAVEN,BLKREG,FALSE

- Set the status of voltage reduction regulators:

SUB,SUB_NAME,VREDSTAT,TRUE_FALSE

For example:

SUB,RAVEN,VREDSTAT,TRUE

- Set the status of disabling the advanced applications:

SUB,SUB_NAME,DISABLEDSO,TRUE_FALSE

For example:

SUB,RAVEN,DISABLEDSO,TRUE

- Enable/disable optimization for all switches in the substation:

SUB,SUB_NAME,ENABLESWITCHUSEROPT,TRUE_FALSE

For example:

SUB,RAVEN,ENABLESWITCHUSEROPT,TRUE

- Enable/disable optimization for all shunt impedances in the substation:

SUB,SUB_NAME,ENABLESHIMPUSEROPT,DSO_USER_ENABLE

For DSO_USER_ENABLE: 1:User enable, 2:User disable, 3:SCADAdecides

For example:

SUB,RAVEN,ENABLESHIMPUSEROPT,1

- Enable/disable optimization for all tap changing transformers in the substation:

SUB,SUB_NAME,ENABLETAPUSEROPT,DSO_USER_ENABLE

For DSO_USER_ENABLE: 1:User enable, 2:User disable, 3:SCADAdecides

For example:

SUB,RAVEN,ENABLETAPUSEROPT,1

- Set high and low voltage limits (per unit) for all loads in the substation:

SUB,SUB_NAME,LOADVLIMHIGH,HIGH_LIMIT,ENABLE_DISABLE

SUB,SUB_NAME,LOADVLIMLOW,LOW_LIMIT,ENABLE_DISABLE

For example:

SUB,RAVEN,LOADVLIMHIGH,1.05,ENABLE

SUB,RAVEN,LOADVLIMLOW,0.95,ENABLE

- Set status for disabling DNAF functions:

SUB,SUB_NAME,DISABLEDNAF,TRUE_FALSE

For example:

SUB,RAVEN,DISABLEDNAF,TRUE

- Set positive and zero sequence equivalent impedance of the transmission system in ohms:

SUB,SUB_NAME,HVEQR1,EQUI_R1

SUB,SUB_NAME,HVEQR0,EQUI_R0

SUB,SUB_NAME,HVEQX1,EQUI_X1

SUB,SUB_NAME,HVEQX0,EQUI_X0

For example:

SUB,RAVEN,HVEQR1,1.51823

SUB,RAVEN,HVEQR0,1.95201

SUB,RAVEN,HVEQX1,6.76062

SUB,RAVEN,HVEQX0,13.76458

- Set the substation LVM target power factor:

SUB,SUB_NAME,VVCMAXPFTARGET,PF_TARGET, ENABLE_DISABLE

SUB,SUB_NAME,VVCMINPFTARGET,PF_TARGET, ENABLE_DISABLE

For example:

SUB,RAVEN,VVCMAXPFTARGET,0.98,ENABLE

SUB,RAVEN,VVCMINPFTARGET,0.95,ENABLE

2. FEEDER

- Set three phase feeder head amps and power factor:

FDR,SUB_NAME,FEEDER_NAME,MANAMPSA,TARGET_AMPS, ENABLE_DISABLE

FDR,SUB_NAME,FEEDER_NAME,MANAMPSB,TARGET_AMPS, ENABLE_DISABLE

FDR,SUB_NAME,FEEDER_NAME,MANAMPSC,TARGET_AMPS, ENABLE_DISABLE

FDR,SUB_NAME,FEEDER_NAME,MANAMPPF,TARGET_PF, ENABLE_DISABLE

For example:

FDR,RAVEN,F8864,MANAMPSA,250,ENABLE

FDR,RAVEN,F8864,MANAMPSB,250,ENABLE

FDR,RAVEN,F8864,MANAMPSC,250,ENABLE

FDR,RAVEN,F8864,MANAMPPF,0.95,ENABLE

- Set the feeder flow limit scaling factor:

FDR,SUB_NAME,FEEDER_NAME,FLLIMMLT,SCAL_FACTOR, ENABLE_DISABLE

For example:

FDR,RAVEN,F8864,FLLIMMLT,1,ENABLE

- Set high and low voltage limits (per unit) for all loads on the feeder:

FDR,SUB_NAME,FDR_NAME,LOADVLIMHIGH,HIGH_LIMIT, ENABLE_DISABLE

FDR,SUB_NAME,FDR_NAME,LOADVLIMLOW,LOW_LIMIT, ENABLE_DISABLE

For example:

FDR,RAVEN,F8864,LOADVLIMHIGH,1.05,ENABLE

FDR,RAVEN,F8864,LOADVLIMLOW,0.95,ENABLE

- Enable the feeder LVM target power factor:

FDR,SUB_NAME,FEEDER_NAME,VVCFDRPFTARGET, TRUE_FALSE

For example:

FDR,RAVEN,F8864,VVCFDRPFTARGET,TRUE

- Set the feeder manual LVM target power factor

FDR,SUB_NAME,FEEDER_NAME,VVCMAXPFTARGET, PF_TARGET, ENABLE_DISABLE

FDR,SUB_NAME,FEEDER_NAME,VVCMINPFTARGET, PF_TARGET,ENABLE_DISABLE

For example:

FDR,RAVEN,F8864,VVCMAXPFTARGET,0.95,ENABLE

FDR,RAVEN,F8864,VVCMINPFTARGET,0.98,ENABLE

3. REFERENCE BUS

- Set the reference bus voltage magnitude (percent) and angle (deg)

RFBSIF,SUB_NAME,REFERNCE_BUS_ID,VTARGET, TARGET_VMAG,ENABLE_DISABLE

RFBSIF,SUB_NAME,REFERNCE_BUS_ID,ANTARGET, TARGET_VANG,ENABLE_DISABLE

For example:

RFBSIF,RAVEN,Intfc_RAVEN_XF_2426,VTARGET,100,ENABLE

RFBSIF,RAVEN,Intfc_RAVEN_XF_2426,ANTARGET,0,ENABLE

4. GENERATOR

- Set the generator voltage magnitude regulation target (percent) and real power target (kW)

GEN,SUB_NAME,GEN_ID,VTARGET,TARGET_V, ENABLE_DISABLE

GEN,SUB_NAME,GEN_ID,PTARGET,TARGET_P, ENABLE_DISABLE

For example:

GEN,GREENACRES,101080001,VTARGET,105,ENABLE

GEN,GREENACRES,101080001,PTARGET,30,ENABLE

5. SHUNT IMPEDANCE

- Set the shunt impedance step. If the step > 0, the status is set to On; otherwise, the status is Off:

SHIMP,SUB_NAME,SHUNTIMP_ID,STEP,STEP_NUMBER

For example:

SHIMP,RAVEN,10218784,STEP,1

- Set the shunt impedance on and off level according to control type:

SHIMP,SUB_NAME,SHUNTIMP_ID,ONLEVEL,ON_LEVEL, ENABLE_DISABLE

SHIMP,SUB_NAME,SHUNTIMP_ID,OFFLEVEL,OFF_LEVEL, ENABLE_DISABLE

For example:

SHIMP,RAVEN,10218784,ONLEVEL,200.0,ENABLE

SHIMP,RAVEN,10218784,OFFLEVEL,300.0,ENABLE

- Set the shunt impedance maximum daily number of switching operations:

SHIMP,SUB_NAME,SHUNTIMP_ID,MAXSWIT,MAX_SWIT_NUM, ENABLE_DISABLE

For example:

SHIMP,RAVEN,10218784,MAXSWIT,4,ENABLE

- Set the shunt impedance hours between consecutive switching operations:

SHIMP,SUB_NAME,SHUNTIMP_ID,CONSSWIT, CONS_SWIT_HOUR, ENABLE_DISABLE

For example:

SHIMP,RAVEN,10218784,CONSSWIT,12,ENABLE

- Enable the shunt impedance local control (voltage) override:

SHIMP,SUB_NAME,SHUNTIMP_ID,VLTLOCALOVERRIDE, TRUE_FALSE,ENABLE_DISABLE

For example:

SHIMP,GREENACRES,10218140,VLTLOCALOVERRIDE,TRUE, ENABLE

- Set the shunt impedance local control (voltage) override on and off level:

SHIMP,SUB_NAME,SHUNTIMP_ID,VLTONLEVEL,ON_LEVEL, ENABLE_DISABLE

SHIMP,SUB_NAME,SHUNTIMP_ID,VLOFFLEVEL,OFF_LEVEL, ENABLE_DISABLE

For example:

SHIMP,GREENACRES,10218140,VLTONLEVEL,1.2,ENABLE

SHIMP,GREENACRES,10218140,VLOFFLEVEL,1.4,ENABLE

- Set the user enable/disable optimization flag for a shunt impedance:

SHIMP,SUB_NAME,SHUNTIMP_ID,USERENABLEOPT, TRUE_FALSE

For example:

SHIMP,RAVEN,10218784,USERENABLEOPT,TRUE

- Set the SCADA enable/disable optimization flag for a shunt impedance:

SHIMP,SUB_NAME,SHUNTIMP_ID,SCADAENABLEOPT, TRUE_FALSE

For example:

SHIMP,RAVEN,10218784,SCADAENABLEOPT,TRUE

- Set the shunt impedance regulated branch:

SHIMP,SUB_NAME,SHUNTIMP_ID,RBRANCH,BRANCH_ID

For example:

SHIMP,GREENACRES,10218140,RBRANCH,22417898

- Set the shunt impedance regulated branch direction (1:RegBr_In, 2:RegBr_Out):

SHIMP,SUB_NAME,SHUNTIMP_ID,BRANCHDIR,IN_OUT

For example:

SHIMP,GREENACRES,10218140,BRANCHDIR,1

- Set the shunt impedance regulated branch end (0:RegBr_None, 1:RegBr_From, 2:RegBr_To, 3:RegBr_Te):

SHIMP,SUB_NAME,SHUNTIMP_ID,BRANCHEND,BRANCH_END

For example:

SHIMP,GREENACRES,10218140,BRANCHEND,1

- Set the shunt impedance movement optimization scaling cost:

SHIMP,SUB_NAME,SHUNTIMP_ID,SCALECOSTOPT,scale_cost, ENABLE_DISABLE

For example:

SHIMP,GREENACRES,10218140,SCALECOSTOPT,10,ENABLE

6. Transformer

- Set the flow limits for the transformer. The unit is defined in the transformer model (amps, kVA, or MVA):

TF,SUB_NAME,TF_ID,FLOWLIM,FLOW_LIMIT,ENABLE_DISABLE

For example:

TF,RAVEN,RAVEN_XF_2426,FLOWLIM,1000.0,ENABLE

7. TAP CHANGER

- Set the tap changer control mode for Remote_Local control type tap changer (1:Local control, 2:Remote control, TAP_SIDE: 1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,TAPCNTRL, CNTRL_MODE,ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,TAPCNTRL,1

- Set the initial tap position (TAP_SIDE: 1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,TAPINIT,INIT_POS, ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,TAPINIT,0,ENABLE

- Set the voltage target (1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,VTARGET,VOLT_TARGET, ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,VTARGET,120,ENABLE

- Set the voltage bandwidth (TAP_SIDE: 1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,BANDWIDTH,VOLT_BAND, ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,BANDWIDTH,3,ENABLE

- Set the line drop compensation resistance and reactance in Volts (TAP_SIDE: 1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,LDCCMPRV,R_VOLT, ENABLE_DISABLE

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,LDCCMPXV,X_VOLT, ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,LDCCMPRV,2,ENABLE

TAPCHR,RAVEN,101110916,2,LDCCMPXV,5,ENABLE

- Set the blocked regulator flag: TAP_SIDE (1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,BLOCKON,TRUE_FALSE

For example:

TAPCHR,RAVEN,101110916,2,BLOCKON,FALSE

- Set voltage reduction on or off (TAP_SIDE: 1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,VREDCTON,TRUE_FALSE

For example:

TAPCHR,RAVEN,101110916,2,VREDCTON,FALSE

- Set the voltage deduction value (TAP_SIDE: 1:Primary, 2:Secondary, 3:Tertiary):

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,VREDVAL,VALUE_VOLT, ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,VREDVAL,4,ENABLE

- Set the user enable/disable optimization flag for a tap changing transformer:

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,USERENABLEOPT, TRUE_FALSE

For example:

TAPCHR,RAVEN,101110916,2,USERENABLEOPT,TRUE

- Set the SCADA enable/disable optimization flag for a tap changing transformer:

TAPCHR,SUB_NAME,TF_ID,TAP_SIDE,SCADAENABLEOPT, TRUE_FALSE

For example:

TAPCHR,RAVEN,101110916,2,SCADAENABLEOPT,TRUE

- Set the tap changer movement optimization scaling cost:

TAPCHR,SUB_NAME,TF_ID,TAPP_SIDE,SCALECOSTOPT, scale_cost, ENABLE_DISABLE

For example:

TAPCHR,RAVEN,101110916,2,SCALECOSTOPT,10,ENABLE

8. TEMP GROUND

- Add the ground connection:

TEMPGROUND,SUB_NAME,LINE_ID,GROUND_CONNECTION_ID

For example:

TEMPGROUND,RAVEN,22459228,FAULT123

9. LOAD

- Set the high and low voltage limits (per unit) for a load:

LOAD,SUB_NAME,LOAD_ID,VLIMHIGH,HIGH_LIMIT, ENABLE_DISABLE

LOAD,SUB_NAME,LOAD_ID,VLIMLOW,LOW_LIMIT, ENABLE_DISABLE

For example:

LOAD,RAVEN,17465888XRES,VLIMHIGH,1.06,ENABLE

LOAD,RAVEN,17465888XRES,VLIMLOW,0.94,ENABLE

- Set the Nominal P and Q values for a load (in kW, kVAR):

LOAD,SUB_NAME,LOAD_ID,NOMPA,Nom_P_PhA

LOAD,SUB_NAME,LOAD_ID,NOMQA,Nom_Q_PhA

LOAD,SUB_NAME,LOAD_ID,NOMPB,Nom_P_PhB

LOAD,SUB_NAME,LOAD_ID,NOMQB,Nom_Q_PhB

LOAD,SUB_NAME,LOAD_ID,NOMPC,Nom_P_PhC

LOAD,SUB_NAME,LOAD_ID,NOMQC,Nom_Q_PhC

For example:

LOAD,RAVEN,17465888XRES,NOMPA,50

LOAD,RAVEN,17465888XRES,NOMQA,20

LOAD,RAVEN,17465888XRES,NOMPB,50

LOAD,RAVEN,17465888XRES,NOMQB,20

LOAD,RAVEN,17465888XRES,NOMPC,50

LOAD,RAVEN,17465888XRES,NOMQC,20

10. SWITCH

- Set flow limits for the switch. The unit is defined in the switch model (amps, kVA, or MVA):

SWT,SUB_NAME,SWITCH_ID,FLOWLIM,FLOW_LIMIT, ENABLE_DISABLE

For example:

SWT,RAVEN,22239986,FLOWLIM,100,ENABLE

- Set the user enable/disable optimization flag for a switch:

SWT,SUB_NAME,SWT_ID,USERENABLEOPT,TRUE_FALSE

For example:

SWT,RAVEN,22239986,USERENABLEOPT,TRUE

- Set the SCADA enable/disable optimization flag for a switch:

SWT,SUB_NAME,SWT_ID,SCADAENABLEOPT,TRUE_FALSE

For example:

SWT,RAVEN,22239986,SCADAENABLEOPT,TRUE

11. OTHERS

- Disable all the previously set manual entries:

RESETALL

- Set the status for disabling all advanced functions for all substations:

ALLSUB,DISABLEDSO,TRUE_FALSE

For example:

ALLSUB,DISABLEDSO,TRUE

- Set the status for disabling all DNAF functions for all substations:

ALLSUB,DISABLEDNAF,TRUE_FALSE

For example:

ALLSUB,DISABLEDNAF,TRUE

- Set the load scaling factor for all substations:

ALLSUB,LSCALER,LOAD_SCALER,ENABLE_DISABLE

For example:

ALLSUB,LSCALER,1,ENABLE

- Enable/disable optimization for all shunt impedances in all substations (DSO_USER_ENABLE: 1:User enable, 2:User disable, 3:SCADA decides):

ALLSUB,ENABLESHIMPUSEROPT,DSO_USER_ENABLE

For example:

ALLSUB,ENABLESHIMPUSEROPT,1

- Enable/disable optimization for all tap changers in all substations: DSO_USER_ENABLE (1:User enable, 2:User disable, 3:SCADA decides):

ALLSUB,ENABLETAPUSEROPT,DSO_USER_ENABLE

For example:

ALLSUB,ENABLETAPUSEROPT,1

- Set the feeder inhibit alarm flag for all feeders:

ALLFDR,INHIBITALARM,TRUE_FALSE

For example:

ALLFDR,INHIBITALARM,TRUE

12. DNAF Alarm Format

This section describes the naming convention for DNAF alarms and supported formats for DNAF alarm exception messages.

12.1. DNAF Alarm Exception Naming Convention

The DNAF alarm naming convention is shown below. The exception name consists of an application prefix followed by an identification number. Generally, the identification number is made up of a device identifier, an event type identifier, and a status identifier. Table below shows an example alarm exception.

See the *ADMS Alarmable Events* spreadsheet for a full list of all events.

Table 20. Sample DNAF Alarm Exceptions and Display

New Four Digit Number DNAF Alarm Exception Naming						
Full Name	Application Prefix	Device Digits	Type Digits	Status Digits	Alarm Message Text	Char1 Format
PF141	PF	1 = Station	4 = Flow Violation	1 = Violation	RAVEN FLOW VIOLATION was LEVEL 2, now LEVEL 1, Limit is Long Term	Format 1

12.2. Alarm Log Message (CHAR1) Formats

Two formats are supported for DNAF alarm exception messages. More formats may be added in future releases as new alarm categories and exceptions are added.

Most DNAF alarm messages issued from the DMS/DNAF server use CHAR1 FORMAT 1 (shown in figure below). This format contains the station name, feeder name, and limit (or rating) name for the current violation.

CHAR1 FORMAT 1: (Total Length for this is 79)

Stat ID	Station Name	Feeder ID	ALT FEEDER	Limit Name
8	16	20	15	20

Figure 75. CHAR1 FORMAT 1 Example

For switching devices, CHAR1 FORMAT 2 (show in figure below) is used for protection validation violation alarm messages such as Maximum Interruption, Minimum Phase Pick-Up Violation, or Minimum Ground Pick-Up Violation.

CHAR1 FORMAT 2: (Total Length for this is 95)

Stat ID	Station Name	Feeder ID	Device Type	Device ID
8	16	20	14	36

Figure 76. CHAR1 FORMAT 2 Example

Substation internal devices modeled in Substation Editor normally have a Device name and a 36-character GUID value. For external devices modeled in a GIS, the device ID is usually provided but is optional. Consider using the device name for substation internal devices, and the device ID for GIS provided substation external devices.

13. DNAF Result Files

This section details the contents of the result files for the standard DNAF applications.

The default location of the result files is %IDMS_ROOT%\log\DNAF\. This location can be changed by modifying the parameter DNAF_OUTPUT_PATH in the DNAF or Combo server configuration file.

13.1. DPF Result File

The DPF results for a station are recorded in files named *DPF Results*<StationName>.dat_file. The data in this file is grouped in sections as follows.

Distribution Power Flow Data

This section provides detailed results from the DPF iterations.

- DPF SOLUTION SUMMARY

This section contains triggering information (power flow execution reason, evaluation time, execution time, last converged evaluation time), power flow status, maximum power flow mismatches for real power, reactive power, and voltage; ID and name of the device connected to the largest voltage mismatch node; BLA status for real, reactive, and apparent power; and the number of BLA and DPF iterations.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The time last modified and revision number for DNAF_PARAMS.xml is also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF INPUT PARAMETERS – POWER FLOW

The section details the values for all the parameters pertaining to DPF from the DNAF Configuration Editor (DCE).

- DNAF INPUT PARAMETERS – BUS LOAD ALLOCATION

The section details the values for all the parameters pertaining to BLA from the DNAF Configuration Editor (DCE).

- ALARM CONFIGURATION

The section lists the status (enabled or disabled) of the following alarms: Failed to Complete, Flow Violation, Low Voltage, High Voltage, Flow Imbalance, Voltage Imbalance, Capacitor Position Validation, Feeder Back-Feed State Caused by Distribution Generation, Tap Position Estimation Source Voltage Adjusted, and AMI Measurement Mismatch as well as the RTN (Return to Normal) components of each alarm.

- VIOLATIONS

The section summarizes the violations found in the following categories: Line Flow Limit, Series Reactor Flow Limit, Switch Flow Limit, Transformer Flow Limit, Load Voltage Limit, and Bus Voltage Limit. The voltage limit sections are for both high and low violations. The component ID, nominal feeder, phasing, per phase values, and per phase limit is given for each violation.

- DPF ITERATION INFORMATION

This section details the real power, reactive power, and voltage mismatch as well as the transformer tap and capacitor moves for each power flow iteration.

- MANUAL ENTRY SUMMARIES

The section summarizes the manually entered values from the following displays: Manual Entry Parameters, Substation Input, Feeder Input, Feeder Voltage Limit, Reference Bus Input, Generator Input, Distributed Energy Resource Input, Capacitor Input, Tap Changer Input, Fault Location Input, Switch Input, Power Transformer Input, Disconnected Distribution Transformer, Load Input, Load Category Input, Bus Voltage Limit Input, Line Input, and Series Reactor Input.

- NETWORK SELECTION INFORMATION

This section lists the reference bus stations, adjacent stations, NTP and DPF energizing stations, the number of topology objects in each device category (lines, switches, feeders, loads, distribution transformers, transformers, shunt impedances, distributed energy resources, and distributed generators), and the closed boundary devices within the network selection.

- INITIAL CONDITIONS

This section includes information on initial conditions, including feeder head manual entry flows, load category information, composite switch status, tap advanced regulation information, source voltages (interface voltages), DER and DG generation outputs, and active inverter functions.

- ANALOG MEASUREMENTS

This section includes a summary of analog measurements in the network selection, grouped by substation.

- STATUS MEASUREMENTS

This section includes a summary of status measurements in the network selection, grouped by substation.

- VOLTAGE MEASUREMENTS

This section includes a summary of voltage measurements in the network selection, grouped by substation as well as corresponding value calculated by DPF for each measurement.

- INVALID ANALOG MEASUREMENTS

This section includes a summary of analog measurements in the network selection with invalid status or quality, grouped by substation.

- DER MEASUREMENT INJECTION

This section includes a summary of the DER Gateway and Scada DER measurements that are used by BLA and DPF, grouped by substation.

- SUMMARY RESULTS

This section includes a summary of results for a selection of devices, including capacitors, transformers, feeder heads, station exception devices (as shown on the Station/Feeder Exception displays), interface groups, and suspect devices (loads, capacitors, tap changing transformers, generators, and energy resources).

Bus Load Allocation Data

This section provides detailed results from the BLA iterations.

- MEASUREMENT ISLAND STATISTICS

This section provides a statistical summary of the measurement islands, including the number of station islands, station power transformers, feeder islands, and station feeders in the network selection. DPF and BLA solution convergence statuses are also included.

- ANALOG MEASUREMENT GROUP SUMMARY

This section provides the station and consistency status of each measurement group. For an explanation of consistency checks, see [Analog Measurement Group Consistency Checks](#).

- ANALOG MEASUREMENT GROUPS

This section provides a summary of measurements that form each measurement group, including the device type, measurement ID, value, and quality.

- MEASUREMENT ISLANDS

This section provides the convergence information, load summary, and measurement group information for each measurement island.

- BOUNDING COMPONENTS AND EQUIVALENT MEASUREMENTS

This section lists the bounding components and flow measurements for each measurement island.

- VALIDATION RESULTS

This section includes the effective accuracies, net three-phase flows, and nominal and net apparent power validation results for each measurement island. If validation was performed on an island, then measured and calculated values for high and low real, reactive, and apparent power are provided.

- MISMATCH BY ITERATION

This section provides the power flow mismatch (kW, kVAR, and/or kVA) for each measurement island in each BLA/DPF iteration.

- ISLAND CONVERGENCE RESULTS

This section provides convergence results per phase for real and reactive power or apparent power (dependent on allocation method) for each measurement island.

- DOWNSTREAM ISLAND POST BLA RESULTS

This section provides a summary of calculated and measured values for current, power, voltage, and power factor for merged measurement islands.

- EQUIVALENT LOAD INFORMATION

This section provides a summary of equivalent loads formed for feeders that are not part of the network selection but are part of a station that is part of the network selection. The equivalent real and reactive power load (and capacitor reactive power injection) is provided for each equivalent load.

Improved Bus Load Allocation Results

This section provides detailed results from the BLA iterations in a more organized way than in the sections above. In particular, the subsections of this section provide similar information to the sections above titled ANALOG MEASUREMENT GROUPS, MEASUREMENT ISLANDS, MISMATCH BY ITERATION, and ISLAND CONVERGENCE RESULTS in a more readable and informative fashion.

- MEASUREMENT ISLANDS

This section includes the convergence data (real, reactive power, and apparent power per phase), load summary, upstream and downstream island information, analog measurement groups, island flows, and island BLA parameters for each measurement island.

- MEASUREMENT ISLAND CONVERGENCE DETAILS

This section includes the convergence data (real and reactive power or apparent power per phase) for each measurement island.

- MEASUREMENT ISLAND TARGETS

This section provides the per phase real and reactive power or apparent power targets and measured values for each measurement island.

- ISLAND MISMATCH BY BLA ITERATION

This section provides the per phase real and reactive power mismatch and calculated values, and the reactive power injections by capacitors, generators, and DERs for each measurement island by iteration.

POWER FLOW SOLUTION RESULTS

This section is included in the result file if the "Create Full DPF Result File" option is enabled in the DNAF Configuration Editor.

This section outputs the full power flow results:

- Line Flows
- Series Flows

- Switch Flows
- Transformer Flows
- Distribution Transformer Flows
- Load Information (Flow and Voltage)
- Nodal Voltages

DETAILED BLA MEASUREMENT ISLAND OUTPUT

This section is included in the result file if the "Create Detailed BLA Result Report" option is enabled in the DNAF Configuration Editor.

If the "Only BLA Not Converged Islands" option is enabled, only information on non-converged measurement islands is included.

This section provides detailed BLA measurement island output:

- MEASUREMENT ISLAND BLA SCALING FACTOR

This section provides the initial and final total load in kVA and the final BLA scaling factor for each measurement island.

- DETAILED LOAD INFORMATION

This section provides the load category, multipliers, scheduled and scaled real and reactive power loads, and power factor for each load.

- DETAILED BLA MEASUREMENT ISLAND OUTPUT

This section provides the detailed measurement island output, including the imbalance ratio; total, three-phase, single-phase, unbalanced, and open Y delta real power values; and mismatch allocation kW/kVAR values for each measurement island by iteration.

- DOWNSTREAM DER/DG INJECTIONS AT SWITCHES

This section provides DER/DG real and reactive power values per phase for switching devices.

13.2. PRV Result File

The PRV results for a station are recorded in files named *PRV Results_<StationName>.dat*. The data in this file is grouped in sections as follows:

- PRV EXECUTION SUMMARY

This section shows the substation ID, substation name, substation area (service center), and substation permission area.

- PRV SOLUTION SUMMARY

This section contains triggering information (execution reason, evaluation time, and execution time), execution status, number of interruption violations, and number of phase or ground

pick-up violations.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are `DNAF_PARAMS.xml`, `alg.dat`, `msgdef.dat`, `optalg.dat`, `phid.dat`, and `tftype.dat`. The time last modified and revision number for `DNAF_PARAMS.xml` is also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF INPUT PARAMETERS – PROTECTION VALIDATION

The section details the values for all the parameters pertaining to PRV from the DNAF Configuration Editor.

- ALARM CONFIGURATION

The section lists the status (enabled or disabled) of the following alarms: Failed to Complete Event, Protection Issue Detected Event, and the RTN (Return to Normal) component for each alarm.

- MANUAL ENTRY SUMMARIES

This section lists manually entered values for the substation.

- PROTECTION VALIDATION RESULTS

This section includes the following information:

- Equivalent system impedance values (the equivalent impedance upstream of the distribution station in R and X for the positive, negative, and zero sequence components).
- A list of the protection zones within the network selection (includes detailed information on the protection device for each zone including its ID, name, minimum and maximum fault nodes, application protection zone bounded self-protected power transformer, minimum and maximum trip and interrupt ratings, and maximum neutral trip rating). Zones are ordered from upstream to downstream, and zone order in branches is based on the location of the protective device.
- Summary of the protection zone fault types, including fault phases, node, voltage level, and phase currents for each PRV protection zone.
- Reference bus current and voltage summary for each PRV protection zone including reference bus ID, name, voltage level, and PRV currents, pre-fault DPF currents, and PRV voltages per phase.
- Validation status in each PRV protection zone including maximum and minimum fault currents per phase, maximum interruption violation, minimum phase pick-up violation, minimum ground pick-up violation, phase fault sensitivity, and ground fault sensitivity.

13.3. SCC Result File

The SCC results for a station are recorded in files named *SCC Results_<StationName>.dat*. The data in

this file is grouped in sections as follows:

- SCC EXECUTION SUMMARY

This section shows the substation ID, substation name, substation area (service center), and substation permission area.

- SCC SOLUTION SUMMARY

This section contains the substation name, evaluation time, execution time, status, and triggering information for the SCC solution.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The time last modified and revision number for DNAF_PARAMS.xml is also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF INPUT PARAMETERS – SHORT CIRCUIT CALCULATION

The section details the values for all the parameters pertaining to SCC from the DNAF Configuration Editor.

- MANUAL ENTRY SUMMARIES

This section lists manually entered values for the substation.

- GENERATOR IMPEDANCE

This section includes ID, name, and reactance for generators that contribute to the fault current in the SCC solution area.

- ENERGY RESOURCE FAULT CURRENT LEVEL

This section includes ID, name, voltage range, and fault current level for energy resources that contribute to the fault current in the SCC solution area.

- SHORT CIRCUIT CALCULATION RESULTS

This section lists the faulted device ID, type, node, and node voltage, as well as the fault type, fault impedance type, and current and voltage by phase for each type of fault that could occur at that location given the phasing of the device. This includes all line-to-line and line-to-ground combinations, of which there are eleven total possibilities.

- DEBUG INFORMATION (Appears for diverged solutions)

This section provides details for BLA iterations.

For diverged solutions, three additional debug files are created:

- upf_export.txt
- algin_export_<StationName>_SCC_<TimeStamp>.dat
-

upfin_export_<StationName>_SCC_<TimeStamp>.dat

13.4. FL Result File

The FL results for a station are recorded in files named *FLResults_<StationName>_<FeederName>_<DateTimeStamp>.dat*. The result files' structure depends on the FL solution methodology (legacy or enhanced). The data in these files is grouped in sections as follows.

Legacy Methodology (Radial-Only Network)

- FL SOLUTION SUMMARY

This section contains triggering information (execution reason, device name and ID, and execution time), fault type, fault current, and execution time.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are *DNAF_PARAMS.xml*, *alg.dat*, *msgdef.dat*, *optalg.dat*, *phid.dat*, and *tftype.dat*. The time last modified and revision number for *DNAF_PARAMS.xml* are also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF APPLICATION PARAMETERS

This section details the values for all the parameters pertaining to FL from the DNAF Configuration Editor.

- BREAKER STATUS

Shows the associated breaker status (open or closed).

- EQUIVALENT SYSTEM IMPEDANCE VALUES

This section includes the system equivalent impedance values in R and X for the positive, negative, and zero sequence components.

- UPSTREAM TRANSFORMER

This section provides details for the upstream transformer, including the transformer characteristics, effective impedances, and voltages in primary and secondary windings.

- ESTIMATED FAULT LOCATIONS

This section provides details for estimated fault locations, including the node ID, branch ID, nominal feeder name, main/lateral flag, per phase fault currents, applied calculation method, location range, and coordinate data.

- TAP CHANGER INFORMATION - FROM LAST DPF SOLUTION

This section provides details for tap changers, including the associated transformer ID and name, tap end, and current, minimum, and maximum tap positions.

- **FAULT TYPE**

This section provides fault details, including the node ID, branch name and ID, nominal feeder name, main/lateral flag, fault type, per-phase fault current, per-phase voltages, and coordinate data.

- **IMPEDANCE DATA**

This section provides impedance information, including the line length and per phase line and total resistance and reactance.

Enhanced Methodology (Network with DERs)

- **FL SOLUTION SUMMARY**

This section contains execution start and end time, fault type, the number of short circuit executions, the number of search area nodes and branches, and the number of found fault locations.

- **FAULT LOCATION PARAMETERS**

This section details the values for the fault location parameters pertaining to FL from the DNAF Configuration Editor.

- **POWER FLOW PARAMETERS**

This section details the values for the power flow convergence parameters pertaining to FL from the DNAF Configuration Editor.

- **SHORT CIRCUIT PARAMETERS**

This section details the values for the short circuit calculation parameters pertaining to FL from the DNAF Configuration Editor.

- **FAULT CURRENT MEASUREMENTS**

This section provides details for faulted devices, including device ID, type, and fault current by phase.

- **FAULT INDICATORS**

This section provides details for fault indicators, including device ID, name, state, type, and fault current by phase.

- **ANALYSIS SECTIONS**

This section provides analysis details, including analysis type, summary, the number of branches, names of the top and bottom branches, and infeeds for the top and bottom branches.

- **FAULT LOCATIONS AND MAIN INFEED**

This section provides details for potential fault locations, including exact location flag, branch ID, name, band, and to-node and from-node fault currents per phase.

- DETAILS FOR POTENTIAL FAULT ON <BRANCH NAME>

This section provides details for a potential fault location, including branch ID, name, type, fault location between from and to nodes, total fault currents per phase, and fault location voltages per phase.

This section also provides details for the fault location's infeeds, including infeed ID, name, type, current per phase, and faulted voltage per phase.